MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

Capstone Project Document

PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching

GFA26SE111	
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Capstone Project code	PKV

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Finally, we thank our families and friends for their patience and support during the many evenings of development and testing.

Definition and Acronyms

The definitions and acronyms used in this document are listed below. Pickleball terms and product-specific terms are also explained in Part III, §5.4.

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BR	Business Rule
CI/CD	Continuous Integration / Continuous Delivery
CMS	Content Management System (lesson, item and mission management in the admin web)
DAU	Daily Active Users
Dedicated server	Headless Unity server that runs an online match with authority over the result
Drill	Practice exercise with a fixed number of shots and a target
Elo	Rating system used for ranked matches (K-factor 32, floor 100)
ERD	Entity Relationship Diagram
EV	Evidence item of a test run (Report 5, §5.1)
FE	Major feature (FE-01 to FE-10)
FR	Functional Requirement (FR-01 to FR-33)
Fusion	Photon Fusion 2, the real-time networking library and cloud
GUI	Graphical User Interface
HMAC	Hash-based Message Authentication Code
HUD	Head-Up Display (in-match information)
JWT	JSON Web Token
Kitchen / NVZ	Non-volley zone, the area within 2.13 m of the net where volleys are not allowed
LI	Limitation or exclusion (Report 1, §6.2)
NFR	Non-Functional Requirement (NFR-01 to NFR-12)
OTP	One-Time Password
PB	Product backlog item
RACI	Responsibility matrix: Do, Review, Support, Inform

RTT	Round-Trip Time
S3	Amazon Simple Storage Service API, used for replay storage
SDD	Software Design Description
SPMP	Software Project Management Plan
SRS	Software Requirement Specification
TC	Test Case
Tokenpad	In-game currency earned by playing
UAT	User Acceptance Test
UC	Use Case (UC-01 to UC-68)
UI	User Interface
W01–W04	End-to-end review workflows committed in the Review 2 form BM04
WBS	Work Breakdown Structure
XP	Experience points

About this document. This final report consolidates the six capstone reports of the project in their latest versions: Part I – Project Introduction (Report 1), Part II – Project Management Plan (Report 2), Part III – Software Requirement Specification (Report 3), Part IV – Software Design Description (Report 4), Part V – Software Testing Documentation (Report 5) and Part VI – Release Package & User Guides (Report 6). A reference of the form "Report n" points to Part n of this document. Figures and tables are numbered separately within each part. The test workbooks, the project tracking workbook and the schedule are delivered as separate files (Part VI, §1).

I. Project Introduction

1. Overview

1.1 Project Information

Table 1. Project information

Item	Value
Project name (English)	PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching
Project name (Vietnamese)	PickleVerse – Game Huấn Luyện Kỹ Thuật và Thi Đấu Pickleball Nhiều Người Chơi Tích Hợp AI
Project code	PKV
Group name	GFA26SE111
Software type	3D multiplayer sports game for Windows desktop and Android, supported by a web API, a real-time dedicated game server, an AI coaching service and an administration web application
Duration	14 weeks of development, from 07/09/2026 (Week 1) to 13/12/2026 (Week 14); final defense in Week 15 (14–20/12/2026)
Source repository	https://github.com/Tad-Min/PickleVerse (private; access is granted to the supervisor and the review committee on request)
Staging environment	Admin web: https://tad-min.io.vn – Web API: https://tad-min.io.vn/api (HTTPS since 13/11/2026; HTTP on port 8080 before) – Dedicated game server: tad-min.io.vn, UDP 3669

1.2 Project Team

The team consists of four software engineering students supervised by one lecturer. Every member owns one end-to-end review workflow, as committed in the Review 2 form BM04 (v1.0 of 07/11/2026, reviewers DucDNM2 and TruongLV11; the form records that it has not yet been approved at Review 2), and one or more technical areas; the leader additionally coordinates planning, meetings and submissions. Workflow ownership is separate from implementation effort: according to the repository history, Nguyễn Gia Huy and Nguyễn Văn Thành Đạt implemented the largest share of the code (Report 2, §4).

Table 2. Project team

Full Name	Student ID	Role	Main responsibility	Email	Mobile
Lại Đức Hùng	–	Supervisor	Supervises scope, schedule and report quality	Hungld5@fe.edu.vn	–
Huỳnh Trung Kiên	SE184454	Leader	Planning and tracking; admin web application; music and sound effects; owner of workflow W04 (review a report and publish a lesson)	kienht7704@gmail.com	–
Nguyễn Văn Thành Đạt	SE184450	Member	Web API and database; online multiplayer and dedicated server; AI coaching service; CI/CD; owner of workflow W03 (play online and save the result); support for W02	nguyenvanthanhdat44@gmail.com	–
Nguyễn Gia Huy	SE184439	Member	Unity gameplay, ball physics and rules; training and tutorial; bot behaviour; 3D characters, animation, courts and in-game UI; owner of workflow W02 (play a bot and get a tip); support for W01 and W03	huyngse184439@fpt.edu.vn	–

Full Name	Student ID	Role	Main responsibility	Email	Mobile
Trần Nguyễn Hiếu Thuận	SE184471	Member	Unity user interface (login, loading, settings); audience and character assets; test lead; owner of workflow W01 (learn and sync a drill); support for W04	hieuthuan833@gmail.com	–

Mobile numbers are kept in the team's internal contact list and shared with the supervisor directly; they are not printed in this report because the report is distributed to reviewers outside the team.

2. Product Background

Pickleball has become one of the fastest-growing recreational sports in Vietnam. Local press reported that the number of regular players nearly doubled during 2024 to about 30,000 [1], that more than 200 new courts opened in Hanoi alone in the first seven months of 2024 [2], and that Vietnam was listed among the ten fastest-growing pickleball nations by registered-player growth on the DUPR rating platform [3]. The sport attracts office workers, university students and older adults because the court is small and the basic strokes are easy to start with.

The growth, however, is not matched by structured learning. Most newcomers learn by imitating other players on the court and receive no feedback on their technique. Qualified coaches are scarce and private lessons are expensive, so beginners often reinforce incorrect fundamentals – an overhand or late serve, dinks that float too high, volleys taken inside the non-volley zone (the "kitchen") – which are difficult to correct later. Casual players also find it hard to meet partners of a similar level, and they cannot practise when a court is unavailable because of weather, cost or schedule. Recent reports that some courts in central Hanoi are now under-used [4] suggest that players who do not progress tend to drop the sport.

The project idea was proposed by the team members, who are themselves beginner and casual players, and was registered as a student-initiated capstone topic under the supervision of Mr. Lại Đức Hùng. The proposal identifies Vietnamese beginners and casual players as the primary users and pickleball clubs, academies and community sport centres as potential organisational customers. PickleVerse addresses the problems above with a 3D game in which a player learns the rules and controls, repeats short graded drills, receives technique feedback computed from in-game telemetry, and applies the skills in matches against bots or other players.

3. Existing Systems

The team studied three products that either simulate pickleball or help players improve their technique. They were selected because together they cover the three ideas that PickleVerse combines: an accessible pickleball game, a realistic training simulator, and AI-based technique analysis.

3.1 Pickleball Smash

Link: https://www.xbox.com/games/store/pickleball-smash/9N4DL8WL5DV2 [5]

Overview: An arcade pickleball game developed by 1PXL Games and published by GameMill Entertainment on 24 October 2023 for Nintendo Switch, PlayStation 4/5, Xbox One/Series X|S and PC (Steam) [5], [6].

Actors: Player (single player against the computer, or up to four players sharing one console).

Main features: Singles and doubles matches against computer opponents; local multiplayer; mini-games that each practise one shot; several playable characters and courts.

Pros: Simple controls and short sessions; mini-games show that a single activity can teach a single skill; available on many platforms.

Cons: Arcade physics rather than real pickleball technique; no lesson structure, scoring of technique or feedback; local multiplayer requires all players on the same console; no Vietnamese interface; no mobile version.

3.2 Eleven Pickleball

Link: https://www.meta.com/experiences/eleven-pickleball [7]

Overview: A virtual-reality pickleball simulator by For Fun Labs, the studio behind Eleven Table Tennis, released for Meta Quest headsets [7].

Actors: Player (single player against AI, or online against other players).

Main features: Physics-based paddle control with motion controllers; a programmable ball machine whose speed, spin and placement can be set for each shot; matches against AI bots; online 1v1 and 2v2 matches.

Pros: Realistic ball physics and paddle motion; the ball machine supports deliberate repetition of one shot; real online play.

Cons: Requires a VR headset, which few Vietnamese beginners own; no guided curriculum or star scoring; no technique diagnosis or recommendation; English interface.

3.3 PB Vision

Link: https://apps.apple.com/us/app/pb-vision/id6467020610 [8]

Overview: A mobile and web service that analyses videos of real pickleball games with computer vision and machine learning [8], [9].

Actors: Player (uploads a recorded game and reviews the analysis); club (records courts with an installed camera).

Main features: Shot speed, placement and error statistics; rally and shot breakdowns; heat maps and highlights; skill ratings.

Pros: Feedback is based on the player's real games; detailed statistics; integrates with club court cameras.

Cons: A full game must be filmed from a fixed angle with all four court corners visible; feedback arrives after the game, not during practice; no practice content when no court is available; advanced analysis is a paid service.

3.4 Comparison

Table 3 compares the three products with PickleVerse against the criteria that matter to the target users.

Table 3. Comparison of existing systems with PickleVerse

Criterion	Pickleball Smash	Eleven Pickleball	PB Vision	PickleVerse
Device needed	Console or PC	VR headset	Phone and camera, real court	Windows PC or Android phone
Structured lessons with graded drills	No (mini-games only)	No (free ball machine)	No	Yes – tutorial and 15 lessons scored 0–3 stars
Technique feedback	No	No	Yes, after a filmed game	Yes, after each drill and match, from in-game telemetry
Practice without a court or partner	Yes	Yes	No	Yes, including offline training
Online play and rating	No online play	Yes	Rating only	Yes – ranked singles with Elo, custom singles and doubles rooms
Vietnamese interface	No	No	No	Yes

Criterion	Pickleball Smash	Eleven Pickleball	PB Vision	PickleVerse
Administration tools for an organisation	No	No	Club integration	Yes – user, content, match, moderation and analytics tools

4. Business Opportunity

The comparison shows a gap between entertainment games, which are accessible but do not teach technique, and analysis tools, which give useful feedback but require a real court, a recorded game and in many cases payment. A Vietnamese beginner who wants to learn the fundamentals at home, at low cost and in Vietnamese currently has no suitable product.

PickleVerse targets this gap. It runs on hardware the users already own (a Windows PC or an Android phone), teaches the official rules through a tutorial and graded drills, and turns each drill and match into feedback without any camera. Because the feedback is computed from the game's own telemetry, the cost of a coaching session is close to zero, which makes the product attractive to price-sensitive students and young workers. Online ranked matches with an Elo rating and a leaderboard keep players engaged after the learning phase and help them find opponents of a similar level, which is a common complaint in local player communities.

For organisations, the administration web application lets a club or academy manage accounts, publish its own lessons and technique references, monitor matches and handle player reports. This creates a business-to-business direction described in the proposal: clubs could use PickleVerse as an inexpensive digital training and community tool for new members. Cosmetic items are already bought with an in-game currency; real-money payment through local wallets such as MoMo or VNPay is a post-capstone option and is excluded from this project (LI-02).

5. Software Product Vision

For Vietnamese pickleball beginners and casual players **who** want to learn the rules and fundamental strokes without a coach, a partner or a court, **PickleVerse** is a 3D pickleball game for Windows and Android **that** teaches the rules through an interactive tutorial, trains each core technique with graded drills, gives technique feedback computed from gameplay telemetry, and lets players apply their skills in matches against adaptive bots and other players online. **Unlike** arcade pickleball games, which do not teach technique, and video-analysis services, which need a filmed game on a real court, **our product** combines practice, feedback and competition in one Vietnamese-language application that runs on devices the players already own, and gives clubs a web dashboard to manage their members and training content.

6. Project Scope & Limitations

6.1 Major Features

The ten major features below define the scope of version 1.0. Report 3 refines each feature into use cases and functional requirements (FR-01 to FR-33), and the Review 2 commitment features F01–F18 are traced to them in Report 3, Appendix A.

Table 4. Major features of PickleVerse 1.0

ID	Feature	Description
FE-01	Account and onboarding	Register with an email address, verify it with a six-digit one-time password (OTP), sign in and out, recover a forgotten password, edit the display name, choose a male or female character and choose a starting level (beginner or experienced) once. Configure audio, graphics, key bindings and the touch layout.

ID	Feature	Description
FE-02	Tutorial and technique training	A 17-step interactive tutorial on the court, the rules and the controls. Fifteen lessons in four groups (basic, control, advanced and doubles) train twelve techniques: serve, drive, dink, volley, power and cross-court modifiers, third-shot drop, reset, smash, footwork, approach, and doubles coverage and transition. Each drill is scored with 0–3 stars. Training also works offline after the first sign-in, and results are synchronised once when the game is online again.
FE-03	Bot practice	Singles matches against a computer opponent at three fixed levels (Easy, Medium, Hard) or in an adaptive mode whose behaviour-tree bot scales its shot choice, aggression and error rate to the player's Elo tier. The player chooses a professional or outdoor court and day or night lighting.
FE-04	Online matches	Ranked singles quick match that pairs players of similar Elo, with a 10-second ready check and a bot opponent after 180 seconds without a human match; quick doubles, an unrated 2v2 search for a solo player or a party of two that is filled only with human players; private singles and doubles rooms joined by room code or friend invitation. Matches run on an authoritative dedicated game server and support reconnection.
FE-05	AI coaching	After a drill or match, the AI coaching service analyses the shot telemetry and returns a summary, detected errors (for example a late serve or a dink that is too high) and at most three recommended drills.
FE-06	Progress, rewards and shop	Experience points and levels, lesson stars, best accuracy and the four most recent accuracy results of each lesson, technique mastery, a daily play streak (rewarded through the achievement for a 5-day streak and a weekly streak mission), daily and weekly missions, achievements, and the Tokenpad currency. Players buy and equip cosmetic paddles in the shop and view owned items in the collection.
FE-07	Social, history and ranking	Send, accept and decline friend requests; remove and block players; view match history and replays (pause, seek, 0.5×–2× speed); view the Elo leaderboard.
FE-08	Reports, restrictions and appeals	Report another player after a match. A restricted (suspended or banned) player can see the sanction decision and submit one appeal.
FE-09	Administration web application	Administrators manage accounts, roles and sanctions, reset player progress by group, manage lessons, technique references, cosmetics and missions, monitor and stop live matches, review player reports and appeals, and view analytics (active users, retention, popular lessons, technique errors and matchmaking health).
FE-10	Expert review of AI results	Administrators acting as pickleball experts label recorded shots and export a labelled dataset. This tool is used to evaluate the AI coaching quality (Report 3, NFR-06) and is an addition to the proposal.

6.2 Limitations & Exclusions

The following items are deliberately excluded from version 1.0, or deviate from the registered proposal. Each deviation is recorded with its reason so that requirement changes can be evaluated against it.

Table 5. Limitations and exclusions

ID	Limitation or exclusion
LI-01	Coaching uses in-game telemetry only. Analysis of real swing video, augmented reality and a separate coach portal are excluded.
LI-02	Ranked seasons, tournaments, spectator mode, club management, voice or text chat and real-money payments are excluded. The Tokenpad currency is earned only by playing. The Elite Token currency exists in the data model but has no earning source in version 1.0, so items priced only in Elite Token cannot be bought.
LI-03	Only Ranked Singles and Ranked Bot matches change the Elo rating (K-factor 32, floor 100). Doubles and private rooms are unrated. The proposal described an Elo update after every match; rating only one-to-one ranked results keeps the rating meaningful.

ID	Limitation or exclusion
LI-04	Offline mode covers training only, after at least one online sign-in on the device. Bot matches, online matches, the shop and the administration tools need a connection.
LI-05	The proposal's "self-assessment with a starting-tier classifier" is implemented as a one-time choice between Beginner and Experienced. The trained tier model in the AI service is gated: its estimate is shown only when the model passes the accuracy target in NFR-06; otherwise the rule-based result is used.
LI-06	Release platforms are Windows 10 or later and Android 8.0 (API 26) or later on ARM64. An iOS build is outside the delivered scope because the team has no Apple build machine and developer account; the project settings keep iOS 15.0 as the minimum for future work (the proposal named iOS 14).
LI-07	The staging deployment runs one dedicated game-server container, so one server-authoritative match can run at a time; when the slot is busy players are told that the server is busy. The proposal target of 500 concurrent matches requires a server pool and is treated as a scalability goal, verified by load testing of the API (Report 3, NFR-04).
LI-08	Technology deviations from the proposal: the web API uses ASP.NET Core (.NET 10) instead of Spring Boot or FastAPI because the team shares C# with Unity; real-time networking uses Photon Fusion 2 (shared lobby plus a dedicated server) instead of Netcode for GameObjects, after Netcode with Relay showed unstable connections during the October prototype; the game interface uses uGUI instead of UI Toolkit; ball flight uses a custom fixed-step simulation instead of PhysX for deterministic replay and server authority. The reasons are detailed in Report 4, §1.4.
LI-09	The game interface is in Vietnamese only. The administration web application is in Vietnamese, and project documents are in English.
LI-10	The proposal stated that stored data would be encrypted with AES-256. Version 1.0 protects secrets by one-way hashing instead (BCrypt for passwords, SHA-256 for refresh tokens, HMAC for verification and reset codes); the database files, backups and replay objects are not encrypted at rest. Disk encryption of the production host is planned before release (Report 4, §3.10).
LI-11	The proposal targeted 60 FPS on all devices. The SRS keeps 60 FPS for the reference PC but sets an average of 30 FPS for a mid-range Android phone (NFR-03), because mobile rendering removes HDR, MSAA and most shadows to stay within the phone's thermal budget; the game still requests up to 60 FPS on Android. The Android APK was built on 12/11/2026, but on 13/11/2026 it has not yet been measured or smoke-tested on a physical device (test cases TC-N03 and TC-N10 are pending in Report 5).

7. References

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II. Project Management Plan

1. Overview

1.1 Scope & Estimation

The scope of version 1.0 is the ten major features FE-01 to FE-10 defined in Report 1, plus the technical work that the features depend on: the gameplay core, the networking layer, art and audio assets, deployment and testing. Table 1 decomposes this scope into a two-level work breakdown structure (WBS). Each function is classified as Simple, Medium or Complex and estimated in man-days, where one man-day is eight hours of work by one member.

Table 1. Work breakdown structure and effort estimation

#	WBS Item	Complexity	Est. Effort (man-days)	Owner
1	**Project management and documentation**		**28**	
1.1	Planning, scheduling and weekly tracking	Medium	5	Kiên
1.2	Report 1 and Report 2	Simple	3	Huy, Kiên
1.3	Report 3 – Software Requirement Specification	Complex	8	Huy
1.4	Report 4 – Software Design Document	Complex	7	Đạt
1.5	Reports 5–7, user guides and defense slides	Medium	5	Huy, Thuận
2	**Foundation and architecture**		**20**	
2.1	Unity project, scenes and assembly structure	Simple	3	Huy
2.2	Layered Web API, EF Core model and database schema (51 tables)	Complex	9	Đạt
2.3	Docker Compose environment with PostgreSQL, Redis and S3 storage	Medium	4	Đạt
2.4	CI/CD pipeline (GitHub Actions, Docker Hub, SSH deployment)	Medium	4	Đạt
3	**Gameplay core**		**30**	
3.1	Ball flight and bounce simulation at 120 Hz	Complex	8	Huy
3.2	Paddle contact, shot input and seven shot kinds	Complex	7	Huy
3.3	Singles rules engine (scoring, serve, faults, non-volley zone)	Complex	7	Huy
3.4	Character movement and animation controller	Medium	5	Huy
3.5	Match flow, HUD, intro cinematic and result screen	Simple	3	Thuận
4	**FE-01 Account and onboarding**		**16**	
4.1	Registration and OTP email verification	Medium	4	Đạt
4.2	Sign in, sign out, JWT access and refresh tokens	Medium	4	Đạt
4.3	Password recovery	Simple	2	Thuận
4.4	Profile, character and onboarding level	Simple	2	Thuận
4.5	Settings: audio, graphics, key binding, touch layout	Medium	4	Thuận
5	**FE-02 Tutorial and technique training**		**23**	
5.1	Interactive tutorial (17 steps)	Medium	5	Huy
5.2	Lesson and drill framework with star scoring	Complex	7	Huy
5.3	Content of 15 lessons and 12 techniques	Medium	5	Huy
5.4	Offline training and one-time synchronisation	Complex	6	Đạt
6	**FE-03 Bot practice**		**15**	
6.1	Fixed-level bot (Easy, Medium, Hard)	Medium	5	Huy
6.2	Adaptive behaviour-tree bot with Elo tiers	Complex	7	Đạt
6.3	Bot match setup, court and lighting options, rewards	Simple	3	Thuận
7	**FE-04 Online matches**		**36**	
7.1	Photon Fusion session, lobby and room codes	Complex	7	Đạt
7.2	Ranked matchmaking, ready check and bot fallback	Complex	7	Đạt
7.3	Dedicated server build, signed tickets, lease and result	Complex	9	Đạt
7.4	State synchronisation, input prediction and reconnection	Complex	9	Đạt
7.5	Elo rating and match result persistence	Medium	4	Đạt
8	**FE-05 AI coaching**		**18**	
8.1	Telemetry capture, local spool and upload	Medium	5	Đạt
8.2	FastAPI rule engine and gated tier model	Complex	7	Đạt
8.3	Hybrid analysis worker, analysis jobs and coaching panel	Medium	6	Huy
9	**FE-06 Progress, rewards and shop**		**15**	

#	WBS Item	Complexity	Est. Effort (man-days)	Owner
9.1	XP, levels, technique mastery and progress dashboard	Medium	5	Huy
9.2	Daily and weekly missions, achievements, streak rewards	Medium	5	Đạt
9.3	Currencies, shop and collection	Medium	5	Huy
10	**FE-07 Social, history and ranking**		**12**	
10.1	Friends, blocking and room invitations	Medium	4	Đạt
10.2	Match history, replay recording and replay viewer	Complex	6	Đạt
10.3	Elo leaderboard with cache	Simple	2	Kiên
11	**FE-08 Reports, restrictions and appeals**		**6**	
11.1	Player report after a match	Simple	2	Kiên
11.2	Restricted access, decision view and appeal	Medium	4	Kiên
12	**FE-09 Administration web application**		**24**	
12.1	Admin authentication, layout and protected routing	Simple	2	Kiên
12.2	User management, sanctions and progress reset	Complex	6	Kiên
12.3	Content management: lessons, references, cosmetics, missions	Medium	6	Kiên
12.4	Live match monitoring and stopping	Medium	4	Đạt
12.5	Moderation of player reports and appeals	Medium	3	Kiên
12.6	Analytics dashboard	Medium	3	Kiên
13	**FE-10 Expert review**		**4**	
13.1	Shot labelling and dataset export	Medium	4	Đạt
14	**Art, audio and interface assets**		**15**	
14.1	Characters, 20 animation clips, courts and stadium	Complex	8	Huy
14.2	Music and sound effects	Simple	2	Kiên
14.3	Front-end screens, loading artwork and Vietnamese fonts	Medium	5	Thuận
15	**Testing and release**		**18**	
15.1	Unit tests (Unity EditMode, Python) and coverage	Medium	5	Thuận
15.2	Integration and system testing on staging	Medium	5	Thuận
15.3	Acceptance tests, playtests and defect fixing	Medium	6	All
15.4	Staging deployment and release package	Simple	2	Đạt
	Total Estimated Effort (man-days)		**280**	

The estimate covers 56 functions (11 Simple, 28 Medium and 17 Complex) and totals **280 man-days**. The available capacity is four members × 14 weeks × 5 working days = 280 man-days, so the plan uses the whole capacity and has no reserve. The team accepts this because Complex networking items were estimated from the actual effort of the October prototypes, and AI coding assistants are used for boilerplate code and first drafts of documents; every generated output is reviewed by the member who owns the item. If an iteration overruns, the first item to be deferred is FE-10 (expert review), which is not part of the core workflows.

The total fits the team capacity, but the owners in Table 1 are the members who actually built each item (repository history), and Table 2 shows that the load per member does not fit. Each member has 14 × 5 = 70 man-days.

Table 2. Estimated load per member

Member	Estimated effort of owned items (man-days)	Capacity (man-days)	Load
Huỳnh Trung Kiên	38.0	70	54%
Nguyễn Văn Thành Đạt	119.5	70	171%
Nguyễn Gia Huy	89.5	70	128%
Trần Nguyễn Hiếu Thuận	33.0	70	47%
Total	**280**	**280**	**100%**

Nguyễn Văn Thành Đạt and Nguyễn Gia Huy carry about 171% and 128% of their capacity, while Huỳnh Trung Kiên and Trần Nguyễn Hiếu Thuận carry about 54% and 47%. Until Iteration 4 the two

overloaded members worked beyond the planned hours, which is one reason why documents were late. **Corrective action for Iterations 5 and 6:** new work is assigned first to Kiên and Thuận – system and acceptance testing, test data and seed accounts, the remaining UI defects, the admin web and the documents – while Đạt and Huy review it and fix the defects in the modules only they know; pair-programming sessions on the API and Photon Fusion (risk 1) let the other members take over online defects. The load is re-checked at each iteration planning in the WBS sheet of the tracking workbook.

1.2 Project Objectives

The overall objective is to deliver, by Week 14 (13/12/2026), a deployed and tested version of PickleVerse in which a beginner can learn the rules, train the core techniques with graded drills and feedback, and play bot and online matches, while administrators manage accounts, content, matches and moderation from the web application. The quality, time and cost objectives are measurable as follows.

Quality. Table 3 states the planned test coverage and the expected number of defects at each testing stage. The defect estimate (about 120 defects in total) is based on the defects already found in the October prototypes and the 10/11/2026 playtest; it is used to judge whether a stage is finding fewer defects than expected, which usually means that testing is too shallow.

Table 3. Quality objectives by testing stage

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Reviewing	100% of reports, diagrams and database changes are reviewed by at least one other member	20	17%	Document and design review before each submission
2	Unit Test	100% of public methods in the rule, scoring, Elo, input and economy classes; ≥70% line coverage of core gameplay assemblies	30	25%	Unity EditMode and Python pytest; run before every merge
3	Integration Test	All 129 API endpoints exercised at least once; all database write paths	25	21%	IntegrationChecks harness and API contract checks in the CI validate job (twelve check groups; the job passes since 11/11/2026, Report 5 EV-08)
4	System Test	100% of the 118 test cases in Report 5 executed on staging	35	29%	Includes network, device and AI quality tests
5	Acceptance Test	8 demo scenarios covering workflows W01–W04	10	8%	Executed with the supervisor before the final review
	Total		**120**	**100%**	

In addition, the release criteria for version 1.0 are: at least 90% of system test cases passed, no open Critical or High defect, at least 50% of test cases executed automatically, and every functional requirement covered by at least one passed test case.

Milestone timeliness. Up to 11/11/2026, six milestones (M01–M06) were due and one of them (M06) was delivered on time, a timeliness of 17%. Reports 1–4 were first submitted on 06/11/2026 because the team prioritised the playable prototype in Weeks 5–8 (§2.1). The target for the remaining milestones M07–M10 is 100% on time, and the corrective actions are: one document owner per report, a weekly tracking meeting with the leader, and submission one day before each deadline.

Cost (allocated effort). The 280 man-days are distributed over the project activities as shown in Table 4. Feature and technical items include their own design (15%), coding (70%) and unit testing (15%); documentation items are counted under requirements, design or project management.

Table 4. Allocated effort by activity

Activity	Source in the WBS	Effort (man-days)	Share
Requirements	1.2, 1.3	11	3.9%
Design	1.4 + 15% of items 2–14	42	15.0%
Coding	70% of items 2–14	164	58.6%
Testing	15% of items 2–14 + item 15	53	18.9%
Project management and documentation	1.1, 1.5	10	3.6%
Total		**280**	**100%**

1.3 Project Risks

Risks are reviewed in the weekly tracking meeting. A risk that occurs is closed in this table and recorded as an issue in the Issues sheet of the project tracking workbook.

Table 5. Project risks

#	Risk Description	Impact	Possibility	Trigger	Response Plans	Owner
1	Key-person dependency: most of the networking, backend and AI code has been written by one member, so the online features stall if that member is unavailable.	High	Medium	Owner unavailable for more than 3 days, or one member has more than 60% of the commits in an iteration	Each member owns and demonstrates one workflow; pair-programming sessions on Fusion and the API; handover notes in Doc/; code walkthrough at the end of each iteration	Kiên
2	Leaked credentials: the SMTP password and JWT signing key were committed in appsettings.json and could be misused.	High	High	Secret found in a tracked file (occurred; see issue I-03)	Move secrets to environment variables and the server .env file; rotate the leaked values; add a secret scan to CI	Đạt
3	One dedicated game-server slot limits concurrent online matches and the scalability target.	High	High	"Server busy" responses during tests with more than two players	Bot fallback and a clear busy message; load-test the API separately; document a server pool as future work	Đạt
4	The Photon free plan limits concurrent users, which can block group playtests or the demo.	Medium	Medium	Connection refused when more than 20 clients connect	Use the dedicated server path for matches; schedule playtests; upgrade the plan before the defense if needed	Đạt
5	Unity tests are not run in GitHub Actions, so automation evidence depends on manual Unity CLI runs. The workflow defines a GameCI build of the dedicated server with a Unity licence but no test step.	Medium	High	CI shows no Unity test step	Add a GameCI unity-test-runner EditMode step that reuses the licence secrets of the server build (the pipeline passes since 13/11/2026); until then attach Unity CLI result files with the commit hash	Thuận
6	Resetting the development database deletes test accounts and data.	Medium	Medium	Schema change that requires a reset	Keep seed accounts and content in Database.sql; back up with pg_dump before a reset; re-run seed checks	Đạt
7	Documents or code produced with AI assistants contain statements that do not match the implementation.	High	Medium	A reviewer finds a claim without evidence in code or tests	The owner of each section verifies it against code and test results; cross-document consistency check before submission	Huy
8	Too little real player data to validate the AI tier model.	Medium	High	Fewer than 200 labelled sessions by Week 12	Keep the tier model gated; use rule-based findings; collect labels with the expert review tool during playtests	Đạt
9	High latency or packet loss between Vietnamese players and the single staging host degrades matches.	Medium	Medium	Round-trip time above 100 ms or rubber-banding reported in playtests	Measure RTT in each playtest; tune interpolation; reconnection within 30 seconds	Đạt

#	Risk Description	Impact	Possibility	Trigger	Response Plans	Owner
10	Schedule slip and scope creep (milestones M01–M05 were late).	High	High	A milestone is not ready two days before its date	Freeze scope after Review 3; weekly tracking; defer FE-10 first	Kiên
11	The staging host fails and data is lost (single server, no replica).	High	Low	Host unreachable or disk failure	Since be556a1 (13/11/2026) the deployment takes a pg_dump before every release and checks it with pg_restore, but the dumps stay on the same host and there is no daily job: add a daily pg_dump with off-host copies (PB-21); redeploy from the Docker Hub images; keep a local Docker environment as fallback	Đạt
12	The game does not reach the frame-rate target on mid-range Android phones, or the Android build fails on a real device (the APK was built on 12/11/2026 but has not yet been installed on a phone).	Medium	Medium	Average below 30 FPS on the test phone, or a crash in the smoke test TC-N10	Rebuild the APK from main @ d9200cc, where D-21 is fixed; borrow a reference phone by 18/11/2026; profile with the Unity Profiler; quality presets; reduce stadium detail and shadows on mobile	Huy
13	The CI/CD pipeline does not run, so builds, checks and deployments depend on manual steps. Occurred from 05/11 to 13/11/2026: 29 of the 32 runs of dockerhub.yml did not pass (two configuration failures on 05/11; 18 runs did not start while the GitHub account was locked for a billing issue from 06/11 to 11/11; six dedicated-server build failures on 11–12/11; two failed deployment steps and one failed FastAPI test on 13/11). Closed on 13/11/2026: runs 27, 28 and 30 passed every job (issue I-11).	High	Low (was High)	A push to main without a green run (occurred 05/11–13/11/2026)	Done: the leader had the billing lock lifted on 11/11/2026 and Đạt repaired the dedicated-server build and the deployment script on 12–13/11/2026. From now on a red run is fixed before the next merge; the manual release script stays as the fallback	Kiên

2. Management Approach

2.1 Project Process

The team follows an iterative and incremental process with two-week iterations, adapted from Scrum for a student team. The leader acts as Scrum Master and product owner on behalf of the supervisor; each member owns a technical area and one end-to-end workflow. Every iteration starts with planning from the product backlog in the tracking workbook and ends with a demonstration to the team and, at review weeks, to the supervisor. Figure 1 shows the cycle.

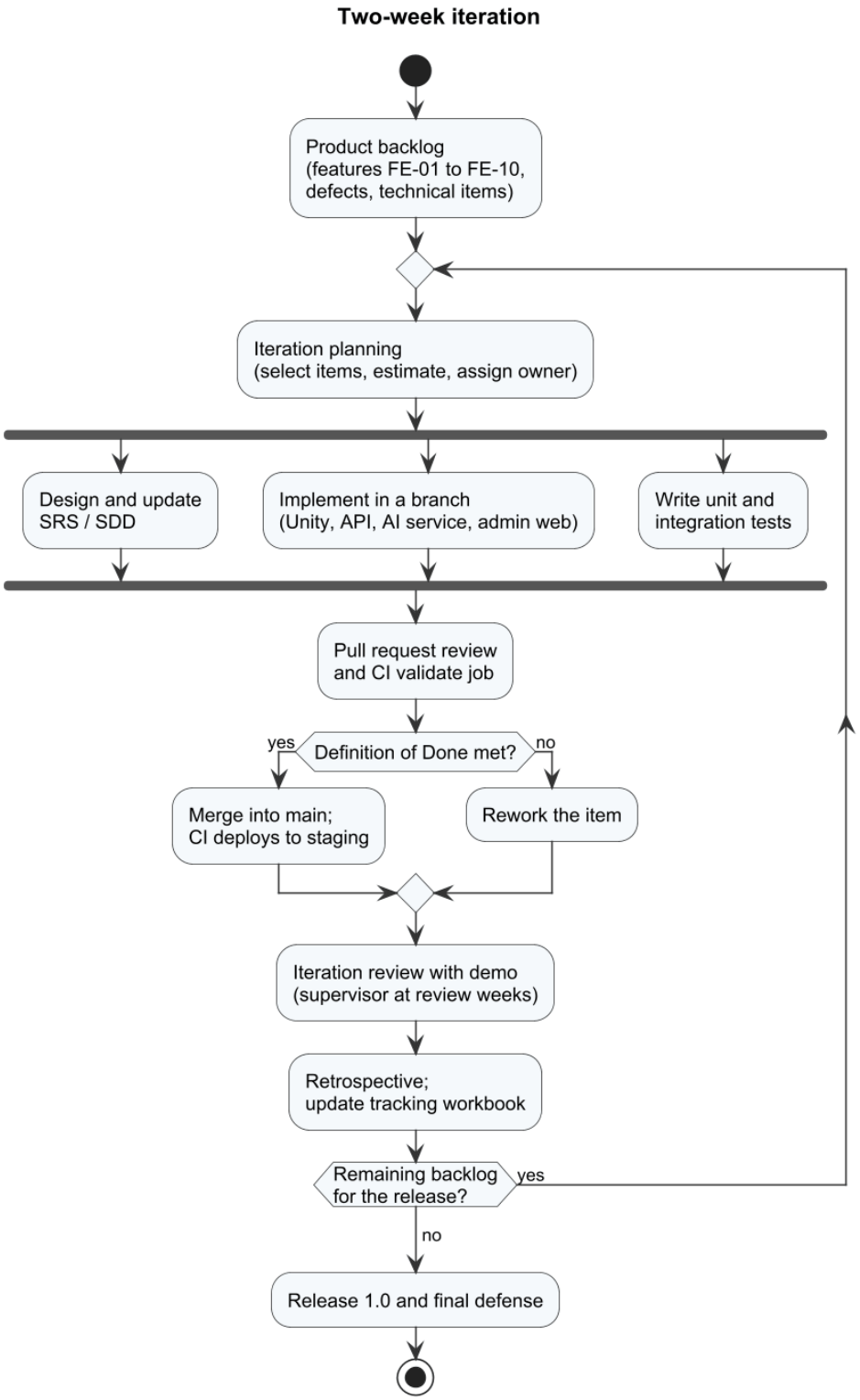


Figure 1. Iterative development process used by the team

Definition of Done. A backlog item is done when (1) the code is merged into the main branch, (2) its unit tests pass and the existing EditMode and Python suites still pass, (3) the related API, database and UI changes are consistent with Database.sql and the API client, (4) it works on the staging server, and (5) the SRS, SDD and test cases are updated.

The actual iterations, reconstructed from the repository history, are listed in Table 6. The architecture changed three times, which explains why the early milestones were late: the online layer moved from a Netcode for GameObjects prototype to Photon Fusion in shared mode and then to an authoritative dedicated server, and player data moved from local JSON files to the Web API and PostgreSQL.

Table 6. Iterations and actual progress

Iteration	Weeks and dates	Main results	Status
It-0 Inception	W1–W2, 07/09–20/09	Proposal and team set-up; repository created (first commit 23/08); Unity project initialised (17/09)	Done
It-1 Prototype	W3–W4, 21/09–04/10	Unity project structure and camera; 3D character models generated and rigged (03/10)	Done
It-2 Single-player core	W5–W6, 05/10–18/10	20 animation clips; BotMatch singles with rules (07/10); mode selection; music and SFX (09/10); shop and collection (10/10); tutorial (11/10); training mode (12/10); history, missions, Elo and ranks (11/10); friends and leaderboard (12/10); replay and streaks (13/10); Netcode for GameObjects prototype (07/10–21/10)	Done
It-3 Online and backend	W7–W8, 19/10–01/11	Loading scene and touch controls; Web API and database (20/10); move to Photon Fusion (20/10); Unity–API integration (21–23/10); admin web (22–24/10); OTP email (24/10); doubles and custom rooms (25/10); Review 2 package (27/10); AI service and Unity Behavior bot (27/10); normalised schema and ApiResponse refactor (31/10); login scene (01/11)	Done
It-4 Dedicated server and coaching	W9–W10, 02/11–15/11	Dedicated server, accounts and coaching (03/11); Docker deployment (04/11); CI/CD workflow written (05–06/11; first passing run on 13/11 – risk 13); admin tools and response training (06/11); reports v0.2 (06/11); manual staging releases (08–10/11); reconnect and offline fixes (10/11); playtest (10/11); 515 EditMode tests passed (11/11); UI and camera fixes D-03, D-04, D-05, D-07, D-19 and 516 EditMode tests passed (11/11); reports v1.0–v1.2 (11–12/11); pipeline repaired with the first automatic deployment, HTTPS for the staging API and admin web, credentials removed from the settings file, Android APK built, reports v1.3 (12–13/11)	In progress
It-5 Stabilisation	W11–W12, 16/11–29/11	Fix the open defects of test Round 1 (Report 5) and retest the fixed ones; rotation of the exposed credentials; daily off-host database backup; Unity tests in CI; pull-request workflow; Android device test; system and load tests; Report 5 results; Review 3	Planned
It-6 Release	W13–W14, 30/11–13/12	Acceptance tests and demo scenarios; release package; final updates of Reports 6 and 7 (v1.0 written on 11/11/2026); defense slides and rehearsal (W15)	Planned

2.2 Quality Management

Defect prevention: Coding rules are written in AGENTS.md and enforced in review: the layered backend architecture (API → Services → Repositories → Entities), access to data only through the unit of work, ApiResponse<T> envelopes, a single schema source (src/Database.sql) and no secrets in source files. Gameplay rules are implemented in engine-independent C# classes so that they can be unit-tested.

Reviewing: Every report section is written by its owner and reviewed by one other member against the code and test results before submission (Table 9). Until 12/11/2026 code was committed directly to the main branch without pull requests; from Iteration 5 every change is merged through a pull request that needs one reviewer and, once the pipeline runs again, a passing CI run (§6.2).

Unit testing: Unity EditMode tests (516 tests in 10 assemblies, all passed on 12/11/2026 at commit 5963501) cover the rules, scoring, Elo, input, economy, replay and multiplayer protocol classes; pytest

covers the AI service. The suite is run with the Unity CLI before each merge and the result file is stored with the commit hash.

Integration testing: The IntegrationChecks console program and the API contract checks verify the API, the database writes and the dedicated-server ticket flow against a PostgreSQL and Redis instance. They run in the CI pipeline for every push that changes the backend, the Unity scripts or the AI service; they pass since 11/11/2026 and their results are linked to test cases in Report 5 (EV-08).

System testing: The 118 test cases in Report 5 are executed on the staging server for each package milestone; defects are logged in the tracking workbook with severity and retested before closing.

Acceptance testing: The eight demo scenarios of the Review 2 commitment are run with the supervisor, and the results are recorded in Report 5.

2.3 Training Plan

Table 7. Training plan

Training Area	Participants	When, Duration	Waiver Criteria
Git and GitHub flow (branches, pull requests, reviews)	All members	W1, 2 hours; refresher W11, 1 hour	Mandatory
Unity 6, C# gameplay scripting, Input System	Thuận, Kiên	W2–W3, 12 hours	Has completed a Unity game project before
Photon Fusion 2 and dedicated server builds	Huy, Thuận	W8–W9, 8 hours (taught by Đạt)	Has run and debugged a networked match on the staging server
ASP.NET Core Web API, EF Core and PostgreSQL	Kiên, Huy	W6–W7, 8 hours (taught by Đạt)	Has implemented an API endpoint with a repository and an EF Core query
React 18, TypeScript and React Query	Kiên, Thuận	W7, 6 hours	Has built a React application with TypeScript
Python FastAPI and scikit-learn	Đạt, Huy	W8, 4 hours	Has deployed a FastAPI service
Docker, Docker Compose and GitHub Actions	All members	W9, 4 hours	Can start the local stack and read container logs
Blender and character animation import	Huy, Thuận	W4, 6 hours	Has rigged and imported a character into Unity
UML modelling with PlantUML and draw.io	All members	W3, 3 hours	Mandatory

3. Project Deliverables

Week 1 starts on Monday 07/09/2026; each milestone is due on the Sunday of its week. Actual dates are taken from the repository history and submission records.

Table 8. Project deliverables and milestones

#	Deliverable	Due Date	Actual Date	Status	Notes
M01	Report 1 – Project Introduction	13/09/2026	06/11/2026	Done, late	Proposal registered before Week 1; v1.0 on 11/11/2026
M02	Report 2 – Project Management Plan, schedule, tracking workbook	20/09/2026	06/11/2026	Done, late	v1.0 on 11/11/2026
M03	Report 3 – Software Requirement Specification	27/09/2026	06/11/2026	Done, late	v1.0 on 11/11/2026
M04	Report 4 – Software Design Document; Report 5 test plan	11/10/2026	06/11/2026	Done, late	v1.0 on 11/11/2026
M05	Software Package 1 – single-player core, training, first online build (Review 2)	25/10/2026	27/10/2026	Done, 2 days late	Review 2 forms BM01–BM04
M06	Software Package 2 – Web API, admin web, AI service, dedicated server, CI/CD	08/11/2026	06/11/2026	Done, on time	Staging on tad-min.io.vn; the CI/CD workflow was written on time but first passed on 13/11/2026, so the releases of 08–10/11 were manual (risk 13)

#	Deliverable	Due Date	Actual Date	Status	Notes
M07	Software Package 3 – defect fixes, system and load tests (Review 3)	22/11/2026	–	In progress	Report 5 results due with this package
M08	Full software package with test results and open defects	06/12/2026	–	Planned	
M09	Report 6 – Software User Guides	13/12/2026	11/11/2026	Done (v1.0)	Installation, player and administrator guides; updated with the release build
M10	Report 7 – Final Project Report, release package and defense	20/12/2026	–	In progress	Report 7 v1.0 on 11/11/2026; slides and defense in Week 15

Figure 2 places the iterations and milestones on the calendar; the red line marks the status date of the schedule (11/11/2026).

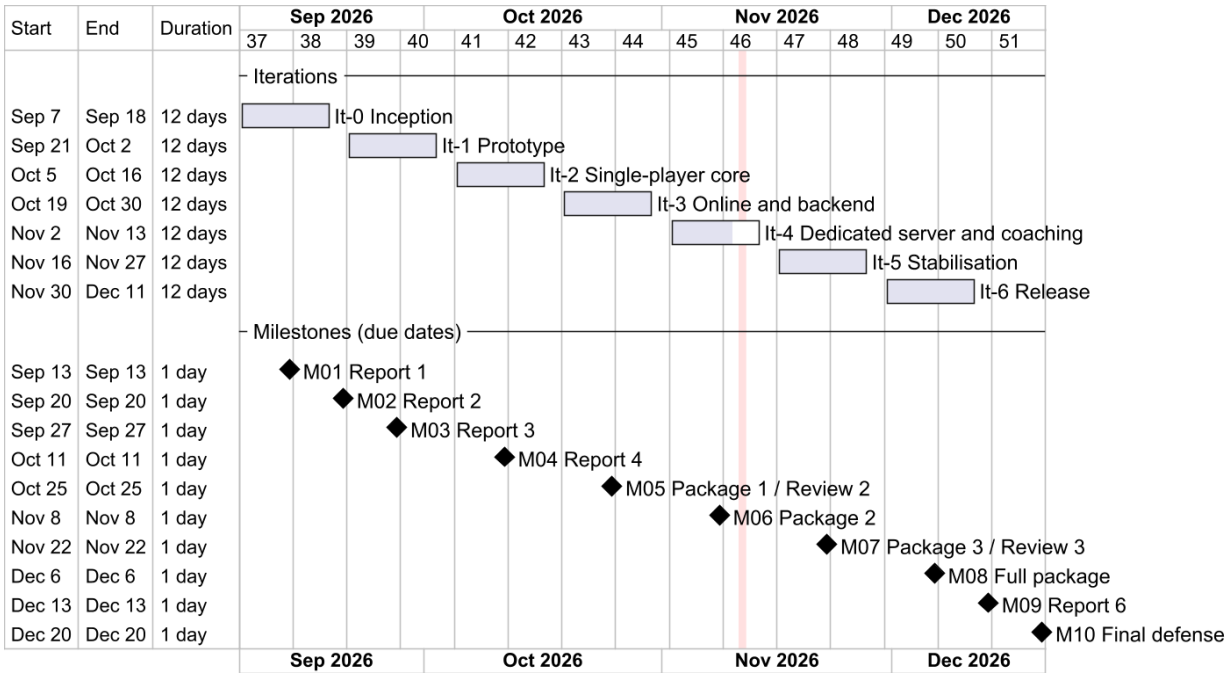


Figure 2. Master schedule: iterations and milestones

4. Responsibility Assignments

D~Do; R~Review; S~Support; I~Informed; <blank>~Omitted. Every row has exactly one D. Each member owns one end-to-end workflow, which that member demonstrates at the reviews; the workflow owners and supporting members follow the Review 2 commitment form BM04 (v1.0 of 07/11/2026, not yet approved by the reviewers). Workflow ownership is an accountability for acceptance, not a measure of implementation effort: the technical rows show who built each part, and the repository history shows that Nguyễn Gia Huy and Nguyễn Văn Thành Đạt carried the largest share of the code (Table 2). Huỳnh Trung Kiên is the D for the admin web because he wrote most of it (about 6,800 added lines under src/Frontend).

Table 9. Responsibility assignment matrix (RACI)

Responsibility	KienHT SE184454	DatNVT SE184450	HuyNG SE184439	ThuanTNH SE184471
Project planning and tracking	D	R	S	S
Report 1 – Project Introduction	R	S	D	I
Report 2 – Project Management Plan	D	S	R	I
Report 3 – SRS: overview, use cases, screens	R	S	D	S

Responsibility	KienHT SE184454	DatNVT SE184450	HuyNG SE184439	ThuanTNH SE184471
Report 3 – SRS: functional and non-functional requirements	S	R	D	S
Report 4 – SDD: architecture, database, API, security	I	D	R	S
Report 4 – SDD: gameplay, bot and client design	I	R	D	S
Report 5 – Test plan, test cases and test report	R	S	S	D
Report 6 – User guides	R	S	D	S
Report 7 – Final report	R	R	D	S
Defense slides and presentation	D	S	S	S
Gameplay core, training and tutorial	I	R	D	S
Online matches, dedicated server and netcode	I	D	R	S
Web API, database and AI coaching service	S	D	R	I
Admin web application	D	R	I	S
Unity user interface, settings and loading	S	I	R	D
3D art, animation and courts	I	I	D	S
Music and sound effects	D	I	R	I
CI/CD and staging deployment	I	D	S	R
Workflow W01 – Learn and sync a drill (F01–F04, F13, F14)	R	I	S	D
Workflow W02 – Play a bot and get a tip (F05, F06, F11)	R	S	D	I
Workflow W03 – Play online and save the result (F07–F10, F12)	R	D	S	I
Workflow W04 – Review a report and publish a lesson (F15–F18)	D	R	I	S

5. Project Communications

Table 10. Project communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Supervisor meeting	Team and Mr. Lại Đức Hùng	Demonstrate progress, agree on scope changes, record decisions and answers	Weekly, 60 minutes; extra meeting before each review	Online or on-campus meeting; minutes and decisions in the Q&A sheet
Iteration planning	All members	Select backlog items, estimate and assign owners	First day of each two-week iteration	Meeting; Iteration backlog in the tracking workbook
Progress check	All members	Report done work, next work and blockers	Twice a week (Tuesday and Friday evening)	Group chat and short call
Iteration review and retrospective	All members, supervisor at review weeks	Demonstrate the increment; decide improvements	Last day of each iteration	Meeting; notes in the tracking workbook
Code review	Author and one reviewer	Check correctness, tests and architecture rules	Each pull request (from Iteration 5)	GitHub pull request with CI result
Defect and issue updates	Owner of the defect or issue	Keep status, severity and owner current	When status changes; reviewed weekly	Tracking workbook (Defects and Issues sheets)
Report submission	Leader	Submit reports and packages	One day before each deadline	FPT University LMS

6. Configuration Management

6.1 Document Management

- All project documents are stored in the Git repository under `Doc/`: the proposal in `Doc/Proposal`, review forms in `Doc/Project_report/Review*`, and Reports 1–7 with the tracking workbook and schedule in `Doc/Project_report/Report`.
- File names follow the course template: `Report<n>_<Title>.docx`, `PickleVerse_Project_Tracking.xlsx` and `PickleVerse_Project_Schedule.xml`.
- Each document records its versions in the Record of Changes table: v0.x for drafts and v1.0, v1.1 and so on for reviewed versions submitted to the supervisor.
- Diagrams are written as PlantUML sources in `Doc/Project_report/Report/diagrams/src` and rendered to PNG, so every figure can be regenerated and reviewed as text; the use case and conceptual ERD drawings are also kept as draw.io files.
- A document is submitted only after it has been reviewed by the member marked R in Table 9 and the leader; the PDF exported from the reviewed version is uploaded to the LMS.

6.2 Source Code Management

- Repository: GitHub `Tad-Min/PickleVerse`, one repository for the Unity client (`My project/`), the backend and admin web (`src/`), deployment files (`deploy/`, Docker Compose files) and documents.
- Practice until 13/11/2026: members committed directly to `main` and merged local work with merge commits; the coaching work used the branch `realtime-AI-response`. No pull requests or release tags were used, so reviews were informal. On 13/11/2026 `main` has 272 commits (at d9200cc) and no tags.
- Practice from Iteration 5 (16/11/2026): every change is made on a branch named `feature/<item-id>-<short-name>` or `fix/<defect-id>`, merged into `main` through a pull request with one reviewer and a passing CI run. Review packages will be tagged `review-3` and `v1.0`, and the Review 2 package will be tagged retrospectively at the commit of 27/10/2026.
- Commit messages follow the Conventional Commits style (`feat:`, `fix:`, `chore:`, `docs:`) and reference the backlog item or defect ID. The rule was not enforced before: only 117 of the 272 commits on `main` (43%) use the style. From Iteration 5 the reviewer rejects a pull request whose commits do not follow it.
- Large binary sources (Blender work files, local Unity work folders) are excluded by `.gitignore`; only exported models, textures and animation clips are committed.
- Secrets: on staging, connection strings, the JWT signing key, the SMTP password, Photon keys and the dedicated-server service key are supplied through environment variables and the server `.env` file. Until 13/11/2026 the repository also contained development credentials in `appsettings.json` and a fallback JWT key in `Program.cs` (defect D-17, High). Commit be556a1 emptied these values and the API now refuses to start without a JWT key of at least 32 bytes; the published images are in addition built from sanitised settings (`deploy/api/sanitize_settings.py`). The old values are still in the Git history, so they must be rotated on staging and the rotation recorded. The rule from Iteration 5 is that no secret is committed, and a secret scan will be added to the pipeline.
- Delivery: on each push to `main` the GitHub Actions workflow `dockerhub.yml` reads the release that is deployed, plans which checks and images the changed files need, runs the IntegrationChecks, the FastAPI tests and the admin web type check, builds the Unity Linux dedicated server, publishes the changed images to Docker Hub (`tadminm/pickelverse`, tags `<component>-<commit>-<run>-<attempt>`), smoke-tests the four image digests and

deploys them to the staging host over SSH with Docker Compose after a verified database dump. The pipeline first passed on 13/11/2026 (runs 27, 28 and 30 of 32; risk 13). Before that the staging stack was built and pushed by hand as four releases tagged `api-`, `ai-`, `web-` and `server-manual-<date-time>` (08/11–10/11/2026). A commit that changes only documents builds and deploys nothing; the team also marks such commits `[skip ci]`.

6.3 Tools & Infrastructures

Table 11. Tools and infrastructure

Category	Tools / Infrastructure
Technology	Unity 6000.6.2f1 with C# (game client and dedicated server); Photon Fusion 2.1.3; Unity Behavior 1.0.16; ASP.NET Core on .NET 10 (Web API); Python 3.12 with FastAPI and scikit-learn (AI service); React 18, TypeScript 5, Vite 5 and React Query (admin web)
Database	PostgreSQL 16 (Npgsql Entity Framework Core); Redis 7.4 (matchmaking queue, leases, presence, cache); S3-compatible object storage (replays)
IDEs/Editors	Unity Editor and Unity CLI, Visual Studio Code, Visual Studio, pgAdmin 4, Docker Desktop, Photon Dashboard
Art and audio	Blender (rigging and animation), Meshy AI (base character meshes), Figma (interface mock-ups)
Diagramming	PlantUML with Graphviz, draw.io
Documentation	Microsoft Office (Word, Excel, Project-compatible XML), Markdown
Version Control	Git and GitHub (source code and documents)
CI/CD and deployment	GitHub Actions (four jobs, passing since 13/11/2026), Docker and Docker Compose, Docker Hub; staging host tad-min.io.vn (Ubuntu 24.04, Caddy reverse proxy with HTTPS, deploy/linux/Caddyfile)
Testing	Unity Test Framework (EditMode), Unity CLI, pytest, IntegrationChecks console harness, Swagger UI; k6 is planned for the API load test (NFR-04)
Project management	Tracking workbook (backlog, WBS, issues, defects, Q&A) and the MS Project XML schedule; GitHub for code and pull requests
AI assistants	OpenAI Codex and Anthropic Claude for code suggestions and first drafts of documents; every output is reviewed by the item owner

III. Software Requirement Specification

1. Product Overview

PickleVerse is a new 3D pickleball game that teaches beginners the rules and core techniques of pickleball and lets them practise against bots and other players online. Release 1.0 consists of five deployable parts that together form the system under specification: the Unity game client for Windows and Android, the Unity dedicated game server that runs online matches with server authority, the ASP.NET Core Web API with its PostgreSQL database and Redis cache, the Python AI coaching service, and the React administration web application. The Web API hosts the background workers for matchmaking, dedicated-server leases, AI analysis and moderation expiry.

Figure 1 shows the context of release 1.0. The system exchanges data with two human actors and three external systems. Photon Cloud provides the real-time session and relay service used by the game client and the dedicated server; Gmail SMTP delivers one-time passwords; S3-compatible object storage keeps match replays. Real-money payment, social login and push notifications are outside the boundary of this release.

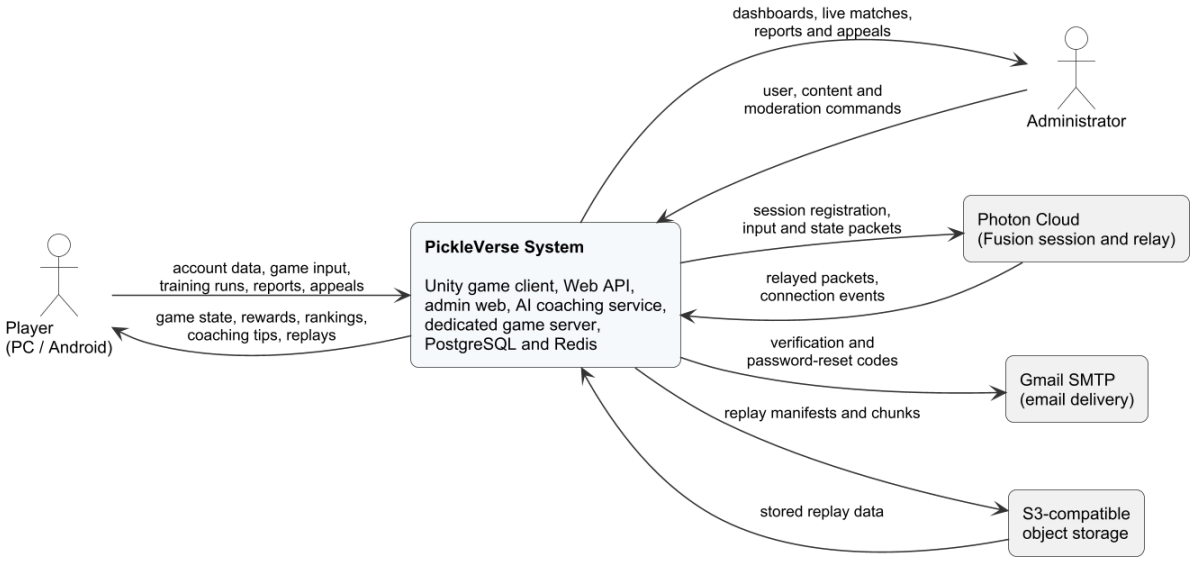


Figure 1. Context diagram of PickleVerse release 1.0

Table 1. External entities of the context diagram

External entity	Type	Data exchanged with PickleVerse
Player	Human (Windows PC or Android phone)	Sends account data, game input, training attempts, reports and appeals; receives game state, rewards, rankings, coaching tips and replays
Administrator	Human (web browser)	Sends user, content, match and moderation commands; receives dashboards, live match status, reports, appeals and analytics
Photon Cloud	External real-time service	Receives session registration and input/state packets from the client and the dedicated server; returns relayed packets and connection events
Gmail SMTP	External e-mail service	Receives verification and password-reset messages to deliver to the player's mailbox
S3-compatible object storage	Storage service	Stores replay manifests and compressed replay chunks; returns them for the replay viewer and moderation

2. User Requirements

2.1 Actors

Table 2. Actors

#	Actor	Description
1	Visitor	A person who uses the game client without being signed in. A visitor can register, verify an e-mail address, sign in, reset a forgotten password and, on a device that has a cached profile, start offline training.
2	Player	A signed-in user with the Player role and an active account. A player inherits every use case of a visitor and can learn, train, play bot and online matches, view progress, use the shop and the social features, and report other players.
3	Restricted Player	A player whose account is suspended or banned. After signing in, a restricted player is limited to viewing the sanction decision and submitting one appeal for it.
4	Administrator	A user with the Admin role who signs in to the administration web application to manage accounts, content, matches, moderation and analytics, and to label coaching sessions as an expert reviewer.
5	Email Service	Secondary actor (Gmail SMTP) that delivers verification and password-reset codes.
6	Photon Cloud	Secondary actor that hosts the real-time sessions through which the game client, other players and the dedicated server exchange match packets.

2.2 Use Cases

2.2.1 Diagram(s)

The 68 use cases are grouped into eight packages. Figure 2 gives the package overview; Figures 3 to 11 show each package with its actors and the «include» and «extend» relationships (package P4 is drawn in two parts for readability). Player and Restricted Player are specialisations of Visitor, so they inherit the visitor use cases.

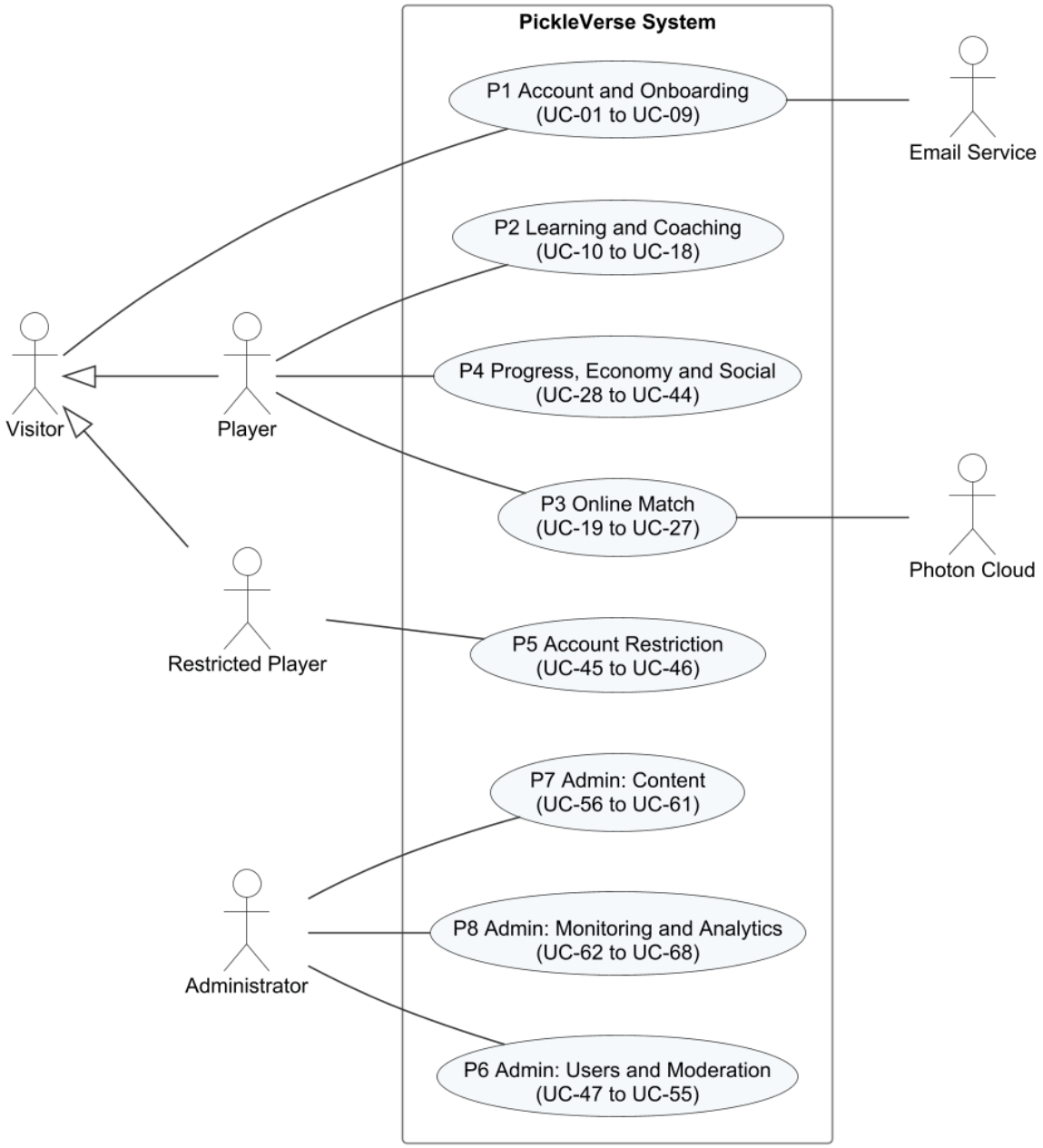


Figure 2. Use case overview by package

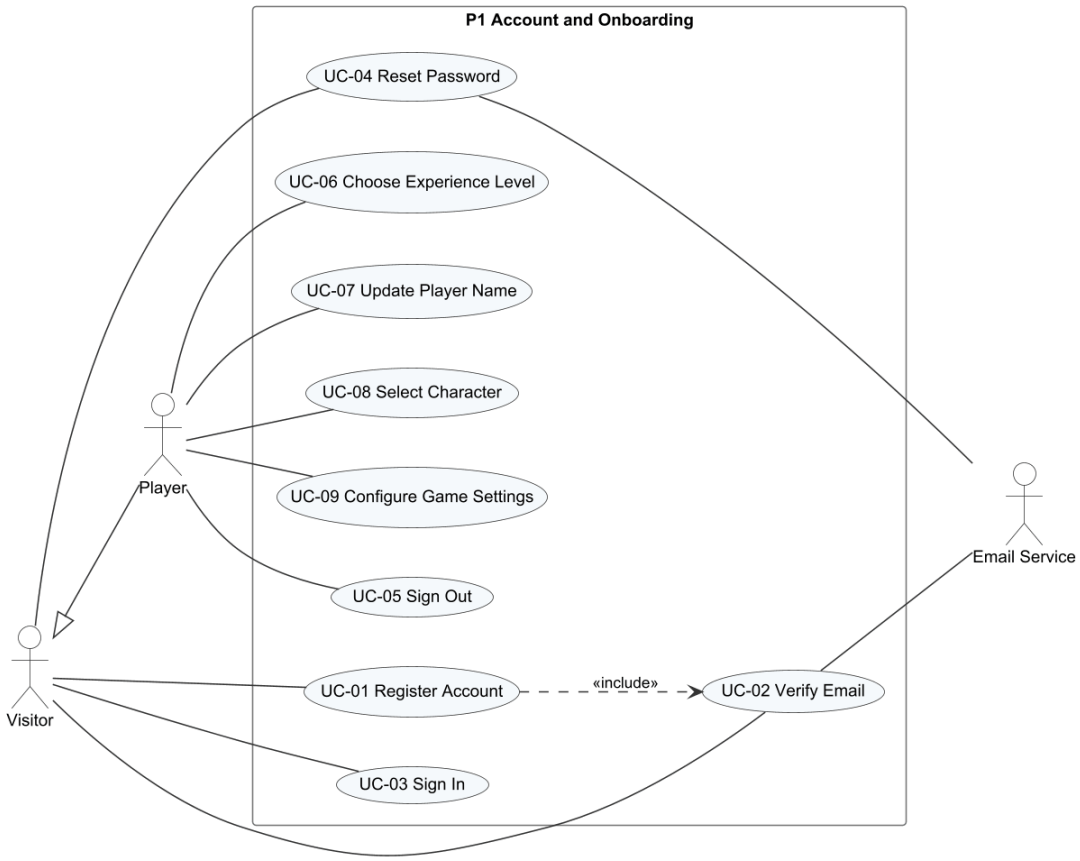


Figure 3. Use case diagram P1 – Account and Onboarding

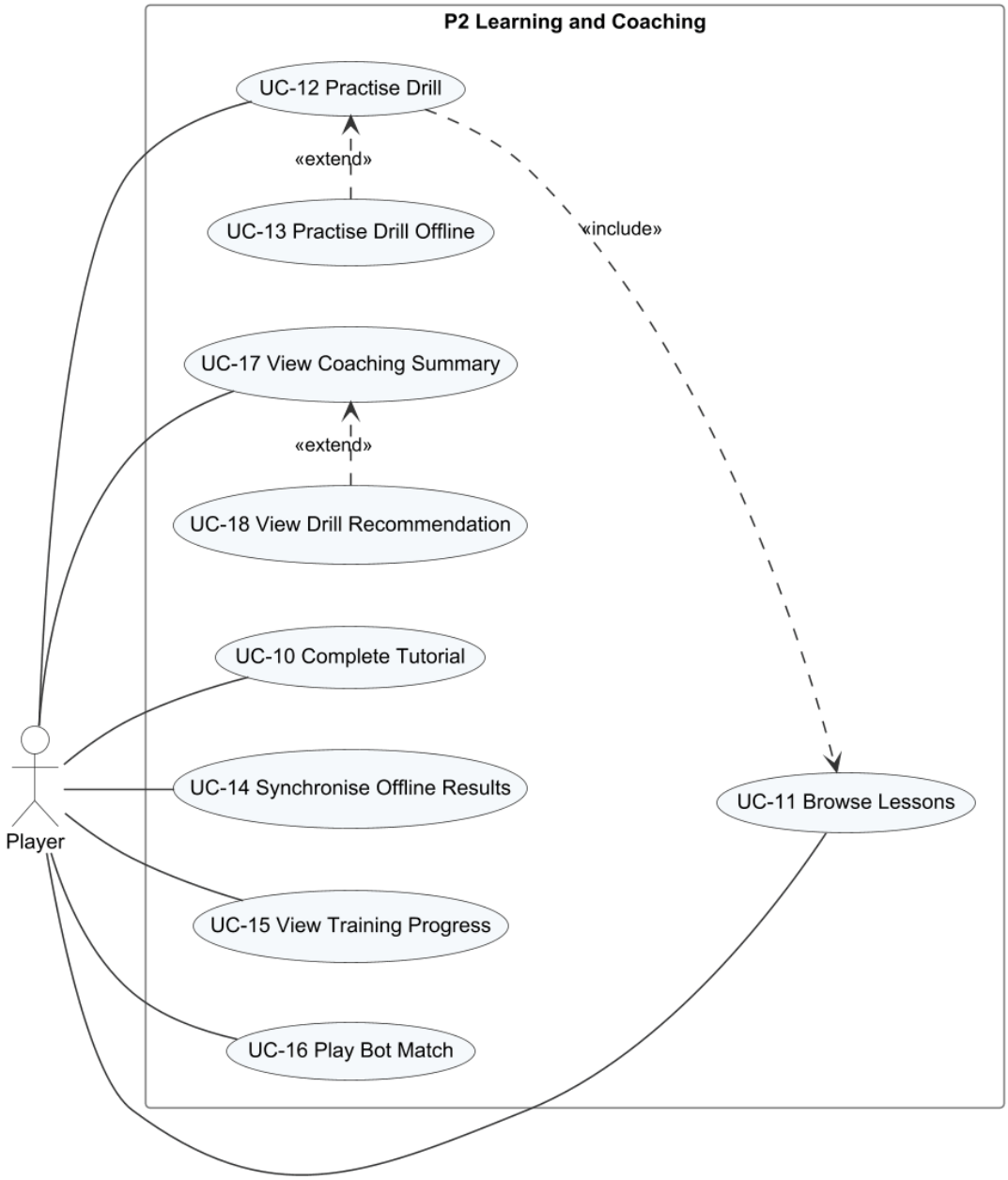


Figure 4. Use case diagram P2 – Learning and Coaching

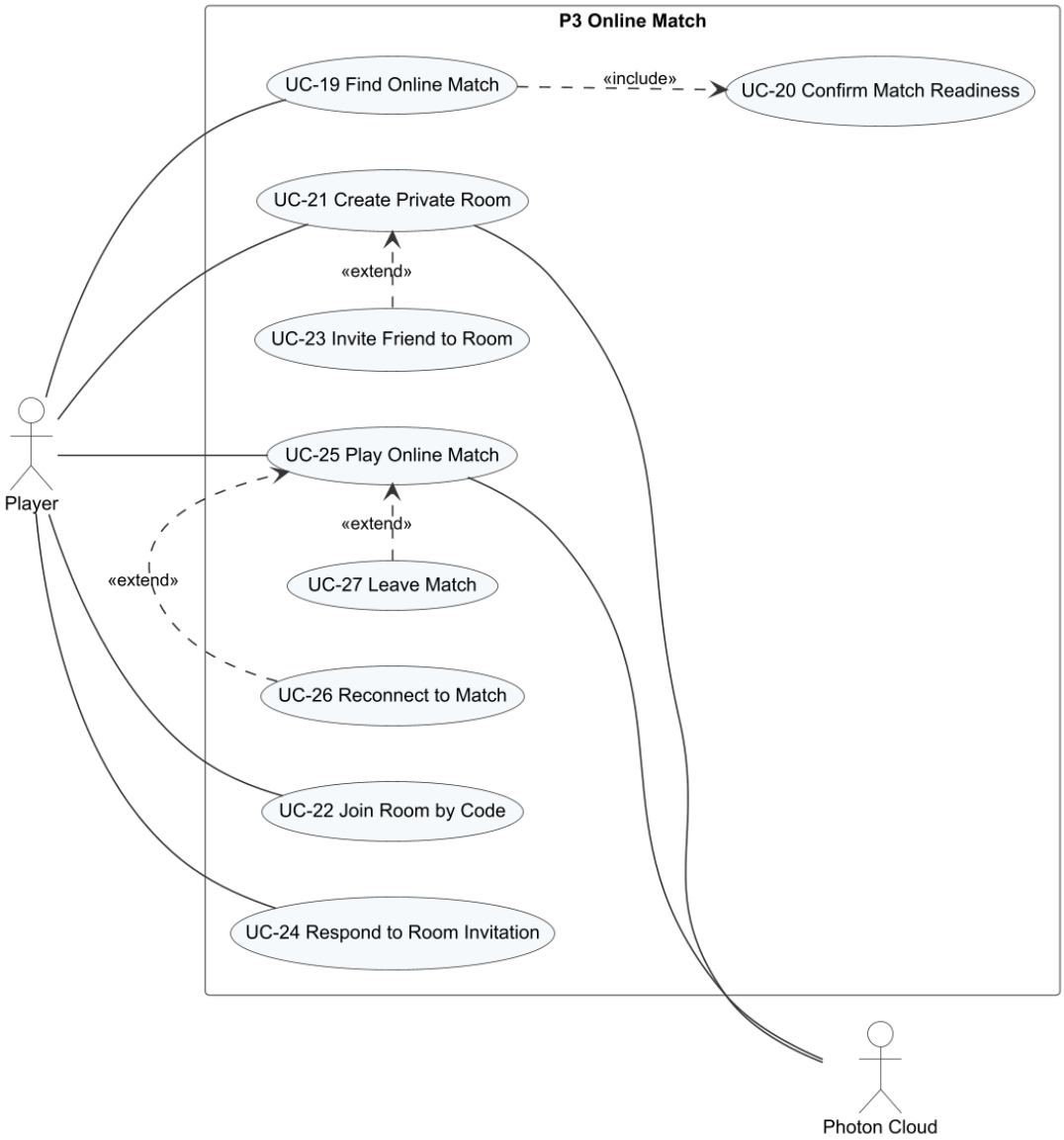


Figure 5. Use case diagram P3 – Online Match

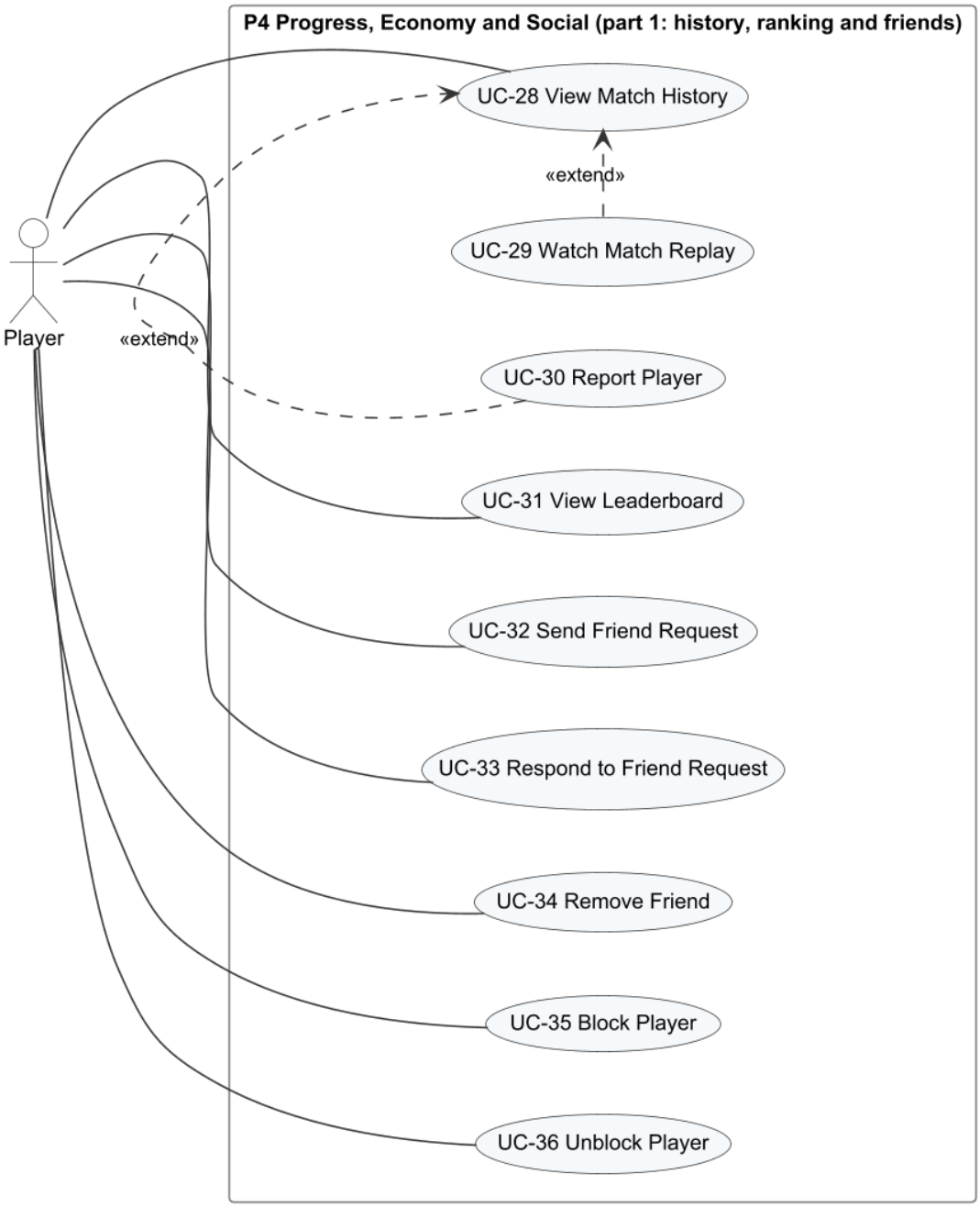


Figure 6. Use case diagram P4 (part 1) – History, Ranking and Friends

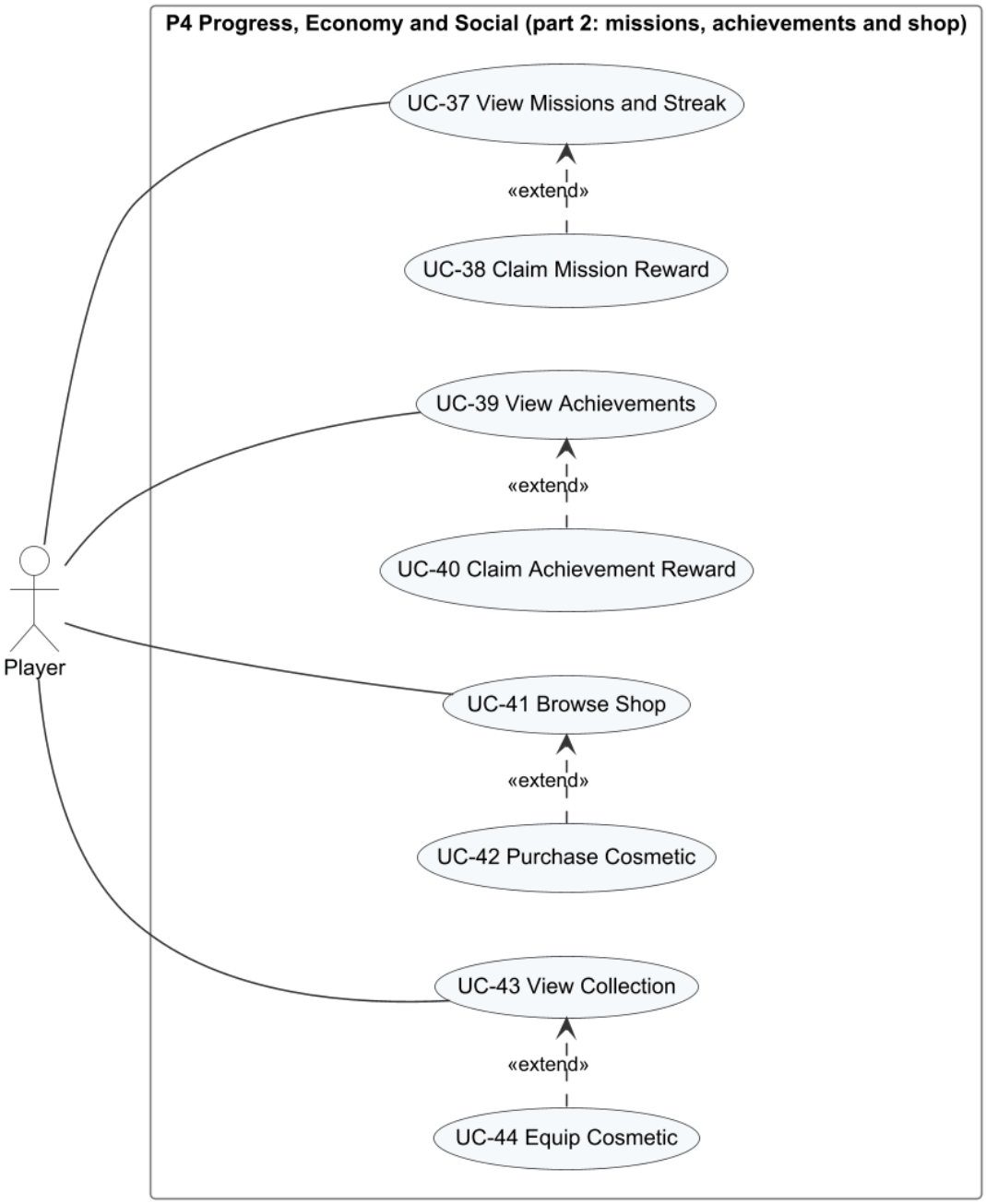


Figure 7. Use case diagram P4 (part 2) – Missions, Achievements and Shop

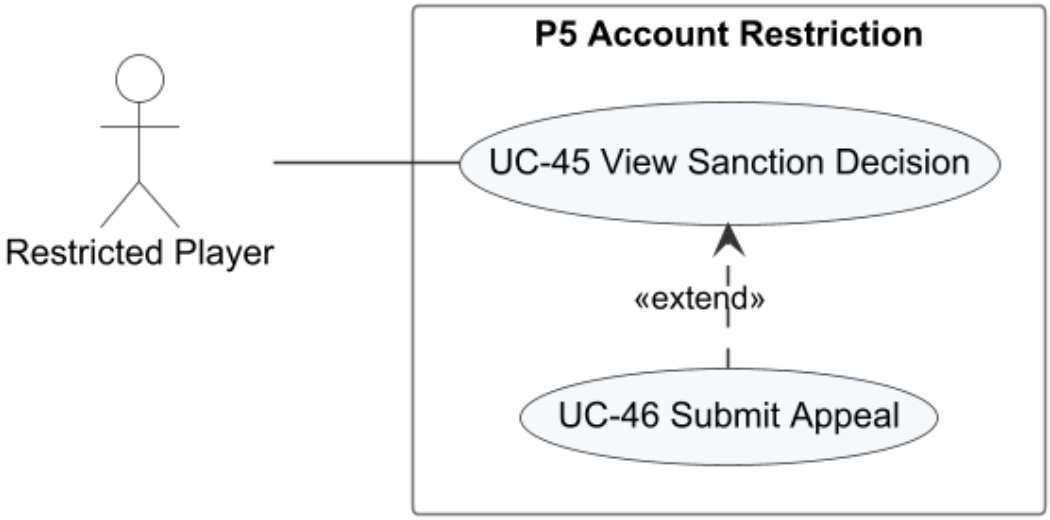


Figure 8. Use case diagram P5 – Account Restriction

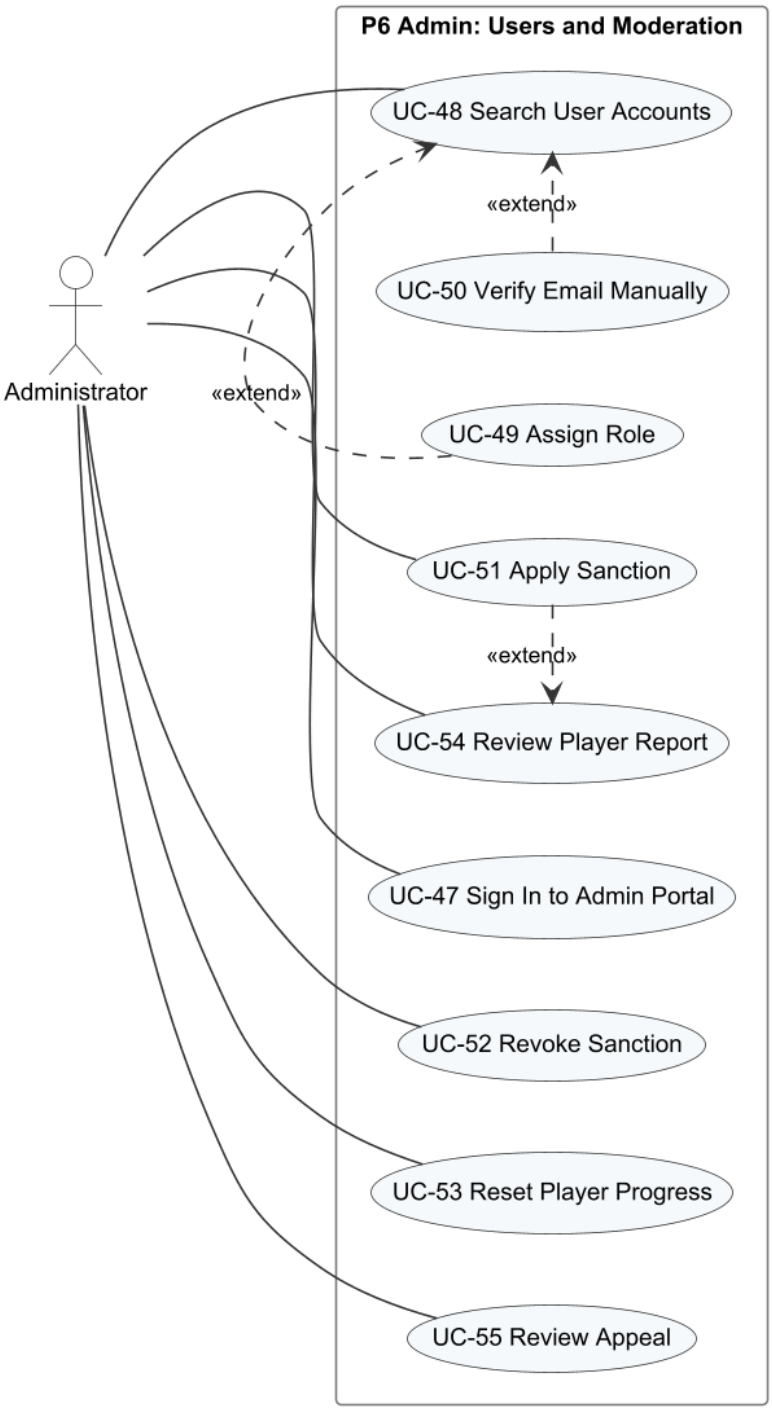


Figure 9. Use case diagram P6 – Admin: Users and Moderation

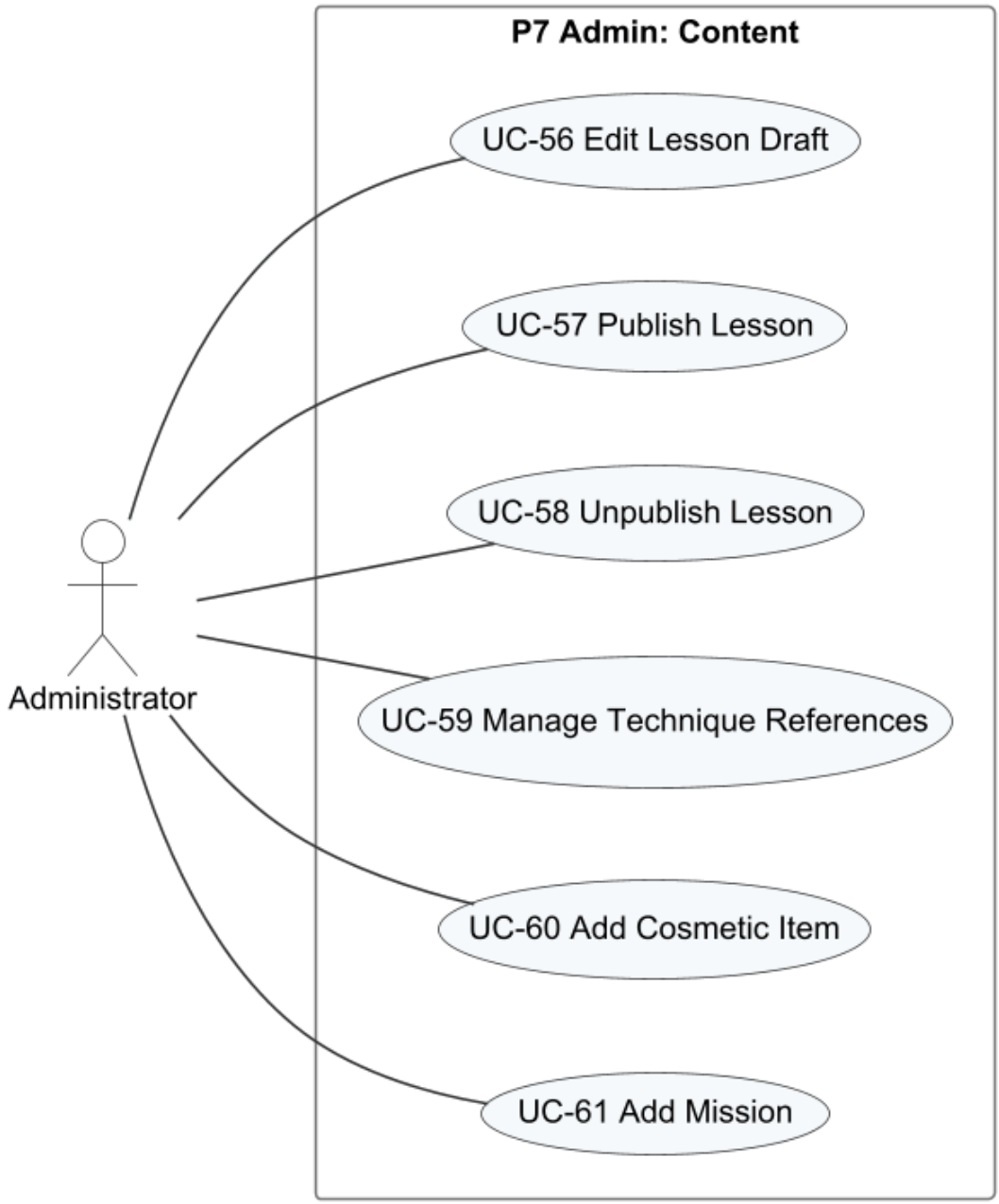


Figure 10. Use case diagram P7 – Admin: Content

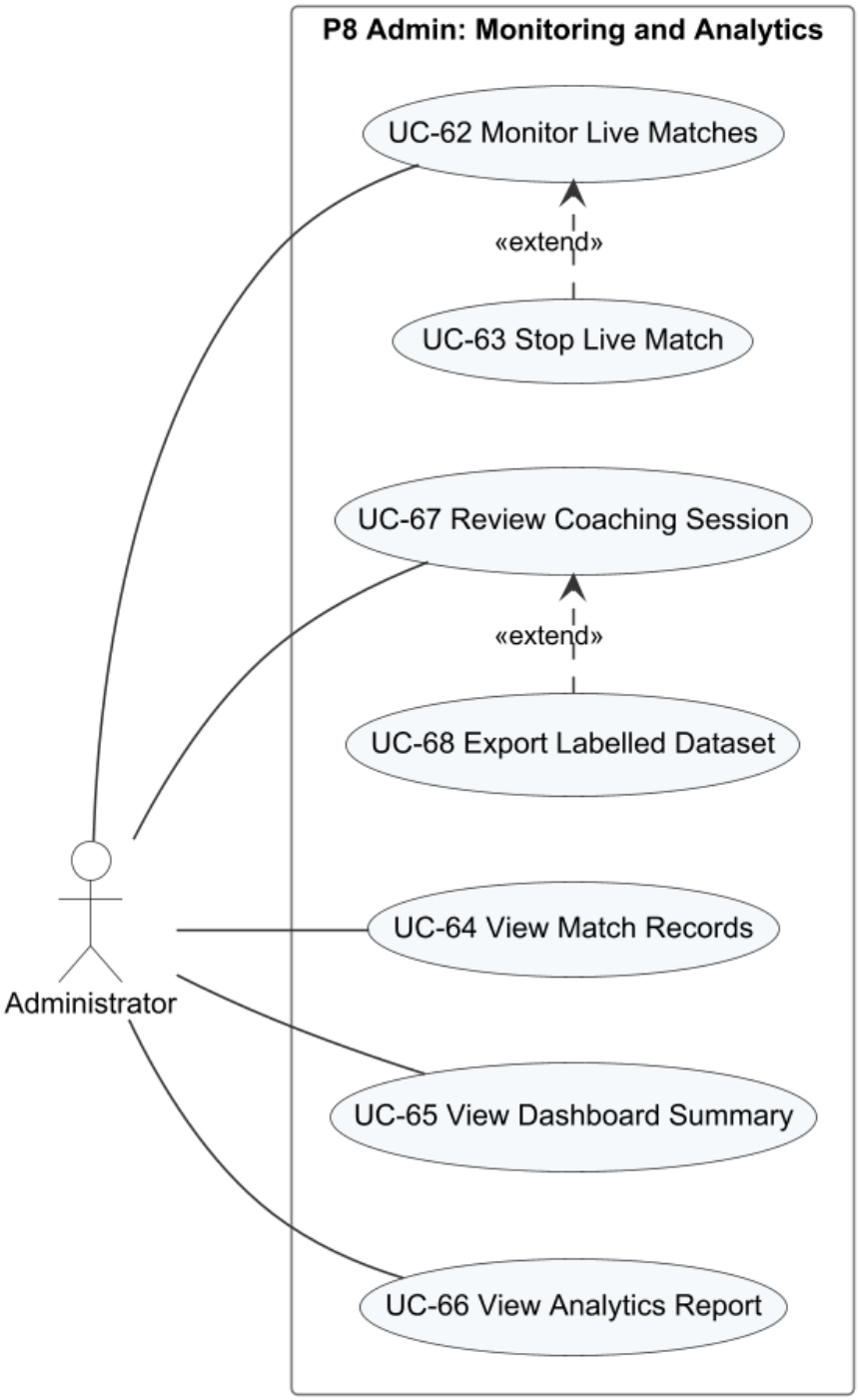


Figure 11. Use case diagram P8 – Admin: Monitoring and Analytics

2.2.2 Descriptions

Table 3. Use case descriptions

ID	Use Case	Actors	Use Case Description	FR
UC-01	Register Account	Visitor	The visitor creates an account with an e-mail address and a password; the system sends a six-digit code to that address.	FR-01
UC-02	Verify Email	Visitor, Email Service	The visitor enters the code received by e-mail to activate the account, or asks for a new code.	FR-01
UC-03	Sign In	Visitor	The visitor signs in with e-mail and password and is taken to the menu with the saved profile.	FR-02

ID	Use Case	Actors	Use Case Description	FR
UC-04	Reset Password	Visitor, Email Service	The visitor receives a reset code by e-mail and sets a new password; all existing sessions are revoked.	FR-03
UC-05	Sign Out	Player	The player ends the session on the device.	FR-02
UC-06	Choose Experience Level	Player	At the first sign-in the player states whether they are new to pickleball, which decides whether the tutorial starts.	FR-04
UC-07	Update Player Name	Player	The player changes the display name shown to other players.	FR-04
UC-08	Select Character	Player	The player chooses the male or female character used in matches.	FR-04
UC-09	Configure Game Settings	Player	The player adjusts audio, graphics, key bindings and the touch layout.	FR-05
UC-10	Complete Tutorial	Player	The player follows the 17-step tutorial on the court, rules and controls.	FR-06
UC-11	Browse Lessons	Player	The player views the lesson catalogue by group, with stars and lock status.	FR-07
UC-12	Practise Drill	Player	The player plays the drill of an unlocked lesson and receives a score of 0 to 3 stars.	FR-08
UC-13	Practise Drill Offline	Player	Without a connection, the player practises a cached drill; the result is stored for later synchronisation.	FR-09
UC-14	Synchronise Offline Results	Player	When online again, the stored offline results are sent and applied exactly once.	FR-09
UC-15	View Training Progress	Player	The player reviews passed lessons, stars, best and recent accuracy of each lesson, practice time, common errors and post-session feedback.	FR-10
UC-16	Play Bot Match	Player	The player configures and plays a singles match against a computer opponent.	FR-11, FR-12, FR-13
UC-17	View Coaching Summary	Player	After a drill or match the player reads the analysis of their shots and the main errors.	FR-19
UC-18	View Drill Recommendation	Player	The player opens a recommended drill that targets a detected weakness.	FR-19
UC-19	Find Online Match	Player	The player searches a ranked singles match (opponent of similar Elo, or a bot after 180 seconds) or a quick doubles match (unrated 2v2, alone or with a party of two, human players only).	FR-14
UC-20	Confirm Match Readiness	Player	Each paired player accepts the match within 10 seconds before it starts.	FR-14
UC-21	Create Private Room	Player, Photon Cloud	The player creates a singles or doubles room identified by a room code.	FR-15
UC-22	Join Room by Code	Player	The player joins a friend's room by typing its code.	FR-15
UC-23	Invite Friend to Room	Player	The host invites an online friend into the room.	FR-15
UC-24	Respond to Room Invitation	Player	The invited player accepts or declines the invitation.	FR-15
UC-25	Play Online Match	Player, Photon Cloud	The players play a match whose rules and score are decided by the dedicated server.	FR-16
UC-26	Reconnect to Match	Player	After a disconnection the player rejoins the same match within the reconnection window.	FR-17
UC-27	Leave Match	Player	The player leaves a match before it ends and accepts the consequences.	FR-17
UC-28	View Match History	Player	The player views the last 20 matches with result, score and Elo change.	FR-24
UC-29	Watch Match Replay	Player	The player watches the replay of one of their own matches with pause, seek and speed control.	FR-24
UC-30	Report Player	Player	The player reports an opponent from a finished match with a reason.	FR-27
UC-31	View Leaderboard	Player	The player views players ordered by Elo and their own position.	FR-26
UC-32	Send Friend Request	Player	The player searches by name or ID and sends a friend request.	FR-25
UC-33	Respond to Friend Request	Player	The player accepts or declines a received request.	FR-25
UC-34	Remove Friend	Player	The player removes a friend from the friend list.	FR-25
UC-35	Block Player	Player	The player blocks another player, which removes friendship and prevents requests and invitations.	FR-25

ID	Use Case	Actors	Use Case Description	FR
UC-36	Unblock Player	Player	The player removes a block.	FR-25
UC-37	View Missions and Streak	Player	The player views daily and weekly missions, their progress and the daily streak.	FR-21
UC-38	Claim Mission Reward	Player	The player claims the reward of a completed mission or mission set once.	FR-21
UC-39	View Achievements	Player	The player views achievement progress.	FR-21
UC-40	Claim Achievement Reward	Player	The player claims the reward of a completed achievement once.	FR-21
UC-41	Browse Shop	Player	The player views cosmetic items with their prices and currencies.	FR-22
UC-42	Purchase Cosmetic	Player	The player buys an item with in-game currency.	FR-22
UC-43	View Collection	Player	The player views owned cosmetic items.	FR-23
UC-44	Equip Cosmetic	Player	The player equips an owned paddle, which is then shown in matches.	FR-23
UC-45	View Sanction Decision	Restricted Player	The restricted player reads the restriction type, reason and expiry.	FR-28
UC-46	Submit Appeal	Restricted Player	The restricted player submits one written appeal against a sanction.	FR-28
UC-47	Sign In to Admin Portal	Administrator	The administrator signs in to the web application; only the Admin role is accepted.	FR-29
UC-48	Search User Accounts	Administrator	The administrator searches and filters accounts by text and status.	FR-29
UC-49	Assign Role	Administrator	The administrator changes the role of an account between Player and Admin.	FR-29
UC-50	Verify Email Manually	Administrator	The administrator activates an account whose verification e-mail could not be delivered.	FR-29
UC-51	Apply Sanction	Administrator	The administrator suspends or bans an account with a reason, optionally from a report.	FR-29, FR-30
UC-52	Revoke Sanction	Administrator	The administrator lifts an active sanction with a reason.	FR-29
UC-53	Reset Player Progress	Administrator	The administrator previews and resets selected progress groups of a player.	FR-29
UC-54	Review Player Report	Administrator	The administrator examines a report with match evidence and accepts or declines it.	FR-30
UC-55	Review Appeal	Administrator	The administrator accepts or declines an appeal; acceptance revokes the related sanction.	FR-30
UC-56	Edit Lesson Draft	Administrator	The administrator creates or edits a lesson and saves it as a draft.	FR-31
UC-57	Publish Lesson	Administrator	The administrator publishes the saved draft so that new attempts use it.	FR-31
UC-58	Unpublish Lesson	Administrator	The administrator hides a published lesson from players.	FR-31
UC-59	Manage Technique References	Administrator	The administrator maintains the reference values used to judge each technique.	FR-31
UC-60	Add Cosmetic Item	Administrator	The administrator adds an item to the shop with type, price and currency.	FR-31
UC-61	Add Mission	Administrator	The administrator creates a daily or weekly mission with target and reward.	FR-31
UC-62	Monitor Live Matches	Administrator	The administrator watches active matches with score and connection status refreshed every 5 seconds.	FR-32
UC-63	Stop Live Match	Administrator	The administrator stops a dedicated-server match with a reason.	FR-32
UC-64	View Match Records	Administrator	The administrator browses finished and interrupted matches.	FR-32
UC-65	View Dashboard Summary	Administrator	The administrator sees today's key figures and items that need action.	FR-32
UC-66	View Analytics Report	Administrator	The administrator views active users, retention, lesson pass rates, technique errors and matchmaking health for a date range.	FR-32
UC-67	Review Coaching Session	Administrator	Acting as expert, the administrator inspects shot evidence of a training session and labels the player level and errors.	FR-33
UC-68	Export Labelled Dataset	Administrator	The administrator exports the labelled sessions used to evaluate the AI coaching quality.	FR-33

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

Figure 12 shows the screens of the game client and how a player moves between them; Figure 13 shows the administration web application. Rounded white boxes marked [popup], [dialog], [drawer] or [overlay] are windows opened above another screen; the other boxes are full scenes or pages.

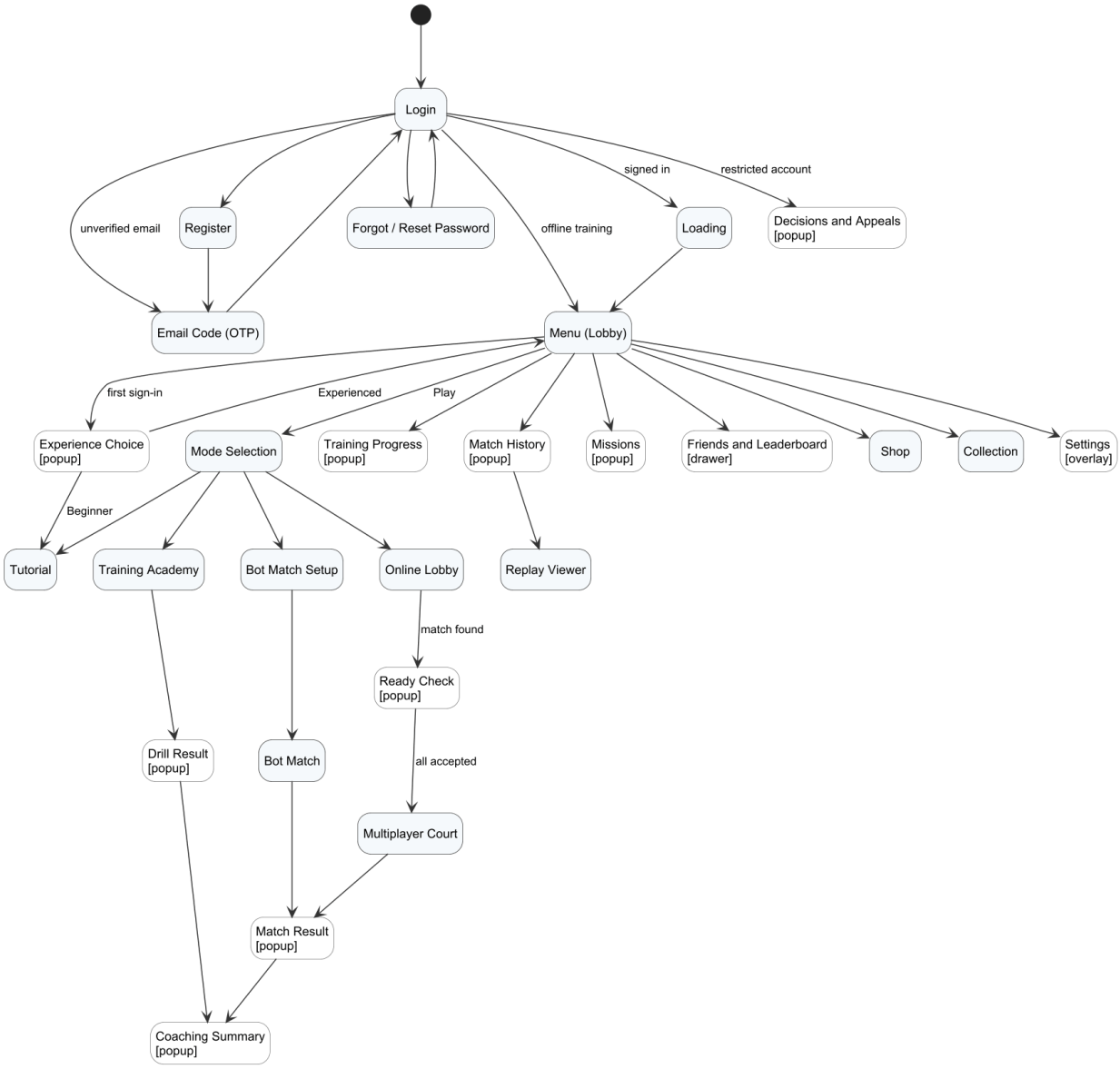


Figure 12. Screen flow of the game client

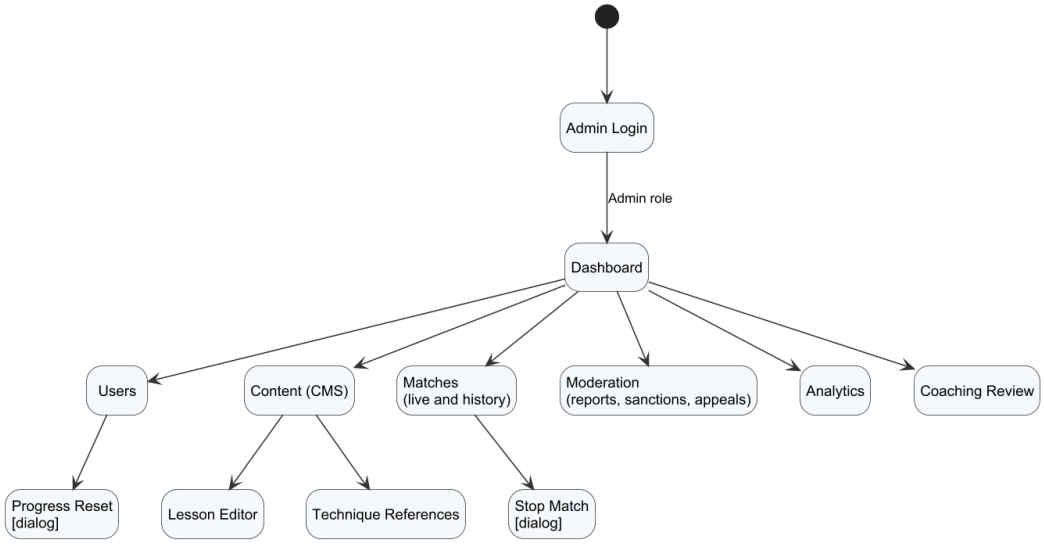


Figure 13. Screen flow of the administration web application

3.1.2 Screen Descriptions

Table 4. Screen descriptions

#	Feature	Screen	Description
1	FE-01	Login	E-mail and password fields with a show/hide control, the Sign in button, links to e-mail verification, password recovery and registration, and an Offline training button when a profile is cached on the device.
2	FE-01	Register	E-mail, password and password confirmation fields with inline validation and the Create account button.
3	FE-01	Email Verification (OTP)	Six-digit code field, Verify and Resend code buttons, and a link back to sign-in.
4	FE-01	Forgot / Reset Password	Step 1 asks for the e-mail address; step 2 asks for the reset code, the new password and its confirmation.
5	FE-01	Loading	Animated artwork while the profile, wallet, content catalogue and settings are loaded.
6	FE-01	Experience Choice [popup]	Asked once at the first sign-in: "Bạn đã quen với pickleball chưa?" with Beginner and Experienced options.
7	FE-01	Menu (Lobby)	3D court with the player's character; name tag with level, rank and Elo; Tokenpad and Elite Token balances; male/female selector; day/night toggle; music player; Play button (Enter); one row of seven tiles – Progress, Decisions and Appeals ("!"), History, Missions, Collection, Shop and Settings; friend drawer on the left edge.
8	FE-01	Settings [overlay]	Tabs Overview (with Sign out), Audio, Graphics and Controls (key binding and touch layout); Restore defaults, Undo and Apply and save buttons.
9	FE-02	Tutorial	Seventeen steps on the court; each step has a title, an explanation and a demonstration, with Continue, Replay and Later buttons.
10	FE-02	Mode Selection	Four cards – Bot match, Online match, Training and Create room – and a Tutorial strip.
11	FE-02	Training Academy	Lesson catalogue with group tabs (All, Basic, Control, Advanced, Doubles); lesson briefing with target, reward and star rules; practice HUD with shot counter, timing feedback and key hints.
12	FE-02	Drill Result [popup]	Stars, accuracy (hits in target / shots), mean timing error, practice time, most frequent error with a tip, and Retry, Next lesson and Lesson list buttons.
13	FE-02	Training Progress [popup]	Passed lessons out of 15, next lesson, post-session feedback, practice time, most frequent error, and for each lesson the stars, best accuracy and the four most recent accuracy results ("Gần đây") with a Retry button.
14	FE-03	Bot Match Setup	Difficulty (Easy, Medium, Hard, Adaptive – shown as "Tự động"), time of day (day, night) and court (professional, outdoor), with a summary line and the Enter court button.
15	FE-03	Bot Match	Court seen from behind the player's baseline with both players; score panel with serving side and serve timer; rule hint; fault banners; shot cue under the player with the key combination (touch buttons on phones); Settings (Esc) and help (F1).

#	Feature	Screen	Description
16	FE-03	Match Result [popup]	Winner, game scores, Tokenpad and XP earned, Elo change for ranked matches, and Play again, Mode selection and View analysis buttons.
17	FE-05	Coaching Summary [popup]	Counts of shots, contacts and serves; main errors with their frequency; up to three recommended drills with a Practise now link.
18	FE-04	Online Lobby	Singles 1v1 and Doubles 2v2 tabs, room code with Copy, Join by code, roster slots with Invite buttons, a collapsible friends column listing online friends who can be invited, and the Find match / Cancel button with an elapsed-time counter.
19	FE-04	Ready Check [popup]	Match found message with Accept and Decline buttons and a 10-second countdown.
20	FE-04	Multiplayer Court	Online match on the dedicated server with score, connection state, reconnecting overlay and leave control.
21	FE-07	Match History [popup]	Level card and the last 20 matches with opponent, mode, Vietnam time, result, game score, Elo change, and Report and Replay buttons.
22	FE-07	Replay Viewer	Playback of a recorded match with play/pause, a seek bar and speeds from 0.5× to 2×.
23	FE-06	Missions [popup]	Daily and weekly tabs, mission list with progress and reward, set bonus, daily streak, reset countdown, and Claim buttons.
24	FE-07	Friends and Leaderboard [drawer]	Friends tab with search by name or ID and sub-tabs Friends, Find friends and Blocked; Leaderboard tab with own position.
25	FE-06	Shop	Featured and all items, item preview, price in Tokenpad and/or Elite Token, wallet balances and the Buy button.
26	FE-06	Collection	Owned paddles with preview and the Equip button.
27	FE-08	Decisions and Appeals [popup]	List of sanctions on the account with type, reason and expiry; appeal form and appeal status.
28	FE-08	Report Player [dialog]	Opened from a history entry: reported player, match, reason field and Send button.
29	FE-09	Admin Login	Account or e-mail and password fields and the Authenticate button; only Admin accounts are accepted.
30	FE-09	Dashboard	Total users, daily active users, matches played, online time, pending reports, technique-error heat map and quick links.
31	FE-09	Users	Search and status filter; table of player and e-mail, role, status, Elo, play time and creation date; actions to assign a role, sanction, verify e-mail and reset progress.
32	FE-09	Progress Reset [dialog]	Selection of LEARNING, RANKING, MISSIONS and ECONOMY groups, an impact preview and the Reset button.
33	FE-09	Content (CMS)	Lesson manager (draft, publish, unpublish), drill form, shop item form, mission form and technique references.
34	FE-09	Matches	Live matches refreshed every 5 seconds with score and connection state, match history, and a Stop match dialog that requires a reason.
35	FE-09	Moderation	Reports, sanctions and appeals with match evidence and replay download, decision form and decision reason.
36	FE-09	Analytics	Date range in Vietnam time; daily active players, retention cohorts, popular lessons with pass rate, technique-error heat map and matchmaking health.
37	FE-10	Expert Review	Training sessions with shot-by-shot evidence, level and error labels, data group (training, calibration, holdout) and export.

3.1.3 Screen Authorization

X means that the role may open the screen or perform the activity. Visitor is a user who has not signed in; Restricted Player is a signed-in player with a suspended or banned account. The Web API enforces the same rules: a missing or expired token returns HTTP 401, and a valid token without the required role or scope returns HTTP 403.

Table 5. Screen authorization

Screen / Activity	Visitor	Player	Restricted Player	Administrator
Login, Register, Email Verification, Forgot Password	X			
Start offline training from a cached profile	X	X		
Menu, Settings, Mode Selection, Tutorial		X		
Update name, select character, sign out		X		

Screen / Activity	Visitor	Player	Restricted Player	Administrator
Training Academy, Drill Result, Training Progress		X		
Bot Match Setup, Bot Match, Match Result, Coaching Summary		X		
Online Lobby, Ready Check, Multiplayer Court		X		
Create room, invite friend, start a custom match (host only)		X		
Match History, Replay Viewer		X		
Query own matches and replays only		X		
Report a player of an own finished match		X		
Missions, Shop, Collection		X		
Claim rewards, purchase and equip own items		X		
Friends and Leaderboard		X		
Decisions and Appeals		X	X	
Submit an appeal for an own sanction			X	
Admin Login				X
Dashboard, Analytics				X
Users				X
Query all accounts				X
Assign role, verify e-mail, apply and revoke sanctions (not own account)				X
Preview and reset player progress				X
Content (CMS)				X
Add and update lessons, references, items and missions; publish and unpublish				X
Matches				X
Stop a live dedicated-server match				X
Moderation				X
Decide reports and appeals				X
Expert Review				X
Save expert labels and export the dataset				X

3.1.4 Non-Screen Functions

Table 6. Non-screen functions

#	Feature	System Function	Description
1	FE-01	Send verification and reset e-mails	Generates a six-digit code, stores only its HMAC, and sends the message through Gmail SMTP (STARTTLS) from a background mail queue.
2	FE-01	Refresh access token	Exchanges a valid refresh token for a new access token (2 hours) and a rotated refresh token (14 days); revoked or reused tokens are rejected.
3	FE-01	Record activity days	Records each player's login day and active day (Vietnam time) and online presence in Redis for analytics and streaks.
4	FE-02	Apply training results	Validates each training attempt from its telemetry, applies stars and first-pass rewards, and returns a receipt APPLIED, DUPLICATE or REJECTED for each request ID.
5	FE-04	Matchmaking	Keeps search tickets in a Redis queue. Ranked singles: pairs tickets within the Elo window and assigns a ranked bot after 180 seconds. Quick doubles: fills two teams of two from solo players and parties of two in the same region and protocol version, without Elo or bot fallback. Both create the ready check.
6	FE-04	Dedicated server allocation	Issues HMAC-signed match tickets, leases the dedicated server slot, receives heartbeats every 8 seconds and releases expired leases (DedicatedLeaseMonitor).
7	FE-04	Settle match result	Saves the game scores once, applies Elo for ranked matches, grants rewards and updates missions and achievements in one transaction.
8	FE-04	Store replay	Uploads the replay manifest and compressed chunks to object storage and links them to the match.
9	FE-05	Ingest telemetry	Receives shot telemetry parts from the client (or the dedicated server for online matches), stores them and creates an analysis job.

#	Feature	System Function	Description
10	FE-05	Hybrid AI worker	Sends pending analysis jobs to the AI coaching service, stores the coaching result and retries failed jobs (HybridAiWorker).
11	FE-05	AI coaching service	POST /v1/analyze-session: rule-based findings, summary, at most three drill recommendations and, when the gated model is approved, a tier estimate.
12	FE-06	Mission cycle	Opens new daily missions at 07:00 Vietnam time each day and new weekly missions each week, and updates the daily streak.
13	FE-07	Leaderboard cache	Caches the Elo leaderboard in Redis for 15 seconds.
14	FE-08	Sanction expiry	Restores suspended accounts whose sanction has expired (ModerationExpiryWorker).
15	FE-09	Analytics projection	Aggregates daily active players, retention cohorts, lesson statistics and matchmaking events (AnalyticsProjectionWorker).
16	FE-09	Admin audit	Writes an audit record for every administrative change with actor, action, target and time.

3.1.5 Entity Relationship Diagram

The conceptual ERD is drawn in Chen notation and split into six subject areas so that every diagram stays readable. Entities are business concepts rather than database tables: junction tables appear as relationships with their own attributes (for example PARTICIPATES carries Team and Outcome), and purely technical tables such as refresh tokens and telemetry parts are omitted. The logical design of all 51 tables is given in Report 4, §2.

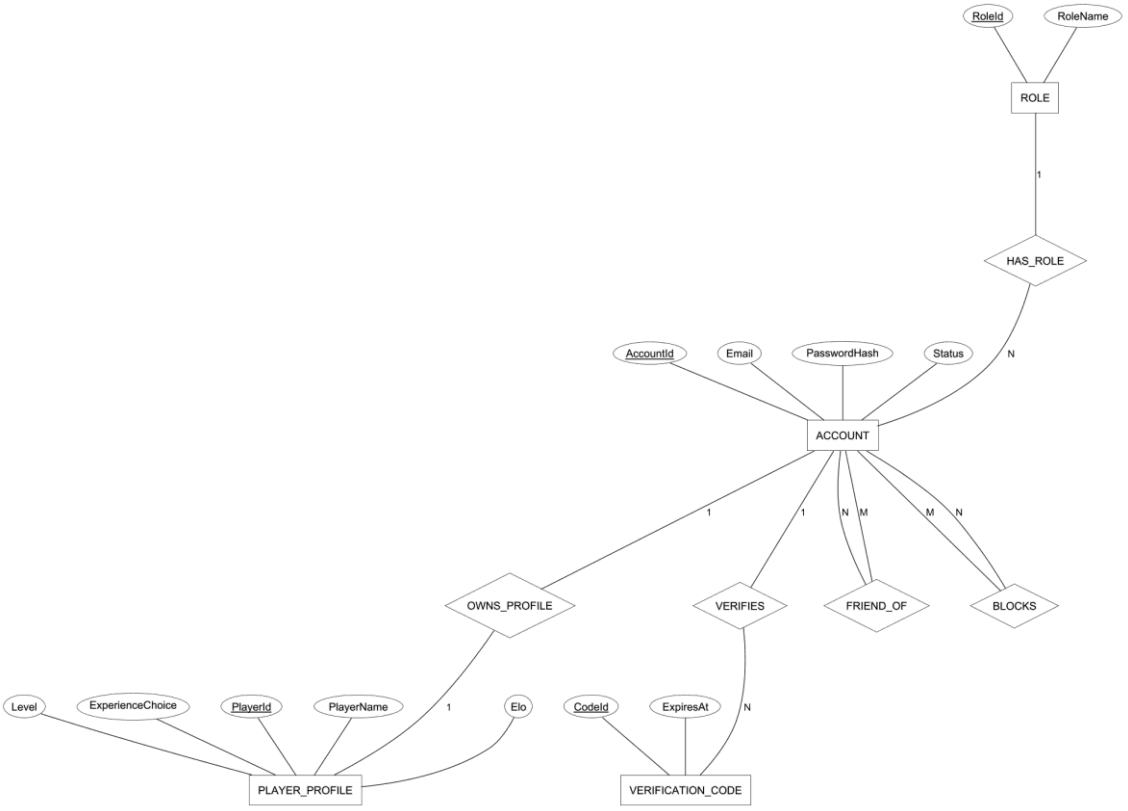


Figure 14. Conceptual ERD – accounts, profiles and social relations

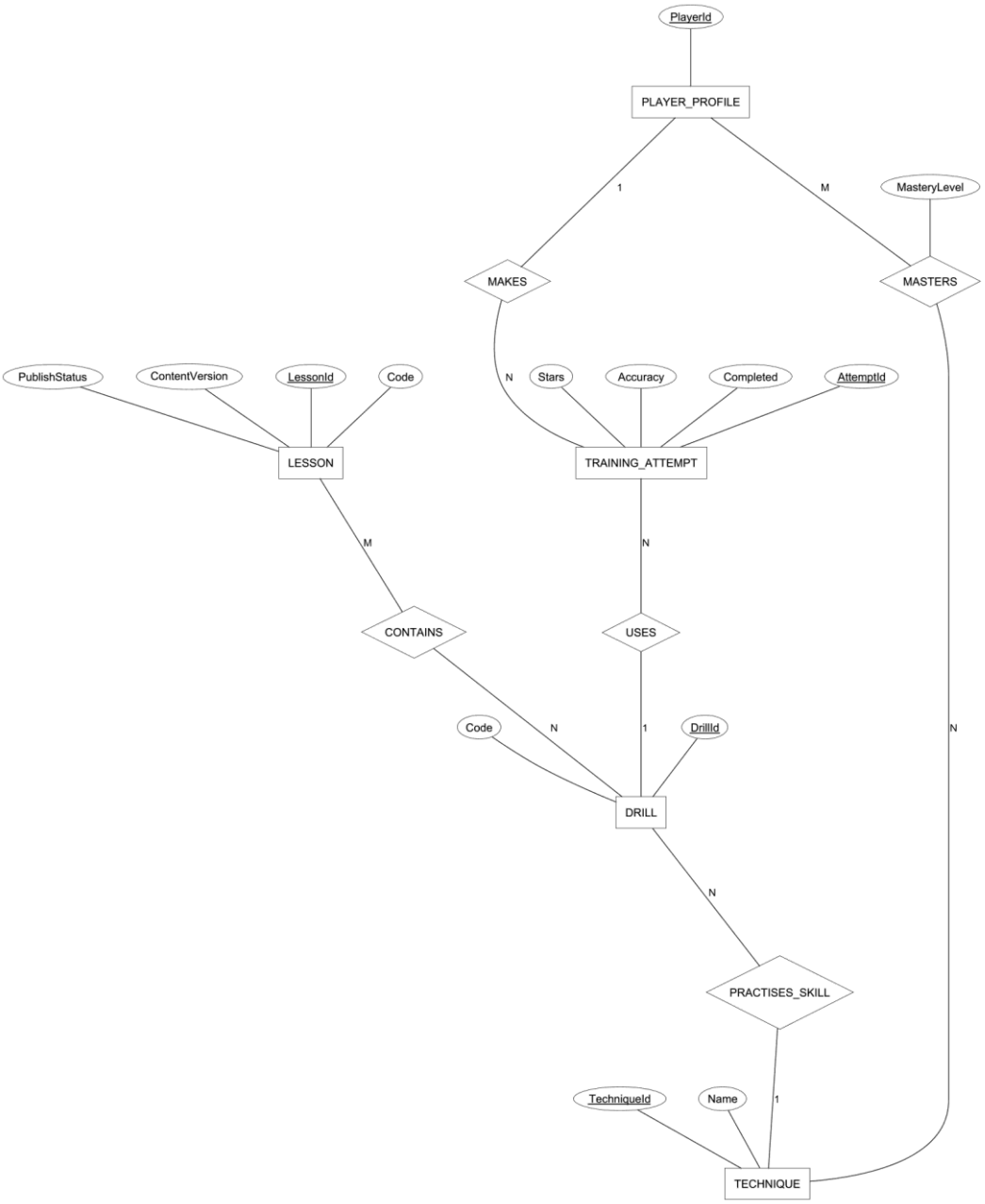


Figure 15. Conceptual ERD – techniques, lessons and training

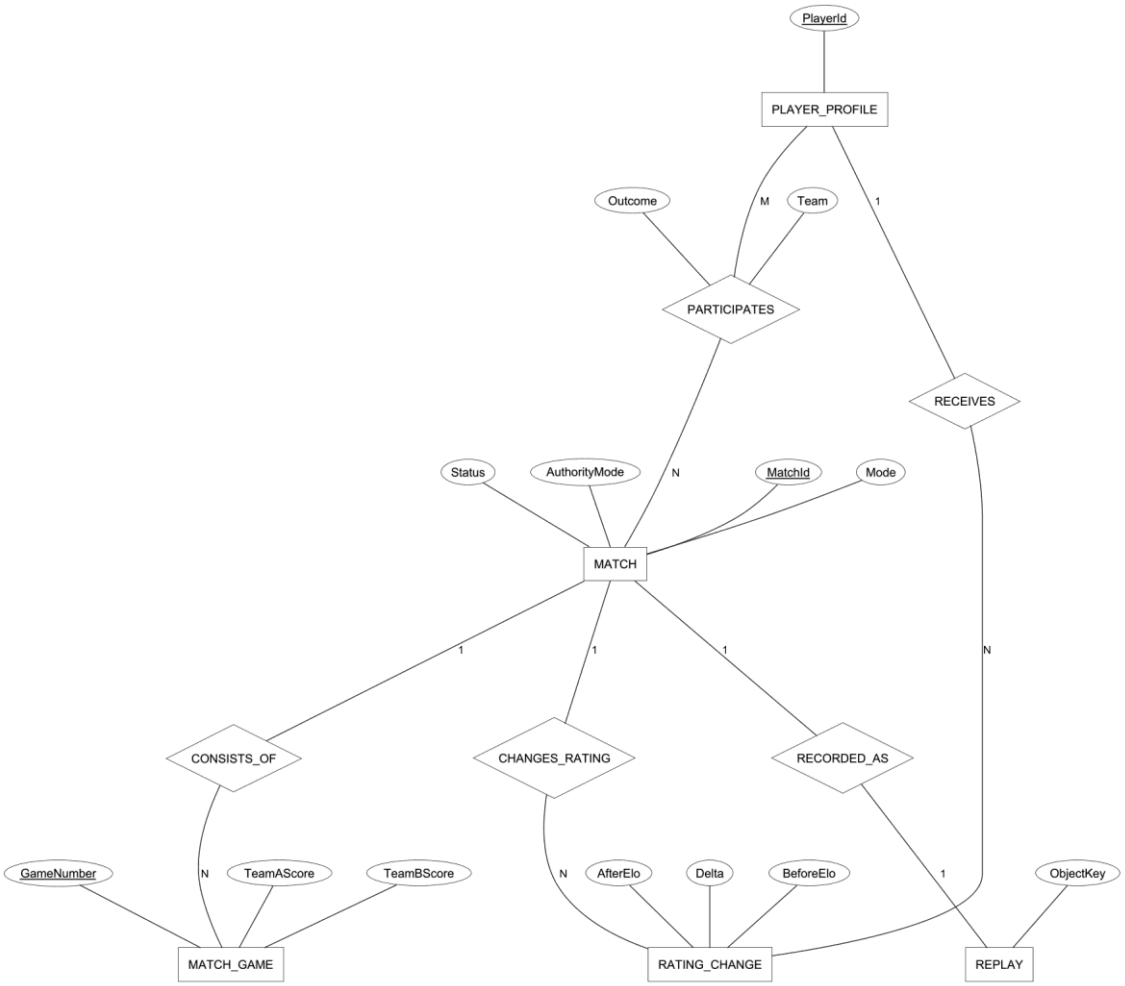


Figure 16. Conceptual ERD – matches, ratings and replays

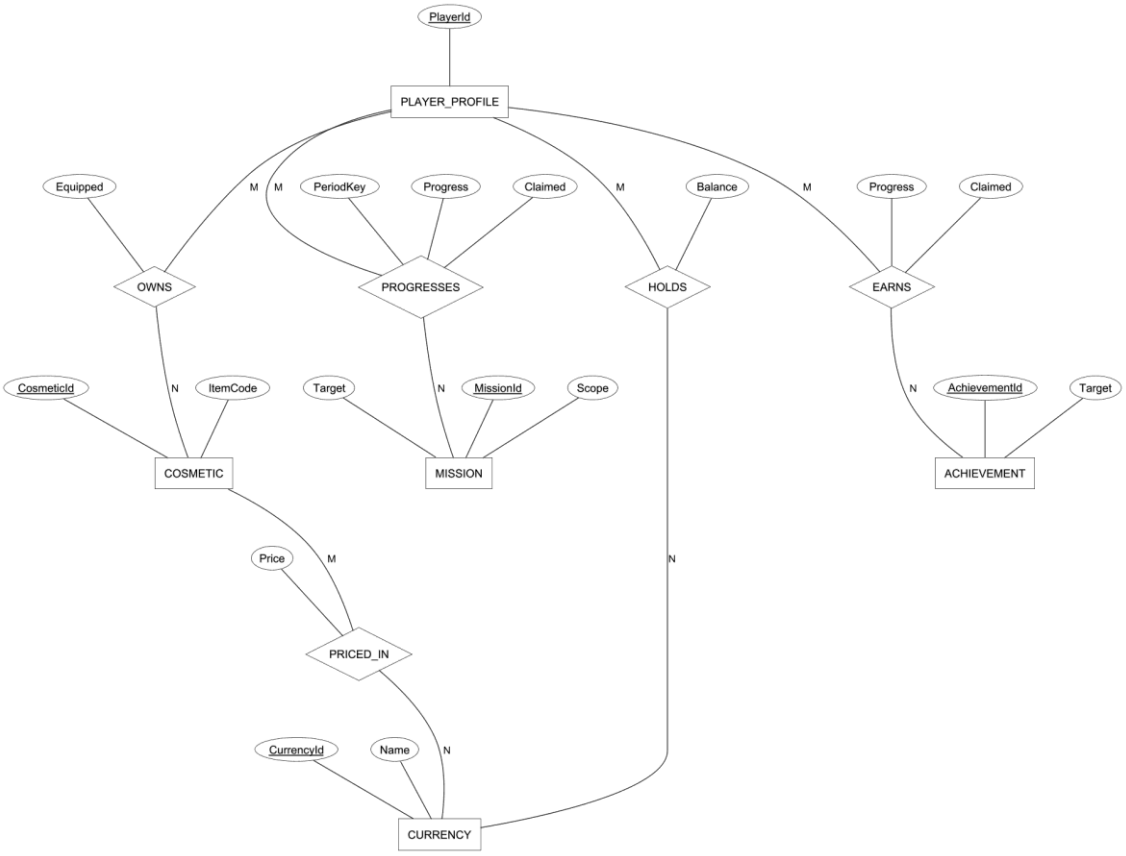


Figure 17. Conceptual ERD – currencies, cosmetics, missions and achievements

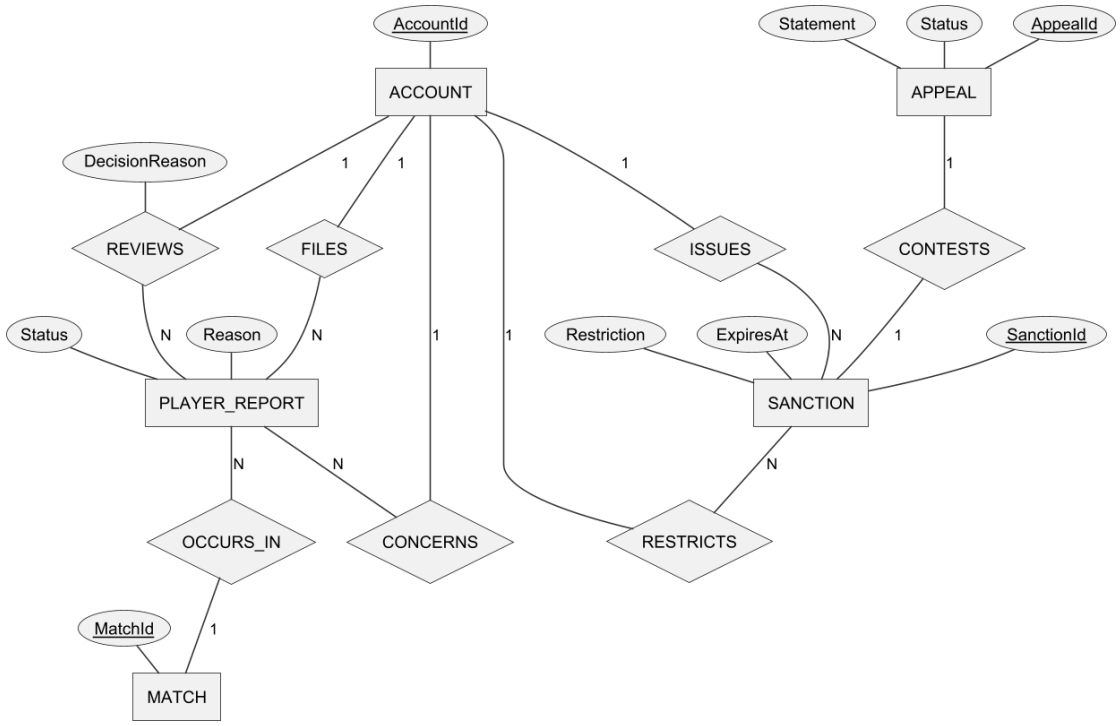


Figure 18. Conceptual ERD – reports, sanctions and appeals

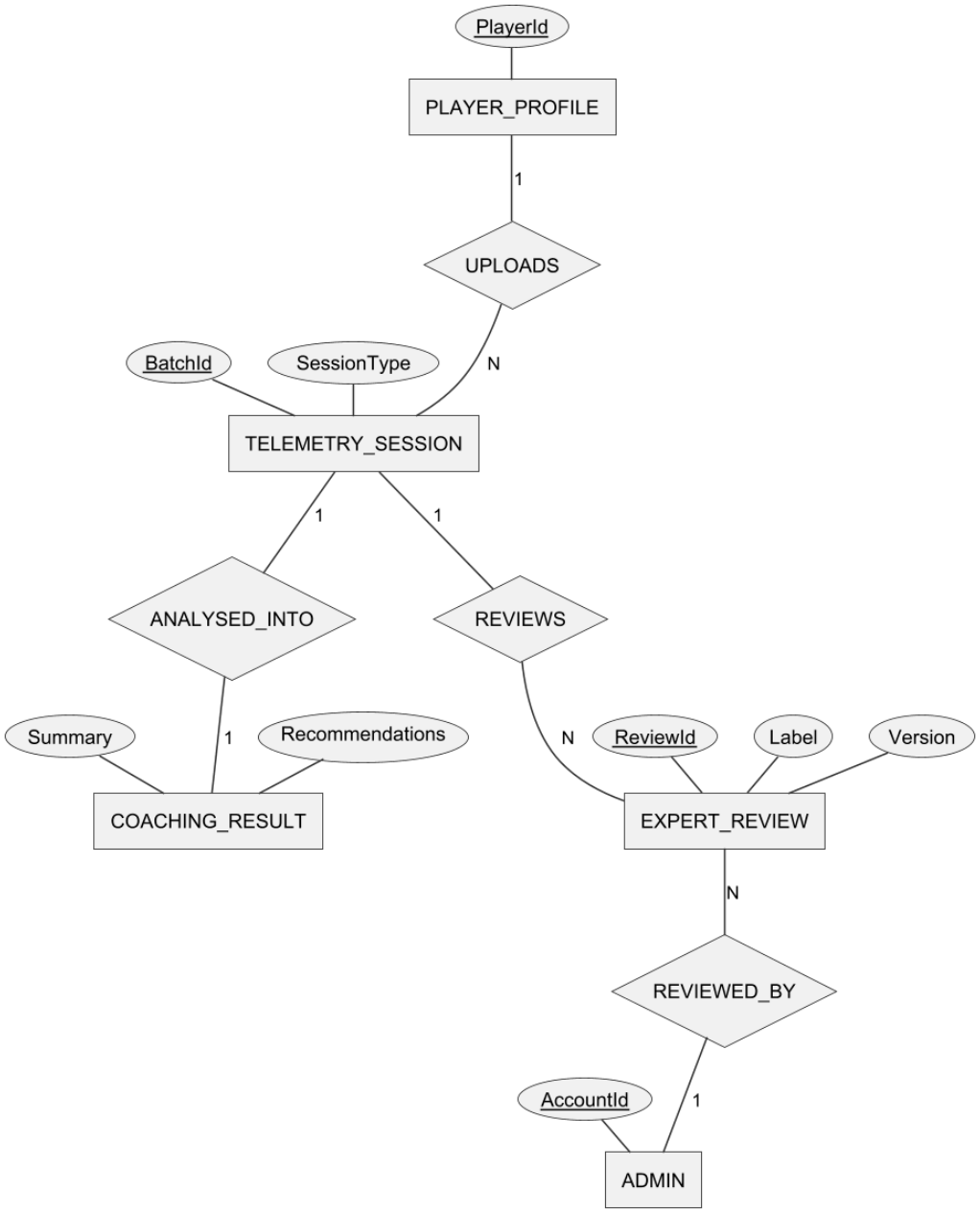


Figure 19. Conceptual ERD – telemetry, coaching and expert review

Entities Description

Table 7. Entities description

#	Entity	Description
1	ACCOUNT	A registered user identified by e-mail; holds the password hash, status (ACTIVE, SUSPENDED, BANNED) and verification state.
2	ROLE	Player or Admin; each account has exactly one role.
3	VERIFICATION_CODE	A one-time code for e-mail verification or password reset, with expiry and remaining attempts.
4	PLAYER_PROFILE	Game identity of an account: display name, character, level, XP, Elo, onboarding choice and statistics.
5	TECHNIQUE	A pickleball technique to be learnt (serve, drive, dink, volley, drop, reset, smash, footwork, approach, doubles coverage and transition, modifiers).
6	LESSON	A unit of the curriculum with code, group, order, target, reward, prerequisite and publication status.
7	DRILL	An exercise that practises one technique; a lesson contains one or more drills.

#	Entity	Description
8	TRAINING_ATTEMPT	One run of a drill by a player, with shots, hits, timing, stars and synchronisation receipt.
9	MATCH	A bot, ranked, matched doubles or custom match with mode, kind, authority mode and status.
10	MATCH_GAME	One game (1 to 3) of a match with the score of each team; a weak entity identified by match and game number.
11	RATING_CHANGE	The Elo before, change and after of one player caused by one ranked match.
12	REPLAY	The stored recording of a match (object key of its manifest).
13	PLAYER_REPORT	A complaint by one player about another player in a given match, with reason, status and decision.
14	SANCTION	A suspension (with expiry) or ban applied to an account, with reason, actor and revocation data.
15	APPEAL	A restricted player's written objection to one sanction, with status and decision.
16	CURRENCY	In-game currency: Tokenpad (earned by playing) and Elite Token.
17	COSMETIC	A purchasable or granted item (paddle, outfit, hat) with price and currency.
18	MISSION	A daily or weekly goal with target and reward.
19	ACHIEVEMENT	A long-term goal (wins, matches, points, streaks) with a one-time reward.
20	TELEMETRY_SESSION	The shot-level data captured during a drill or match and uploaded for analysis.
21	COACHING_RESULT	The AI analysis of a telemetry session: findings, summary, recommendations and optional tier.
22	EXPERT_REVIEW	An expert's labels for a telemetry session (level, errors, data group) used to evaluate the AI.
23	ADMIN	An account with the Admin role, shown as a separate box in the coaching diagram for readability.

3.2 Account and Onboarding (FE-01)

FR-01 Register Account and Verify Email

Related use cases: UC-01 Register Account, UC-02 Verify Email

Function trigger: Login screen → link "Đăng ký" (Register). The verification page also opens after a successful registration, or from the link "Bạn cần xác thực email?".

Function description: A visitor creates an account with an e-mail address and a password. The client validates the form, the Web API creates the account in the unverified state, stores a hash of a six-digit one-time password (OTP) and queues an e-mail through Gmail SMTP. The visitor enters the OTP to activate the account; after activation the account can sign in.

Screen layout

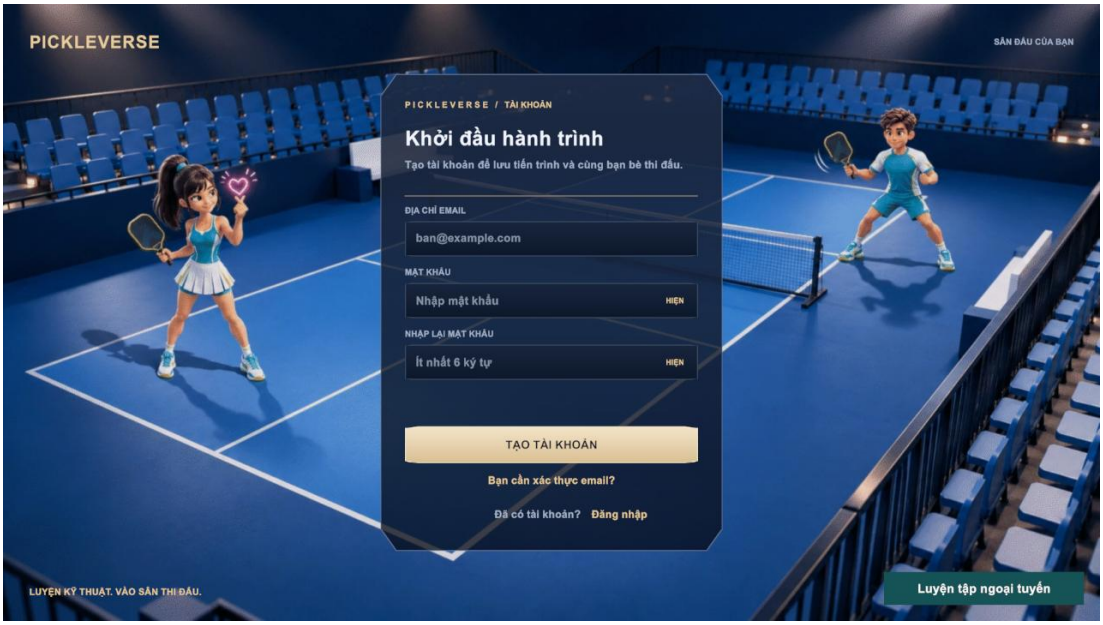


Figure 20. Register screen

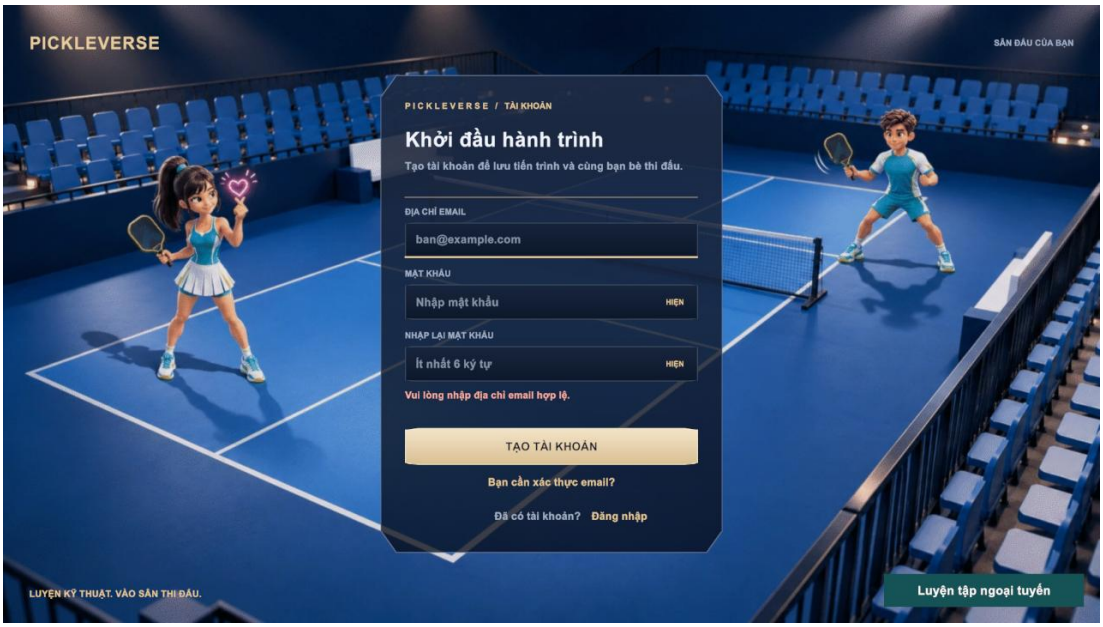


Figure 21. Register screen with an inline validation message

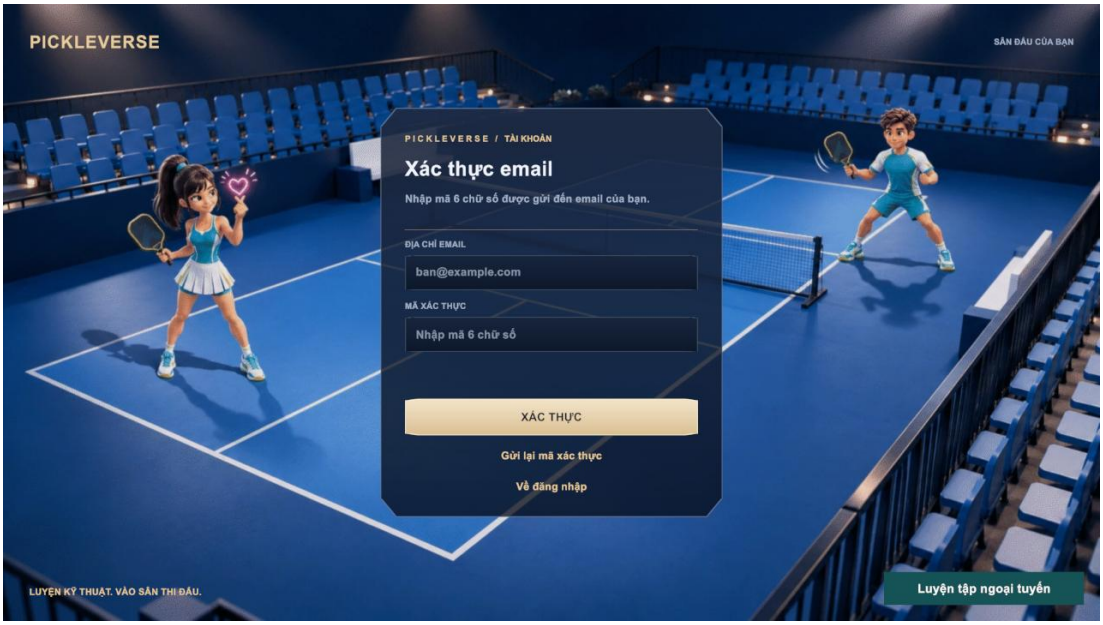


Figure 22. E-mail verification screen

Function details

Table 8. FR-01 data fields

Field	Type	Required	Validation and constraints
Email	Text (≤254)	Yes	Valid e-mail format (MSG01, MSG02); not already registered (MSG06)
Password	Password	Yes	6–128 characters (MSG03, MSG04, MSG17); show/hide toggle
Confirm password	Password	Yes	Must equal Password (MSG05)
Verification code	Text (6 digits)	Yes	Exactly six digits (MSG08); valid for 10 minutes; at most 5 wrong attempts

Business rules: BR-01, BR-02, BR-03, BR-04.

Normal flow:

1. The visitor enters e-mail, password and confirmation and selects **Tạo tài khoản**.
2. The client validates the fields; the API creates an ACTIVE but unverified account with the Player role and a player profile.
3. The API sends the OTP e-mail and the client shows MSG07 and the verification page.
4. The visitor enters the six-digit code and selects **Xác thực**.
5. The API checks the code, marks the e-mail as verified and the client returns to sign-in.

Alternative and exception flows:

- E1 – The e-mail is already registered: MSG06; an unverified owner can use **Gửi lại mã xác thực**.
- E2 – Resend is requested within 60 seconds of the previous code: HTTP 429 and MSG10.
- E3 – Wrong or expired code: the attempt is counted; after 5 wrong attempts the code is invalid and a new code must be requested.
- E4 – The e-mail cannot be delivered (SMTP not available): MSG22; an administrator can verify the account manually (FR-29).
- E5 – The e-mail is already verified: the client asks the visitor to sign in.

FR-02 Sign In and Sign Out

Related use cases: UC-03 Sign In, UC-05 Sign Out

Function trigger: Login screen → **Đăng nhập**; Settings → Overview → **Đăng xuất**. Token refresh runs automatically before the access token expires.

Function description: A visitor signs in with e-mail and password. The API verifies the BCrypt hash, issues a JWT access token (2 hours) and a refresh token (14 days, stored as a SHA-256 hash) and returns the profile summary. The client loads the profile, wallet, catalogue and settings on the Loading screen and opens the Menu. A restricted account receives a token limited to the appeal scope and is taken to Decisions and Appeals. Signing out revokes the refresh token on the server and clears the session on the device.

Screen layout

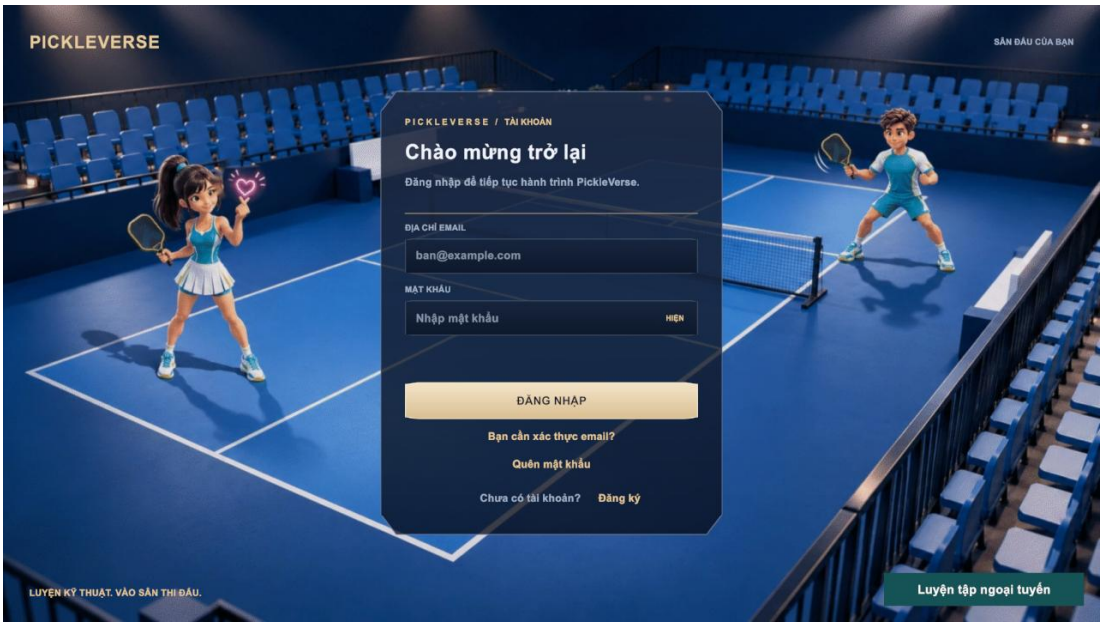


Figure 23. Login screen of the game client

Function details

Table 9. FR-02 data fields

Field	Type	Required	Validation and constraints
Email	Text	Yes	Not empty (MSG01), valid format (MSG02)
Password	Password	Yes	Not empty (MSG03)

Business rules: BR-04, BR-05.

Normal flow:

1. The visitor enters e-mail and password and selects **Đăng nhập**.
2. The API validates the credentials and returns tokens and the profile; the client shows MSG13 and the Loading screen.
3. On the first sign-in the client asks for the experience level (FR-04); otherwise it opens the Menu.
4. To sign out, the player opens Settings → Overview and selects **Đăng xuất**; the client returns to the Login screen.

Alternative and exception flows:

- E1 – Wrong e-mail or password: MSG12 (the message does not reveal which field is wrong).
- E2 – The e-mail is not verified: MSG11 and a link to the verification page.
- E3 – The account is suspended or banned: MSG14 and the Decisions and Appeals popup (FR-28).
- E4 – A system bot account tries to sign in: MSG65.
- E5 – The refresh token is expired or revoked: MSG18 and the Login screen.
- A1 – No network and a cached profile exists: the visitor can select **Luyện tập ngoại tuyến** (FR-09).

FR-03 Reset Password

Related use cases: UC-04 Reset Password

Function trigger: Login screen → **Quên mật khẩu**.

Function description: A visitor who has forgotten the password requests a reset code. To prevent account enumeration the API always answers with the same neutral message; a code is sent only when the address belongs to a verified account. The visitor enters the code and a new password; the API stores the new hash and revokes every refresh token of the account.

Screen layout

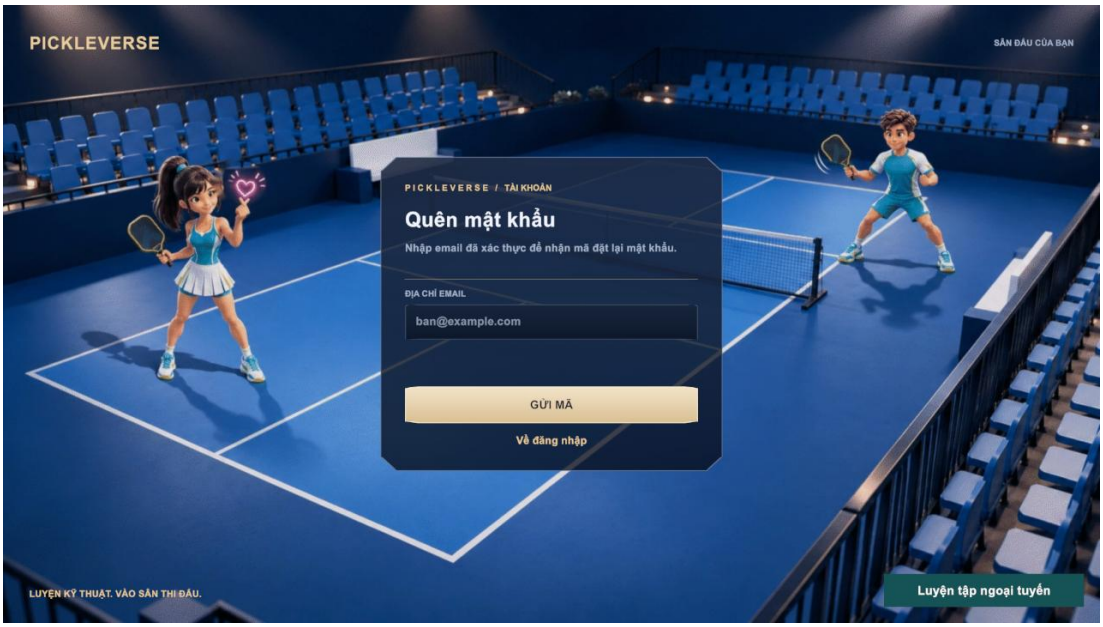


Figure 24. Forgot password screen

Function details

Table 10. FR-03 data fields

Field	Type	Required	Validation and constraints
Email	Text	Yes	Valid format (MSG02)
Reset code	Text (6 digits)	Yes	Valid for 10 minutes; at most 5 attempts (MSG16)
New password / confirmation	Password	Yes	6–128 characters; equal (MSG04, MSG05, MSG17)

Business rules: BR-02, BR-03, BR-05.

Normal flow:

1. The visitor enters the e-mail address and selects **Gửi mã**; the client shows MSG15.
2. The visitor enters the code, the new password and its confirmation.
3. The API changes the password, revokes all sessions and the client returns to sign-in.

Alternative and exception flows:

- E1 – Wrong or expired code: MSG16.
- E2 – Resend within 60 seconds: MSG10.

FR-04 Onboarding and Player Profile

Related use cases: UC-06 Choose Experience Level, UC-07 Update Player Name, UC-08 Select Character

Function trigger: First sign-in (experience popup); Menu → name tag or profile card edit icon (name); Menu → **Nam** / **Nữ** selector (character).

Function description: At the first sign-in the player answers MSG21. **Người mới** (Beginner) opens the tutorial; **Đã thành thạo** (Experienced) goes to the Menu. The answer is saved once. The player can later change the display name and choose the male or female character; the character is used in the menu, training and matches.

Screen layout



Figure 25. Menu (lobby) of a signed-in player

Function details

Table 11. FR-04 data fields

Field	Type	Required	Validation and constraints
Experience level	Choice	Yes (once)	Beginner or Experienced; cannot be changed afterwards
Display name	Text	Yes	3–18 characters, no control characters, unique (MSG19, MSG20)
Character	Choice	Yes	Male or Female; default Male

Business rules: BR-06, BR-07, BR-08.

Normal flow:

1. The player answers the experience question; the choice is saved.
2. The player edits the name and confirms; the API checks length and uniqueness and saves it.
3. The player selects **Nam** or **Nữ**; the menu character changes immediately and the choice is saved.

Alternative and exception flows:

- E1 – Name outside 3–18 characters: MSG19.
- E2 – Name already used: MSG20.
- E3 – The player is in a room, searching or in a match: profile changes are disabled until the activity ends.

FR-05 Configure Game Settings

Related use cases: UC-09 Configure Game Settings

Function trigger: Menu → **Cài đặt**, or Esc during a match or drill.

Function description: The player adjusts audio (music and effects volume), graphics (the quality options of the Graphics tab) and controls (keyboard bindings for PC and the touch layout for phones). Changes are previewed and stored on the device when the player selects **Áp dụng và lưu**.

Screen layout

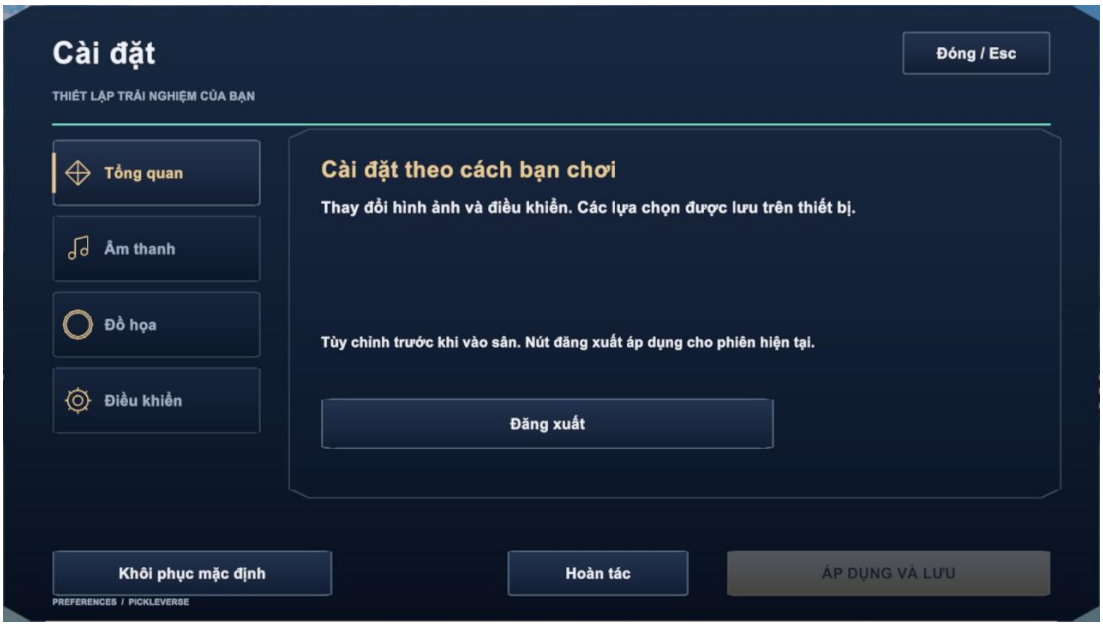


Figure 26. Settings – Overview tab

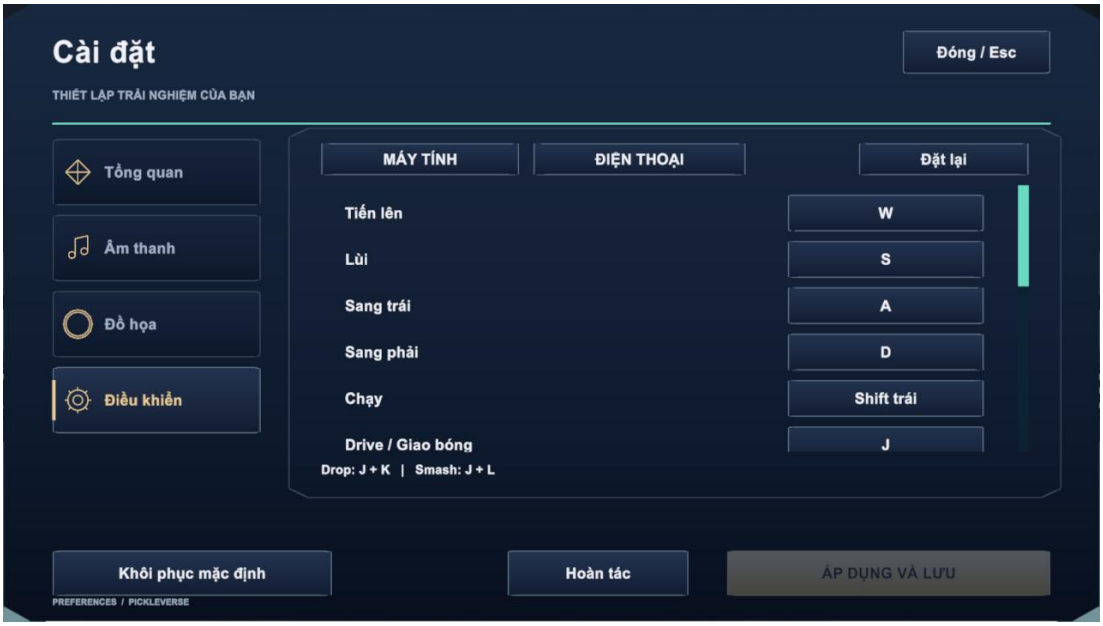


Figure 27. Settings – Controls tab with key bindings

Function details

Table 12. FR-05 data fields

Field	Type	Required	Validation and constraints
Music / effects volume	Slider 0–100	No	Default from the build
Graphics options	Choice	No	Applied after Apply and save
Key binding	Key	No	One key per action; Restore defaults resets all (defaults: W/A/S/D move, Left Shift run, J drive and serve, K dink, L volley, U power, I cross)

Business rules: BR-17.

Normal flow:

1. The player changes values; **Áp dụng và lưu** becomes active.
2. The player saves; settings apply immediately and persist on the device.

Alternative and exception flows:

- A1 – **Hoàn tác** restores the last saved values; **Khôi phục mặc định** restores factory values.

3.3 Tutorial and Technique Training (FE-02)

FR-06 Complete Tutorial

Related use cases: UC-10 Complete Tutorial

Function trigger: Automatically after choosing Beginner (FR-04); Mode Selection → **Tutorial / Hướng dẫn cơ bản**.

Function description: The tutorial teaches the court, the rules and the controls in 17 short steps (about 2–3 minutes). Each step shows a title, an explanation and a demonstration on the court, for example the court dimensions, a ball that does not cross the net, the two-bounce rule and the non-volley zone. Completion is saved to the profile.

Screen layout

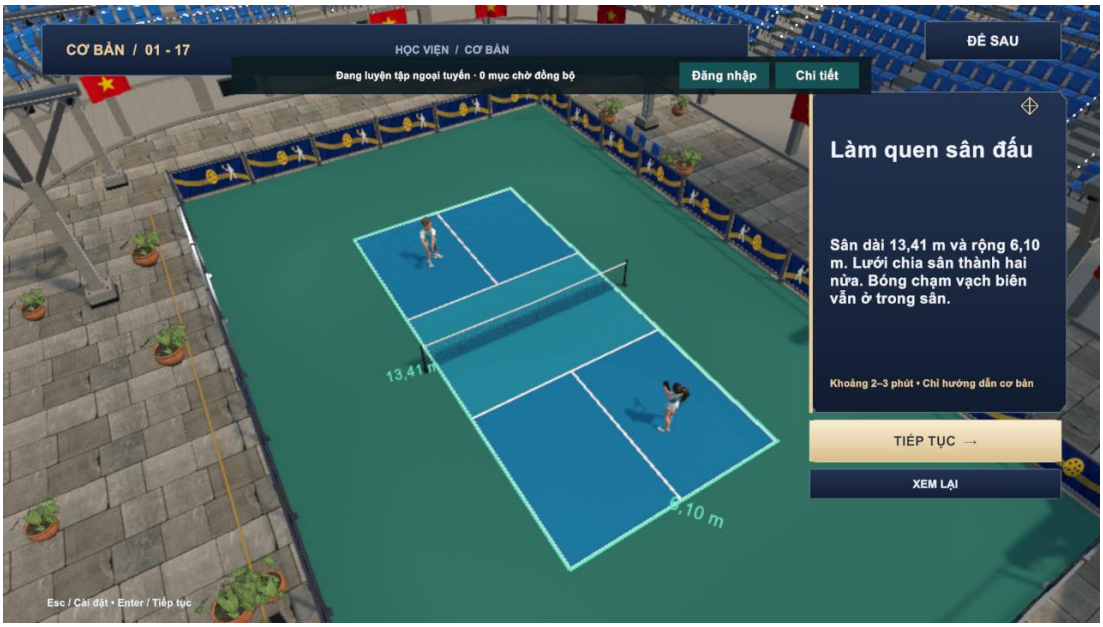


Figure 28. Tutorial step 1 – getting to know the court

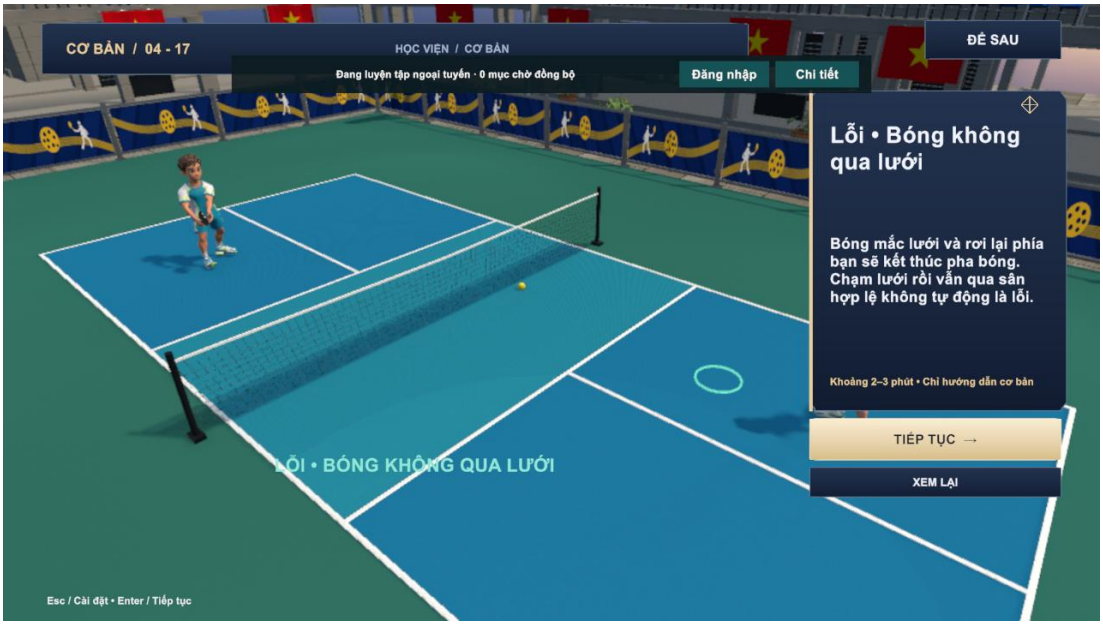


Figure 29. Tutorial step 4 – fault: the ball does not cross the net

Function details

Business rules: BR-09 to BR-14.

Normal flow:

1. The tutorial opens at step 1.
2. The player reads each step and selects **Tiếp tục** (Enter) to continue or **Xem lại** to replay the demonstration.
3. After step 17 the completion is saved and the Menu opens.

Alternative and exception flows:

- A1 – **Để sau** leaves the tutorial; it can be reopened from Mode Selection.

FR-07 Browse Lessons

Related use cases: UC-11 Browse Lessons

Function trigger: Mode Selection → **Luyện tập**; Training Progress → **Tiếp tục luyện**.

Function description: The academy lists the 15 published lessons grouped as Basic, Control, Advanced and Doubles. Each row shows the group, the first-pass reward, the best stars and whether the lesson is unlocked. Selecting a lesson opens its briefing: coach explanation, target, reward and the star rules.

Screen layout

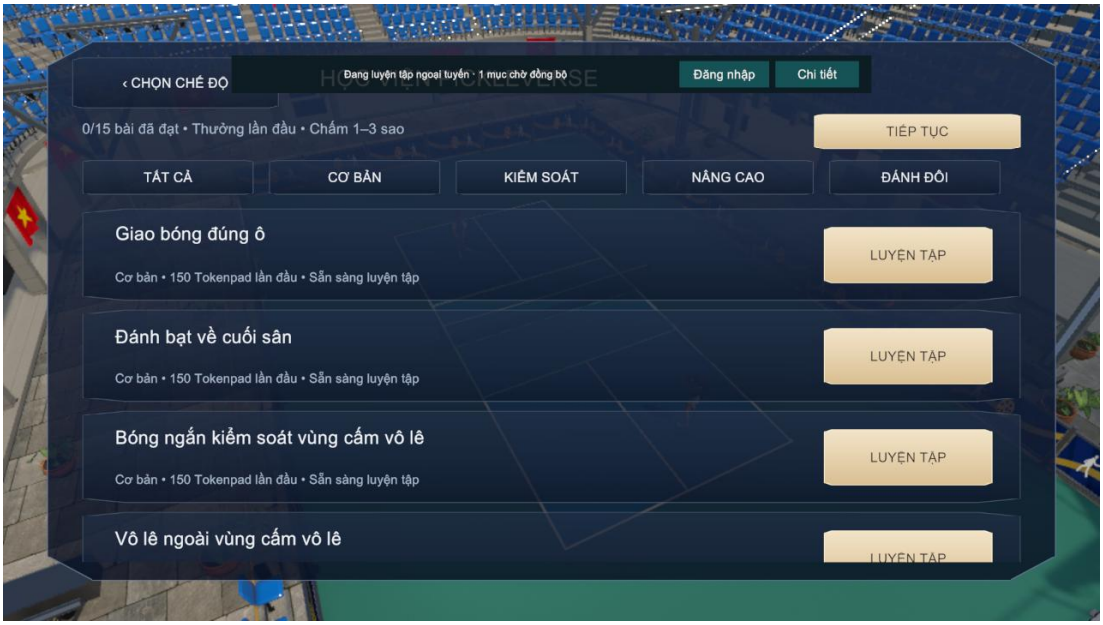


Figure 30. Lesson catalogue with group tabs

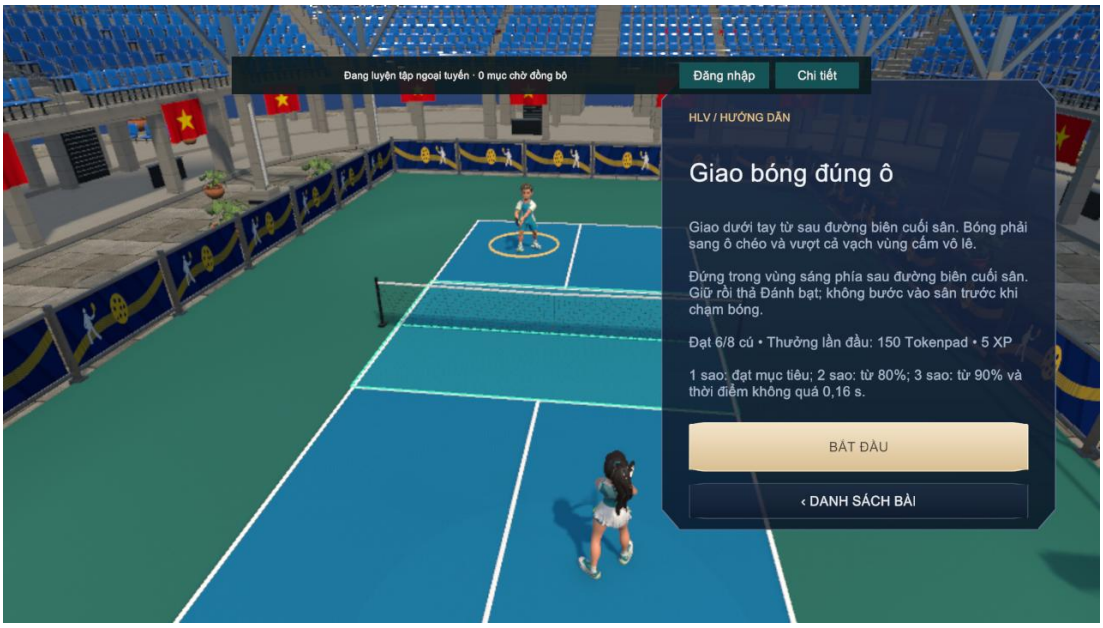


Figure 31. Lesson briefing for "Giao bóng đúng ô" (serve)

Function details

Business rules: BR-18, BR-19, BR-20.

Normal flow:

1. The player opens the academy; the catalogue loads from the server (or the cached catalogue when offline).
2. The player filters by group and selects **Luyện tập** on a lesson.
3. The briefing opens; **Bắt đầu** starts the drill (FR-08).

Alternative and exception flows:

- E1 – The lesson is locked: the start button is disabled and MSG23 is shown.

FR-08 Practise Drill

Related use cases: UC-12 Practise Drill

Function trigger: Lesson briefing → **Bắt đầu**; Drill Result → **Luyện lại**.

Function description: The player performs the drill on the training court, for example eight serves into the highlighted service court. For each shot the game measures whether the ball lands in the target area and the timing error between the key release and the ideal contact. In-drill feedback tells the player whether the release was early or late. At the end the client sends the attempt and its telemetry to the API, which validates it, applies the stars and the first-pass reward and returns a receipt.

Screen layout



Figure 32. Practice HUD of the volley drill: coach panel, feedback after a late hit ("Chưa đúng • Thử lại") and the shot cue L under the player

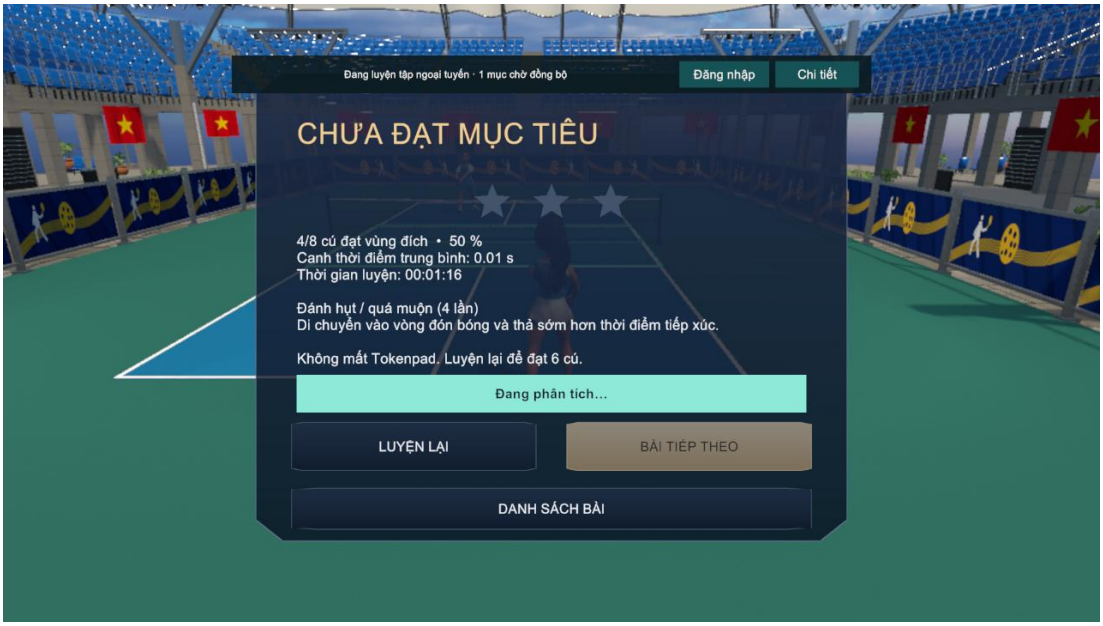


Figure 33. Drill result – target not reached

Function details

Table 13. FR-08 data fields

Field	Type	Required	Validation and constraints
Shots / target	Integer	System	Fixed per lesson (for example 6 of 8 for the serve lesson)
Hits in target	Integer	System	Counted by the rules engine
Mean timing error	Seconds	System	Absolute difference between release and ideal contact
Stars	0–3	System	Computed by BR-18; the best result is kept

Business rules: BR-16, BR-17, BR-18, BR-20.

Normal flow:

1. The drill starts and the HUD shows lesson, shot counter and key hints.
2. The player hits the required shots; feedback (MSG26, MSG27) appears after each shot.
3. After the last shot the Drill Result popup shows stars, accuracy, mean timing, time, the most frequent error and a tip.
4. The API stores the attempt; on the first pass it grants the lesson reward and unlocks the next lesson.
5. The coaching analysis appears when ready (FR-19).

Alternative and exception flows:

- A1 – Target not reached: MSG24 and MSG25; no currency is lost and **Bài tiếp theo** stays disabled.
- A2 – The device is offline: the attempt is queued for synchronisation (FR-09).
- E1 – The attempt is rejected by the server (for example inconsistent telemetry): receipt REJECTED and no reward.

FR-09 Offline Training and Synchronisation

Related use cases: UC-13 Practise Drill Offline, UC-14 Synchronise Offline Results

Function trigger: Login → **Luyện tập ngoại tuyến** when a profile is cached; automatic synchronisation when the connection returns, or banner → **Đăng nhập**.

Function description: A player who has signed in on the device at least once can train without a connection. The client uses the cached catalogue and profile, stores every finished attempt with a unique request ID in a local queue and shows the offline banner MSG28. When the player is online again, the queue is sent; the API applies each request at most once and answers APPLIED, DUPLICATE or REJECTED. Online-only features (bot and online matches, shop) are disabled offline.

Screen layout

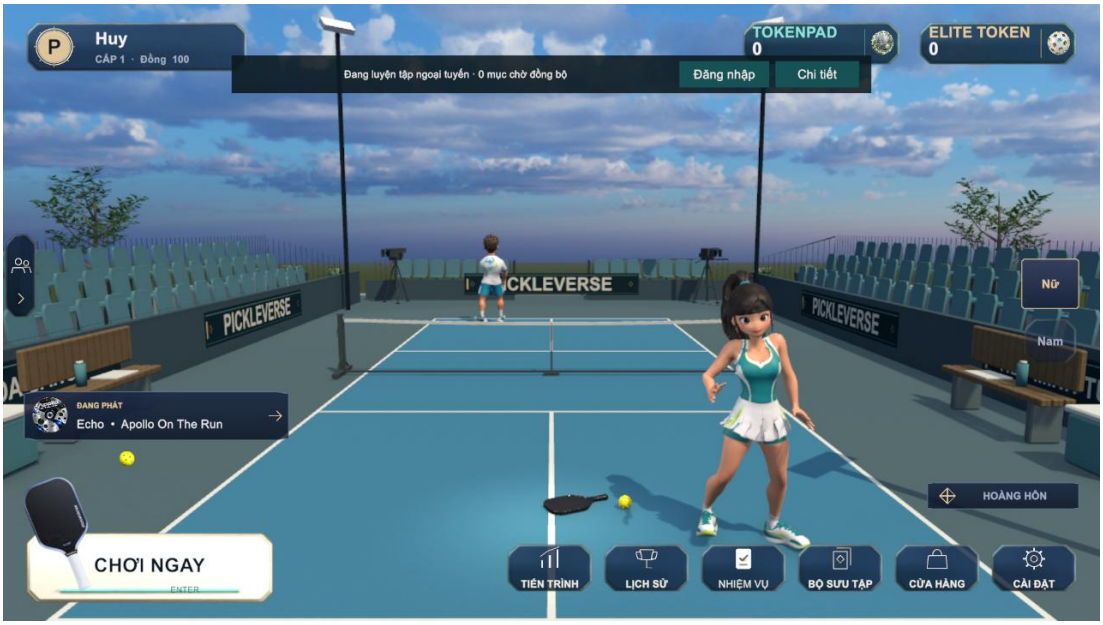


Figure 34. Menu in offline training mode with the offline banner

Function details

Business rules: BR-21.

Normal flow:

1. The player starts offline training from the Login screen.
2. Drills work as in FR-08; each result is added to the queue and the banner shows the pending count.
3. When connected, the client sends the queue; applied items are removed and rewards are added to the profile.

Alternative and exception flows:

- A1 – The same request is sent twice: receipt DUPLICATE; no second reward.
- E1 – No cached profile on the device: the offline button is not shown.
- E2 – A queued item is invalid: receipt REJECTED; the item is removed and the player is informed through **Chi tiết**.

FR-10 View Training Progress

Related use cases: UC-15 View Training Progress

Function trigger: Menu → **Tiến trình**.

Function description: The progress popup summarises the player's learning: passed lessons out of 15, the next lesson, the latest post-session feedback, total practice time, the most frequent error and the most retried lesson. For every lesson it shows the stars, the best accuracy and the four most recent accuracy results ("Gần đây"), so that the player can see the trend, with a Retry shortcut. Technique mastery is updated from the same attempts. Tabs give access to daily and weekly missions and achievements (FR-21).

Screen layout

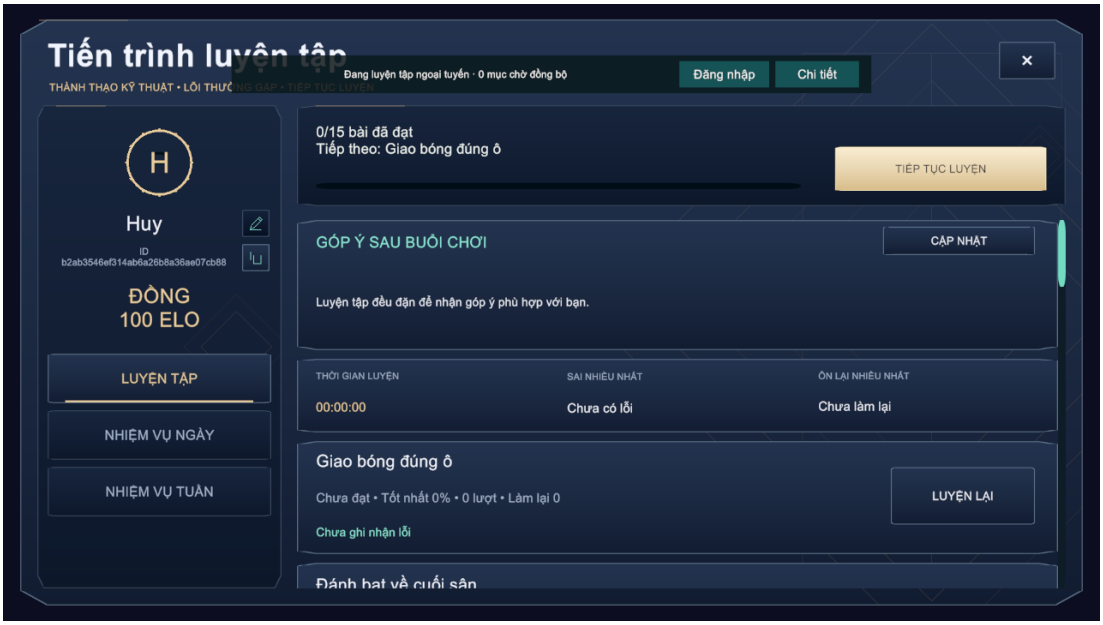


Figure 35. Training progress popup

Function details

Business rules: BR-18.

Normal flow:

1. The player opens Progress; the data is loaded from the profile.
2. The player selects **Luyện lại** on a lesson to open its briefing.

3.4 Core Gameplay and Bot Practice (FE-03)

FR-11 Control the Player and Hit Shots

Related use cases: UC-12, UC-16, UC-25

Function trigger: Any rally in a drill, bot match or online match.

Function description: On PC the player moves with W/A/S/D, runs with Left Shift and hits by holding and releasing shot keys. One of seven shot kinds is chosen from the key combination when the first key is released; U adds power and I aims cross-court. On phones the left joystick moves the player, the CHẠY button is held to run, and the DRIVE, DINK and VOLLEY buttons are held and released; dragging a held button up turns DRIVE into SMASH, DINK into DROP and VOLLEY into RESET, dragging left aims cross-court (CHÉO), dragging right adds power (LỰC) and dragging down cancels (HỦY); the serve uses DRIVE. The paddle contact is accepted only inside the timing window, and the quality of the contact (Perfect, Good, early or late) affects the ball speed and accuracy. The ball flight, bounce and net contact are simulated at 120 Hz. The camera stays behind the player's baseline and keeps the player above the shot cue (PC) or the touch controls (phone).

Screen layout



Figure 36. Bot match: the player is serving; the shot cue under the player shows the key combination J + I (cross-court serve)

Table 14. FR-11 screen layout

Area	Elements and behaviour
Score panel (top left)	Mode and difficulty, game number, games and points of each side, serving side and serve timer
Rule hint (left)	What the player must do now, for example "Bạn giao • giữ rồi thả J" and the score to call
Controls (top right)	Settings (Esc) and help (F1)
Court	Player character, opponent, ball, landing marker and fault banner
Shot cue (bottom centre)	Key combination and name of the shot being prepared; on phones the joystick and shot buttons take this area

Function details

Business rules: BR-16, BR-17.

Normal flow:

1. The player moves to the ball and holds a shot key (J, K or L, alone or combined).
2. The player releases the key close to the contact moment; the game selects the shot kind and checks the timing.
3. The ball is hit towards the target computed from the shot kind and modifiers.

Alternative and exception flows:

- A1 – Release outside ±65 ms of the ideal contact: no contact (miss) and feedback MSG26 or MSG27.
- A2 – Invalid key combination: the input is ignored until all keys are released.

FR-12 Apply Singles Match Rules

Related use cases: UC-16, UC-25

Function trigger: Every rally of a bot or online match.

Function description: The rules engine decides legal serves, faults, points, games and the match winner according to the official singles rules simplified for the game. In online matches the same engine runs on the dedicated server, which is the only authority for the score.

Screen layout

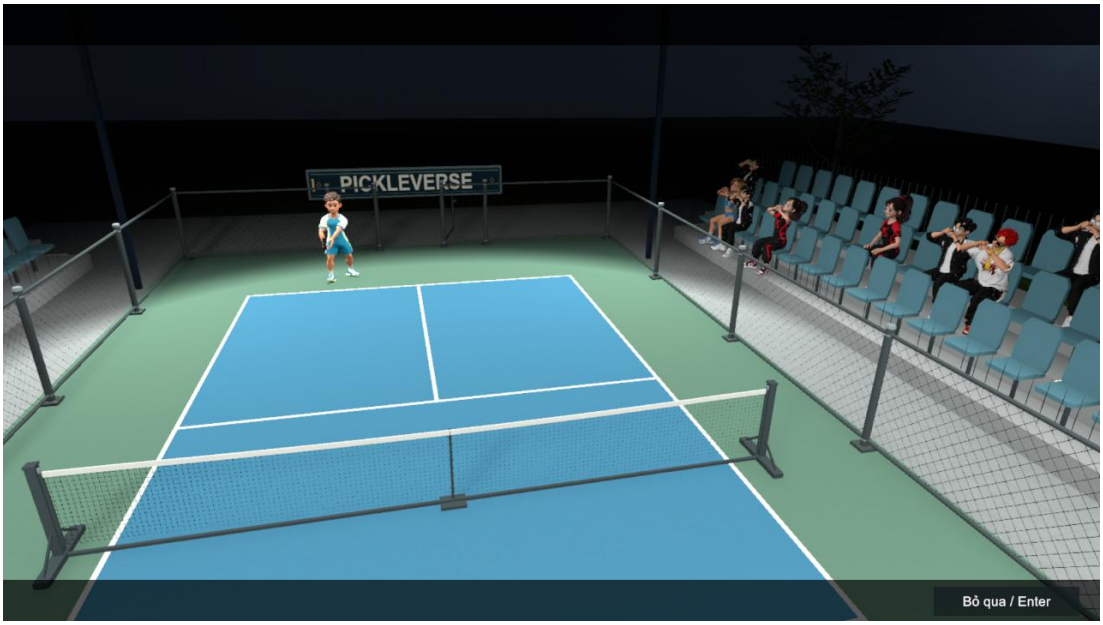


Figure 37. Bot match introduction

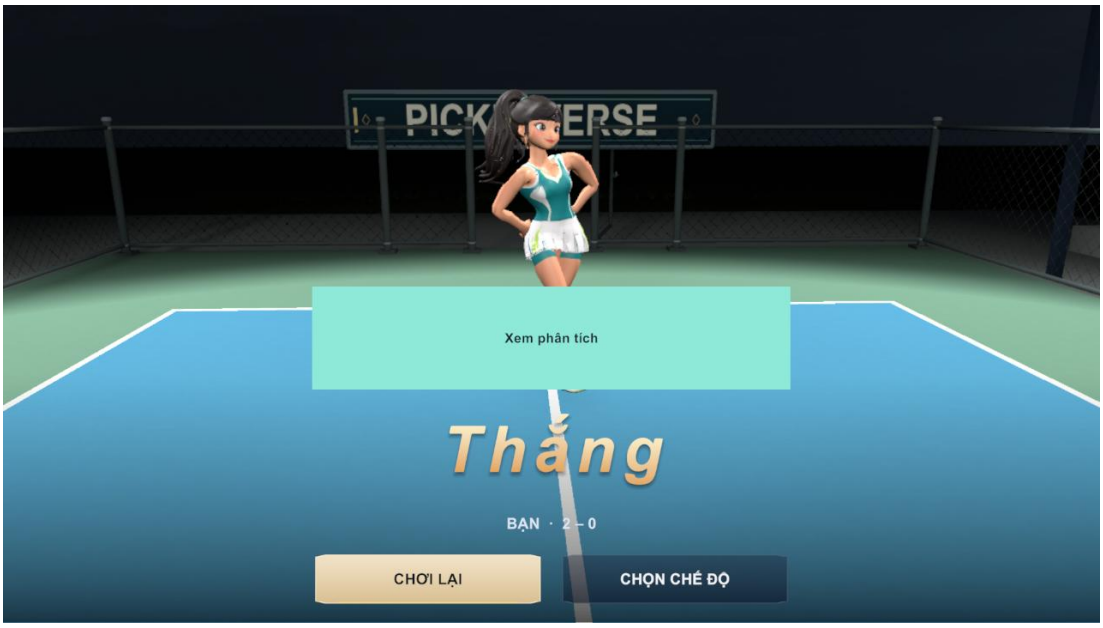


Figure 38. Match result – victory 2–0

Function details

Business rules: BR-09 to BR-15.

Normal flow:

1. The server of the rally serves underhand from behind the baseline into the diagonal service court within 10 seconds.
2. The return and the third shot must bounce; afterwards volleys are allowed outside the non-volley zone.
3. When a fault occurs, the rally ends, a fault banner (MSG30) is shown and the point or side-out is applied.
4. A game ends at 11 points with a 2-point lead; the first side to win two games wins the match (MSG31).

Alternative and exception flows:

- A1 – In the third game the players change ends when a side reaches 6 points.
- E1 – The serve timer expires: fault ServeTimeout and MSG29.

FR-13 Set Up and Play a Bot Match

Related use cases: UC-16 Play Bot Match

Function trigger: Menu → **Chơi ngay** → Mode Selection → **Đấu với bot** → Bot Match Setup → **Vào sân**.

Function description: The player chooses the difficulty, the time of day and the court, then plays a singles match against the computer. Easy, Medium and Hard use fixed reaction and accuracy settings; the adaptive option uses a behaviour-tree bot whose tier follows the player's Elo. After the match the result popup shows the winner, scores and rewards, and the coaching analysis can be opened.

Screen layout

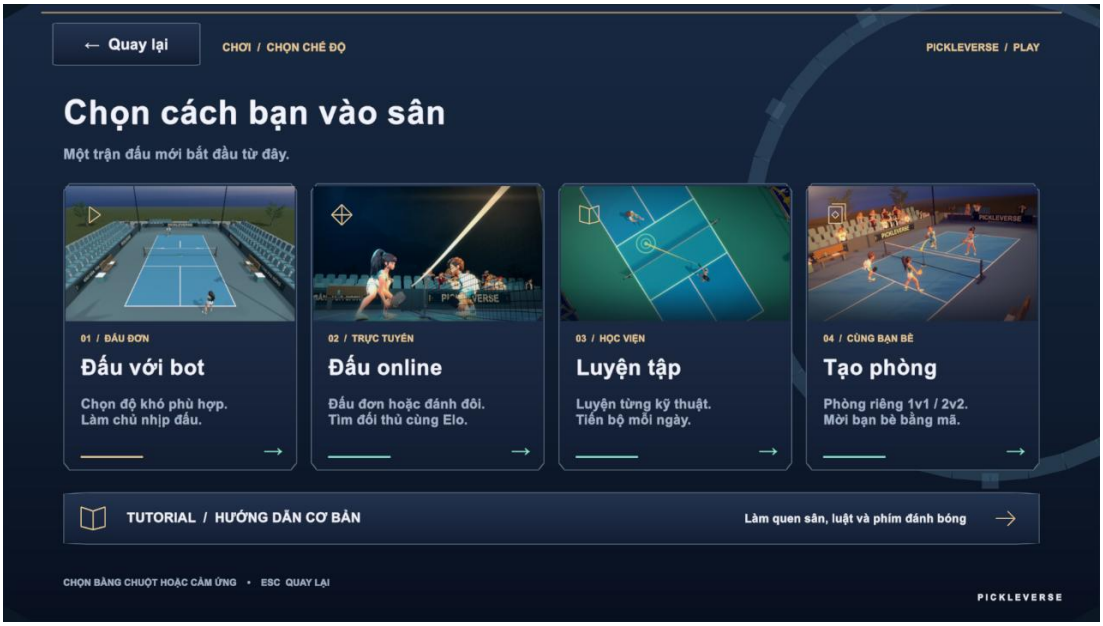


Figure 39. Mode selection

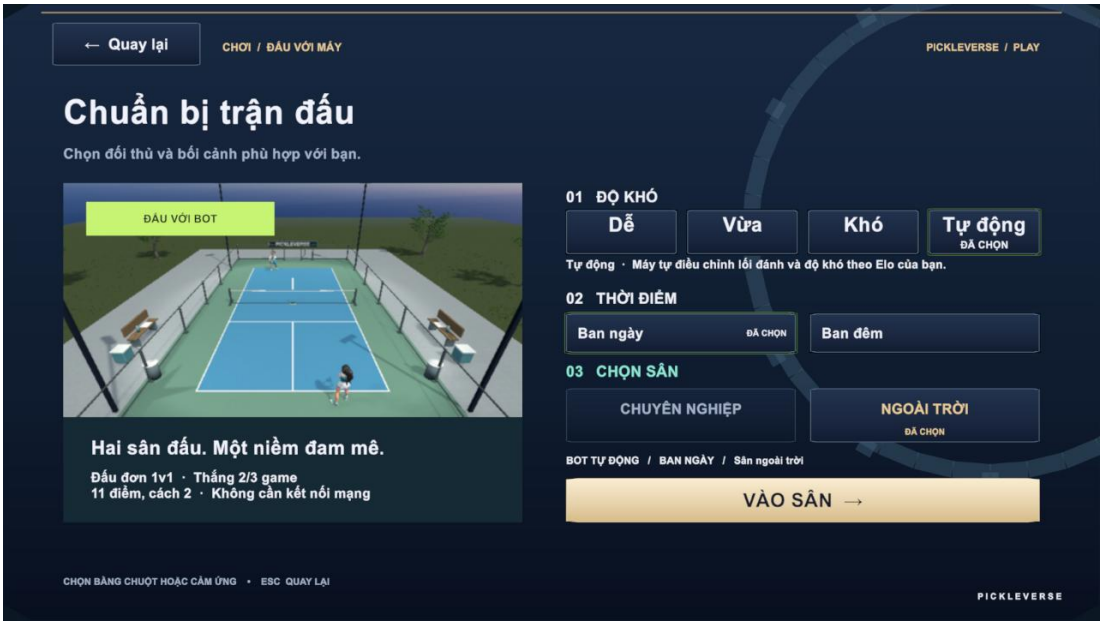


Figure 40. Bot match setup

This function uses the screen already shown in Figure 38; no separate screenshot is repeated here.

Function details

Table 15. FR-13 data fields

Field	Type	Required	Validation and constraints
Difficulty	Choice	Yes	Easy, Medium, Hard or Adaptive
Time of day	Choice	Yes	Day or night
Court	Choice	Yes	Professional or outdoor

Business rules: BR-10 to BR-16, BR-22, BR-31, BR-32.

Normal flow:

1. The player selects the options and **Vào sân**; the intro cinematic plays.
2. The match is played with FR-11 and FR-12.
3. At the end the API settles the result: rewards and XP are added; Elo does not change in a practice bot match.
4. The player selects **Chơi lại**, **Chọn chế độ** or **Xem phân tích**.

Alternative and exception flows:

- A1 – Ranked Bot match assigned by matchmaking (FR-14): Elo changes according to BR-23.
- E1 – The device is offline: the bot match card is disabled.

3.5 Online Matches (FE-04)

FR-14 Find an Online Match

Related use cases: UC-19 Find Online Match, UC-20 Confirm Match Readiness

Function trigger: Mode Selection → **Đấu online** → Online Lobby → **Đấu đơn 1v1** (ranked singles) or **Đấu đôi 2v2** (quick doubles) → **Tìm trận**.

Function description: Ranked singles: the player joins the ranked queue; the matchmaking worker pairs two searching players whose Elo is within the pairing window, and if no human opponent is found within 180 seconds a ranked bot opponent is assigned. **Quick doubles:** a player searches alone or the captain of a party of two (formed by room code or invitation) searches for both; the API fills two teams of two from solo players and parties in the same region, without Elo pairing and without a bot; the

match is unrated. In both modes every player must accept within 10 seconds. When all players accept, the API allocates the dedicated server and issues signed tickets, and the clients load the multiplayer court.

Screen layout

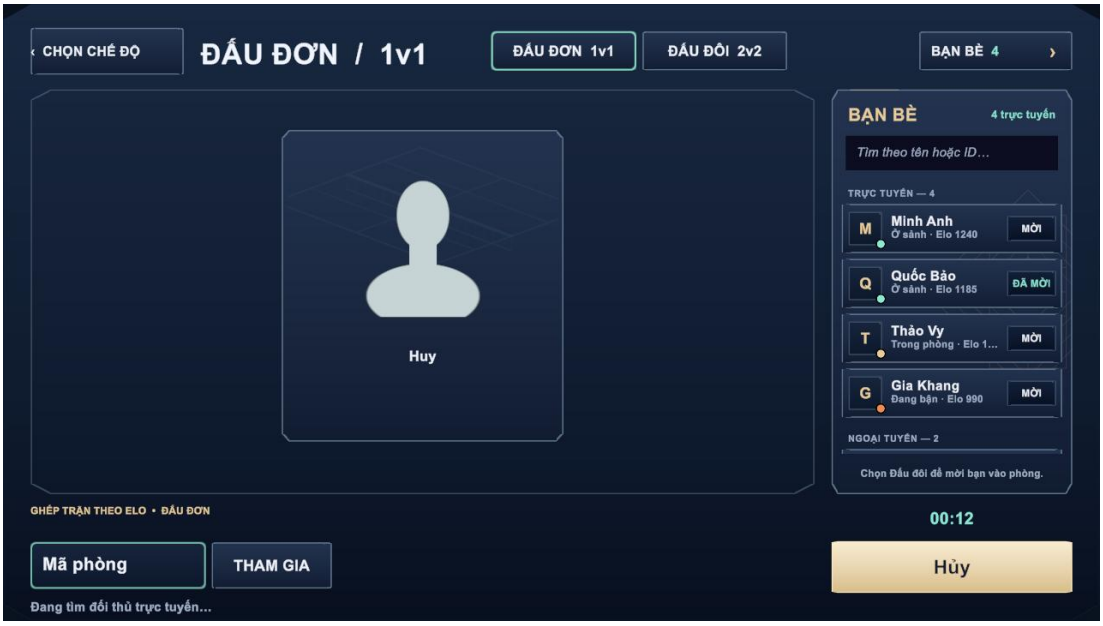


Figure 41. Searching for a singles opponent with the elapsed time (layout preview rendered in the Unity Editor)

Function details

Business rules: BR-22 to BR-27.

Normal flow:

1. The player selects **Tìm trận**; MSG32 and a timer are shown.
2. A pair (or two full teams for doubles) is found; the Ready Check shows MSG33 with a 10-second countdown.
3. All players accept; the server allocates the match and the court loads (FR-16).

Alternative and exception flows:

- A1 – Ranked singles only: no opponent within 180 seconds: a ranked bot is assigned and the match starts on the dedicated server.
- A2 – Quick doubles: no two full teams are found before the search lease ends: MSG34; there is no bot fallback.
- A3 – The player selects **Hủy** while searching: the ticket is cancelled.
- E1 – A player declines or the countdown ends: the offer closes; the decliner sees MSG35 and the other player returns to the queue.
- E2 – The dedicated server is busy: MSG36 and the request waits for the slot.
- E3 – The player already has a search or an unfinished match: MSG41.
- E4 – Search expired: MSG34.

FR-15 Private Rooms and Invitations

Related use cases: UC-21 Create Private Room, UC-22 Join Room by Code, UC-23 Invite Friend to Room, UC-24 Respond to Room Invitation

Function trigger: Mode Selection → **Tạo phòng**; Online Lobby → **Mã phòng** + **Tham gia bằng mã**; friend list → invite.

Function description: A player creates a singles (2 players) or doubles (up to 4 players) room. Other players join with the room code or accept an invitation sent to an online friend. The collapsible friends column of the lobby lists the friends who can be invited; only friends who are in the lobby receive invitations. The host can change teams, remove a guest and start the match. Custom matches are unrated and give no Elo. A doubles room of two players can also be used as a party for quick doubles (FR-14).

Screen layout

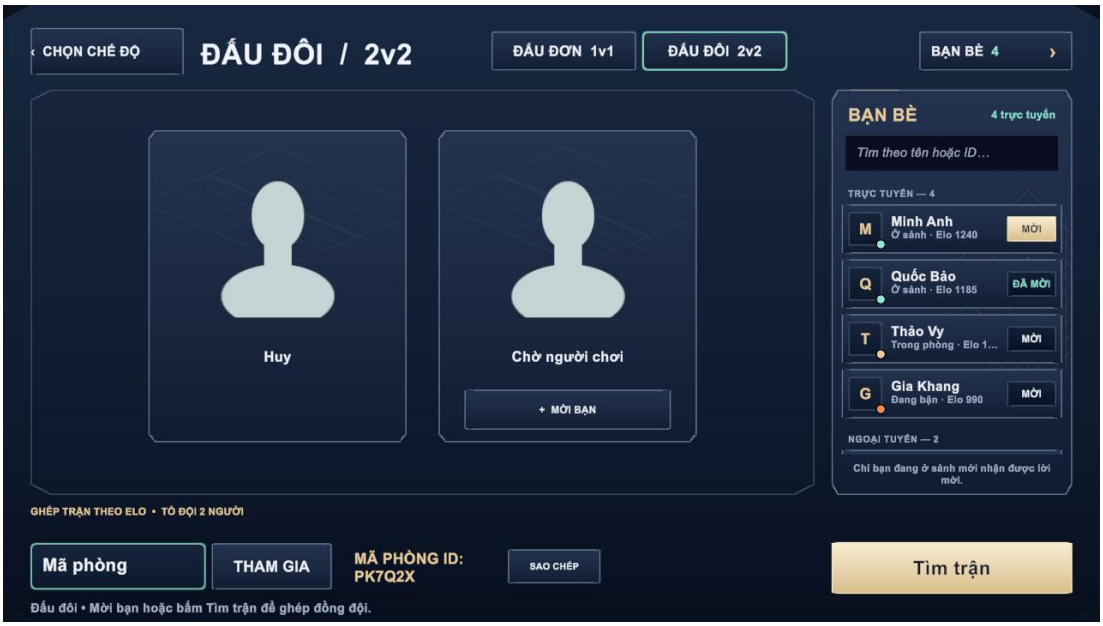


Figure 42. Online lobby – doubles room with the room code and the friends column (layout preview rendered in the Unity Editor with sample friend data)

Function details

Table 16. FR-15 data fields

Field	Type	Required	Validation and constraints
Room code	Text	Yes (to join)	Code of an open room on Photon; wrong code → MSG39
Mode	Choice	Yes	Singles 1v1 or Doubles 2v2

Business rules: BR-22, BR-30, BR-38.

Normal flow:

1. The host creates the room; the lobby shows the room code and roster slots.
2. Guests join by code or by accepting an invitation.
3. The host starts the match; the dedicated server is allocated as in FR-14.

Alternative and exception flows:

- E1 – Room full: MSG37.
- E2 – Room closed or the player was removed: MSG38.
- E3 – Invited player is busy or offline: MSG40.
- E4 – Blocked players cannot invite each other (BR-38).

FR-16 Play an Online Match

Related use cases: UC-25 Play Online Match

Function trigger: Automatically after the ready check (FR-14) or when the host starts a room (FR-15).

Function description: Each client connects to the dedicated server with its signed ticket through Photon Fusion. The server simulates the ball and applies the rules (FR-12); clients send only input and render server state with interpolation. The server sends heartbeats to the API every 8 seconds, records telemetry and the replay, and reports the final result.

Screen layout



Figure 43. Multiplayer court scene

Function details

Business rules: BR-12 to BR-17, BR-28, BR-29.

Normal flow:

1. The court loads and the server starts the match after all players are connected.
2. Players play rallies; scores are shown from the server state.
3. The server ends the match and the result popup appears (FR-18).

Alternative and exception flows:

- E1 – Photon cannot be reached: MSG45.
- E2 – The connection to the match server is lost: MSG42 and FR-17 starts.

FR-17 Reconnect to or Leave a Match

Related use cases: UC-26 Reconnect to Match, UC-27 Leave Match

Function trigger: Connection loss during an online match; pause menu → leave.

Function description: If a player's connection drops, the client shows a reconnecting overlay and retries for up to 30 seconds with the same identity. The dedicated server pauses the match and keeps the player's slot for 30 seconds. A player who reconnects in time continues with the server score. A player who leaves, or who does not return within 30 seconds, is recorded as having forfeited and the server ends the match as interrupted; interrupted matches change no Elo and give no reward. If the dedicated server itself is lost, its lease expires and the API ends the match as interrupted (BR-28).

Screen layout

Table 17. FR-17 screen layout

Area	Elements and behaviour
Reconnecting overlay	Status text and countdown; the court stays visible in the background
Leave dialog	Warning about the consequence and Confirm / Cancel buttons

Function details

Business rules: BR-28, BR-29.

Normal flow:

1. The client detects the loss and starts reconnecting.
2. The player rejoins with the same ticket identity and control is restored.

Alternative and exception flows:

- E1 – Reconnection fails within 30 seconds: MSG43; the result is shown when the server finishes processing.
- A1 – The player confirms leaving: MSG44 and the Menu opens.

FR-18 Record the Match Result and Update Elo

Related use cases: UC-16, UC-25 (non-screen settlement)

Function trigger: End of a bot match (reported by the client) or of an online match (reported by the dedicated server).

Function description: The API settles each match exactly once in a single transaction: it stores the game scores, marks the match FINISHED, applies Elo for ranked matches, grants Tokenpad and XP, and updates missions, achievements and streaks. The result popup and the history (FR-24) show the outcome.

Screen layout

This function uses the screen already shown in Figure 38; no separate screenshot is repeated here.

Function details

Business rules: BR-22, BR-23, BR-29, BR-32 to BR-35.

Normal flow:

1. The API receives the result with the match identifier.
2. It validates that the match is still PLAYING and that the reporter is allowed to report it.
3. It saves scores, Elo, rewards and progress, then returns the summary to the clients.

Alternative and exception flows:

- A1 – The result is sent again: the stored result is returned and nothing is applied twice.
- E1 – The match was stopped by an administrator before the result: no Elo or reward (BR-48).

3.6 AI Coaching (FE-05)

FR-19 View Coaching Summary and Drill Recommendations

Related use cases: UC-17 View Coaching Summary, UC-18 View Drill Recommendation

Function trigger: Drill Result → analysis panel; Match Result → **Xem phân tích**.

Function description: The client uploads the shot telemetry of the drill or match (for online matches the dedicated server uploads it). The hybrid AI worker sends the session to the AI coaching service, which detects common errors with rules (for example late contact, dink too high, entering the non-volley zone after a volley), writes a short summary and recommends at most three drills. When the gated tier model is approved, it also estimates the player's tier. The coaching popup shows the counts of shots, contacts and serves, the main errors with their frequency, and the recommended drills.

Screen layout

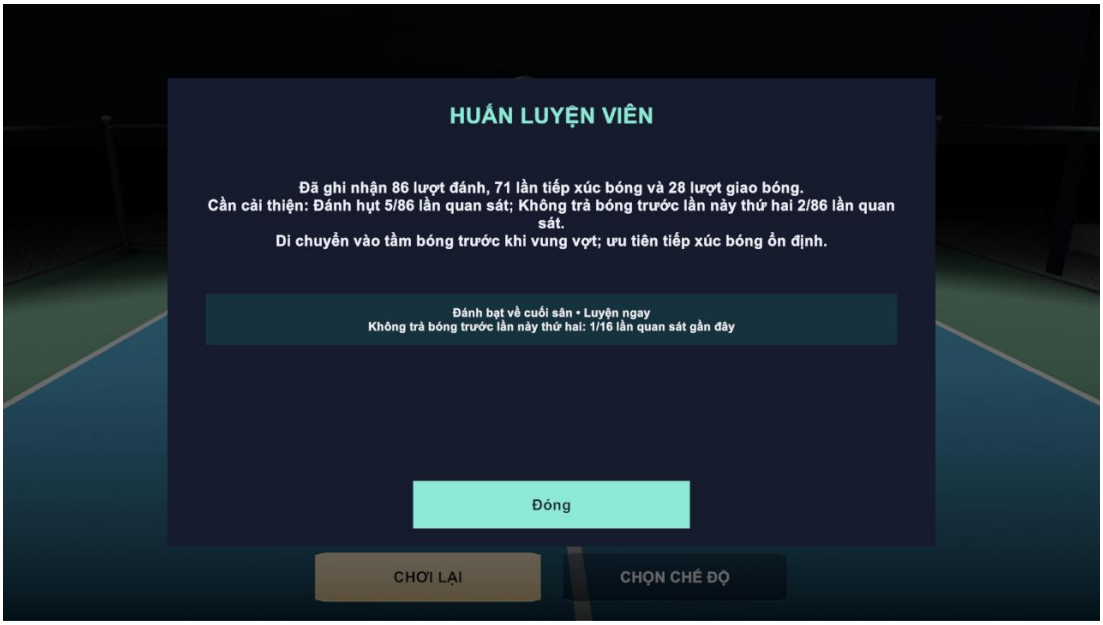


Figure 44. Coaching summary after a bot match

Function details

Business rules: BR-50, BR-51.

Normal flow:

1. The player opens the analysis; the client polls the analysis job.
2. The summary and recommendations appear when the job is completed.
3. **Luyện ngay** opens the recommended lesson briefing.

Alternative and exception flows:

- A1 – The analysis is still running: the panel shows "Đang phân tích…".
- E1 – Too few shot opportunities: no finding is reported for that error.
- E2 – AI service unavailable: the job is retried by the worker; the player can open the analysis later.

3.7 Progress, Rewards and Shop (FE-06)

FR-20 Earn Experience, Levels and Progression Rewards

Related use cases: UC-15, UC-16 (non-screen progression)

Function trigger: Settlement of a drill (FR-08) or match (FR-18); claiming mission rewards (FR-21).

Function description: Players earn XP from matches, lessons and missions. Reaching the XP threshold raises the level, which is shown in the name tag and history card. Two items are granted automatically once: the THOT paddle after 10 passed lessons and the GEMSTONE paddle at level 25.

Screen layout

This function uses the screen already shown in Figure 25; no separate screenshot is repeated here.

Function details

Business rules: BR-20, BR-32, BR-33, BR-37.

Normal flow:

1. XP is added by the settlement.
2. If XP reaches level × 200, the level increases and the remaining XP carries over.
3. Progression grants are checked and recorded in the reward log so that each is given once.

FR-21 Missions, Streak and Achievements

Related use cases: UC-37 to UC-40

Function trigger: Menu → **Nhiệm vụ**; Progress → **Nhiệm vụ ngày / tuần**.

Function description: Seven daily and ten weekly missions track matches, wins, points, valid returns and serves, bot matches and human matches. Completing all daily or weekly missions unlocks a set bonus. The daily streak counts consecutive days with at least one match; it has no reward of its own but is rewarded through the weekly mission for a 3-day streak (350 Tokenpad, 20 XP) and the achievement for a 5-day streak (500 Tokenpad, 30 XP). Ten achievements reward long-term goals such as 5, 25 and 100 wins.

Screen layout

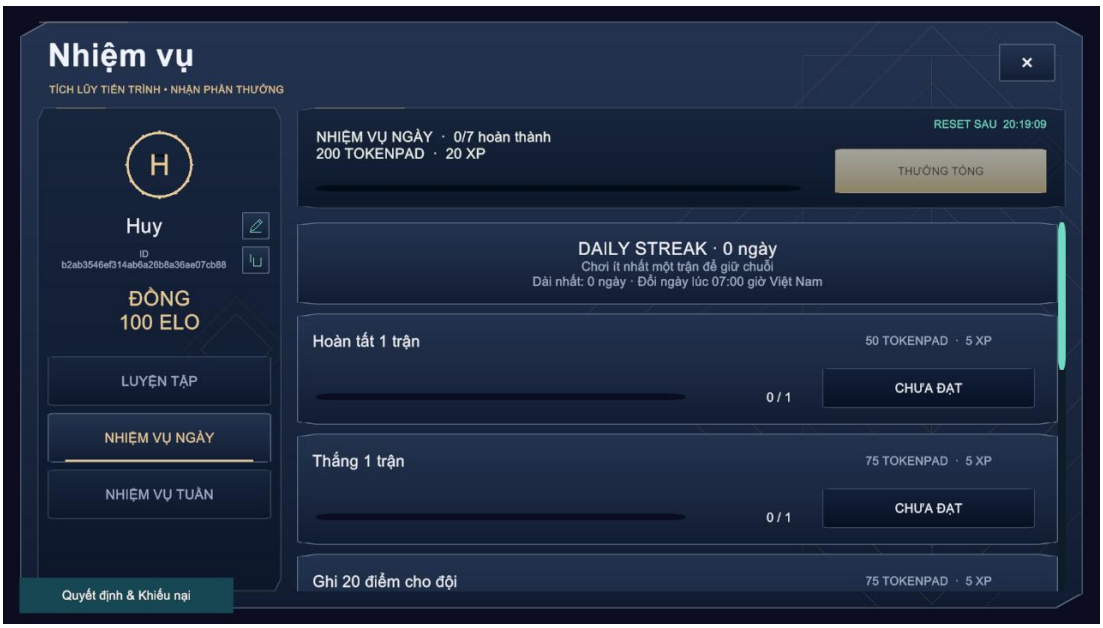


Figure 45. Missions popup with daily streak and reset countdown

Function details

Business rules: BR-34, BR-35.

Normal flow:

1. The player opens Missions; progress and the reset countdown are shown.
2. A completed mission shows **NHẬN THƯỞNG** (or **NHẬN THƯỞNG TỔNG** for the set bonus); the player claims it and the reward is added.
3. Achievement rewards are claimed in the same way (MSG49).

Alternative and exception flows:

- A1 – A reward is claimed twice: the second claim adds nothing.
- A2 – Missions renew at 07:00 Vietnam time; unclaimed progress of the old cycle is closed.

FR-22 Browse Shop and Purchase Cosmetic

Related use cases: UC-41 Browse Shop, UC-42 Purchase Cosmetic

Function trigger: Menu → **Cửa hàng**.

Function description: The shop lists cosmetic paddles with preview, description and price. An item can have a Tokenpad price, an Elite Token price or both. The purchase debits the wallet and adds the item to the collection in one transaction.

Screen layout



Figure 46. Shop with featured GEMSTONE and THOT paddles

Function details

Business rules: BR-36.

Normal flow:

1. The player selects an item and its currency and confirms the purchase.
2. The API checks ownership and balance, debits the price and grants the item.
3. The wallet and the collection are refreshed.

Alternative and exception flows:

- E1 – Not enough balance: MSG46.
- E2 – Item already owned: MSG47.
- E3 – Transaction failure: no balance is changed.

FR-23 View Collection and Equip Cosmetic

Related use cases: UC-43 View Collection, UC-44 Equip Cosmetic

Function trigger: Menu → **Bộ sưu tập**.

Function description: The collection shows all paddles with their status: in use, owned or not owned (preview only). The player equips an owned paddle, which is then used in matches and replays.

Screen layout

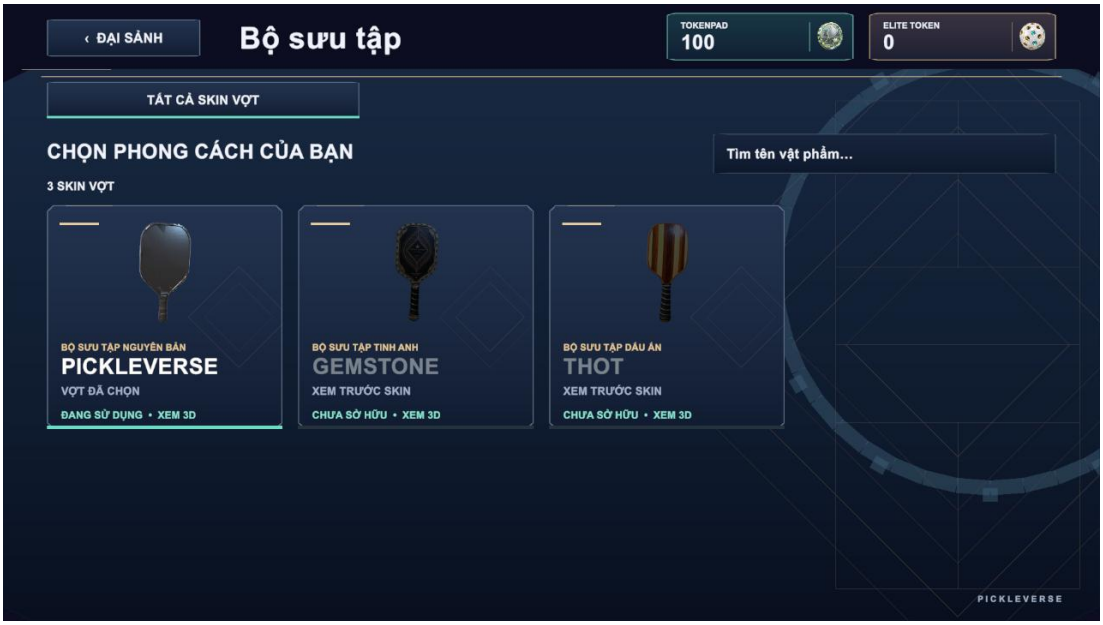


Figure 47. Collection of paddle skins

Function details

Business rules: BR-36.

Normal flow:

1. The player selects an owned paddle and **TRANG BỊ**.
2. The API marks it as the only equipped item; the menu character shows it.

Alternative and exception flows:

- E1 – The item is not owned: MSG48.

3.8 Social, History and Ranking (FE-07)

FR-24 View Match History and Watch Replay

Related use cases: UC-28 View Match History, UC-29 Watch Match Replay

Function trigger: Menu → **Lịch sử**; history entry → replay button.

Function description: The history popup lists the player's last 20 matches in Vietnam time with opponent, mode, result (win, loss or interrupted), game score and Elo change, plus the level card. A match with a stored replay can be watched in the Replay Viewer with pause, seek and speeds from 0.5× to 2×. Only participants can open a replay.

Screen layout



Figure 48. Match history with interrupted ranked matches

Function details

Business rules: BR-29, BR-40.

Normal flow:

1. The player opens History; the list loads from the API.
2. The player selects a replay; the manifest and chunks are downloaded and played.

Alternative and exception flows:

- A1 – No replay stored: the entry shows "Chưa có replay".
- E1 – A player who did not take part requests the replay: HTTP 403.

FR-25 Manage Friends and Blocked Players

Related use cases: UC-32 to UC-36

Function trigger: Menu → friend drawer → **Bạn bè**.

Function description: The player searches other players by name or ID, sends friend requests, accepts or declines received requests, removes friends and blocks or unblocks players. Online friends can be invited to a room (FR-15).

Screen layout



Figure 49. Friends drawer

Function details

Business rules: BR-38.

Normal flow:

1. The player searches by name or ID and sends a request.
2. The receiver accepts or declines it under the requests list.
3. The player can remove a friend or block a player from the list.

Alternative and exception flows:

- E1 – No match for the search: MSG51.
- E2 – The other player has already sent a request: MSG50.
- E3 – A request to oneself or to a blocked player is rejected.

FR-26 View Leaderboard

Related use cases: UC-31 View Leaderboard

Function trigger: Menu → friend drawer → **Bảng xếp hạng**.

Function description: The leaderboard lists players by Elo in descending order with rank tier, and highlights the player's own position, which can be located with **Vị trí của tôi**.

Screen layout



Figure 50. Elo leaderboard

Function details

Business rules: BR-27, BR-39.

Normal flow:

1. The player opens the leaderboard; the cached list is loaded.
2. The player searches by name or jumps to the own position.

3.9 Reports, Restrictions and Appeals (FE-08)

FR-27 Report a Player

Related use cases: UC-30 Report Player

Function trigger: Match History → **Tố cáo** on a finished match.

Function description: A player reports an opponent from one of their own matches with a reason. The report is stored as PENDING for administrators (FR-30).

Screen layout

This function uses the screen already shown in Figure 48; no separate screenshot is repeated here.

Function details

Table 18. FR-27 data fields

Field	Type	Required	Validation and constraints
Reported player	Choice	Yes	A participant of the match other than the reporter (MSG52)
Reason	Text (≤254)	Yes	Not empty

Business rules: BR-41, BR-44.

Normal flow:

1. The player selects **Tố cáo**, the opponent and enters the reason.
2. The API stores the report; the button changes to the reported state.

Alternative and exception flows:

- E1 – The player has already reported this player in this match: the duplicate is rejected.
- E2 – No eligible player: MSG52.

FR-28 View Sanction Decision and Submit Appeal

Related use cases: UC-45 View Sanction Decision, UC-46 Submit Appeal

Function trigger: Sign-in of a restricted account (automatic); Menu → **Quyết định & Khiếu nại**.

Function description: A restricted player sees each sanction with its type (suspended or banned), reason and expiry, and can submit one appeal per sanction. The appeal status (PENDING, CHECKING, ACCEPTED, DECLINED) and the administrator's decision reason are shown when available.

Screen layout

Table 19. FR-28 screen layout

Area	Elements and behaviour
Decision list	Restriction type, reason, start and expiry ("Không thời hạn" for an indefinite ban)
Appeal form	Statement field (10–4000 characters) and Send button; hidden when an appeal already exists
Appeal status	Status badge and decision reason

Function details

Table 20. FR-28 data fields

Field	Type	Required	Validation and constraints
Statement	Text	Yes	10–4000 characters (MSG53)

Business rules: BR-42, BR-43, BR-44.

Normal flow:

1. The restricted player opens the popup and reads the decision.
2. The player writes the statement and sends it; the status becomes PENDING.

Alternative and exception flows:

- E1 – Statement shorter than 10 characters: MSG53.
- E2 – No sanction to appeal: MSG54.
- E3 – An appeal already exists for the sanction: the form is not shown.

3.10 Administration Web Application (FE-09)

FR-29 Admin Sign-in and User Management

Related use cases: UC-47 to UC-53

Function trigger: Browser → admin web URL → Login; sidebar → **Quản lý người dùng** (/users).

Function description: An administrator signs in with an Admin account; other roles are rejected. On the Users page the administrator searches and filters accounts, assigns the Player or Admin role, verifies an e-mail manually when delivery failed, applies or revokes a sanction with a reason, and resets selected progress groups of a player after previewing the impact. Every change is written to the audit log.

Screen layout

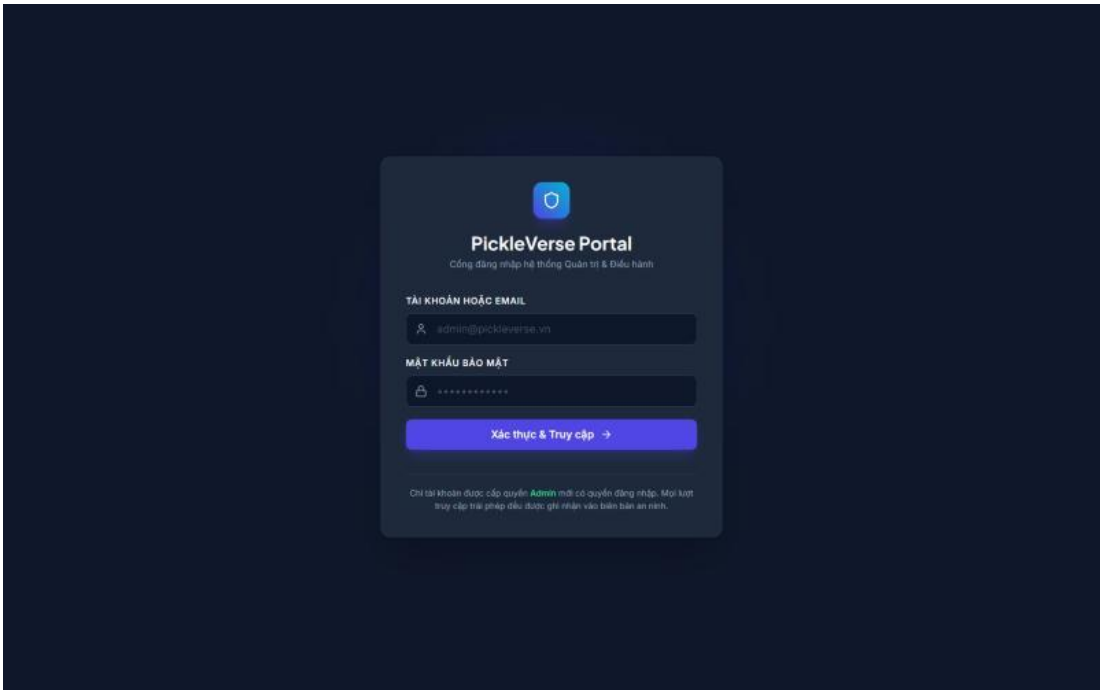


Figure 51. Admin web login page

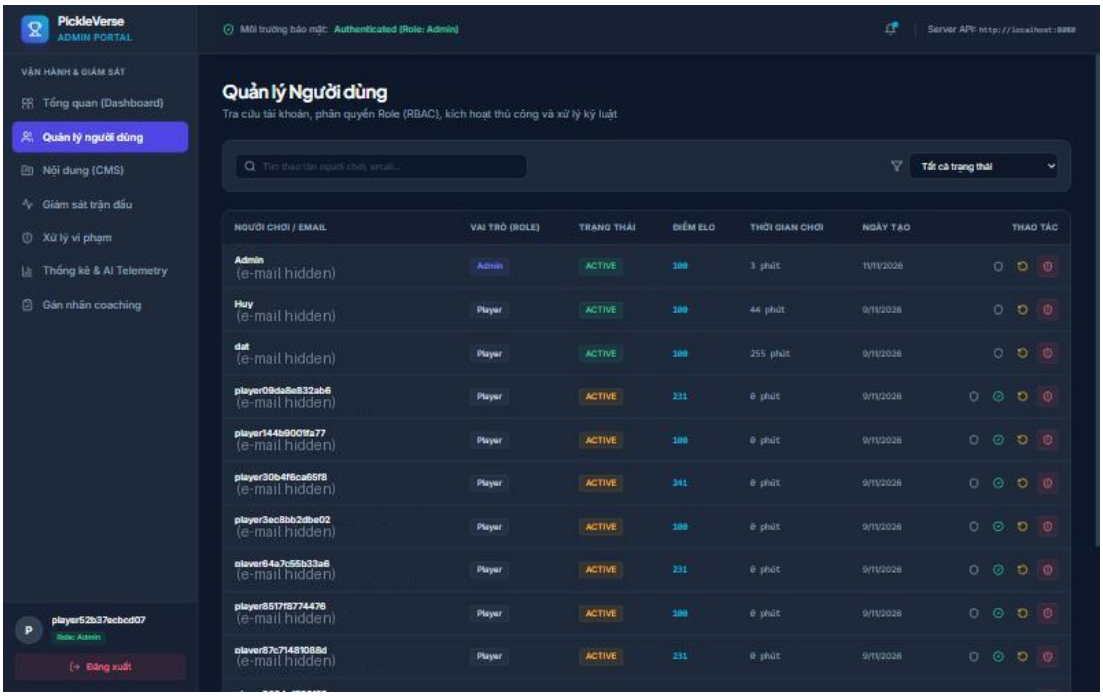


Figure 52. Users page of the admin web application

Function details

Table 21. FR-29 data fields

Field	Type	Required	Validation and constraints
Search text	Text	No	Matches name, e-mail or ID
Restriction	Choice	Yes (sanction)	SUSPENDED or BANNED
Expiry	Date-time (UTC)	Suspend: yes	Must be in the future (MSG56)
Reason	Text	Yes	3–2000 characters (MSG55)

Field	Type	Required	Validation and constraints
Reset groups	Multi-choice	Yes (reset)	At least one of LEARNING, RANKING, MISSIONS, ECONOMY

Business rules: BR-42, BR-45, BR-46.

Normal flow:

1. The administrator signs in and opens Users.
2. The administrator searches, selects an account and an action.
3. The administrator confirms; the API applies the change, writes the audit record and the table refreshes.

Alternative and exception flows:

- E1 – Non-admin account at login or on any admin API: HTTP 403.
- E2 – Sanction on the own account or a system account: MSG57.
- E3 – Reset while the player is searching or in a match: MSG58.
- E4 – Invalid reason or expiry: MSG55, MSG56.

FR-30 Review Player Reports and Appeals

Related use cases: UC-54 Review Player Report, UC-55 Review Appeal, UC-51 Apply Sanction

Function trigger: Sidebar → **Xử lý vi phạm** (/moderation).

Function description: The administrator opens a pending report, marks it CHECKING, examines the match evidence and replay, and accepts or declines it with a reason; accepting may apply a sanction to the reported player. Appeals are handled in the same way; accepting an appeal revokes only the sanction it concerns.

Screen layout

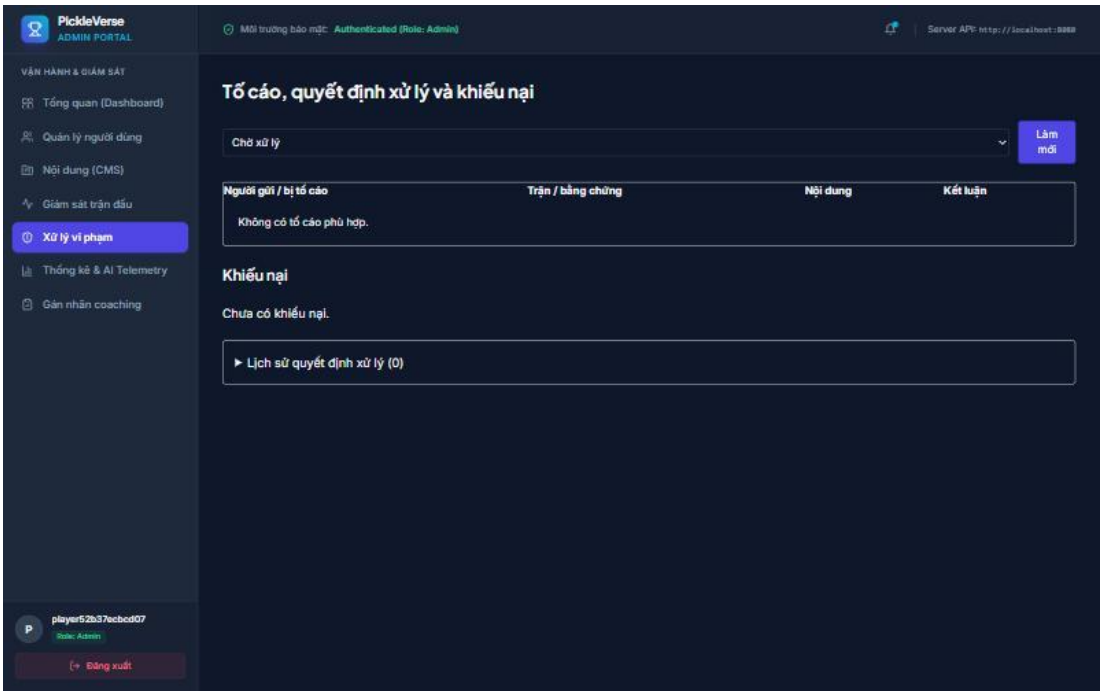


Figure 53. Moderation page

Function details

Table 22. FR-30 data fields

Field	Type	Required	Validation and constraints
Decision	Choice	Yes	ACCEPTED or DECLINED

Field	Type	Required	Validation and constraints
Decision reason	Text	Yes	3–2000 characters (MSG55)

Business rules: BR-42, BR-43, BR-44, BR-45.

Normal flow:

1. The administrator selects a report and reviews evidence.
2. The administrator enters the decision and reason and saves.
3. The report status changes; a sanction, if chosen, is applied and the player is restricted.

Alternative and exception flows:

- E1 – The report or appeal was changed by another administrator: the latest version is reloaded.
- E2 – A non-admin token: MSG63 (HTTP 403).

FR-31 Manage Training Content, Items and Missions

Related use cases: UC-56 to UC-61

Function trigger: Sidebar → **Nội dung (CMS)** (/cms).

Function description: The administrator creates and edits lessons as drafts, publishes a saved draft so that new attempts use it, and unpublishes a lesson to hide it from players; attempts already in progress keep their version. The administrator also maintains technique reference values, adds shop items and creates missions.

Screen layout

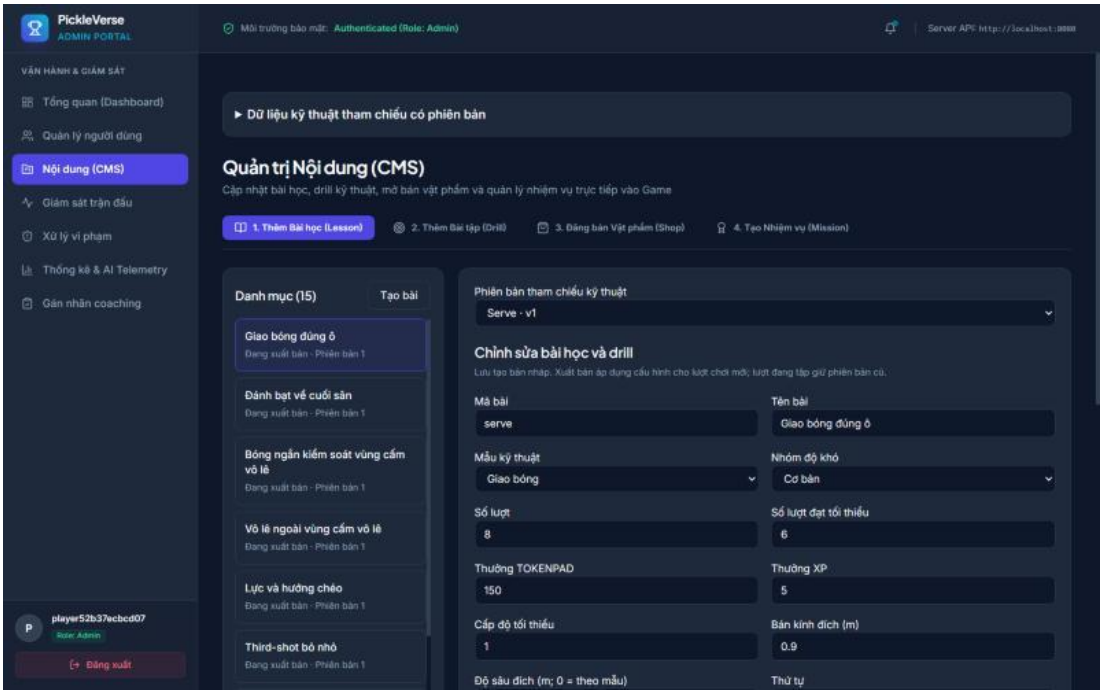


Figure 54. Content management page

Function details

Table 23. FR-31 data fields

Field	Type	Required	Validation and constraints
Lesson code	Text	Yes	Unique (MSG59)
Target	Integer	Yes	1 ≤ target ≤ number of shots
Prerequisite	Lesson	No	Must not create an unlock cycle
Price	Integer	Yes (item)	≥ 0, with a currency

Business rules: BR-19, BR-45, BR-47.

Normal flow:

1. The administrator edits a lesson and selects **Lưu bản nháp**.
2. The administrator selects **Xuất bản bản đã lưu**; the lesson becomes PUBLISHED.
3. **Gỡ xuất bản** sets it to UNPUBLISHED.

Alternative and exception flows:

- E1 – Duplicate lesson code: MSG59.
- E2 – The content changed since it was loaded: MSG60.
- E3 – The prerequisite creates a cycle: the save is rejected.

FR-32 Monitor Matches and View Analytics

Related use cases: UC-62 to UC-66

Function trigger: Sidebar → **Tổng quan (Dashboard)**, **Giám sát trận đấu** (/matches), **Thống kê & AI Telemetry** (/analytics).

Function description: The dashboard summarises today's activity and pending work. The Matches page shows live matches with the score and connection state reported by the dedicated server and lets the administrator stop a live match with a reason. The Analytics page reports daily active players, retention cohorts, popular lessons with pass rate (passes divided by started attempts), the technique-error heat map and matchmaking health for a chosen date range.

Screen layout

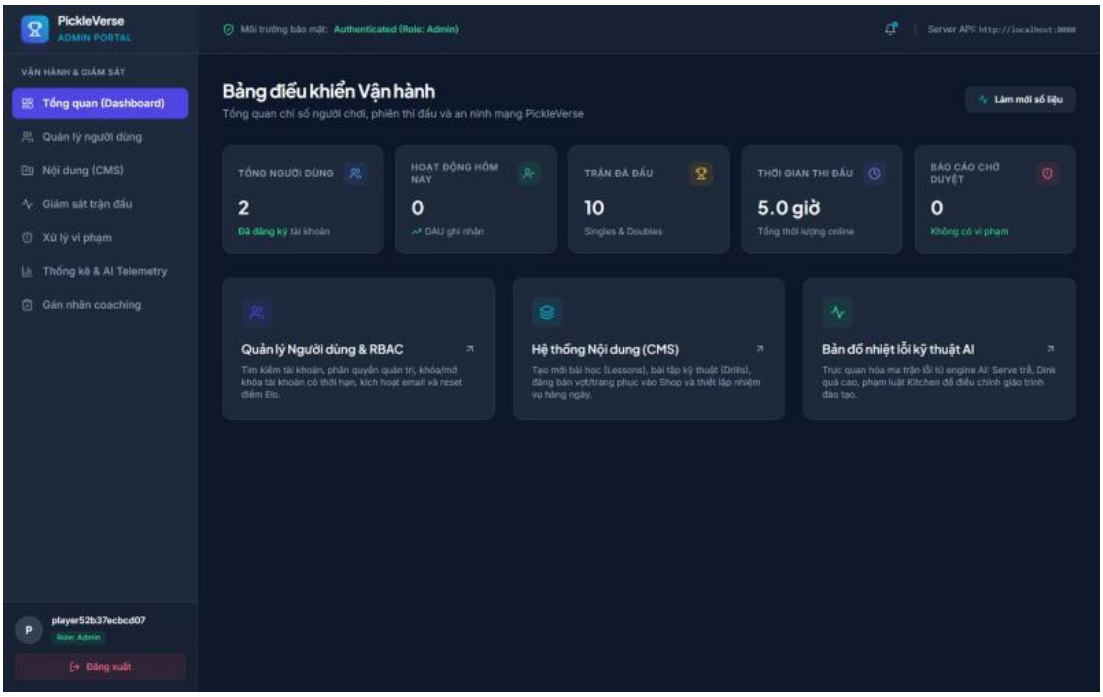


Figure 55. Admin dashboard

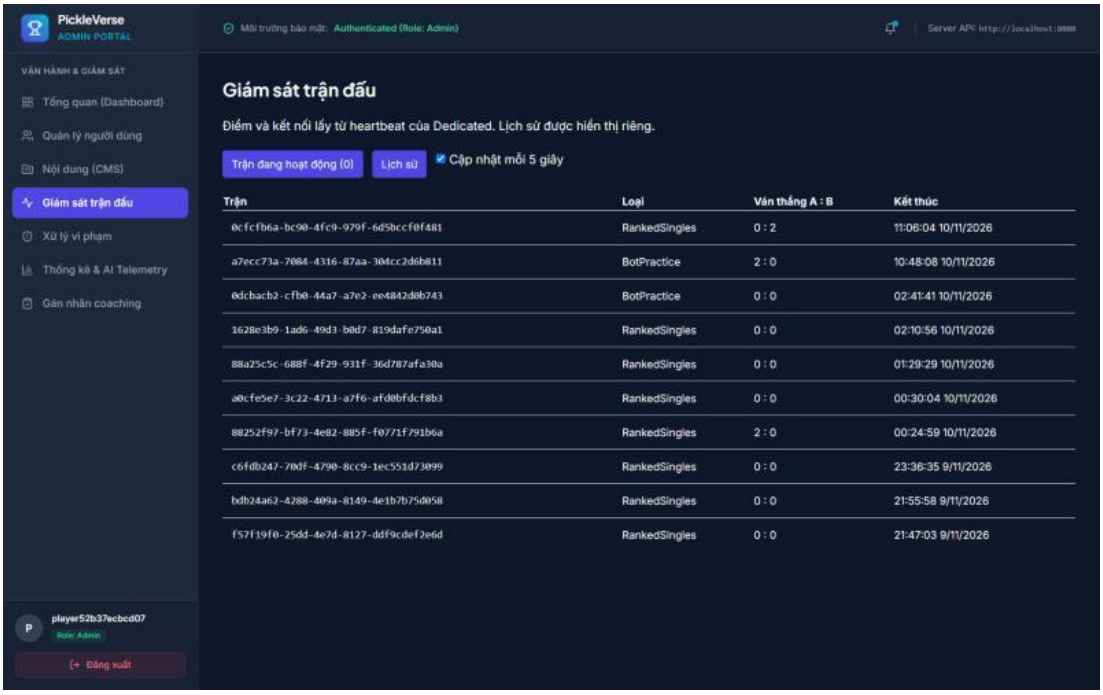


Figure 56. Live matches page

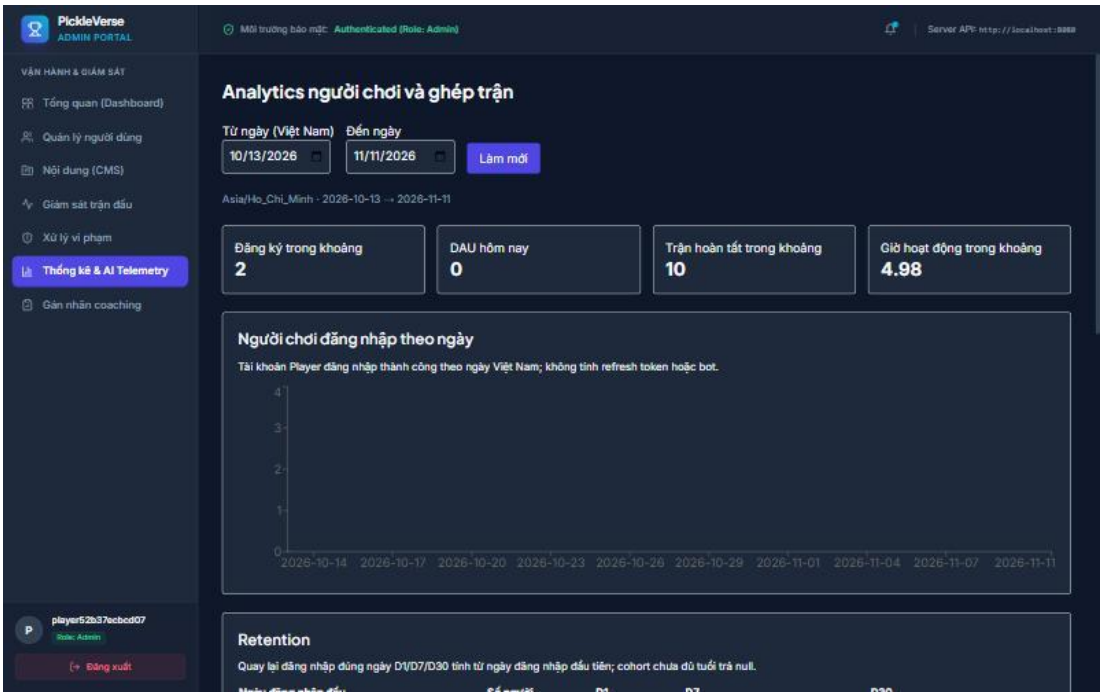


Figure 57. Analytics page

Function details

Table 24. FR-32 data fields

Field	Type	Required	Validation and constraints
Stop reason	Text	Yes	3–2000 characters
From / To date	Date	Yes	From ≤ To; at most 366 days (MSG62)

Business rules: BR-45, BR-48, BR-49.

Normal flow:

1. The administrator opens Matches; the list refreshes every 5 seconds.
2. To stop a match the administrator enters a reason and confirms MSG61; the server ends the match as interrupted.
3. The administrator opens Analytics, sets the dates and reads the reports.

Alternative and exception flows:

- E1 – The match already finished: the stop is ignored and the final result is kept.
- E2 – Invalid date range: MSG62.

3.11 Expert Review (FE-10)

FR-33 Review Coaching Sessions and Export Dataset

Related use cases: UC-67 Review Coaching Session, UC-68 Export Labelled Dataset

Function trigger: Sidebar → **Gán nhãn coaching** (/coaching).

Function description: An administrator acting as a pickleball expert inspects the shot-level evidence of a real training session, assigns the player level and confirms or rejects each error, and assigns the data group. All sessions of one player must use the same data group. The labelled sessions are exported to calibrate the rule thresholds and to evaluate the tier model against NFR-06.

Screen layout

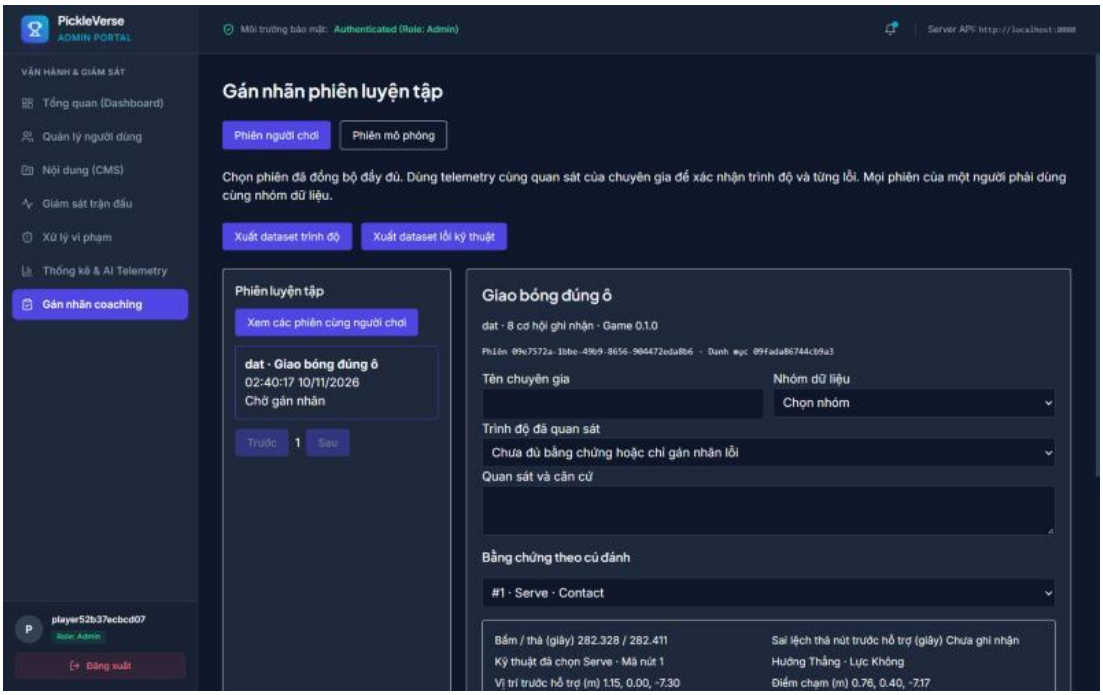


Figure 58. Expert review page

Function details

Table 25. FR-33 data fields

Field	Type	Required	Validation and constraints
Level	Choice	Yes	Beginner, intermediate or advanced
Error labels	Per error code	At least one	Present or absent with evidence (MSG64)
Data group	Choice	Yes	Training, calibration or holdout; consistent for a player

Business rules: BR-45, BR-50.

Normal flow:

1. The administrator selects a session and reviews the evidence.

2. The administrator sets the labels and saves the review.
3. The administrator exports the labelled dataset.

Alternative and exception flows:

- E1 – Missing confirmation, evidence or labels: MSG64.
- E2 – The session already has a newer review: the evidence must be reloaded.
- E3 – The player's sessions are in different data groups: HTTP 409.

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces

- The game interface is in Vietnamese with the project's Vietnamese fonts; it scales for 16:9 (1920×1080, 1280×720) and 20:9 phone screens; every screen can be operated with mouse and keyboard on PC and with touch on Android; Esc closes overlays and opens Settings, and Enter confirms.
- In-match information (score, serving side, timers, fault banners) is placed at the screen edges so that it does not cover the court.
- The administration web application is a responsive single-page application for desktop browsers with a fixed sidebar, tables with paging and confirmation dialogs for every destructive action.

4.1.2 Hardware Interfaces

- PC: keyboard and mouse; game controller input is not supported in release 1.0.
- Android: multi-touch screen for the virtual joystick and shot buttons.
- Server: Linux host with Docker; the dedicated game-server container is limited to 2 CPUs and 2 GB RAM by default.

4.1.3 Software Interfaces

Table 26. Software interfaces

Interface	Description
Web API (REST/JSON)	129 endpoints under /api (Report 4, §3.9), JSON bodies wrapped in the ApiResponse<T> envelope {success, message, data}; error responses carry {success, message, code, errors}; JWT bearer authentication; Swagger/OpenAPI documentation
AI coaching service	Internal HTTP API POST /v1/analyze-session protected by the X-AI-Key header; called only by the Web API worker
Dedicated server ↔ Web API	HTTP calls for lease, heartbeat, telemetry, replay and result, authenticated with a service key; match tickets signed with HMAC-SHA256
Photon Fusion 2.1.3	Real-time session service (protocol version 18) used by clients and the dedicated server
PostgreSQL 16 / Redis 7.4	Persistent storage through Npgsql Entity Framework Core; Redis for matchmaking queue, leases, presence and caches
S3-compatible storage	Replay manifests and chunks through the S3 API

4.1.4 Communication Interfaces

- Client and admin web to Web API: HTTP/1.1 with JSON over TLS in production (port 443); the staging environment serves the API and the admin web over HTTPS on port 443 since 13/11/2026 (HTTP on port 8080 until then).
- Game traffic: UDP through Photon Cloud and to the dedicated server on UDP port 3669.
- E-mail: SMTP with STARTTLS on port 587 to Gmail.
- All timestamps are exchanged in UTC (ISO 8601) and displayed in Vietnam time (UTC+7).

4.2 Quality Attributes

Each quality attribute below is measurable and is verified by the test case named in the last column (Report 5).

4.2.1 Usability

Table 27. Usability requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-01	A new player can learn the game without outside help.	At least 8 of 10 first-time testers complete the 17-step tutorial in 5 minutes or less and then pass the first lesson (serve) within three attempts, without help from the team.	Moderated usability test with 10 students who have not played PickleVerse (Report 5, TC-N01)

4.2.2 Reliability

Table 28. Reliability requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-07	Online services are available when players want to play.	The API health endpoint /health/ready is available ≥ 99.5% of the time measured every 5 minutes by an external probe over the 14 days before the final defense (at most 100 minutes of downtime).	Uptime probe report (TC-N07)
NFR-08	No result or reward is lost or duplicated.	Every match result, training receipt and purchase is applied exactly once under retries; after a container failure the service is restored from Docker Hub images within 1 hour (MTTR) with at most 24 hours of data loss from the daily database backup. No open Critical defect is allowed at release (Critical = loss of player data or inability to sign in or play).	Retry tests on settlement and sync APIs; restore drill (TC-N08)
NFR-11	Training is possible without a connection.	After one online sign-in, all published lessons can be practised offline, and 100% of offline attempts are synchronised exactly once (no duplicate reward) when the device reconnects.	Offline–online test with 20 queued attempts (TC-N11)

4.2.3 Performance

Table 29. Performance requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-02	Online play is responsive for players in Vietnam.	Round-trip time between the client and the staging game server is at most 100 ms at the 95th percentile over 10 online matches on home broadband in Ho Chi Minh City.	Fusion RTT statistics logged by the client (TC-N02)
NFR-03	Matches are smooth on mid-range devices.	Average frame rate ≥ 60 FPS (1% low ≥ 45 FPS) during a bot match on a reference PC (Intel Core i5 10th generation, GTX 1650, 16 GB RAM, 1920×1080, default quality); average ≥ 30 FPS on a reference Android phone (Snapdragon 7-series, 6 GB RAM).	Unity Profiler capture of a full bot match on each device (TC-N03)
NFR-04	The Web API supports the target player population.	With 2,000 concurrent virtual users performing a read-mostly mix (profile, missions, leaderboard, history, catalogue) the API keeps the 95th-percentile response time ≤ 500 ms and the error rate < 1% for 10 minutes. The release capacity of online matches equals the number of dedicated server slots (one in the staging deployment, LI-07 in Report 1).	k6 load test against staging (TC-N04)
NFR-05	Coaching feedback arrives quickly.	The AI coaching service returns the analysis of one drill or match in ≤ 1 second at the 95th percentile once the telemetry has been received; the coaching popup shows the result within 5 seconds of the end of the drill when online.	Timing of 50 analysis requests from the worker log (TC-N05)

4.2.4 Accuracy of AI Coaching

Table 30. Accuracy of AI Coaching requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-06	Coaching findings are trustworthy.	Each rule-based finding reaches precision ≥ 0.80 and recall ≥ 0.70 on the expert-labelled holdout sessions. The tier model is shown to players only when it is approved: real expert data, at least 10 holdout players per tier, macro-F1 ≥ 0.70 and at least 0.05 above the majority baseline, prediction confidence ≥ 0.65 and at least 20 shot opportunities.	Calibration and training reports of the AI service (TC-N06)

4.2.5 Security

Table 31. Security requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-09	Accounts and match results are protected.	Passwords are stored with BCrypt; access tokens are signed JWTs valid for 2 hours and refresh tokens are stored as SHA-256 hashes; OTPs and dedicated-server tickets are protected with HMAC; authentication endpoints are rate-limited; every admin endpoint requires the Admin role; secrets are supplied by environment variables and never committed; production traffic uses HTTPS (TLS 1.2+); match outcomes are decided only by the dedicated server.	Security checklist, API negative tests and secret scan (TC-N09)

4.2.6 Compatibility

Table 32. Compatibility requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-10	The game runs on the target platforms.	The client runs on Windows 10/11 64-bit and on Android 8.0 (API 26) or later on ARM64, at 16:9 and 20:9 aspect ratios with keyboard or touch input; the admin web supports the latest Chrome and Edge at 1366×768 or larger.	Installation and smoke test on each platform (TC-N10)

4.2.7 Maintainability and Privacy

Table 33. Maintainability and Privacy requirements

ID	Requirement	Measure and acceptance criterion	Verification
NFR-12	The system can be changed safely and protects personal data.	The backend keeps the layered dependency rule (API → Services → Repositories → Entities) with no reverse reference; core rule classes are covered by automated unit tests that run before release; other players can see only the display name, player ID, rank and Elo, never the e-mail address, which is visible only to administrators.	Architecture check of project references, test run report and API response review (TC-N12)

5. Requirement Appendix

5.1 Business Rules

Table 34. Business rules

ID	Group	Rule Definition
BR-01	Account	An e-mail address can be used by only one account.
BR-02	Account	A password has 6 to 128 characters and is stored only as a BCrypt hash.
BR-03	Account	A verification or reset code has six digits, is valid for 10 minutes, can be re-sent only after 60 seconds and allows at most 5 wrong attempts.
BR-04	Account	Only an account with a verified e-mail can sign in. System bot accounts can never sign in. A suspended or banned account can sign in only with the appeal scope.
BR-05	Account	An access token is valid for 2 hours and a refresh token for 14 days; a refresh token is replaced each time it is used. Resetting the password or signing out revokes the refresh tokens concerned.

ID	Group	Rule Definition
BR-06	Account	A display name has 3 to 18 characters, contains no control characters and is unique among players.
BR-07	Account	The experience level (Beginner or Experienced) is chosen once, at the first sign-in.
BR-08	Account	The display name and character cannot be changed while the player is in a room, searching for a match or playing a match.
BR-09	Gameplay	The court is 13.41 m × 6.10 m; the non-volley zone extends 2.13 m from the net on each side. A ball touching a line is in.
BR-10	Gameplay	A singles match is the best of three games. A game is won by the first side to reach 11 points with a lead of at least 2. Only the serving side scores a point; a fault by the server gives the serve to the opponent (side-out).
BR-11	Gameplay	The serve is hit underhand, with both feet behind the baseline, into the diagonal service court and beyond the non-volley zone, within 10 seconds after the server is ready. The server serves from the right court when the server's score is even and from the left when it is odd.
BR-12	Gameplay	Two-bounce rule: the return of serve and the third shot must bounce once before they are hit.
BR-13	Gameplay	A volley may not be hit while standing in the non-volley zone, and the player's momentum after a volley may not carry them into the zone.
BR-14	Gameplay	The rally ends with a fault of one of 17 types: serve foot, serve motion, serve timeout, short serve, serve to the wrong court, missed serve, out, ball into own court, two-bounce rule, double bounce, double hit, kitchen volley, kitchen momentum, net touch, body hit, net plane and permanent object.
BR-15	Gameplay	In the third game the players change ends when a side first reaches 6 points. Time-outs are not offered in release 1.0.
BR-16	Gameplay	A paddle contact is accepted only when the key is released within ±65 ms of the ideal contact time. A contact is Perfect when the contact error is ≤ 0.145 m and the timing error is within ±30 ms.
BR-17	Gameplay	Shot keys: J drive (and serve), K dink, L volley, J+K drop, J+L smash, K+L reset; U adds power and I aims cross-court. The shot is decided once, when the first base key is released.
BR-18	Training	Drill score: 0 stars below the target; 1 star when the target is reached; 2 stars at ≥ 80% hits; 3 stars at ≥ 90% hits with a mean timing error ≤ 0.16 s. The best result of each lesson is kept.
BR-19	Training	A lesson is unlocked when its prerequisite lesson has been passed; a locked lesson cannot be started. Prerequisites cannot form a cycle.
BR-20	Training	The first pass of a lesson grants 150, 225, 300 or 450 Tokenpad and 5, 7, 10 or 15 XP according to the lesson group; later passes and failed attempts grant no currency and cost nothing.
BR-21	Training	Offline training is allowed only for a profile cached by a previous online sign-in. Each attempt carries a unique request ID and is applied at most once (receipt APPLIED, DUPLICATE or REJECTED).
BR-22	Match	Match kinds are Bot Practice, Ranked Bot, Ranked Singles, Matched Doubles and Custom. Only Ranked Singles and Ranked Bot change Elo.
BR-23	Match	A new player starts at 100 Elo, which is also the minimum. After a ranked match Elo changes by round(32 × (S − E)), where S is 1 for a win and 0 for a loss and E = 1 / (1 + 10^((R_opponent − R_own)/400)); the result is never below 100.
BR-24	Match	Ranked pairing uses a window of 120 Elo. During the first 30 seconds the window extends only upward (only downward for a player with 5 or more losses); after 30 seconds it extends 120 in both directions.
BR-25	Match	When a match is found, every player must accept within 10 seconds; a decline or a timeout cancels the offer.
BR-26	Match	Ranked singles only: if no human opponent is found within 180 seconds, a ranked bot is assigned with Elo = own + 120 (own − 100 for a player with 5 or more losses). Quick doubles has no bot fallback; its search ends when the search lease expires.
BR-27	Match	Rank tiers by Elo: Đồng < 500, Bạc < 1000, Vàng < 2000, Bạch Kim < 3500, Kim Cương < 5000, Cao Thủ < 7000, Huyền Thoại ≥ 7000.
BR-28	Match	A disconnected client tries to reconnect for 30 seconds, and the dedicated server keeps the player's slot for 30 seconds while the match is paused; a player who does not return is recorded as having forfeited and the match ends as interrupted. Clients send a heartbeat to the API every 8 seconds; in shared-mode matches (local development only) the API interrupts the match when a player has sent no heartbeat for 35 seconds. If the dedicated server stops renewing its lease, the lease monitor ends the match as interrupted.
BR-29	Match	A match result is saved exactly once. An interrupted or stopped match gives no Elo and no reward.
BR-30	Match	A singles room holds 2 players and a doubles room up to 4. The host starts a custom match and may start a doubles room before all four slots are filled; custom matches are unrated.
BR-31	Match	Bot levels Easy, Medium and Hard use fixed reaction, accuracy and error settings. The adaptive bot selects its tier from the player's Elo: below 500 beginner, below 2000 intermediate, otherwise advanced.

ID	Group	Rule Definition
BR-32	Economy	Match rewards: bot match win 100 / loss 50 Tokenpad; online match win 150 / loss 75 Tokenpad; every finished match gives 10 XP for a win and 5 XP for a loss.
BR-33	Economy	Reaching level n+1 requires n × 200 XP at level n; surplus XP carries over.
BR-34	Economy	There are 7 daily and 10 weekly missions. Daily missions renew at 07:00 Vietnam time (UTC+7) and weekly missions at the start of each mission week. Completing all daily missions gives 200 Tokenpad + 20 XP and all weekly missions 1,000 Tokenpad + 50 XP. The daily streak grows by one for each day with at least one finished match; streaks are rewarded by the weekly mission "3-day streak" (350 Tokenpad + 20 XP) and the achievement "5-day streak" (500 Tokenpad + 30 XP).
BR-35	Economy	Each mission, mission set and achievement reward can be claimed only once.
BR-36	Economy	A purchase requires enough balance in the chosen currency and is refused if the item is already owned; the debit and the grant are one transaction. Exactly one paddle is equipped at a time. Elite Token has no earning source in release 1.0.
BR-37	Economy	The THOT paddle is granted once after 10 passed lessons and the GEMSTONE paddle once at level 25, unless already owned.
BR-38	Social	A player cannot send a friend request to themself, send a duplicate request, or send one to a player who has already sent them a request. Blocking removes the friendship and prevents friend requests and room invitations in both directions.
BR-39	Social	The leaderboard is ordered by Elo descending and is cached for 15 seconds.
BR-40	Social	A match history and its replay can be read only by the players who took part and by administrators.
BR-41	Moderation	A player can report another participant of their own finished match once per match; self-reporting is not allowed; the reason has at most 254 characters.
BR-42	Moderation	A sanction is SUSPENDED (expiry required and in the future) or BANNED (expiry optional). The reason has 3 to 2,000 characters. An administrator cannot sanction their own account or a system account. An expired suspension ends automatically.
BR-43	Moderation	A restricted player can submit one appeal per sanction, with a statement of 10 to 4,000 characters. Accepting an appeal revokes only the sanction it concerns.
BR-44	Moderation	Reports and appeals move from PENDING to CHECKING to ACCEPTED or DECLINED; a decision requires a reason.
BR-45	Administration	Only the Admin role can use the administration application and API (otherwise HTTP 403). Every administrative change is recorded in the audit log with actor, action, target and time.
BR-46	Administration	Progress reset works on the groups LEARNING, RANKING, MISSIONS and ECONOMY after an impact preview, and is refused while the player is searching or in a match. It keeps the account, friends, match history, decisions and reward log; rewards already granted are not granted again.
BR-47	Administration	A lesson moves DRAFT → PUBLISHED → UNPUBLISHED (and can be published again). Publishing applies to new attempts only. A save is refused when the content changed after it was loaded.
BR-48	Administration	Stopping a live match requires a reason; when the stop is recorded before the final result the match is interrupted without Elo or reward.
BR-49	Administration	Analytics dates are calendar days in Vietnam time; the start date is not after the end date and a range covers at most 366 days.
BR-50	AI coaching	A coaching finding is reported only when the session contains at least 5 opportunities for that error; at most 3 drills are recommended; the tier estimate is shown only under the approval conditions of NFR-06.
BR-51	AI coaching	Telemetry and replays of dedicated-server matches are submitted only by the dedicated server; a client can submit telemetry only for its own drills and bot matches.

5.2 Common Requirements

- Language: all player-facing and administrator-facing text is Vietnamese; documents and source code identifiers are English.
- Date and time: stored in UTC and shown in Vietnam time (dd/MM/yyyy HH:mm); daily cycles (missions, streaks, analytics days) start at 07:00 or 00:00 Vietnam time as stated in the related rule.
- API responses: every JSON response uses the ApiResponse envelope; HTTP 400 validation, 401 not signed in, 403 forbidden, 404 not found, 409 conflict, 429 too many requests, 503 dependency unavailable; HTTP 500 responses never expose internal details.
- Idempotency: commands that change progress, wallet or match state carry a client request ID; repeating a request returns the first result.
- Concurrency: administrative edits use a version check and ask the user to reload when the data changed.

- Lists: account lists are paged (20 rows per page by default in the API); the game history shows the last 20 matches.
- Logging: the API, workers and dedicated server write structured logs without passwords, tokens or codes.

5.3 Application Messages List

Table 35. Application messages

#	Message code	Message Type	Context	Content
1	MSG01	In red, under the text box	E-mail field is empty	Vui lòng nhập email.
2	MSG02	In red, under the text box	E-mail format is invalid	Vui lòng nhập địa chỉ email hợp lệ.
3	MSG03	In red, under the text box	Password field is empty	Vui lòng nhập mật khẩu.
4	MSG04	In red, under the text box	Password shorter than 6 characters	Mật khẩu cần ít nhất 6 ký tự.
5	MSG05	In red, under the text box	Confirmation differs from the password	Mật khẩu xác nhận chưa khớp.
6	MSG06	In red, under the form	E-mail already registered	Email đã được đăng ký. Nếu chưa xác thực, hãy dùng chức năng gửi lại mã.
7	MSG07	In line, information	Registration succeeded	Đã đăng ký. Hãy nhập mã được gửi tới email để kích hoạt tài khoản.
8	MSG08	In red, under the text box	Code is not six digits	Mã xác thực gồm 6 chữ số.
9	MSG09	In line, information	A new code was sent	Mã xác thực đã được gửi. Hãy kiểm tra cả thư rác.
10	MSG10	In red, under the form	Code re-sent within 60 seconds (HTTP 429)	Vui lòng chờ ít nhất 60 giây trước khi gửi lại mã.
11	MSG11	In red, under the form	Sign-in with an unverified e-mail	Email chưa được xác thực. Hãy nhập OTP hoặc yêu cầu gửi lại mã.
12	MSG12	In red, under the form	Wrong e-mail or password	Email hoặc mật khẩu không chính xác.
13	MSG13	Toast message	Sign-in succeeded	Đăng nhập thành công.
14	MSG14	Popup	Sign-in of a suspended or banned account	Tài khoản đang bị hạn chế. Bạn có thể xem quyết định và gửi khiếu nại.
15	MSG15	In line, information	Password reset requested (same text whether or not the e-mail exists)	Nếu email thuộc tài khoản đã xác thực, mã đặt lại mật khẩu sẽ được gửi. Hãy kiểm tra cả thư rác.
16	MSG16	In red, under the form	Reset code wrong or expired	Mã đặt lại mật khẩu không hợp lệ hoặc đã hết hạn.
17	MSG17	In red, under the form	Password rule rejected by the API	Mật khẩu cần từ 6 đến 128 ký tự và xác nhận phải khớp.
18	MSG18	Popup	Refresh token expired or revoked	Phiên đăng nhập không hợp lệ hoặc đã hết hạn.
19	MSG19	In red, under the text box	Display name length invalid	Tên người chơi cần từ 3 đến 18 ký tự.
20	MSG20	In red, under the text box	Display name already used	Tên người chơi đã được sử dụng. Hãy chọn tên khác.
21	MSG21	Popup question	First sign-in experience choice	Bạn đã quen với pickleball chưa? – Người mới sẽ được hướng dẫn luật và điều khiển. Nếu đã thành thạo, bạn có thể vào sảnh ngay.
22	MSG22	In red, under the form	Verification e-mail could not be sent	Chưa gửi được email xác thực. Vui lòng thử gửi lại mã sau ít nhất 60 giây.
23	MSG23	In line	Starting a locked lesson	Bài tập chưa mở khóa.
24	MSG24	Popup title	Drill target not reached	CHƯA ĐẠT MỤC TIÊU
25	MSG25	In line	Drill failed	Không mất Tokenpad. Luyện lại để đạt {target} cú.
26	MSG26	HUD feedback	Key released late	Bạn thả nút hơi muộn. Hãy chuẩn bị vợt và thả nút sớm hơn.
27	MSG27	HUD feedback	Key released early	Bạn thả nút hơi sớm. Hãy đợi bóng tiến gần hơn.

#	Message code	Message Type	Context	Content
28	MSG28	Banner	Offline training mode	Đang luyện tập ngoại tuyến · {n} mục chờ đồng bộ
29	MSG29	HUD banner	Serve timer expired	Bạn đã hết 10 giây giao bóng. Hãy chuẩn bị và giao bóng trước khi hết thời gian.
30	MSG30	HUD banner	Rally ended by a fault (examples)	Lỗi • Bóng ra ngoài / Lỗi • Nảy hai lần / Lỗi • Vô lê trong vùng cấm vô lê / Lỗi • Bóng không qua lưới / Lỗi chân giao bóng
31	MSG31	Result title	Player wins the match	Bạn thắng trận!
32	MSG32	In line	Searching for an opponent	Đang tìm đối thủ trực tuyến…
33	MSG33	Popup	Match found, ready check	Đã tìm thấy trận • Xác nhận để tham gia.
34	MSG34	In line	Search expired	Hết thời gian tìm trận. Hãy thử lại.
35	MSG35	In line	Player declined the match	Bạn đã hủy tham gia. Bấm Tìm trận để tìm lại.
36	MSG36	In line	Dedicated server slot is busy	Máy chủ đang bận • Đang chờ lượt thi đấu…
37	MSG37	In line	Room is full	Phòng đã đủ người.
38	MSG38	In line	Room closed or player removed	Phòng đã đóng hoặc bạn đã được mời ra.
39	MSG39	In line	Joining a room failed (for example wrong code)	Không tham gia được phòng. Vui lòng thử lại.
40	MSG40	In line	Invited player cannot receive invitations	Người chơi chưa sẵn sàng nhận lời mời.
41	MSG41	In line	Player already searching or in an unfinished match	Người chơi đang tìm trận hoặc có trận chưa hoàn tất.
42	MSG42	Overlay	Connection to the match server lost	Mất kết nối máy chủ thi đấu.
43	MSG43	Popup	Reconnection failed	Không thể kết nối lại trận. Kết quả sẽ được cập nhật khi máy chủ xử lý xong.
44	MSG44	Toast message	Player left the match	Bạn đã rời trận.
45	MSG45	Popup	Photon Cloud not reachable	Không kết nối được Photon. Vui lòng kiểm tra mạng và thử lại.
46	MSG46	Popup	Balance too low for a purchase	Số dư tiền tệ không đủ để mua vật phẩm này.
47	MSG47	Popup	Item already owned	Bạn đã sở hữu vật phẩm này rồi.
48	MSG48	Popup	Equipping an item that is not owned	Bạn chưa sở hữu vật phẩm này.
49	MSG49	Toast message	Achievement reward claimed	Mở khóa thành công danh hiệu và phần thưởng: {reward}!
50	MSG50	In line	The other player already sent a request	Người chơi đã gửi lời mời cho bạn. Hãy chấp nhận hoặc từ chối.
51	MSG51	In line	No search result	Không tìm thấy tên hoặc ID phù hợp.
52	MSG52	In line	No player can be reported	Không có người chơi phù hợp để tố cáo
53	MSG53	In red, under the text box	Appeal statement too short	Nội dung khiếu nại cần ít nhất 10 ký tự.
54	MSG54	In line	No decision to appeal	Không có quyết định xử lý để khiếu nại.
55	MSG55	In red, under the text box	Reason length invalid (admin)	Lý do phải có từ 3 đến {max} ký tự.
56	MSG56	In red, under the text box	Suspension without a future expiry	Tạm ngưng cần thời hạn trong tương lai; khóa có thể không có thời hạn.
57	MSG57	Toast message	Admin tries to sanction own account	Không thể tự khóa tài khoản quản trị.
58	MSG58	In line, in the dialog	Reset while player searching or in a match	Người chơi đang tìm trận hoặc có trận chưa hoàn tất. Chưa thể reset.
59	MSG59	In red, under the text box	Lesson code already exists	Mã bài học đã tồn tại.
60	MSG60	Toast message	Content changed since it was loaded	Nội dung đã thay đổi. Hãy tải lại trước khi lưu.
61	MSG61	Warning in the dialog	Confirming a match stop	Trận sẽ bị gián đoạn và không tính Elo/thưởng nếu lệnh dừng được ghi trước kết quả cuối.
62	MSG62	In red, under the date fields	Invalid analytics range	Khoảng ngày thống kê không hợp lệ. / Biểu đồ hỗ trợ tối đa 366 ngày mỗi lần xem.
63	MSG63	Toast message	Non-admin calls a moderation function (HTTP 403)	Bạn không có quyền xử lý vi phạm.
64	MSG64	In red, under the form	Expert review incomplete	Hãy xác nhận phiên người thật, người đánh giá, bằng chứng và ít nhất một nhãn.
65	MSG65	In red, under the form	System account sign-in attempt	Tài khoản hệ thống không được đăng nhập.

5.4 Definitions and Acronyms

Table 36. Definitions and acronyms

Term	Definition
Dink	A soft shot that drops into the opponent's non-volley zone
Drill	A practice exercise with a fixed number of shots and a target
Elo	A rating system in which the rating change depends on the expected result against the opponent
Kitchen / NVZ	Non-volley zone: the area within 2.13 m of the net where volleys are not allowed
Rally	The exchange of shots from the serve until a fault
Side-out	Loss of serve to the opponent after a fault by the serving side
Telemetry	Shot-level data (timing, contact, landing, position) recorded by the game
Tokenpad / Elite Token	In-game currencies; Tokenpad is earned by playing
Dedicated server	A headless Unity server that runs an online match with authority over the result
Photon Fusion	Third-party networking library and cloud used for real-time sessions
API, JWT, OTP, HMAC	Application programming interface; JSON Web Token; one-time password; hash-based message authentication code
BR, FR, NFR, UC, MSG	Business rule; functional requirement; non-functional requirement; use case; message

5.5 Appendix A – Traceability to the Review 2 Commitment

The table below links the 18 features committed in the Review 2 form BM04 (07/11/2026) to the use cases and functional requirements of this specification, with the workflow and the member accountable for each feature at the reviews. Report 5 links each requirement to its test cases.

Table 37. Traceability from Review 2 features to requirements and owners

Review 2 feature	Name	Functional requirements	Use cases	Workflow	Owner (BM04)
F01	Account and first steps	FR-01, FR-02, FR-03, FR-04	UC-01 to UC-08	W01	Trần Nguyễn Hiếu Thuận
F02	Rules and controls	FR-05, FR-06, FR-11	UC-09, UC-10	W01	Trần Nguyễn Hiếu Thuận
F03	Guided drills	FR-07, FR-08	UC-11, UC-12	W01	Nguyễn Gia Huy
F04	Levels and items	FR-20, FR-23	UC-43, UC-44	W01	Nguyễn Văn Thành Đạt
F05	Adaptive bot	FR-12, FR-13	UC-16	W02	Nguyễn Gia Huy
F06	Coaching and tier	FR-19	UC-17, UC-18	W02	Nguyễn Văn Thành Đạt
F07	Quick match	FR-14	UC-19, UC-20	W03	Nguyễn Văn Thành Đạt
F08	Private rooms	FR-15	UC-21 to UC-24	W03	Nguyễn Văn Thành Đạt
F09	Dedicated play	FR-16, FR-17	UC-25 to UC-27	W03	Nguyễn Văn Thành Đạt
F10	Result and Elo	FR-18	UC-25	W03	Nguyễn Văn Thành Đạt
F11	Progress and replay	FR-10, FR-21, FR-24	UC-15, UC-28, UC-29, UC-39	W02	Nguyễn Gia Huy
F12	Friends and ranks	FR-25, FR-26	UC-31 to UC-36	W03	Trần Nguyễn Hiếu Thuận
F13	Daily and streak rewards	FR-21	UC-37 to UC-40	W01	Nguyễn Gia Huy
F14	Offline practice	FR-09	UC-13, UC-14	W01	Trần Nguyễn Hiếu Thuận
F15	Admin and accounts	FR-29	UC-47 to UC-53	W04	Huỳnh Trung Kiên
F16	Lesson publishing	FR-31	UC-56 to UC-61	W04	Huỳnh Trung Kiên
F17	Match care and reports	FR-27, FR-28, FR-30, FR-32	UC-30, UC-45, UC-46, UC-54, UC-55, UC-62 to UC-64	W04	Huỳnh Trung Kiên
F18	Admin charts	FR-32	UC-65, UC-66	W04	Huỳnh Trung Kiên

Review 2 feature	Name	Functional requirements	Use cases	Workflow	Owner (BM04)
–	Shop (not in the Review 2 list)	FR-22	UC-41, UC-42	–	–
–	Expert review (addition to the proposal)	FR-33	UC-67, UC-68	–	–

5.6 References

[1] USA Pickleball, "Official Rulebook 2024." https://usapickleball.org/what-is-pickleball/official-rules/ (accessed 11 Nov. 2026).

[2] K. E. Wiegers and J. Beatty, Software Requirements, 3rd ed. Microsoft Press, 2013.

[3] Photon Engine, "Fusion 2 Documentation." https://doc.photonengine.com/fusion/current (accessed 11 Nov. 2026).

[4] GFA26SE111, "Report 1 – Project Introduction" and "Report 4 – Software Design Document," PickleVerse, 2026.

IV. Software Design Description

1. System Design

1.1 System Architecture

PickleVerse uses a client–server architecture with five deployable sub-systems (Figure 1). The Unity game client contains the presentation, the local gameplay simulation used for training and bot matches, and an offline store. All persistent data and business rules that must be trusted are on the server side: the ASP.NET Core Web API owns accounts, progression, economy, matchmaking and moderation; the Unity dedicated server owns the state of online matches; the AI coaching service analyses shot telemetry; the React administration application is a pure client of the Web API. Background workers run inside the Web API process. PostgreSQL is the system of record, Redis holds short-lived coordination data, and object storage keeps replays.

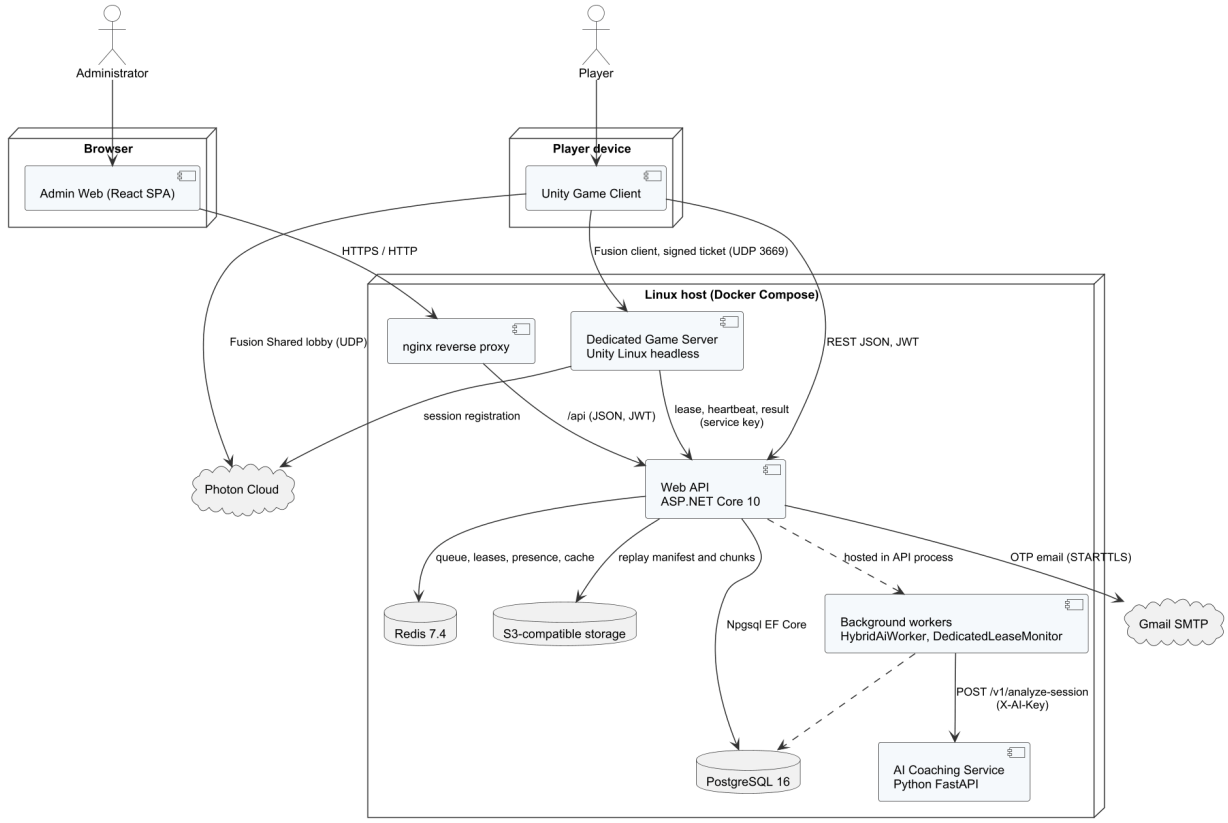


Figure 1. System architecture (component view)

Note: the figure was drawn on 12/11/2026. It names the host proxy nginx and shows the game client calling the API directly; since 13/11/2026 the client and the admin web reach the API through the Caddy HTTPS reverse proxy, as described in Table 2.

Table 1. Architecture components

Component	Technology	Responsibility
Unity game client	Unity 6000.6.2f1, C#, uGUI, Input System	Scenes and UI, local physics and rules for training and bot matches, Photon Fusion client, offline training store, REST client (ApiClient) to the Web API
Dedicated game server	Unity Linux headless build, Photon Fusion 2.1.3 (server mode)	Authoritative simulation of online matches; validates signed tickets; sends heartbeats, telemetry, replay and result to the Web API

Component	Technology	Responsibility
Web API	ASP.NET Core (.NET 10), layered API → Services → Repositories → Entities	REST endpoints, JWT authentication, business rules, transactions through IUnitOfWork and Npgsql EF Core
Background workers	Hosted services in the Web API	HybridAiWorker (analysis jobs), DedicatedLeaseMonitor (expired leases), ModerationExpiryWorker (expired suspensions), AnalyticsProjectionWorker (daily aggregates), PasswordRecoveryMailQueue (e-mail sending)
AI coaching service	Python 3.12, FastAPI, scikit-learn	Rule engine for technique errors, summary and drill recommendations, gated tier model; internal API protected by an API key
Admin web application	React 18, TypeScript 5, Vite 5, React Query, Axios	Administration screens served by nginx; calls the Web API with the administrator's JWT
PostgreSQL 16	Relational database	51 tables: accounts, profiles, content, matches, economy, moderation, telemetry and coaching
Redis 7.4	In-memory store	Matchmaking queue and search state, dedicated-server lease, presence, leaderboard cache (15 s)
Object storage	S3-compatible container	Replay manifests and compressed chunks
Photon Cloud	External service	Fusion session lookup and relay for clients and the dedicated server
Gmail SMTP	External service	Delivery of verification and password-reset e-mails

Online matches support two authority modes recorded in GameMatches.AuthorityMode. In **dedicated** mode, enabled by the configuration flag DedicatedServer:Enabled (true on staging), the dedicated server runs the simulation and is the only authority for points, faults and the result. In **shared** mode, kept for local development without a server, the room host runs the simulation through Photon Fusion shared mode. The staging deployment has one dedicated-server container, so one authoritative match runs at a time; the API reports the slot as busy and the client waits (Report 3, MSG36). Scaling to more matches requires more server containers and is outside release 1.0.

Deployment

Figure 2 shows the staging deployment on the host tad-min.io.vn (Ubuntu 24.04, Docker Engine). On each push to main the GitHub Actions workflow dockerhub.yml first reads the release that is deployed and plans which checks and images the changed files need (plan-release.py). It then runs the selected checks – twelve IntegrationChecks groups on PostgreSQL and Redis, the FastAPI tests and the admin web type check – and builds the Unity Linux dedicated server with GameCI when the Unity project changed. **Publish** (publish-release.sh) removes the credentials from the settings files, builds the changed images among api, ai, web and server, pushes them to Docker Hub with the tag <component>-<commit>-<run>-<attempt> and smoke-tests the four image digests with Docker Compose on the runner. **Deploy** (ssh-deploy.sh) locks the release on the host, takes a PostgreSQL dump and checks it with pg_restore, copies docker-compose.linux.yml and Database.sql, writes the four image digests into the host .env file, runs docker compose up and records the deployed release. **Status on 13/11/2026:** the pipeline passes. Of its 32 runs, 29 did not pass: two configuration failures on 05/11/2026, 18 runs that did not start while the GitHub account was locked for a billing issue (06/11 to 11/11), six failures of the dedicated-server build (11/11 and 12/11), one failed FastAPI test and two failed deployment steps. Runs 27, 28 and 30 of 13/11/2026 completed every job in about 15 minutes each, and run 30 deployed the release that is running now. The four earlier staging releases (Docker Hub tags api-, ai-, web- and server-manual-<date-time>, 08/11 to 10/11/2026) were built and started by hand. Since 13/11/2026 a Caddy reverse proxy on the host serves the admin web and the Web API over HTTPS; Table 2 lists the ports that are reachable from the Internet.

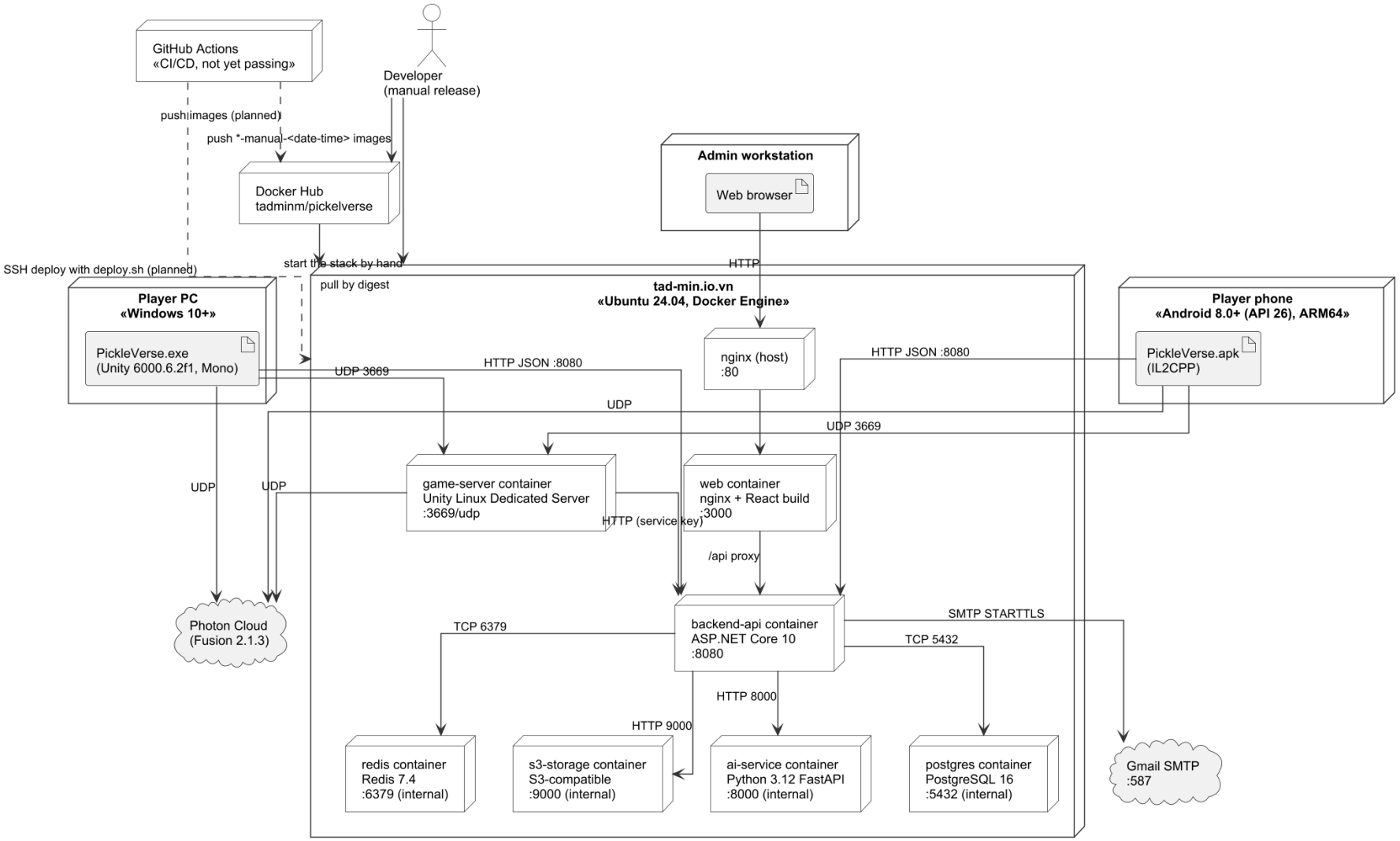


Figure 2. Deployment diagram of the staging environment

Note: the figure was drawn on 12/11/2026. It still shows the pipeline as not passing, the manual release, the former proxy on port 80 and the clients calling port 8080; the paragraph above and Table 2 describe the deployment of 13/11/2026 (pipeline passing, Caddy reverse proxy with HTTPS on port 443 in front of the API and the admin web).

Table 2. Deployment nodes

Node / container	Image or artifact	Port	Notes
Host reverse proxy	Caddy with deploy/linux/Caddyfile	80/tcp, 443/tcp	Port 80 redirects to HTTPS. Port 443 terminates TLS (Let's Encrypt certificate, TLS 1.3) and forwards /api and /health/ready to backend-api and every other path to the web container
web	tadminm/pickelverse:web-<release>	3000/tcp (bound to 127.0.0.1)	nginx serving the React build of the admin web; reached only through the reverse proxy
backend-api	tadminm/pickelverse:api-<release>	8080/tcp (bound to 127.0.0.1)	Web API and background workers; reached from the Internet only through the reverse proxy since 13/11/2026 (port 8080 was public before)
game-server	tadminm/pickelverse:server-<release>	3669/udp	Unity dedicated server; limited to 2 CPUs and 2 GB RAM
ai-service	tadminm/pickelverse:ai-<release>	8000/tcp (internal)	FastAPI coaching service, reachable only from backend-api
postgres	postgres:16-alpine	5432/tcp (internal)	Volume for data; schema and seed from src/Database.sql; dumped before each deployment
redis	redis:7.4-alpine	6379/tcp (internal)	Coordination data (queue, leases, presence, cache) with an append-only file on the host volume
s3-storage	ghcr.io/shyim/local-s3 (S3-compatible server)	9000/tcp (internal)	Bucket pickleverse for replays
Player PC / phone	PickleVerse.exe (Windows, Mono) / PickleVerse.apk (Android 8.0+, IL2CPP, ARM64)	–	Connects to the API, Photon Cloud and the game server

1.2 Package Diagram

The three code bases are organised in layers. Dependencies point only downwards: a package may use the packages below it but never the reverse. Figures 3 to 5 show the packages of the Web API, the Unity client and the admin web with the AI service.

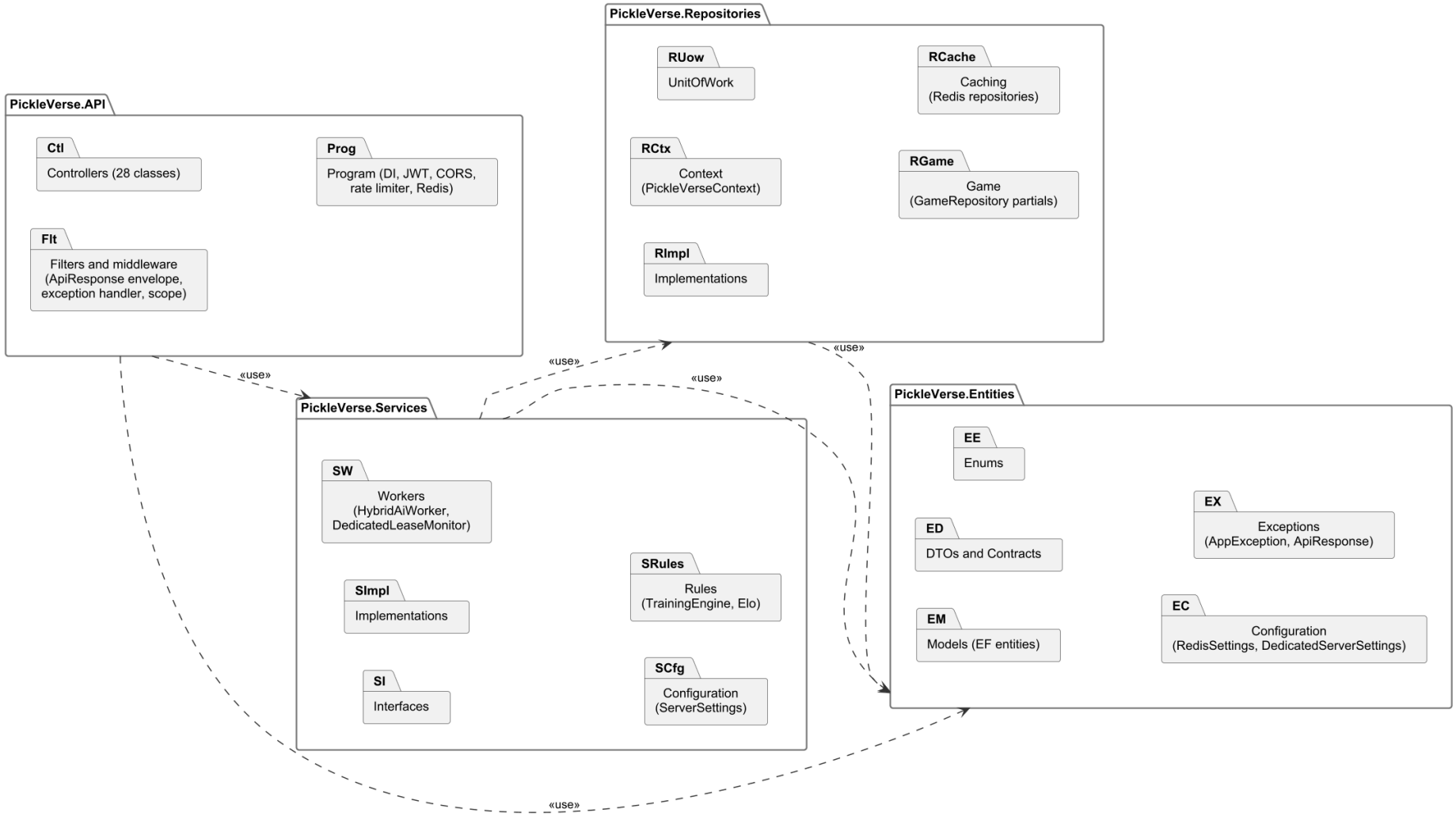


Figure 3. Package diagram of the Web API solution

Table 3. Web API package descriptions

No	Package	Description
01	PickleVerse.API	Entry point: 28 controller classes in 24 files, Program.cs (dependency injection, JWT, CORS, rate limiter, Redis), filters that wrap responses in ApiResponse<T>, the global exception handler that maps service exceptions to HTTP status codes
02	PickleVerse.Services	Business logic: service interfaces (Interfaces) and implementations (Implementations), background workers, typed settings (ServerSettings) and pure rule classes (TrainingEngine, ProgressionEngine, EloPolicy)
03	PickleVerse.Repositories	Data access: PickleVerseContext (Npgsql EF Core mapping of the 51 tables), IUnitOfWork, repositories, GameRepository partial classes for match and progress persistence, Redis repositories for queues, leases and caches
04	PickleVerse.Entities	Shared model: EF entity classes, DTOs and contracts, enums, configuration models (RedisSettings, DedicatedServerSettings) and exception/response types (AppException, ApiResponse)

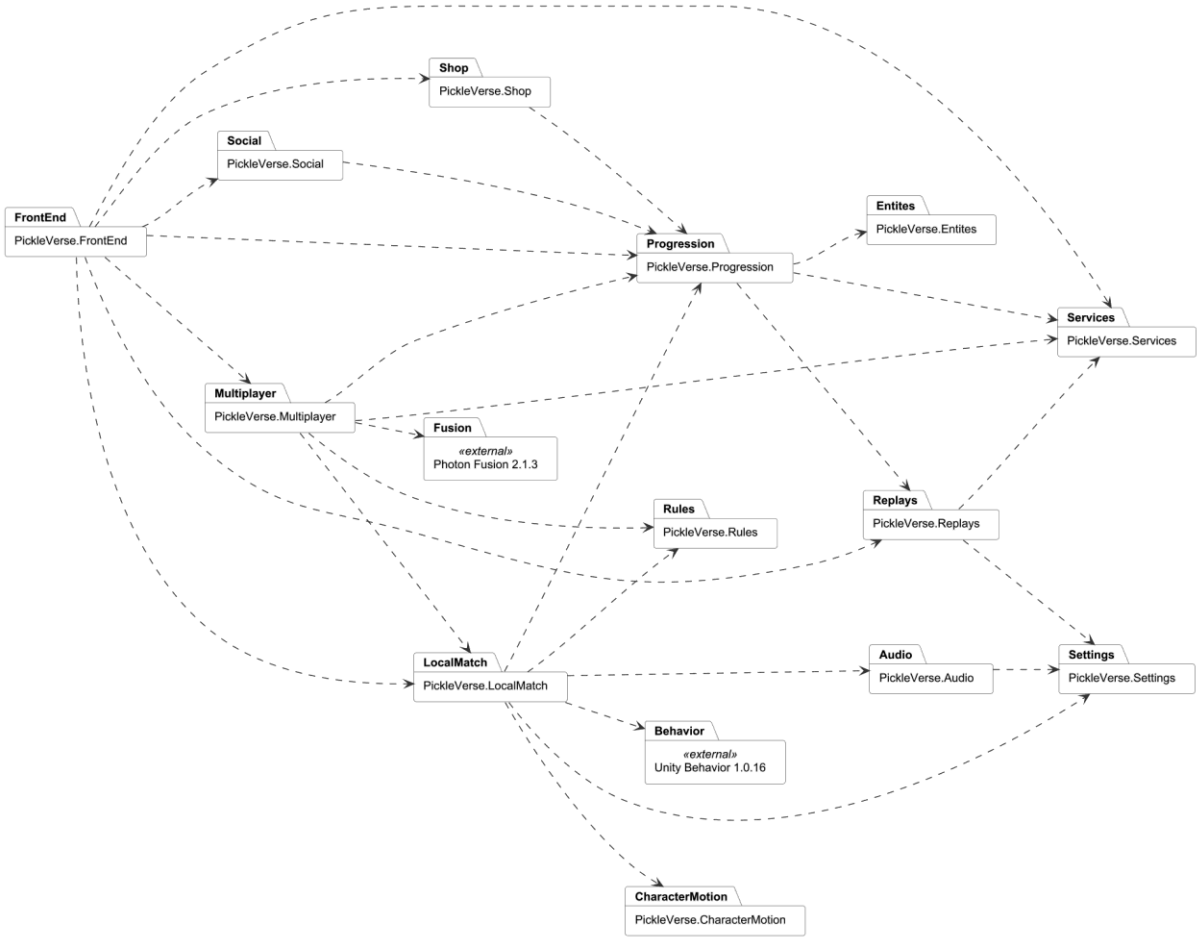


Figure 4. Package diagram of the Unity client (assembly definitions)

Table 4. Unity client package descriptions

No	Package	Description
01	PickleVerse.FrontEnd	Menus and screens: login, loading, menu, mode selection, settings entry, overlays, history and missions popups
02	PickleVerse.Multiplayer	Online flow: MultiplayerApp, PhotonSession, room state, NetworkMatch and NetworkPlayer, dedicated-server client and reconnection
03	PickleVerse.LocalMatch	Local gameplay: BotMatch, MatchActor, ball flight, paddle contact, shot input, LocalBot and BhTreeBot, training and tutorial directors
04	PickleVerse.Rules	Engine-independent rules shared by local and online play (scoring, faults, court geometry)
05	PickleVerse.Progression	ProgressionRuntime, training engine on the client, offline training store, wallet and profile cache

No	Package	Description
06	PickleVerse.Services	ApiClient, AccountService and backend settings
07	PickleVerse.Entites	Data contracts shared by client assemblies (room models, DTOs); the name keeps the original spelling of the assembly
08	PickleVerse.Shop	Shop and collection scenes
09	PickleVerse.Social	Friends, blocking and leaderboard UI
10	PickleVerse.Replays	Replay recording, upload and the replay viewer
11	PickleVerse.Settings	Settings scene: audio, graphics, key binding and touch layout
12	PickleVerse.Audio	Music player and sound effects policy
13	PickleVerse.CharacterMotion	Character animation controller and movement blending

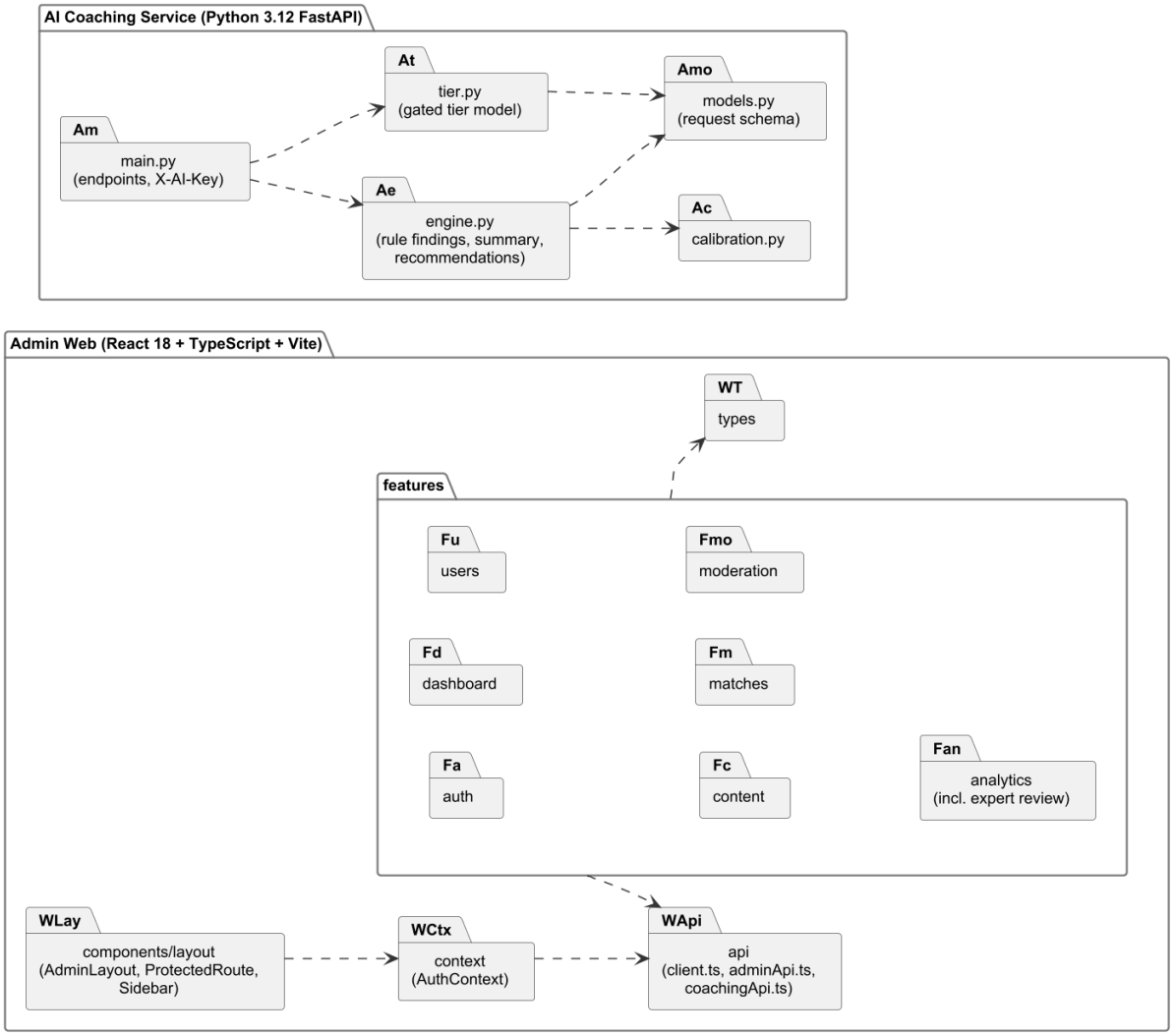


Figure 5. Package diagram of the admin web application and the AI coaching service

Table 5. Admin web and AI service package descriptions

No	Package	Description
01	features (admin web)	One folder per area: auth, dashboard, users, content, matches, moderation, analytics (including expert review); each holds its pages
02	api	Axios client with the JWT interceptor (client.ts) and typed API modules (adminApi.ts, coachingApi.ts)
03	context	AuthContext: stores the session and exposes sign-in and sign-out
04	components/layout	AdminLayout, Sidebar and ProtectedRoute, which redirects users without the Admin role to the login page
05	main.py (AI service)	FastAPI application, request validation and the X-AI-Key check
06	engine.py	Rule-based findings for 15 error codes, summary and at most three recommendations

No	Package	Description
07	tier.py	Gated tier model: 8 features, minimum confidence 0.65, minimum 20 opportunities
08	calibration.py / models.py	Threshold calibration against expert labels; request and response schemas

1.3 Technology Stack

Table 6. Technology stack

Layer	Technology and version	Purpose
Game client	Unity 6000.6.2f1, C#, uGUI, Input System, Unity Behavior 1.0.16, Photon Fusion 2.1.3	Gameplay, UI, bot behaviour graph, networking
Game server	Unity 6000.6.2f1 Linux dedicated server build	Authoritative online matches
Web API	ASP.NET Core on .NET 10, Npgsql EF Core 10, StackExchange.Redis 3.4, BCrypt.Net-Next 4.2, MailKit 4.17, AWSSDK.S3 4.0, JwtBearer 10	REST API, persistence, caching, password hashing, e-mail
AI service	Python 3.12, FastAPI, Pydantic, scikit-learn	Coaching analysis and tier model
Admin web	React 18, TypeScript 5, Vite 5, TanStack React Query 5, Axios 1.6, Recharts 2, Tailwind CSS 3	Administration UI
Data	PostgreSQL 16, Redis 7.4, S3-compatible storage	Persistence, coordination, replays
DevOps	Docker, Docker Compose, Docker Hub, Caddy (HTTPS reverse proxy), Ubuntu 24.04; GitHub Actions workflow with plan, check, build, publish and deploy jobs, passing since 13/11/2026	Build, delivery and hosting

1.4 Architecture Decisions

Table 7 records the main decisions that differ from the registered proposal, the alternatives considered and their consequences. Report 1 (LI-08) refers to this table.

Table 7. Architecture decisions

#	Decision	Alternatives	Reason	Consequence
1	ASP.NET Core Web API (.NET 10)	Spring Boot; FastAPI	The team writes C# for Unity; sharing the language reduces context switching and lets DTOs be checked against the Unity client in IntegrationChecks	The AI service stays in Python; two runtimes on the server
2	Photon Fusion 2 with a dedicated server	Netcode for GameObjects with Relay (prototype 07–21/10/2026); Mirror	The NGO prototype showed unstable relayed connections; Fusion provides input authority, a server mode and tick-based state replication on which client prediction is built	Dependency on Photon Cloud and its CCU plan (Report 2, risk 4)
3	Custom fixed-step ball simulation (120 Hz)	Unity PhysX rigidbody	Deterministic, engine-independent flight and bounce that the server and replays can reproduce exactly	Collision with paddle and net is implemented by the team (sphere sweep against a contact box)
4	uGUI for the game interface	UI Toolkit	Existing art assets and world-space panels were built with uGUI; better support for runtime animation	Two UI systems are not mixed; layout code is in C#
5	Server-authoritative results and rewards	Client-reported results	Prevents cheating on Elo, currency and missions	Bot matches are reported by the client; only Elo-neutral practice uses this path except Ranked Bot, whose result is validated against the server-created match
6	Single schema file (src/Database.sql) without migrations during development	EF Core migrations	The schema changed often; one reviewed file is the single source for schema and seed data	Database resets are allowed only on the development instance (AGENTS.md)

#	Decision	Alternatives	Reason	Consequence
7	Gated AI tier model with rule fallback	Always show the model output	Too little labelled real data; a wrong tier would mislead beginners	The tier is hidden until the acceptance criteria of NFR-06 are met

2. Database Design

The logical database design is derived directly from src/Database.sql (51 tables, 366 columns, 71 foreign keys). To keep the diagrams readable the schema is divided into seven subject areas (Figures 6 to 12). Each diagram uses crow's-foot notation: * marks NOT NULL columns, «PK» and «FK» mark keys, and tables drawn without columns belong to another area and are shown only for their relationships. Table names follow the PascalCase convention of the EF model; the AI tables (ai*) and ratingledger use lower case because they were created for the Python service and the rating ledger and are mapped explicitly in PickleVerseContext.

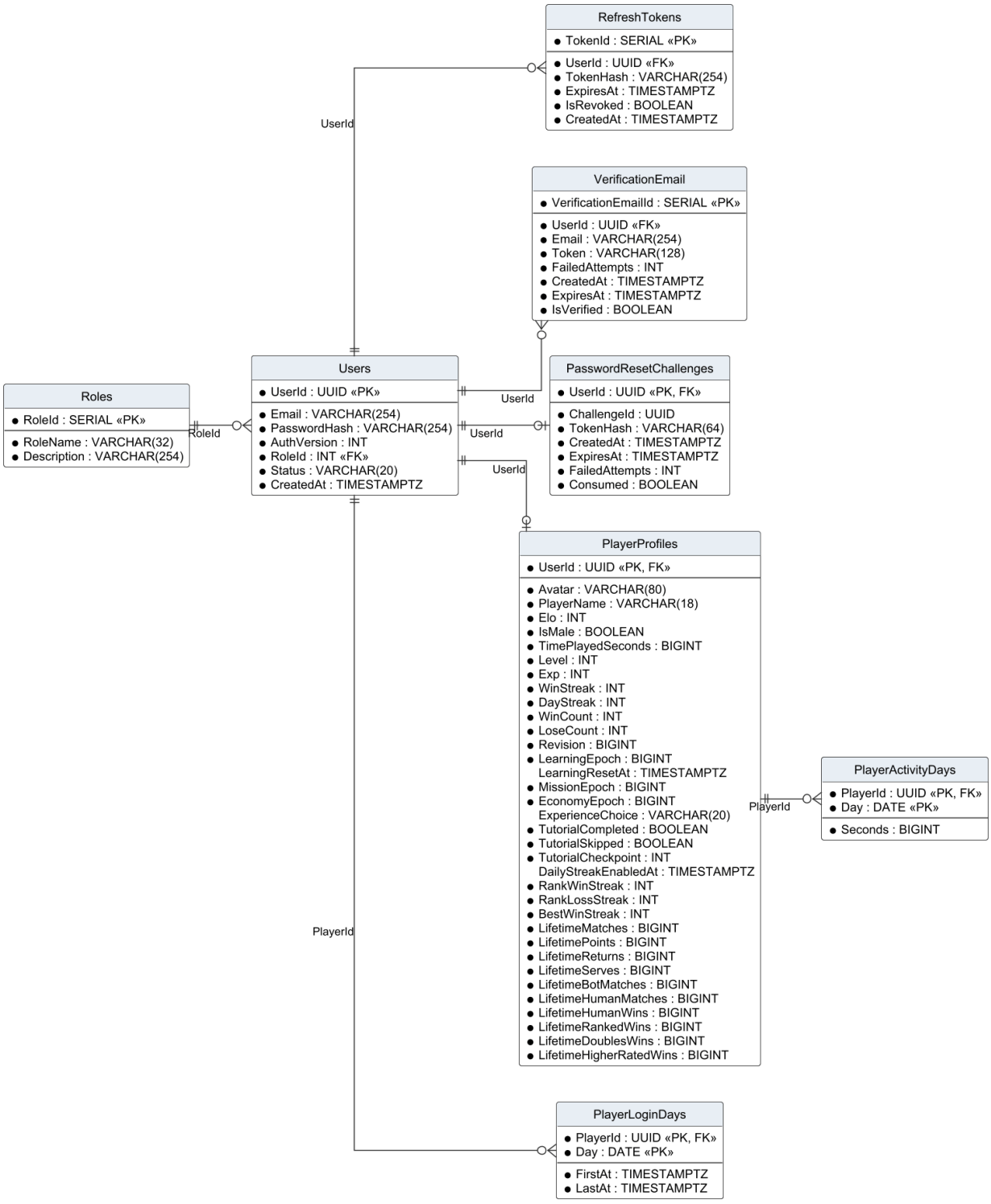


Figure 6. Logical ERD – accounts and profiles

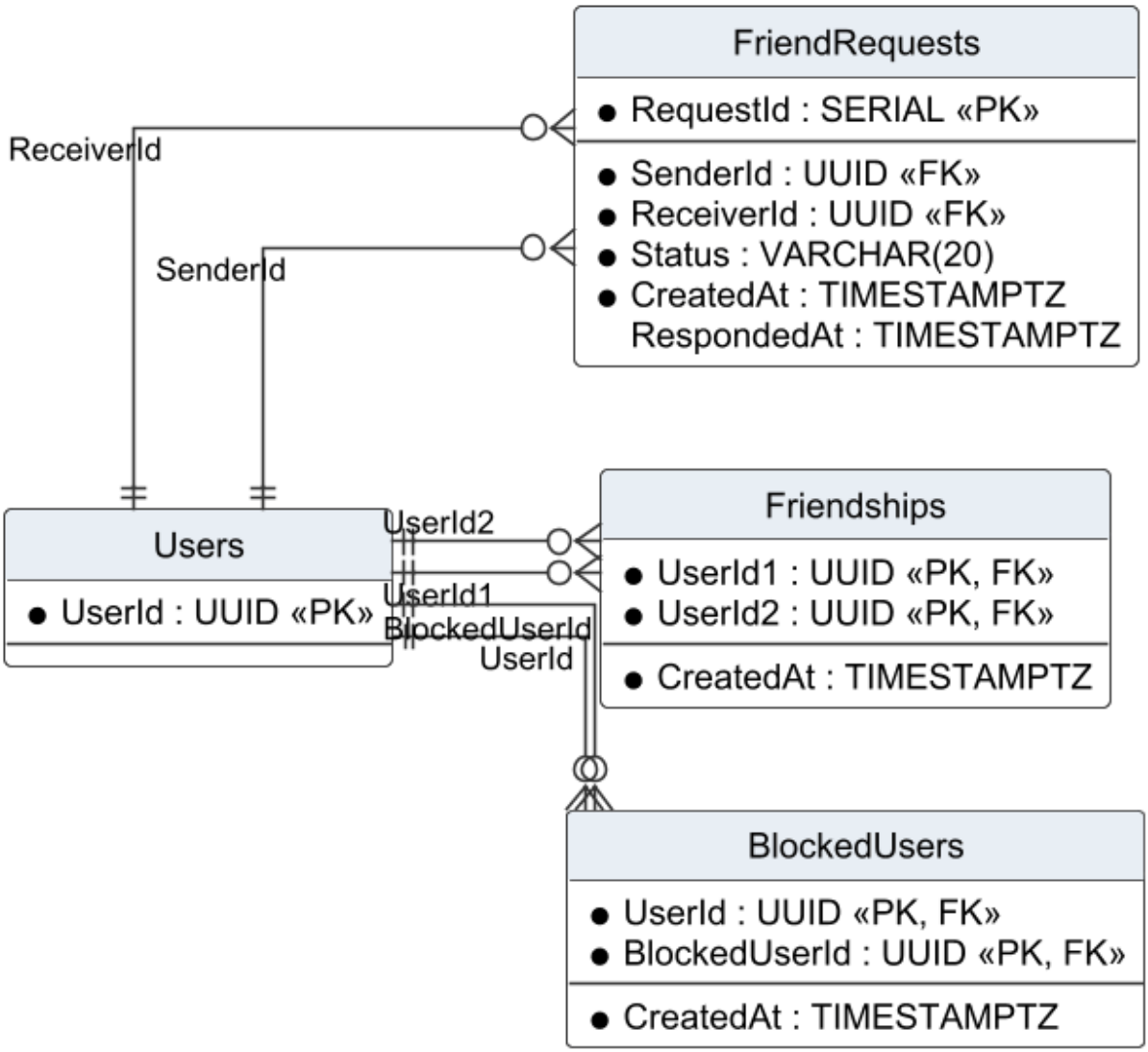


Figure 7. Logical ERD – friends and blocking

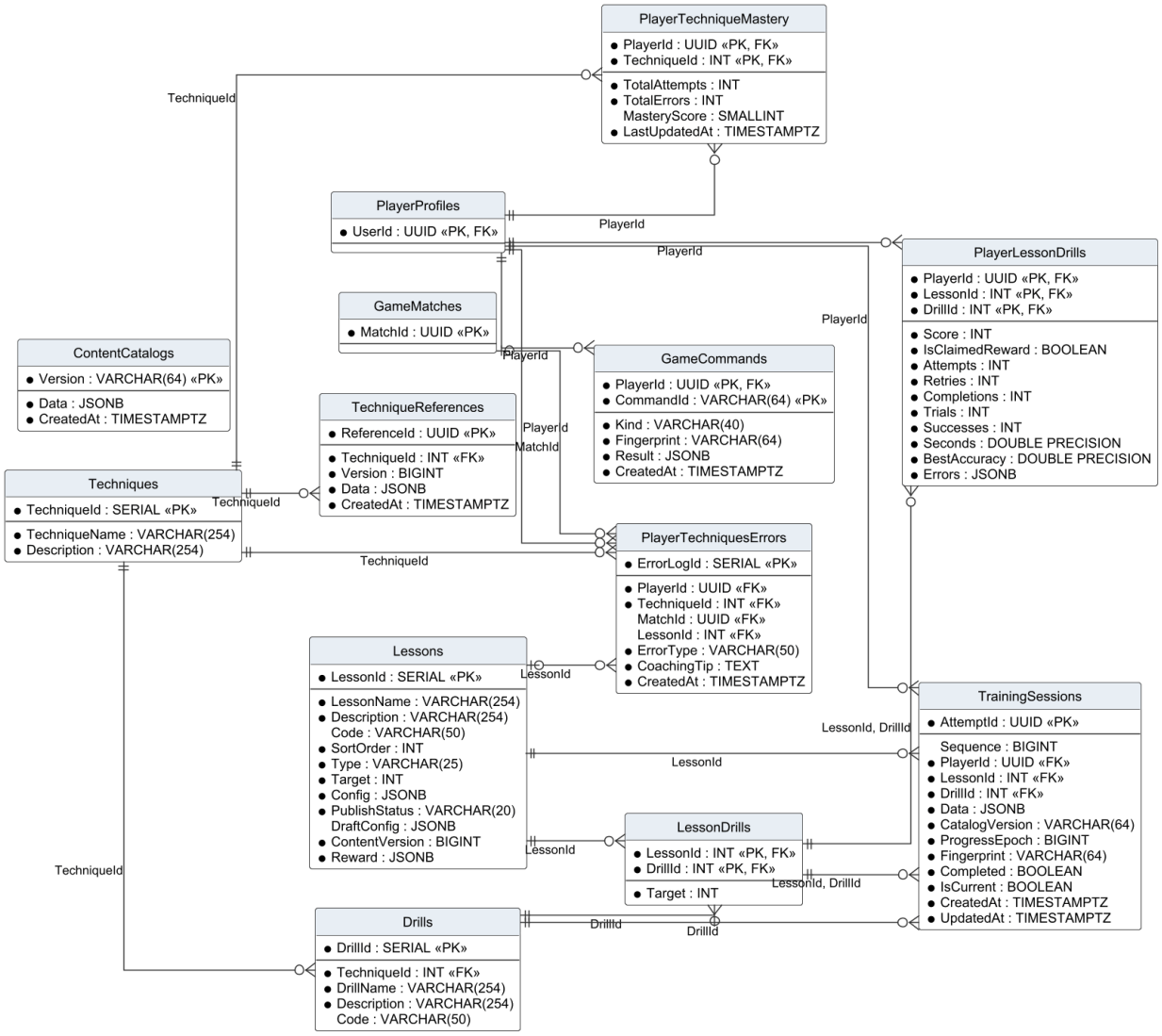


Figure 8. Logical ERD – techniques, lessons and training

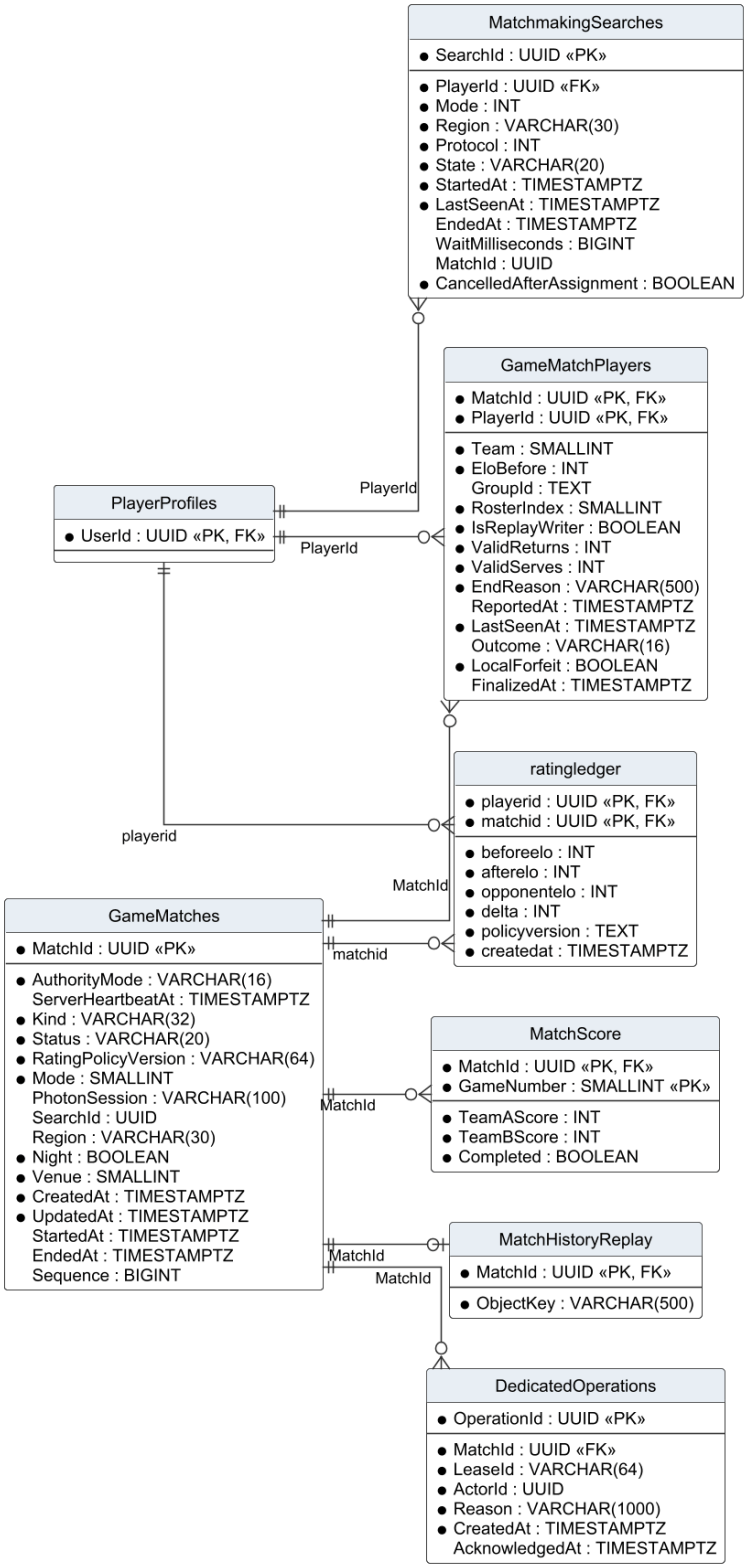


Figure 9. Logical ERD – matches, ratings, replays and matchmaking

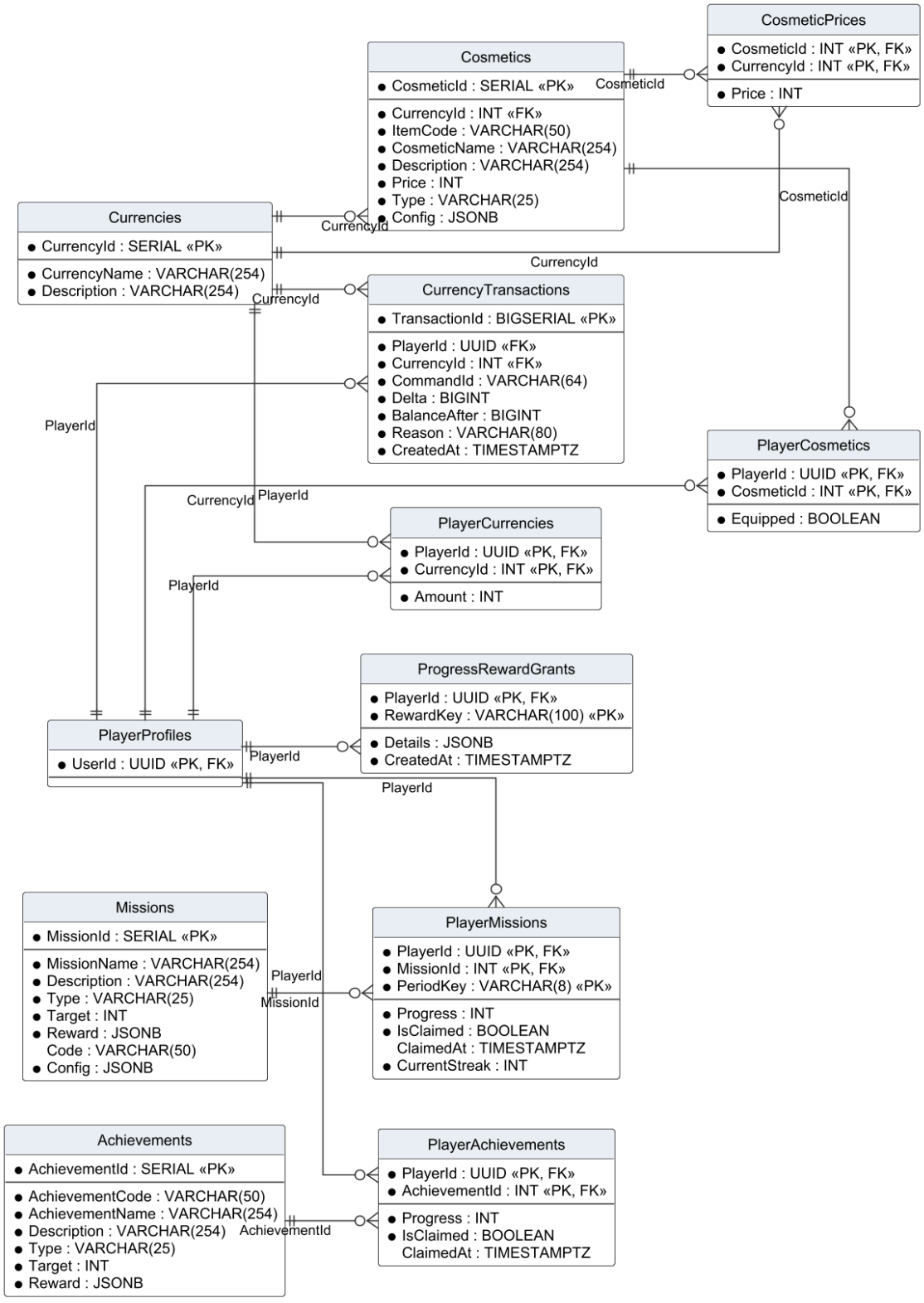


Figure 10. Logical ERD – currencies, cosmetics, missions and achievements

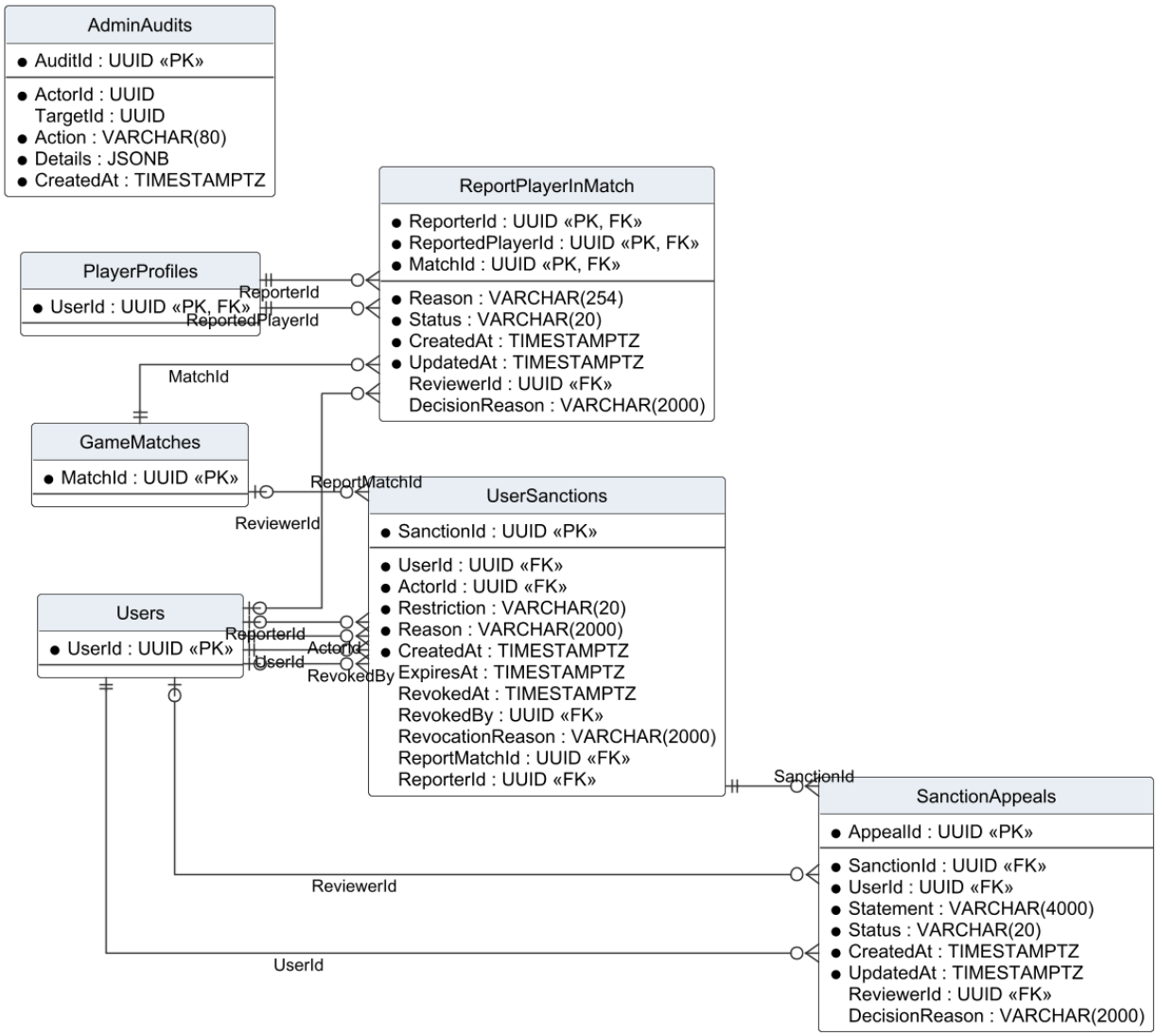


Figure 11. Logical ERD – reports, sanctions, appeals and audit

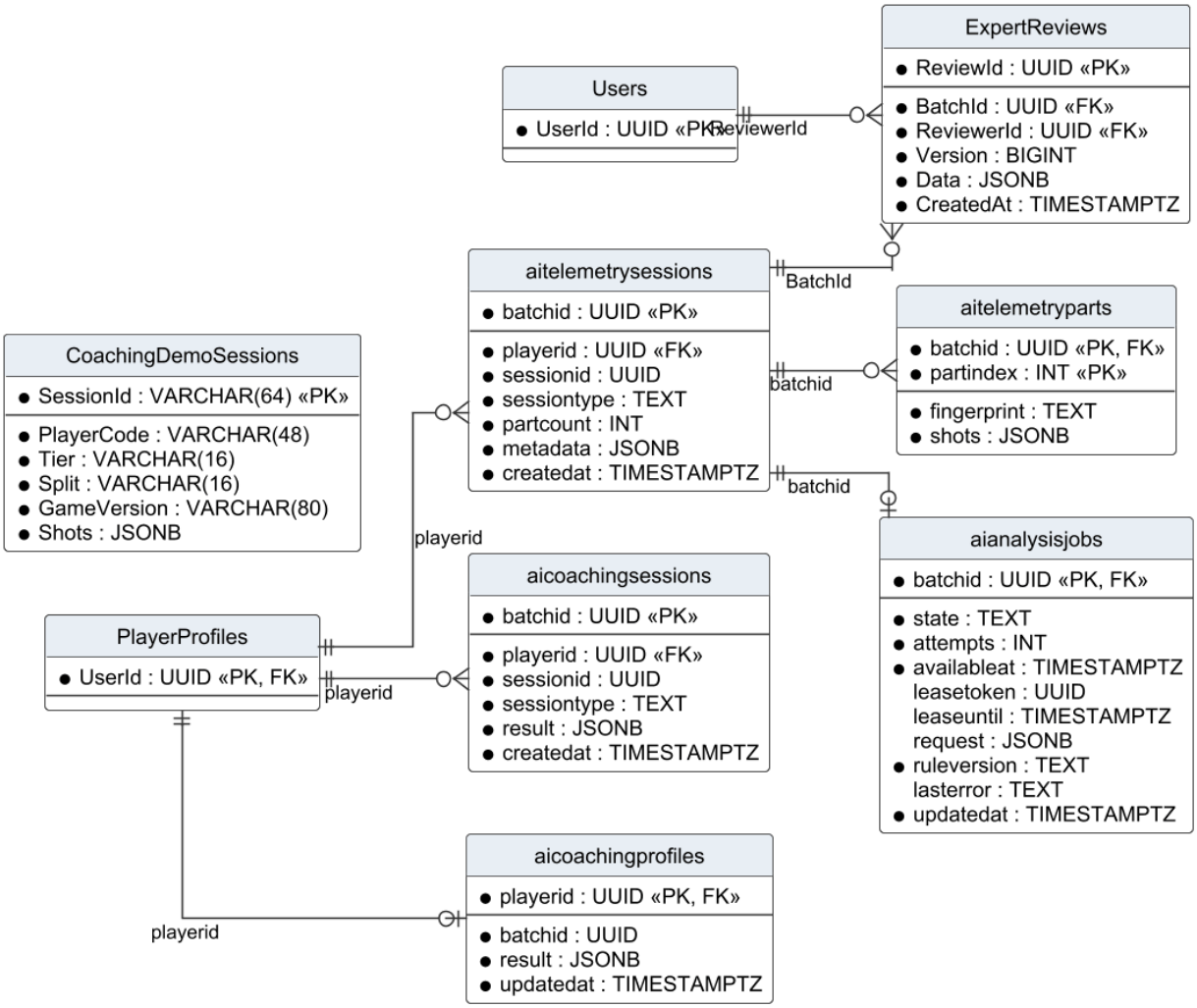


Figure 12. Logical ERD – telemetry, coaching and expert review

Table Descriptions

Table 8. Table descriptions (51 tables)

No	Table	Description
01	Roles	Role catalogue (Player, Admin). - Primary keys: RoleId - Foreign keys: none
02	Users	Account: e-mail, BCrypt password hash, role, status (ACTIVE, SUSPENDED, BANNED), creation time and the auth version used to revoke tokens. Whether the e-mail is verified is read from VerificationEmail.IsVerified. - Primary keys: UserId - Foreign keys: RoleId → Roles
03	RefreshTokens	Refresh tokens stored as SHA-256 hashes with expiry and revocation time. - Primary keys: TokenId - Foreign keys: UserId → Users
04	VerificationEmail	E-mail verification record of an account: HMAC of the current code, expiry, failed attempts and the IsVerified flag. - Primary keys: VerificationEmailId - Foreign keys: UserId → Users
05	PasswordResetChallenges	The current password-reset challenge of an account (one per user). - Primary keys: UserId - Foreign keys: UserId → Users

No	Table	Description
06	PlayerProfiles	Game profile of an account: display name, avatar, gender of the character, onboarding and tutorial state, level, XP, Elo, wins, losses, streaks, reset epochs and lifetime statistics. Equipped items are stored in PlayerCosmetics.Equipped. - Primary keys: UserId - Foreign keys: UserId → Users
07	FriendRequests	Friend requests with status PENDING, ACCEPTED or DECLINED. - Primary keys: RequestId - Foreign keys: SenderId → Users; ReceiverId → Users
08	Friendships	Accepted friendships stored once per pair (UserId1 < UserId2). - Primary keys: UserId1, UserId2 - Foreign keys: UserId1 → Users; UserId2 → Users
09	BlockedUsers	Blocks from one user to another. - Primary keys: UserId, BlockedUserId - Foreign keys: UserId → Users; BlockedUserId → Users
10	GameMatches	Matches of every kind with mode, kind, authority mode, status (PREPARING, PLAYING, FINISHED, CANCELLED), rating policy and timestamps. - Primary keys: MatchId - Foreign keys: none
11	PlayerLoginDays	Days (Vietnam time) on which a player signed in; used for retention. - Primary keys: PlayerId, Day - Foreign keys: PlayerId → Users
12	PlayerActivityDays	Days on which a player was active, with active seconds. - Primary keys: PlayerId, Day - Foreign keys: PlayerId → PlayerProfiles
13	MatchmakingSearches	Durable log of ranked searches with status SEARCHING, MATCHED, BOT, CANCELLED or TIMEOUT. - Primary keys: SearchId - Foreign keys: PlayerId → PlayerProfiles
14	ReportPlayerInMatch	Player reports, one per reporter, reported player and match, with status and decision. - Primary keys: ReporterId, ReportedPlayerId, MatchId - Foreign keys: ReviewerId → Users; ReporterId → PlayerProfiles; ReportedPlayerId → PlayerProfiles; MatchId → GameMatches
15	UserSanctions	Suspensions and bans with actor, reason, expiry and revocation data. - Primary keys: SanctionId - Foreign keys: UserId → Users; ActorId → Users; RevokedBy → Users; ReportMatchId → GameMatches; ReporterId → Users
16	SanctionAppeals	Appeals, at most one per sanction, with statement, status and decision. - Primary keys: AppealId - Foreign keys: SanctionId → UserSanctions; UserId → Users; ReviewerId → Users
17	MatchHistoryReplay	Object-storage key of the replay manifest of a match. - Primary keys: MatchId - Foreign keys: MatchId → GameMatches
18	Currencies	Currency catalogue (TOKENPAD, ELITE TOKEN). - Primary keys: CurrencyId - Foreign keys: none
19	PlayerCurrencies	Wallet balance of a player per currency. - Primary keys: PlayerId, CurrencyId - Foreign keys: PlayerId → PlayerProfiles; CurrencyId → Currencies
20	Cosmetics	Cosmetic items with code, type, default price and currency, and display configuration. - Primary keys: CosmeticId - Foreign keys: CurrencyId → Currencies
21	PlayerCosmetics	Items owned by a player and the equipped flag. - Primary keys: PlayerId, CosmeticId - Foreign keys: PlayerId → PlayerProfiles; CosmeticId → Cosmetics
22	Achievements	Achievement definitions with type, target and reward. - Primary keys: AchievementId - Foreign keys: none
23	PlayerAchievements	Achievement progress and claim time of a player. - Primary keys: PlayerId, AchievementId - Foreign keys: PlayerId → PlayerProfiles; AchievementId → Achievements
24	Missions	Daily and weekly mission definitions with target and reward. - Primary keys: MissionId - Foreign keys: none
25	PlayerMissions	Mission progress of a player in one period (PeriodKey) and the claim time. - Primary keys: PlayerId, MissionId, PeriodKey - Foreign keys: PlayerId → PlayerProfiles; MissionId → Missions
26	Techniques	Techniques taught by the curriculum. - Primary keys: TechniqueId - Foreign keys: none

No	Table	Description
27	TechniqueReferences	Versioned reference values used to judge a technique. - Primary keys: ReferenceId - Foreign keys: TechniqueId → Techniques
28	Lessons	Lessons with code, group, order, target, reward (JSONB), configuration (JSONB), prerequisite and status (DRAFT, PUBLISHED, UNPUBLISHED). - Primary keys: LessonId - Foreign keys: none
29	Drills	Drills that practise one technique. - Primary keys: DrillId - Foreign keys: TechniqueId → Techniques
30	LessonDrills	Drills of a lesson with their target (many-to-many). - Primary keys: LessonId, DrillId - Foreign keys: LessonId → Lessons; DrillId → Drills
31	PlayerTechniqueMastery	Mastery score of a player per technique. - Primary keys: PlayerId, TechniqueId - Foreign keys: PlayerId → PlayerProfiles; TechniqueId → Techniques
32	PlayerTechniquesErrors	Technique errors detected for a player in a lesson or match. - Primary keys: ErrorLogId - Foreign keys: PlayerId → PlayerProfiles; TechniqueId → Techniques; MatchId → GameMatches; LessonId → Lessons
33	PlayerLessonDrills	Best result of a player per lesson drill: stars, accuracy, timing, attempts and first-pass time. - Primary keys: PlayerId, LessonId, DrillId - Foreign keys: PlayerId → PlayerProfiles; LessonId, DrillId → LessonDrills
34	CosmeticPrices	Prices of an item in each accepted currency. - Primary keys: CosmeticId, CurrencyId - Foreign keys: CosmeticId → Cosmetics; CurrencyId → Currencies
35	GameCommands	Idempotency log of client commands (purchase, claim, sync) per player and command ID. - Primary keys: PlayerId, CommandId - Foreign keys: PlayerId → PlayerProfiles
36	TrainingSessions	Each training attempt with lesson, drill, shots, hits, timing, stars and receipt status. - Primary keys: AttemptId - Foreign keys: PlayerId → PlayerProfiles; LessonId → Lessons; DrillId → Drills; LessonId, DrillId → LessonDrills
37	MatchScore	Score of each game of a match (weak entity identified by match and game number). - Primary keys: MatchId, GameNumber - Foreign keys: MatchId → GameMatches
38	GameMatchPlayers	Participants of a match: team, party group, roster slot, Elo before the match, replay-writer flag, valid serves and returns, end reason, last-seen time and outcome (WIN, LOSE, INTERRUPTED). Names, characters and bot accounts are joined from the profile tables. - Primary keys: MatchId, PlayerId - Foreign keys: MatchId → GameMatches; PlayerId → PlayerProfiles
39	CurrencyTransactions	Ledger of every currency change with reason and reference. - Primary keys: TransactionId - Foreign keys: PlayerId → PlayerProfiles; CurrencyId → Currencies
40	ratingledger	Elo before, delta and after for one player in one ranked match. - Primary keys: playerid, matchid - Foreign keys: playerid → PlayerProfiles; matchid → GameMatches
41	aitelemetrysessions	Telemetry batch of a drill or match with source, mode and status. - Primary keys: batchid - Foreign keys: playerid → PlayerProfiles
42	ExpertReviews	Expert labels (level, errors, data group) for a telemetry batch. - Primary keys: ReviewId - Foreign keys: BatchId → aitelemetrysessions; ReviewerId → Users
43	CoachingDemoSessions	Simulated demonstration sessions kept apart from real player data. - Primary keys: SessionId - Foreign keys: none
44	aitelemetryparts	Compressed parts of a telemetry batch. - Primary keys: batchid, partindex - Foreign keys: batchid → aitelemetrysessions
45	aianalysisjobs	Analysis job of a batch with status pending, processing, completed or failed, attempts and error. - Primary keys: batchid - Foreign keys: batchid → aitelemetrysessions
46	aicoachingsessions	Coaching result of a batch: findings, summary and recommendations. - Primary keys: batchid - Foreign keys: playerid → PlayerProfiles
47	aicoachingprofiles	Aggregated coaching profile of a player. - Primary keys: playerid - Foreign keys: playerid → PlayerProfiles

No	Table	Description
48	ContentCatalogs	Published versions of the content catalogue sent to clients. - Primary keys: Version - Foreign keys: none
49	ProgressRewardGrants	Log of one-time progression rewards (for example THOT, GEMSTONE) per player. - Primary keys: PlayerId, RewardKey - Foreign keys: PlayerId → PlayerProfiles
50	DedicatedOperations	Operations on dedicated matches (stop requests and their acknowledgement). - Primary keys: OperationId - Foreign keys: MatchId → GameMatches
51	AdminAudits	Audit trail of administrative actions with actor, action, target and data. - Primary keys: AuditId - Foreign keys: none

2.1 Data Dictionary of the Core Tables

The core tables of the account, match, training and moderation domains are listed column by column. Types are PostgreSQL types; * marks NOT NULL.

Table 9. Data dictionary – Users

Column	Type	Null	Key	Notes
UserId	UUID	*	PK	
Email	VARCHAR(254)	*		
PasswordHash	VARCHAR(254)	*		
AuthVersion	INT	*		
RoleId	INT	*	FK → Roles	
Status	VARCHAR(20)	*		ACTIVE, SUSPENDED, BANNED
CreatedAt	TIMESTAMPTZ	*		

Table 10. Data dictionary – PlayerProfiles

Column	Type	Null	Key	Notes
UserId	UUID	*	PK, FK → Users	
Avatar	VARCHAR(80)	*		
PlayerName	VARCHAR(18)	*		
Elo	INT	*		
IsMale	BOOLEAN	*		
TimePlayedSeconds	BIGINT	*		
Level	INT	*		
Exp	INT	*		
WinStreak	INT	*		
DayStreak	INT	*		
WinCount	INT	*		
LoseCount	INT	*		
Revision	BIGINT	*		
LearningEpoch	BIGINT	*		
LearningResetAt	TIMESTAMPTZ			
MissionEpoch	BIGINT	*		
EconomyEpoch	BIGINT	*		
ExperienceChoice	VARCHAR(20)			
TutorialCompleted	BOOLEAN	*		
TutorialSkipped	BOOLEAN	*		
TutorialCheckpoint	INT	*		
DailyStreakEnabledAt	TIMESTAMPTZ			
RankWinStreak	INT	*		

Column	Type	Null	Key	Notes
RankLossStreak	INT	*		
BestWinStreak	INT	*		
LifetimeMatches	BIGINT	*		
LifetimePoints	BIGINT	*		
LifetimeReturns	BIGINT	*		
LifetimeServes	BIGINT	*		
LifetimeBotMatches	BIGINT	*		
LifetimeHumanMatches	BIGINT	*		
LifetimeHumanWins	BIGINT	*		
LifetimeRankedWins	BIGINT	*		
LifetimeDoublesWins	BIGINT	*		
LifetimeHigherRatedWins	BIGINT	*		

Table 11. Data dictionary – GameMatches

Column	Type	Null	Key	Notes
AuthorityMode	VARCHAR(16)	*		shared or dedicated
ServerHeartbeatAt	TIMESTAMPTZ			
MatchId	UUID	*	PK	
Kind	VARCHAR(32)	*		
Status	VARCHAR(20)	*		PREPARING, PLAYING, FINISHED, CANCELLED
RatingPolicyVersion	VARCHAR(64)	*		
Mode	SMALLINT	*		
PhotonSession	VARCHAR(100)			
SearchId	UUID			
Region	VARCHAR(30)			
Night	BOOLEAN	*		
Venue	SMALLINT	*		
CreatedAt	TIMESTAMPTZ	*		
UpdatedAt	TIMESTAMPTZ	*		
StartedAt	TIMESTAMPTZ			
EndedAt	TIMESTAMPTZ			
Sequence	BIGINT			

Table 12. Data dictionary – GameMatchPlayers

Column	Type	Null	Key	Notes
MatchId	UUID	*	PK, FK → GameMatches	
PlayerId	UUID	*	PK, FK → PlayerProfiles	
Team	SMALLINT	*		
EloBefore	INT	*		
GroupId	TEXT			
RosterIndex	SMALLINT	*		
IsReplayWriter	BOOLEAN	*		
ValidReturns	INT	*		
ValidServes	INT	*		
EndReason	VARCHAR(500)	*		
ReportedAt	TIMESTAMPTZ			
LastSeenAt	TIMESTAMPTZ	*		
Outcome	VARCHAR(16)			

Column	Type	Null	Key	Notes
LocalForfeit	BOOLEAN	*		
FinalizedAt	TIMESTAMPTZ			

Table 13. Data dictionary – MatchScore

Column	Type	Null	Key	Notes
MatchId	UUID	*	PK, FK → GameMatches	
GameNumber	SMALLINT	*	PK	
TeamAScore	INT	*		
TeamBScore	INT	*		
Completed	BOOLEAN	*		

Table 14. Data dictionary – ratingledger

Column	Type	Null	Key	Notes
playerid	UUID	*	PK, FK → PlayerProfiles	
matchid	UUID	*	PK, FK → GameMatches	
beforeelo	INT	*		
afterelo	INT	*		
opponentelo	INT	*		
delta	INT	*		
policyversion	TEXT	*		
createdat	TIMESTAMPTZ	*		

Table 15. Data dictionary – Lessons

Column	Type	Null	Key	Notes
LessonId	SERIAL	*	PK	
LessonName	VARCHAR(254)	*		
Description	VARCHAR(254)	*		
Code	VARCHAR(50)			
SortOrder	INT	*		
Type	VARCHAR(25)	*		
Target	INT	*		
Config	JSONB	*		
PublishStatus	VARCHAR(20)	*		
DraftConfig	JSONB			
ContentVersion	BIGINT	*		
Reward	JSONB	*		

Table 16. Data dictionary – TrainingSessions

Column	Type	Null	Key	Notes
AttemptId	UUID	*	PK	
Sequence	BIGINT			
PlayerId	UUID	*	FK → PlayerProfiles	
LessonId	INT	*	FK → LessonDrills	
DrillId	INT	*	FK → LessonDrills	
Data	JSONB	*		
CatalogVersion	VARCHAR(64)	*		
ProgressEpoch	BIGINT	*		

Column	Type	Null	Key	Notes
Fingerprint	VARCHAR(64)	*		
Completed	BOOLEAN	*		
IsCurrent	BOOLEAN	*		
CreatedAt	TIMESTAMPTZ	*		
UpdatedAt	TIMESTAMPTZ	*		

Table 17. Data dictionary – UserSanctions

Column	Type	Null	Key	Notes
SanctionId	UUID	*	PK	
UserId	UUID	*	FK → Users	
ActorId	UUID	*	FK → Users	
Restriction	VARCHAR(20)	*		BANNED or SUSPENDED; SUSPENDED requires ExpiresAt
Reason	VARCHAR(2000)	*		
CreatedAt	TIMESTAMPTZ	*		
ExpiresAt	TIMESTAMPTZ			
RevokedAt	TIMESTAMPTZ			
RevokedBy	UUID		FK → Users	
RevocationReason	VARCHAR(2000)			
ReportMatchId	UUID		FK → GameMatches	
ReporterId	UUID		FK → Users	

Table 18. Data dictionary – SanctionAppeals

Column	Type	Null	Key	Notes
AppealId	UUID	*	PK	
SanctionId	UUID	*	FK → UserSanctions	UNIQUE: one appeal per sanction
UserId	UUID	*	FK → Users	
Statement	VARCHAR(4000)	*		
Status	VARCHAR(20)	*		PENDING, CHECKING, ACCEPTED, DECLINED
CreatedAt	TIMESTAMPTZ	*		
UpdatedAt	TIMESTAMPTZ	*		
ReviewerId	UUID		FK → Users	
DecisionReason	VARCHAR(2000)			

2.2 Design Notes

- Team game totals are not stored in GameMatches; they are derived from the completed MatchScore rows, which avoids inconsistent copies (AGENTS.md).
- GameMatchPlayers keeps only facts that belong to the participant in that match, including the fixed Elo input needed before a rating ledger row exists; names and cosmetics are joined from the profile tables.
- PlayerProfiles is wide (35 columns) because it is the hot read model of the menu; statistics that change together are kept in one row to avoid joins on every screen.
- TrainingSessions references Lessons, Drills and the composite LessonDrills key: the composite key guarantees that the drill belongs to the lesson, and the single keys keep historical attempts when a lesson is re-configured.
- Lessons.Reward and Lessons.Config are JSONB because each lesson type has different parameters; the API validates them before saving.

- AdminAudits stores the actor ID without a foreign key so that audit records survive the deletion of an administrator account.
- CoachingDemoSessions holds simulated data in its own table so that it can never be mixed with real sessions used for AI evaluation.
- Idempotency: GameCommands, TrainingSessions receipts and the UNIQUE constraints on rewards (ProgressRewardGrants, PlayerMissions period key) make retried requests safe.

3. Detailed Design

Each subsection gives the class diagram of one feature group, the specification of its classes and the sequence diagrams of its main functions. Client classes (Unity) and server classes (Web API) are drawn in the same diagram only when the diagram shows their interaction; the boundary is the HTTP API, marked as a dependency to the controller.

3.1 Account and Authentication

This feature group implements FR-01 to FR-05. The Unity AccountScreen calls AccountService, which uses ApiClient to reach AuthController. AuthService performs the rules of BR-01 to BR-05 through IUnitOfWork and sends e-mails through IEmailSender.

3.1.1 Class Diagram

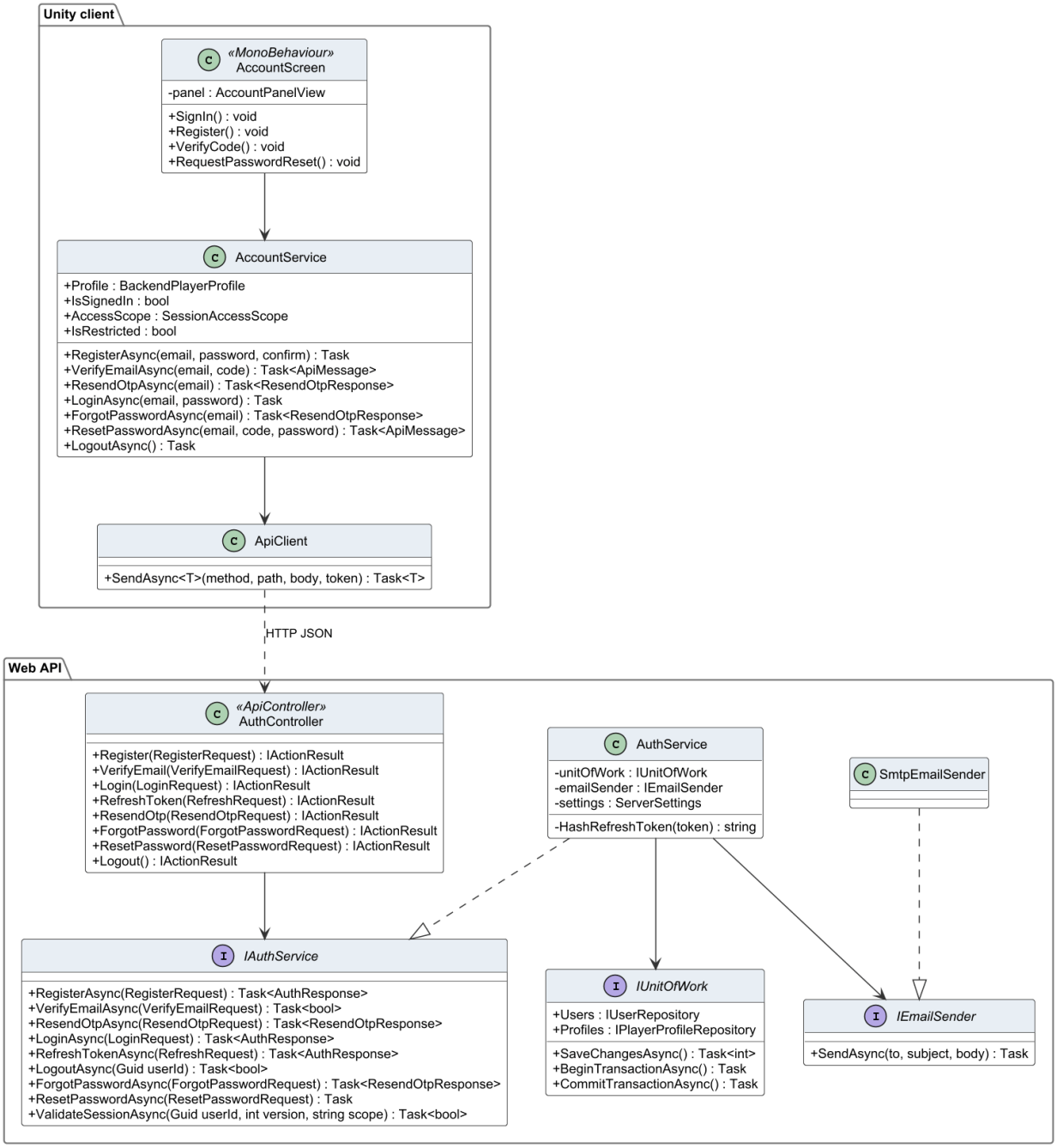


Figure 13. Class diagram – account and authentication

3.1.2 Class Specifications

Table 19. Class specifications – account and authentication

Class	Responsibility	Main attributes	Main operations
AccountScreen «MonoBehaviour»	Login, register, verification and password-reset pages of the Login scene; client-side validation	- panel : AccountPanelView	+ SignIn() : void + Register() : void + VerifyCode() : void + RequestPasswordReset() : void

Class	Responsibility	Main attributes	Main operations
AccountService	Client facade for authentication; keeps the session, scope and profile and refreshes tokens	+ Profile : BackendPlayerProfile + IsSignedIn : bool + AccessScope : SessionAccessScope + IsRestricted : bool	+ RegisterAsync(email, password, confirm) : Task + VerifyEmailAsync(email, code) : Task<ApiMessage> + ResendOtpAsync(email) : Task<ResendOtpResponse> + LoginAsync(email, password) : Task + ForgotPasswordAsync(email) : Task<ResendOtpResponse> + ResetPasswordAsync(email, code, password) : Task<ApiMessage> + LogoutAsync() : Task
ApiClient	Sends HTTP requests with the bearer token and unwraps the ApiResponse envelope	–	+ SendAsync<T>(method, path, body, token) : Task<T>
AuthController «ApiController»	REST endpoints /api/Auth/*; rate-limited	–	+ Register(RegisterRequest) : IActionResult + VerifyEmail(VerifyEmailRequest) : IActionResult + Login(LoginRequest) : IActionResult + RefreshToken(RefreshRequest) : IActionResult + ResendOtp(ResendOtpRequest) : IActionResult + ForgotPassword(ForgotPasswordRequest) : IActionResult + ResetPassword(ResetPasswordRequest) : IActionResult + Logout() : IActionResult
IAuthService «interface»	Contract of the authentication use cases	–	+ RegisterAsync(RegisterRequest) : Task<AuthResponse> + VerifyEmailAsync(VerifyEmailRequest) : Task<bool> + ResendOtpAsync(ResendOtpRequest) : Task<ResendOtpResponse> + LoginAsync(LoginRequest) : Task<AuthResponse> + RefreshTokenAsync(RefreshRequest) : Task<AuthResponse> + LogoutAsync(Guid userId) : Task<bool> + ForgotPasswordAsync(ForgotPasswordRequest) : Task<ResendOtpResponse> + ResetPasswordAsync(ResetPasswordRequest) : Task + ValidateSessionAsync(Guid userId, int version, string scope) : Task<bool>
AuthService	Registration, OTP, sign-in, token issue and rotation, password reset, session validation	- unitOfWork : IUnitOfWork - emailSender : IEmailSender - settings : ServerSettings	- HashRefreshToken(token) : string
IEmailSender «interface»	Abstraction for sending e-mail	–	+ SendAsync(to, subject, body) : Task
SmtpEmailSender	Sends e-mail through Gmail SMTP with STARTTLS	–	–
IUnitOfWork «interface»	Gives services access to repositories and one shared transaction	+ Users : IUserRepository + Profiles : IPlayerProfileRepository	+ SaveChangesAsync() : Task<int> + BeginTransactionAsync() : Task + CommitTransactionAsync() : Task

3.1.3 Register and verify e-mail

The visitor registers; the service hashes the password, stores the HMAC of a six-digit code and queues the e-mail. The alternative fragments show a confirmation mismatch, a registered e-mail, a valid code and an invalid or expired code.

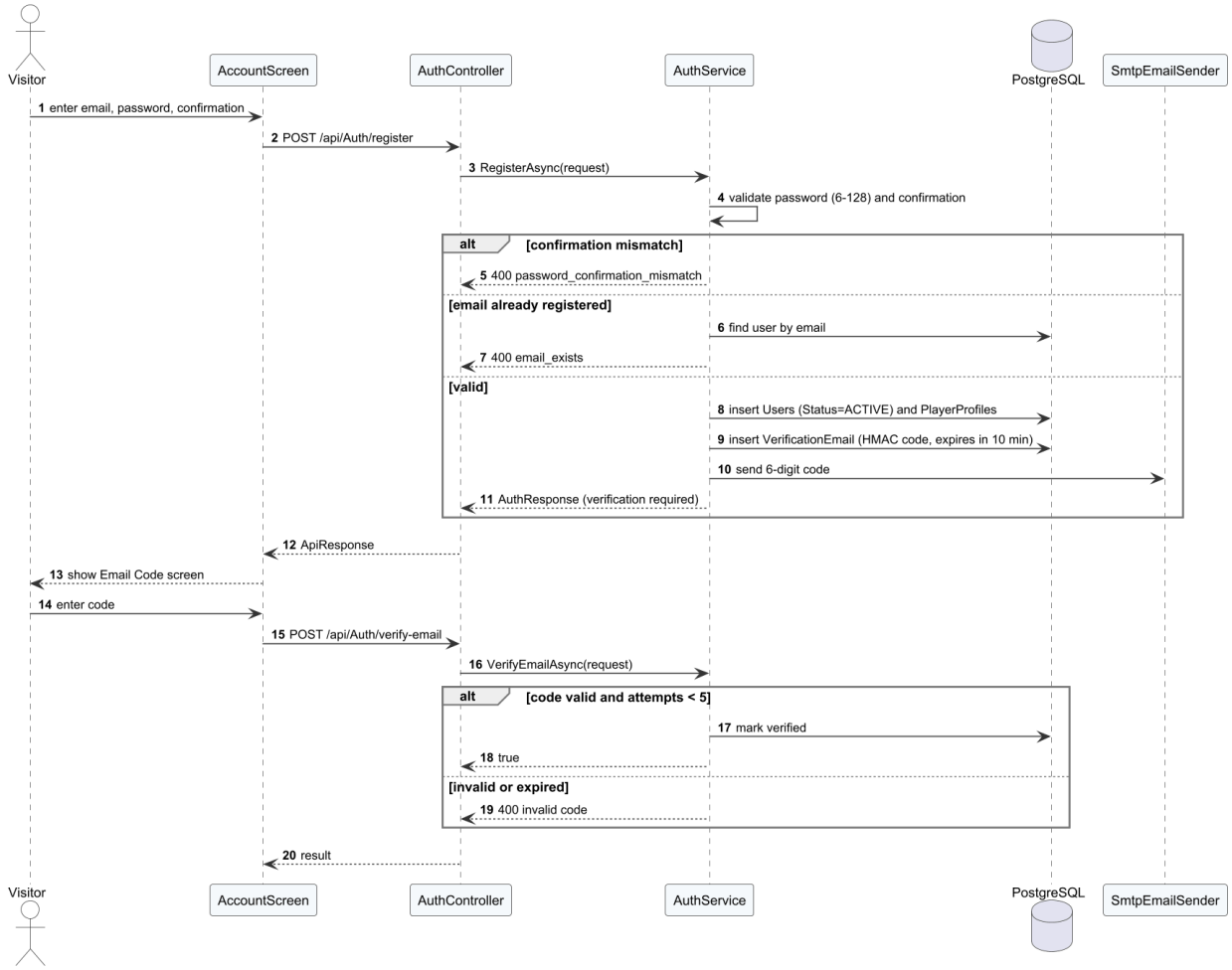


Figure 14. Sequence diagram – Register and verify e-mail

3.1.4 Sign in

The service checks the password hash and the account state. A wrong password, an unverified e-mail and a restricted account return different results; an active account receives the JWT and refresh token, after which the client loads the profile.

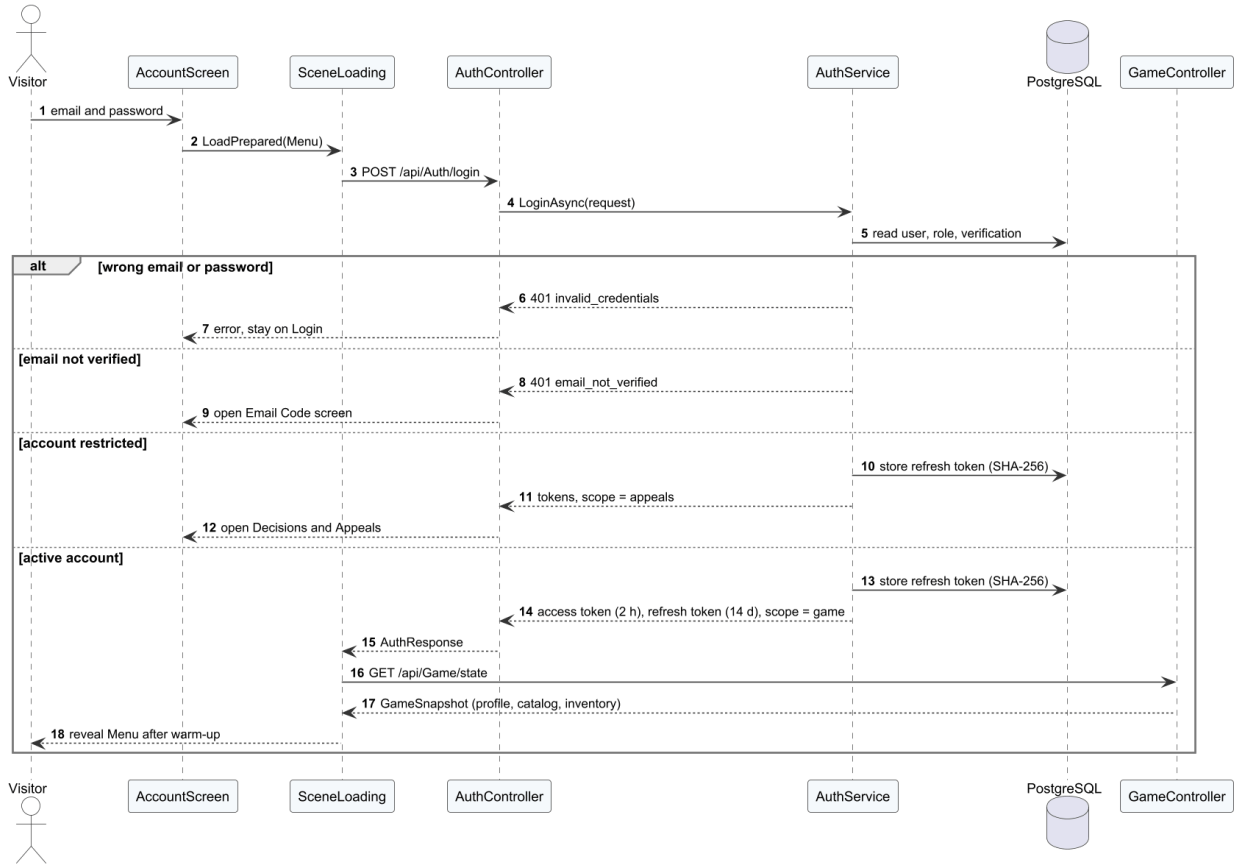


Figure 15. Sequence diagram – Sign in

3.1.5 Reset password

The reset request always returns the same message; the mail queue sends the code only for a verified account. A valid code changes the hash and revokes every refresh token.

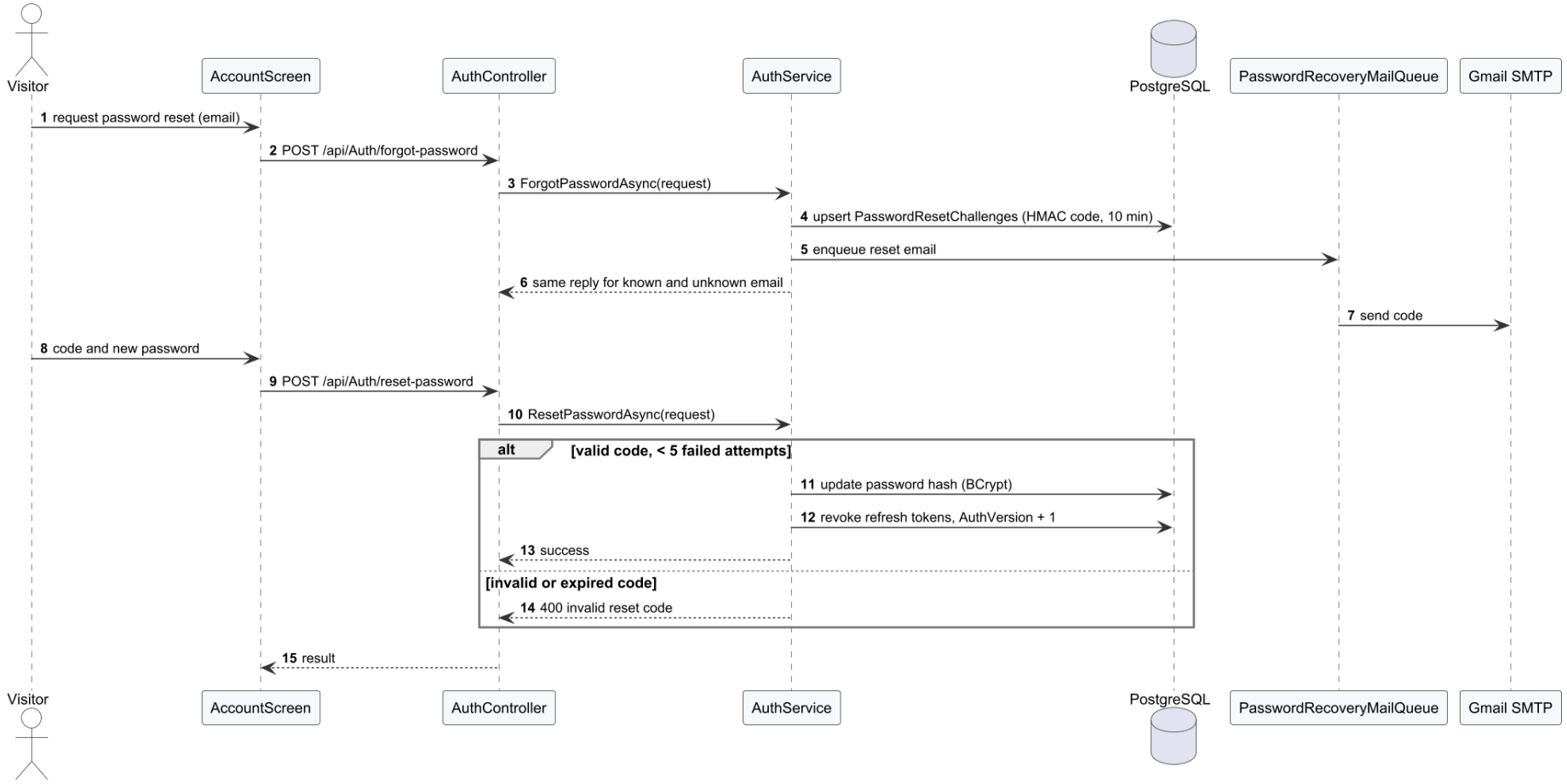


Figure 16. Sequence diagram – Reset password

3.2 Gameplay Core and Ball Physics

The gameplay core implements FR-11 to FR-13. BotMatch drives the match phases. Each MatchActor reads its intent from ShotInput (player) or a bot. PaddleContact accepts the hit, ShotTrajectory computes the launch velocity, BallFlight advances the ball and SinglesRules decides the outcome. The rule and physics classes have no Unity dependencies, so the same code runs in the client, in the dedicated server and in EditMode unit tests. The camera follows the common sports-game convention of a view from behind the player's own baseline: MatchCamera and the online camera both call CourtFollowView, which keeps the player's feet above the bottom HUD band (shot cue on PC, touch controls on Android) and widens the view slightly so that the far baseline stays visible; HudActorClearance moves the shot cue if it would still cover the player (defect D-03).

Table 20. Gameplay and physics parameters

Parameter	Value	Source / meaning
Simulation step	120 Hz (1/120 s)	BallFlight fixed step, independent of frame rate
Gravity	9.81 m/s²	Ball flight
Bounce restitution	0.73	Vertical speed kept after a bounce
Ball radius	0.037 m	Line calls use the ball edge
Court half width / half length / kitchen	3.048 / 6.7056 / 2.1336 m	SinglesRules.Court, official dimensions
Contact window	±65 ms	PaddleContact; outside the window the swing misses
Perfect contact	≤ 0.145 m and ±30 ms	Higher speed and accuracy
Serve timer	10 s	Fault ServeTimeout
Contact delay per shot	serve 0.80 s, dink 0.367 s, drop 0.533 s, smash 0.50 s, volley/reset 0.233 s, drive 0.433 s	ShotIntent.ContactDelay: time from release to the animation's contact frame
Camera height / distance / field of view	5.3 m / 4.25 m / 56°	CourtFollowView: position behind the player's baseline
Feet line	25 % of screen height on PC, 32 % on Android	CourtFollowView.DesktopFeetLine / TouchFeetLine: the feet are never drawn lower
Extra field of view	up to 8°	Added only when needed to keep the far baseline and the opponent's head in view

3.2.1 Class Diagram

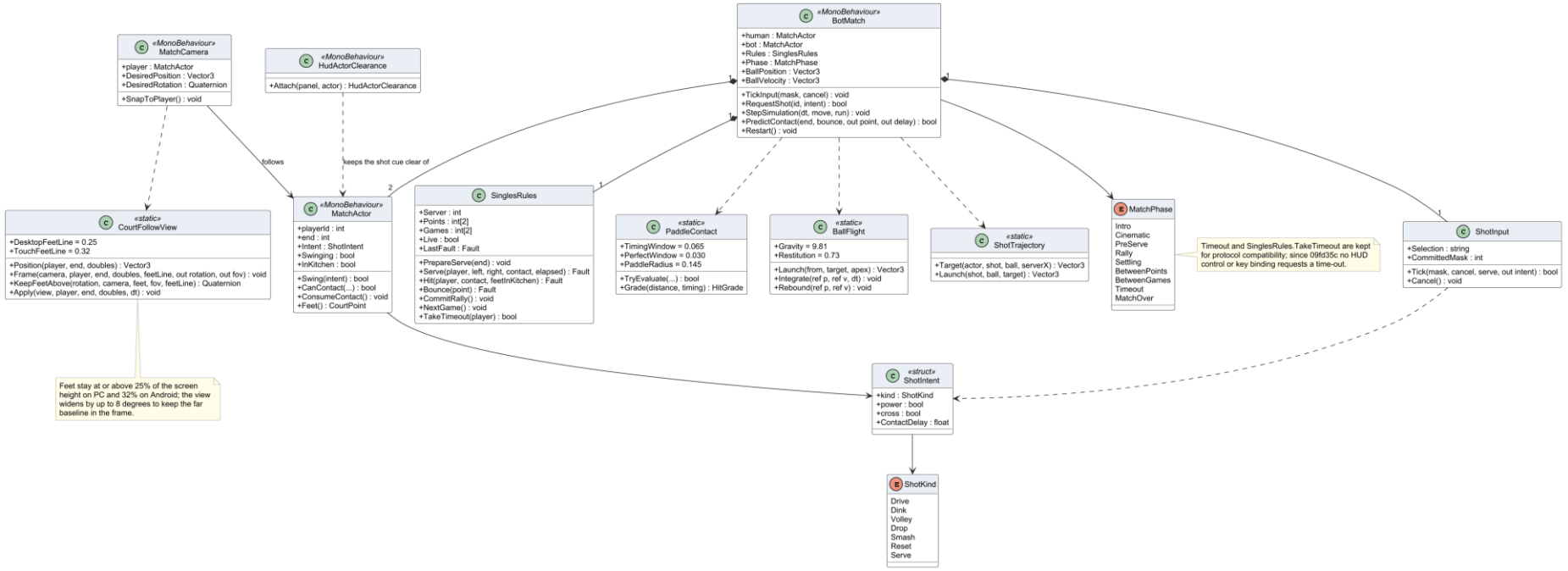


Figure 17. Class diagram – gameplay core (bot match)

3.2.2 Class Specifications

Table 21. Class specifications – gameplay core

Class	Responsibility	Main attributes	Main operations
BotMatch «MonoBehaviour»	Orchestrates a local singles match: phases, serve, rally, scoring and result	+ human : MatchActor + bot : MatchActor + Rules : SinglesRules + Phase : MatchPhase + BallPosition : Vector3 + BallVelocity : Vector3	+ TickInput(mask, cancel) : void + RequestShot(id, intent) : bool + StepSimulation(dt, move, run) : void + PredictContact(end, bounce, out point, out delay) : bool + Restart() : void
MatchActor «MonoBehaviour»	Moves and animates one player and performs hits	+ playerId : int + end : int + Intent : ShotIntent + Swinging : bool + InKitchen : bool	+ Swing(intent) : bool + CanContact(...) : bool + ConsumeContact() : void + Feet() : CourtPoint
SinglesRules	Decides faults, points, side-outs, games and match end (BR-10 to BR-15)	+ Server : int + Points : int[2] + Games : int[2] + Live : bool + LastFault : Fault	+ PrepareServe(end) : void + Serve(player, left, right, contact, elapsed) : Fault + Hit(player, contact, feetInKitchen) : Fault + Bounce(point) : Fault + CommitRally() : void + NextGame() : void + TakeTimeout(player) : bool
ShotInput	Decodes the J/K/L/U/I key state into one shot intent on the first release	+ Selection : string + CommittedMask : int	+ Tick(mask, cancel, serve, out intent) : bool + Cancel() : void
ShotIntent «struct»	Requested shot kind with power and cross modifiers	+ kind : ShotKind + power : bool + cross : bool + ContactDelay : float	–
ShotKind «enum»	Seven shot kinds of the local game	Drive Dink Volley Drop Smash Reset Serve	–
PaddleContact «static»	Checks the contact window (±65 ms) and the contact quality (Perfect, Good)	+ TimingWindow = 0.065 + PerfectWindow = 0.030 + PaddleRadius = 0.145	+ TryEvaluate(...) : bool + Grade(distance, timing) : HitGrade
BallFlight «static»	Fixed-step ball flight: gravity, drag-free arc and bounce with restitution 0.73	+ Gravity = 9.81 + Restitution = 0.73	+ Launch(from, target, apex) : Vector3 + Integrate(ref p, ref v, dt) : void + Rebound(ref p, ref v) : void
ShotTrajectory «static»	Computes launch velocity for a target point and flight time	–	+ Target(actor, shot, ball, serverX) : Vector3 + Launch(shot, ball, target) : Vector3
MatchPhase «enum»	Phases of a match (Intro, Cinematic, PreServe, Rally, Settling, BetweenPoints, BetweenGames, MatchOver); Timeout remains in the enum, with SinglesRules.TakeTimeout, only for protocol compatibility and is no longer entered	Intro Cinematic PreServe Rally Settling BetweenPoints BetweenGames Timeout MatchOver	–
MatchCamera «MonoBehaviour»	Follows the local player behind the baseline and smooths position and rotation	+ player : MatchActor + DesiredPosition : Vector3 + DesiredRotation : Quaternion	+ SnapToPlayer() : void
CourtFollowView «static»	Shared framing rule for local and online matches: keeps the player's feet above the bottom HUD band and the far baseline in view	+ DesktopFeetLine = 0.25 + TouchFeetLine = 0.32	+ Position(player, end, doubles) : Vector3 + Frame(camera, player, end, doubles, feetLine, out rotation, out fov) : void + KeepFeetAbove(rotation, camera, feet, fov, feetLine) : Quaternion + Apply(view, player, end, doubles, dt) : void
HudActorClearance «MonoBehaviour»	Moves the shot-cue panel so that it never covers the player	–	+ Attach(panel, actor) : HudActorClearance

3.3 Training and Offline Synchronisation

Training implements FR-06 to FR-10. TrainingDirector runs a drill locally with the gameplay core and evaluates it with TrainingEngine. Online, the attempt is sent at once; offline, OfflineTrainingStore keeps it until the device reconnects. On the server OnlineGameService re-evaluates the attempt with the same engine and returns a receipt.

3.3.1 Class Diagram

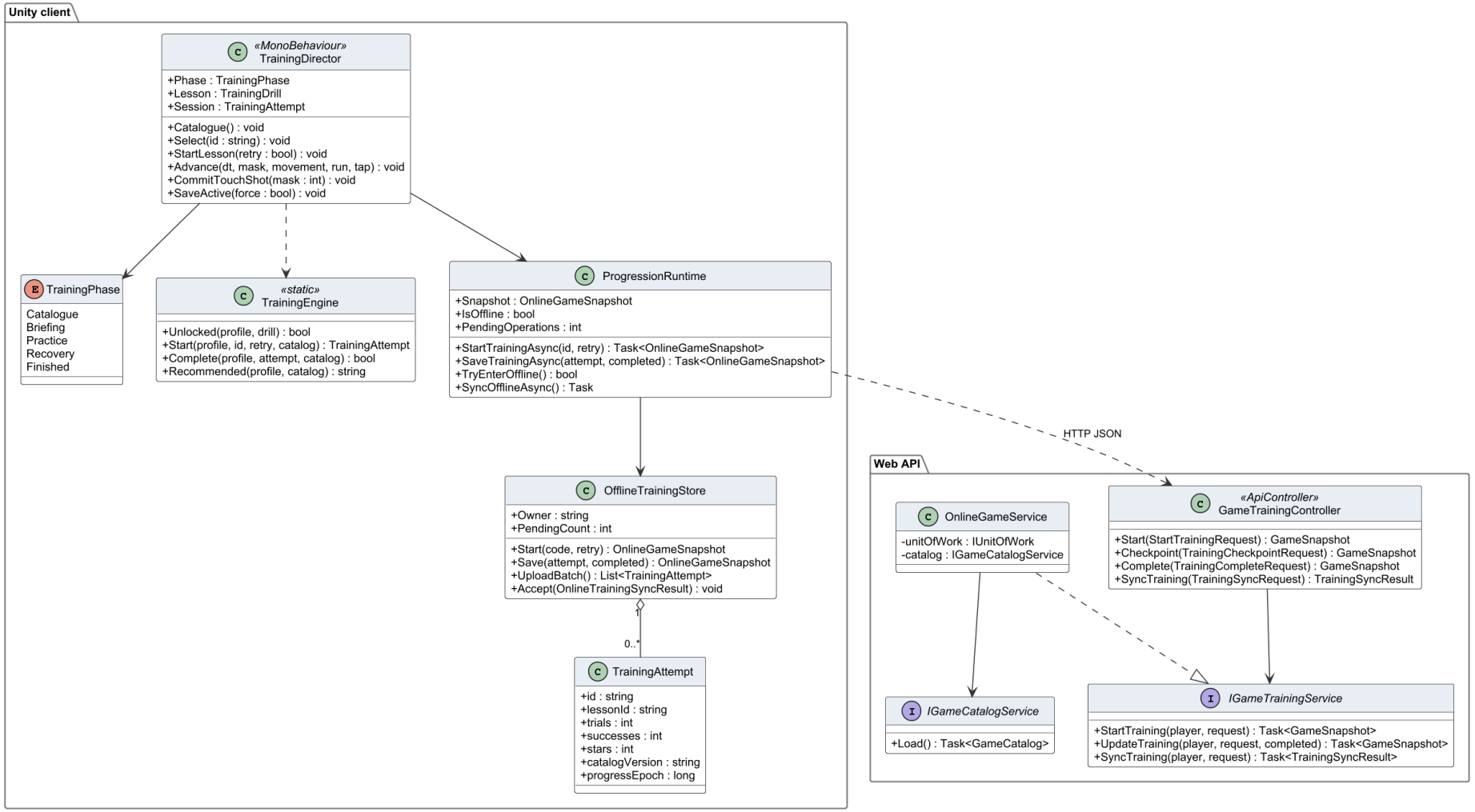


Figure 18. Class diagram – training and synchronisation

3.3.2 Class Specifications

Table 22. Class specifications – training

Class	Responsibility	Main attributes	Main operations
TrainingDirector «MonoBehaviour»	Runs a drill in the Training scene: briefing, shots, feedback and result popup	+ Phase : TrainingPhase + Lesson : TrainingDrill + Session : TrainingAttempt	+ Catalogue() : void + Select(id : string) : void + StartLesson(retry : bool) : void + Advance(dt, mask, movement, run, tap) : void + CommitTouchShot(mask : int) : void + SaveActive(force : bool) : void
TrainingPhase «enum»	Phases of the training screen	Catalogue Briefing Practice Recovery Finished	–
TrainingEngine «static»	Pure rules for drill evaluation: hits, timing and stars (BR-18)	–	+ Unlocked(profile, drill) : bool + Start(profile, id, retry, catalog) : TrainingAttempt + Complete(profile, attempt, catalog) : bool + Recommended(profile, catalog) : string
ProgressionRuntime	Client-side cache of profile, wallet, catalogue and the synchronisation queue	+ Snapshot : OnlineGameSnapshot + IsOffline : bool + PendingOperations : int	+ StartTrainingAsync(id, retry) : Task<OnlineGameSnapshot> + SaveTrainingAsync(attempt, completed) : Task<OnlineGameSnapshot> + TryEnterOffline() : bool + SyncOfflineAsync() : Task
OfflineTrainingStore	Durable local queue of offline attempts with request IDs	+ Owner : string + PendingCount : int	+ Start(code, retry) : OnlineGameSnapshot + Save(attempt, completed) : OnlineGameSnapshot + UploadBatch() : List<TrainingAttempt> + Accept(OnlineTrainingSyncResult) : void
TrainingAttempt	Data of one drill attempt sent to the server	+ id : string + lessonId : string + trials : int + successes : int + stars : int + catalogVersion : string + progressEpoch : long	–
GameTrainingController «ApiController»	REST endpoints for starting, completing and synchronising training	–	+ Start(StartTrainingRequest) : GameSnapshot + Checkpoint(TrainingCheckpointRequest) : GameSnapshot + Complete(TrainingCompleteRequest) : GameSnapshot + SyncTraining(TrainingSyncRequest) : TrainingSyncResult
IGameTrainingService «interface»	Contract of the training use cases	–	+ StartTraining(player, request) : Task<GameSnapshot> + UpdateTraining(player, request, completed) : Task<GameSnapshot> + SyncTraining(player, request) : Task<TrainingSyncResult>
OnlineGameService	Server implementation of training, match settlement, matchmaking and economy use cases	- unitOfWork : IUnitOfWork - catalog : IGameCatalogService	–
IGameCatalogService «interface»	Provides the published content catalogue and its version	–	+ Load() : Task<GameCatalog>

3.3.3 Practise a drill online

The server first checks that the lesson is published and unlocked, then records each trial; at the end it applies stars and the first-pass reward in one transaction.

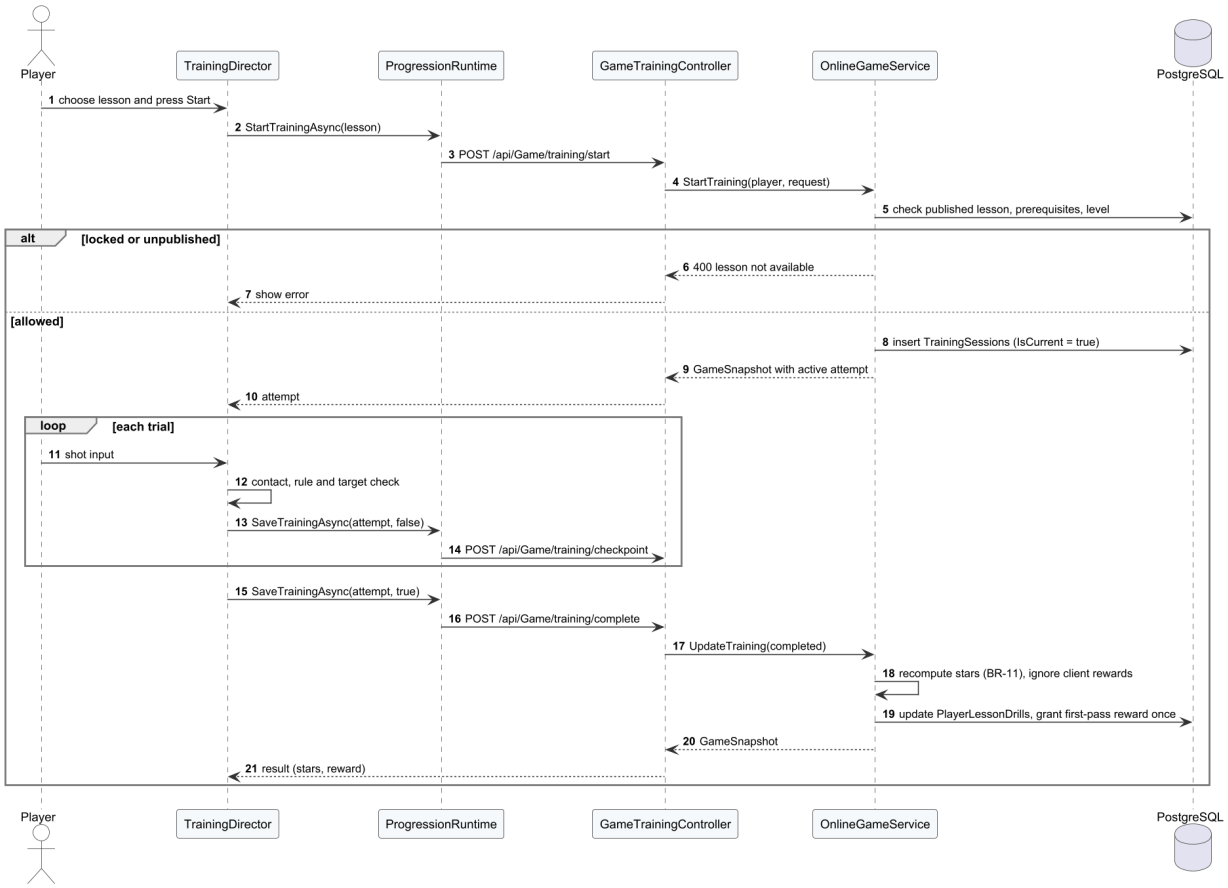


Figure 19. Sequence diagram – Practise a drill online

3.3.4 Synchronise offline attempts

Each queued attempt is sent with its request ID. A new and valid attempt is applied, an already applied one is answered DUPLICATE, and a stale or invalid one is REJECTED.

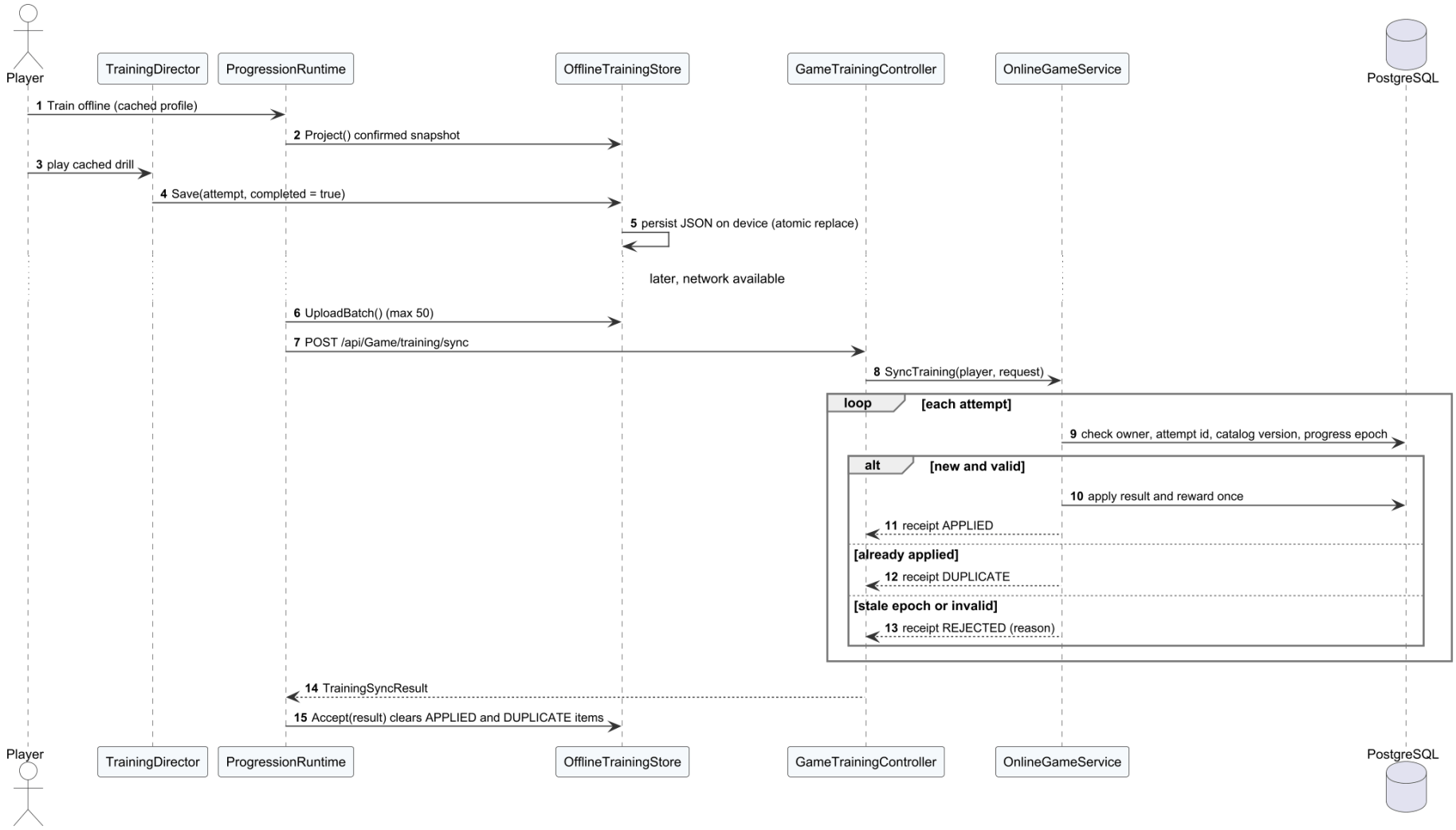


Figure 20. Sequence diagram – Synchronise offline attempts

3.4 Bot AI

The bot design implements FR-13 and BR-31. LocalBot provides three fixed levels. BhTreeBot executes a Unity Behavior graph whose decisions (position, shot choice, aggression, deliberate errors) are parameterised by BotPolicy and a BotTierProfile selected from the opponent's Elo. The IAdaptiveBotContext interface hides whether the bot plays locally or on the dedicated server, so the same bot is used for practice and for the ranked bot fallback.

Table 23. Bot parameters

Parameter	Value	Source / meaning
Fixed levels	Easy, Medium, Hard	LocalBot: fixed reaction and accuracy
Adaptive tiers	Elo < 500 beginner; < 2000 intermediate; ≥ 2000 advanced	BotPolicy tier selection
Ranked bot Elo	own + 120 (own − 100 after 5 losses)	EloPolicy.BotElo (BR-26)
Ranked bot accuracy	min(100, 25 + (Elo − 100) / 120) %	EloPolicy.BotAccuracy

3.4.1 Class Diagram

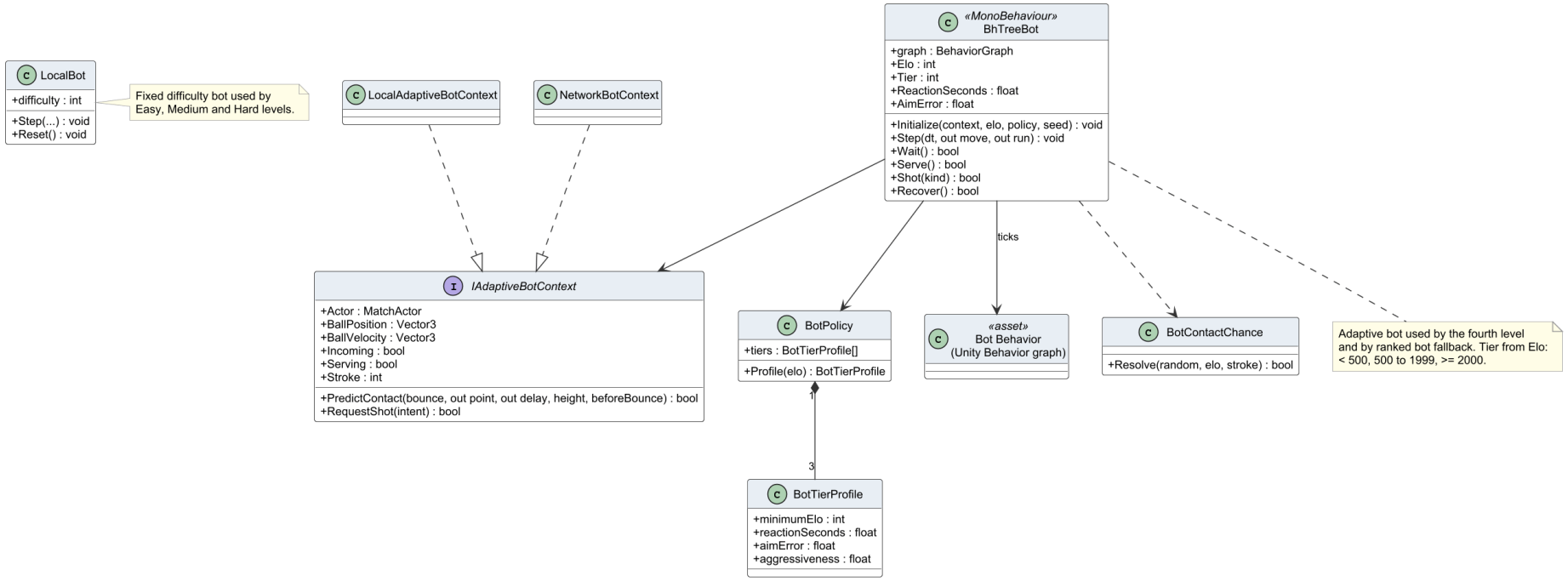


Figure 21. Class diagram – bots

3.4.2 Class Specifications

Table 24. Class specifications – bot AI

Class	Responsibility	Main attributes	Main operations
IAdaptiveBotContext «interface»	Information the bot needs about ball, court and opponent, independent of local or network play	+ Actor : MatchActor + BallPosition : Vector3 + BallVelocity : Vector3 + Incoming : bool + Serving : bool + Stroke : int	+ PredictContact(bounce, out point, out delay, height, beforeBounce) : bool + RequestShot(intent) : bool
LocalAdaptiveBotContext	Bot context for a local bot match	–	–
NetworkBotContext	Bot context for a ranked bot on the dedicated server	–	–
BhTreeBot «MonoBehaviour»	Adaptive bot driven by the Unity Behavior graph; tier chosen from Elo	+ graph : BehaviorGraph + Elo : int + Tier : int + ReactionSeconds : float + AimError : float	+ Initialize(context, elo, policy, seed) : void + Step(dt, out move, out run) : void + Wait() : bool + Serve() : bool + Shot(kind) : bool + Recover() : bool
BotPolicy	Maps a tier to reaction time, accuracy, aggression and error rate	+ tiers : BotTierProfile[]	+ Profile(elo) : BotTierProfile
BotTierProfile	Parameter set of one bot tier	+ minimumElo : int + reactionSeconds : float + aimError : float + aggressiveness : float	–
LocalBot	Fixed-level bot (Easy, Medium, Hard) with deterministic settings	+ difficulty : int	+ Step(...) : void + Reset() : void
Bot Behavior (Unity Behavior graph) «asset»	Behaviour tree asset: position, choose shot, hit, recover	–	–
BotContactChance	Probability that the bot reaches and returns a ball	–	+ Resolve(random, elo, stroke) : bool

3.4.3 Play a bot match

The loop fragment repeats rallies until one side wins two games; at the end the client reports the result to GameMatchesController, which settles rewards.

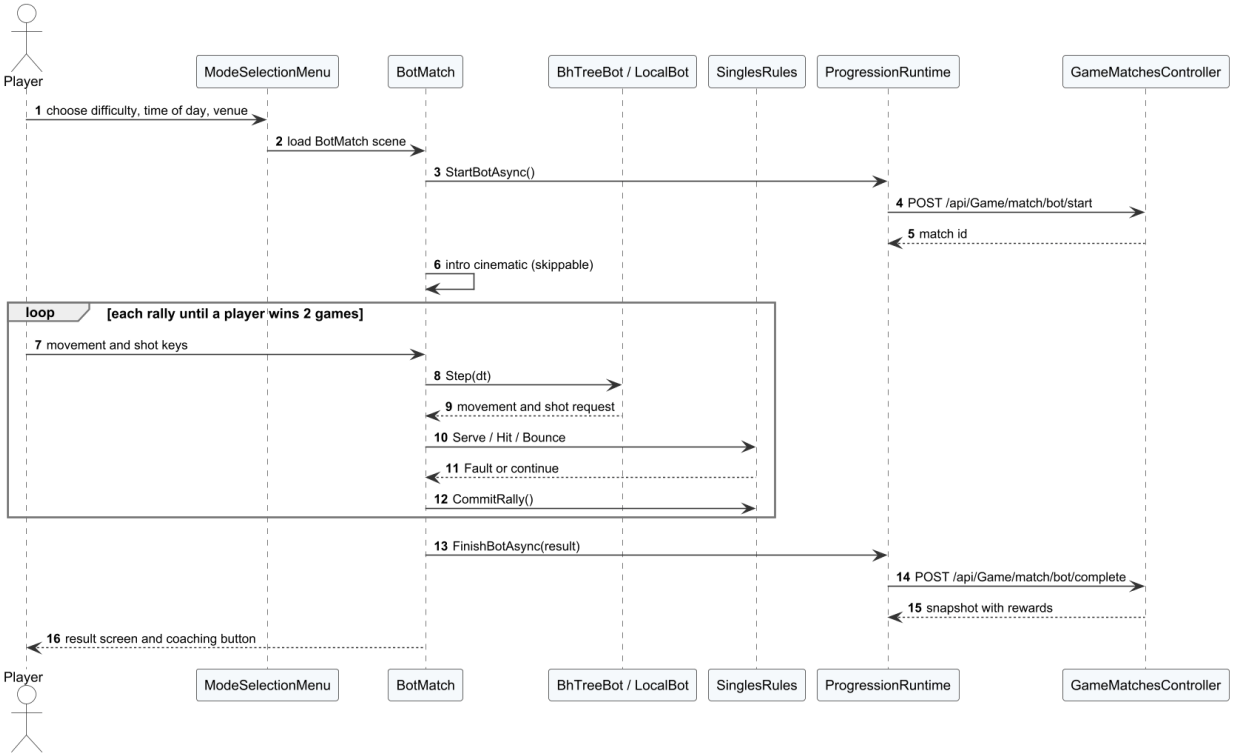


Figure 22. Sequence diagram – Play a bot match

3.5 Online Matchmaking and Netcode

Online play implements FR-14 to FR-18. MultiplayerApp is a state machine over FlowState; PlayMultiplayerPresenter and LobbyFriendsSidebar form the lobby screen. Unity Gaming Services code from the earlier prototype (files guarded by PICKLEVERSE_LEGACY_NGO) is not compiled; the live path uses only Photon Fusion and the Web API. Ranked singles, quick doubles (unrated 2v2, filled from solo players and parties of two, no bot fallback) and private rooms share the same ready check and dedicated-server allocation. PhotonSession connects to Fusion; NetworkMatch and NetworkPlayer replicate state from the dedicated server, which is the only authority. Table below gives the netcode design: clients send input only, predict their own movement and reconcile with the server frame; remote players are interpolated.

Table 25. Netcode design parameters

Parameter	Value	Source / meaning
Authority	Dedicated server (Fusion server mode)	Shared mode is kept only for local development without a server
Client input	DedicatedInput {Movement, Running, Sequence}	Sent every tick; shot commands travel as reliable messages (JSON, at most 16 KB) and are judged on the server
Local prediction	Own player only	The client applies its input at once and keeps the unacknowledged moves in a sequence-numbered list
Server frame	PlayerFrame {position, velocity, sequence, motionEpoch, movement, running}	Networked state of each player; sequence acknowledges the last input applied
Reconciliation	Replay unacknowledged moves from the server state	Acknowledged moves are dropped; the visible correction is smoothed and clamped to 1.5 m; a new motionEpoch (serve reset, teleport) forces a hard reset
Movement integration	120 Hz sub-steps	Same step as the authority, so prediction and server agree
Remote players and ball	Interpolated in Render	Fusion snapshot interpolation
Lag compensation	Disabled	Hits are judged on the server's own state
Tick rate	Fusion project default	No explicit override in the project settings
Protocol version	18	Client and server must match or the connection is refused
Reconnection	client 30 s; server keeps the slot 30 s; API interrupts after 35 s without heartbeat (heartbeat every 8 s)	MultiplayerSettings.reconnectSeconds, NetworkMatch reservations, OnlineMatch settings (BR-28)

3.5.1 Class Diagram

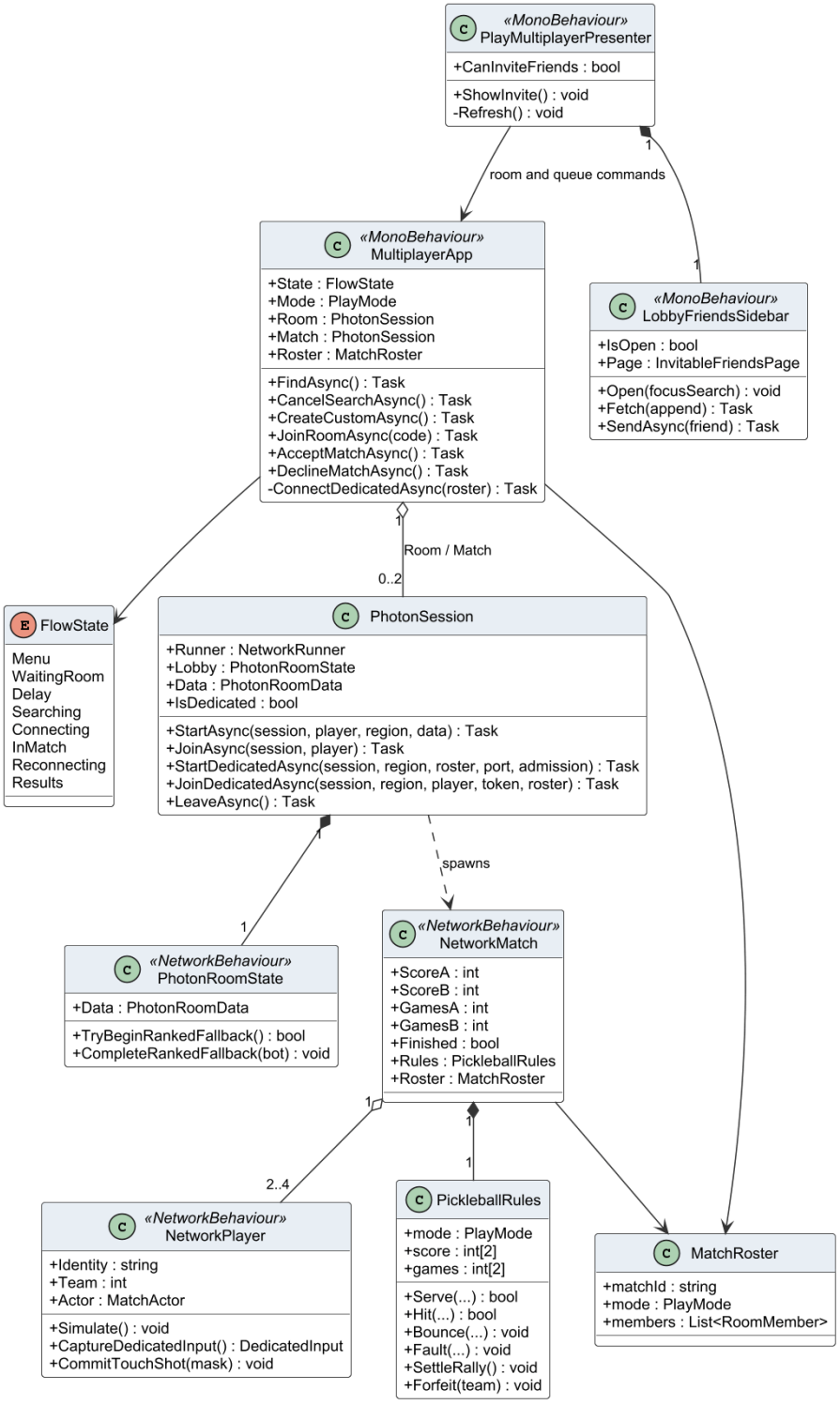


Figure 23. Class diagram – online client

3.5.2 Class Specifications

Table 26. Class specifications – online play

Class	Responsibility	Main attributes	Main operations
MultiplayerApp «MonoBehaviour»	Online flow controller: lobby, search, ready check, loading, match, reconnection and results	+ State : FlowState + Mode : PlayMode + Room : PhotonSession + Match : PhotonSession + Roster : MatchRoster	+ FindAsync() : Task + CancelSearchAsync() : Task + CreateCustomAsync() : Task + JoinRoomAsync(code) : Task + AcceptMatchAsync() : Task + DeclineMatchAsync() : Task - ConnectDedicatedAsync(roster) : Task
FlowState «enum»	States of the online flow (Menu, WaitingRoom, Delay, Searching, Connecting, InMatch, Reconnecting, Results)	Menu WaitingRoom Delay Searching Connecting InMatch Reconnecting Results	–
PhotonSession	Starts and joins Fusion sessions in shared or client mode and handles disconnects	+ Runner : NetworkRunner + Lobby : PhotonRoomState + Data : PhotonRoomData + IsDedicated : bool	+ StartAsync(session, player, region, data) : Task + JoinAsync(session, player) : Task + StartDedicatedAsync(session, region, roster, port, admission) : Task + JoinDedicatedAsync(session, region, player, token, roster) : Task + LeaveAsync() : Task
PhotonRoomState «NetworkBehaviour»	Networked room state: roster, phase and host	+ Data : PhotonRoomData	+ TryBeginRankedFallback() : bool + CompleteRankedFallback(bot) : void
NetworkMatch «NetworkBehaviour»	Networked match state replicated from the authority: score, phase and ball	+ ScoreA : int + ScoreB : int + GamesA : int + GamesB : int + Finished : bool + Rules : PickleballRules + Roster : MatchRoster	–
NetworkPlayer «NetworkBehaviour»	Networked player: input, position and animation	+ Identity : string + Team : int + Actor : MatchActor	+ Simulate() : void + CaptureDedicatedInput() : DedicatedInput + CommitTouchShot(mask) : void
PickleballRules	Rules engine used by the network match (singles and doubles)	+ mode : PlayMode + score : int[2] + games : int[2]	+ Serve(...) : bool + Hit(...) : bool + Bounce(...) : void + Fault(...) : void + SettleRally() : void + Forfeit(team) : void
MatchRoster	Teams, slots, bot flags and mode of a match	+ matchId : string + mode : PlayMode + members : List<RoomMember>	–
PlayMultiplayerPresenter «MonoBehaviour»	Online lobby screen: singles or doubles room, room code, search timer, invite button and ready state	+ CanInviteFriends : bool	+ ShowInvite() : void - Refresh() : void
LobbyFriendsSidebar «MonoBehaviour»	Collapsible friends column of the lobby: paged list of friends who can be invited, search and send invitation	+ IsOpen : bool + Page : InvitableFriendsPage	+ Open(focusSearch) : void + Fetch(append) : Task + SendAsync(friend) : Task

3.5.3 Class Diagram – dedicated server coordination

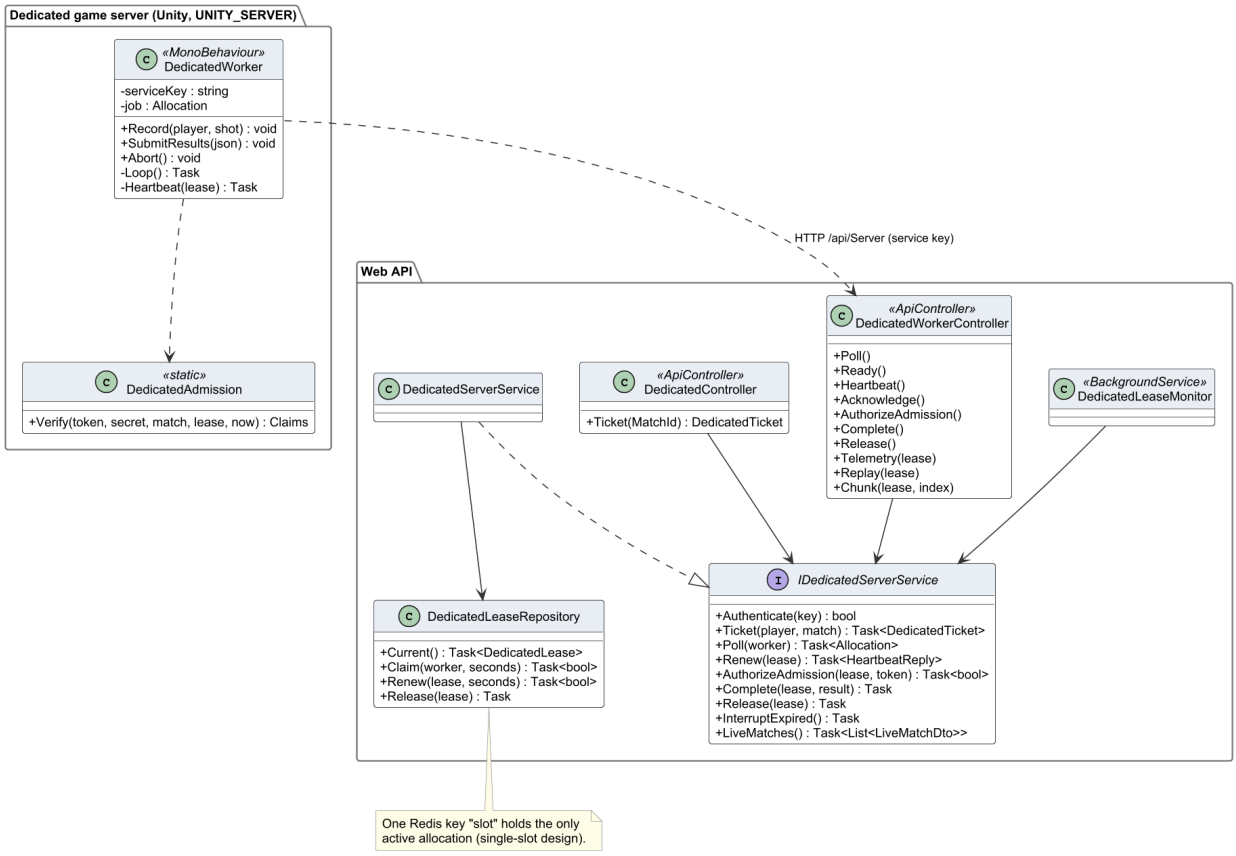


Figure 24. Class diagram – dedicated server coordination

Table 27. Class specifications – dedicated server

Class	Responsibility	Main attributes	Main operations
DedicatedWorker «MonoBehaviour»	Main loop of the dedicated server: lease, admission, heartbeats, telemetry, replay and result	- serviceKey : string - job : Allocation	+ Record(player, shot) : void + SubmitResults(json) : void + Abort() : void - Loop() : Task - Heartbeat(lease) : Task
DedicatedAdmission «static»	Validates HMAC tickets and admits players to the session	–	+ Verify(token, secret, match, lease, now) : Claims
DedicatedWorkerController «ApiController»	Service-key endpoints /api/Server/* used only by the dedicated server: poll, ready, heartbeat, admission, operation acknowledgement, telemetry, replay chunks, result and release	–	+ Poll() + Ready() + Heartbeat() + Acknowledge() + AuthorizeAdmission() + Complete() + Release() + Telemetry(lease) + Replay(lease) + Chunk(lease, index)
DedicatedController «ApiController»	Player endpoint /api/Game/dedicated/ticket: reserves the slot and returns the signed connection ticket	–	+ Ticket(MatchId) : DedicatedTicket

Class	Responsibility	Main attributes	Main operations
IDedicatedServerService «interface»	Contract of dedicated-server coordination	–	+ Authenticate(key) : bool + Ticket(player, match) : Task<DedicatedTicket> + Poll(worker) : Task<Allocation> + Renew(lease) : Task<HeartbeatReply> + AuthorizeAdmission(lease, token) : Task<bool> + Complete(lease, result) : Task + Release(lease) : Task + InterruptExpired() : Task + LiveMatches() : Task<List<LiveMatchDto>>
DedicatedServerService	Allocates the slot, signs tickets, records heartbeats and settles results	–	–
DedicatedLeaseRepository	Redis lease of the single server slot with expiry	–	+ Current() : Task<DedicatedLease> + Claim(worker, seconds) : Task<bool> + Renew(lease, seconds) : Task<bool> + Release(lease) : Task
DedicatedLeaseMonitor «BackgroundService»	Background worker that releases expired leases and interrupts orphan matches	–	–

3.5.4 Find a ranked or quick doubles match

The client polls the search. For singles the alternatives show a human opponent inside the Elo window and the bot fallback after 180 seconds. For quick doubles the captain sends the party members; the API fills two sides of two from parties and solo players, the match is unrated and there is no bot fallback. In both modes all players must accept, and a decline or timeout closes the offer.

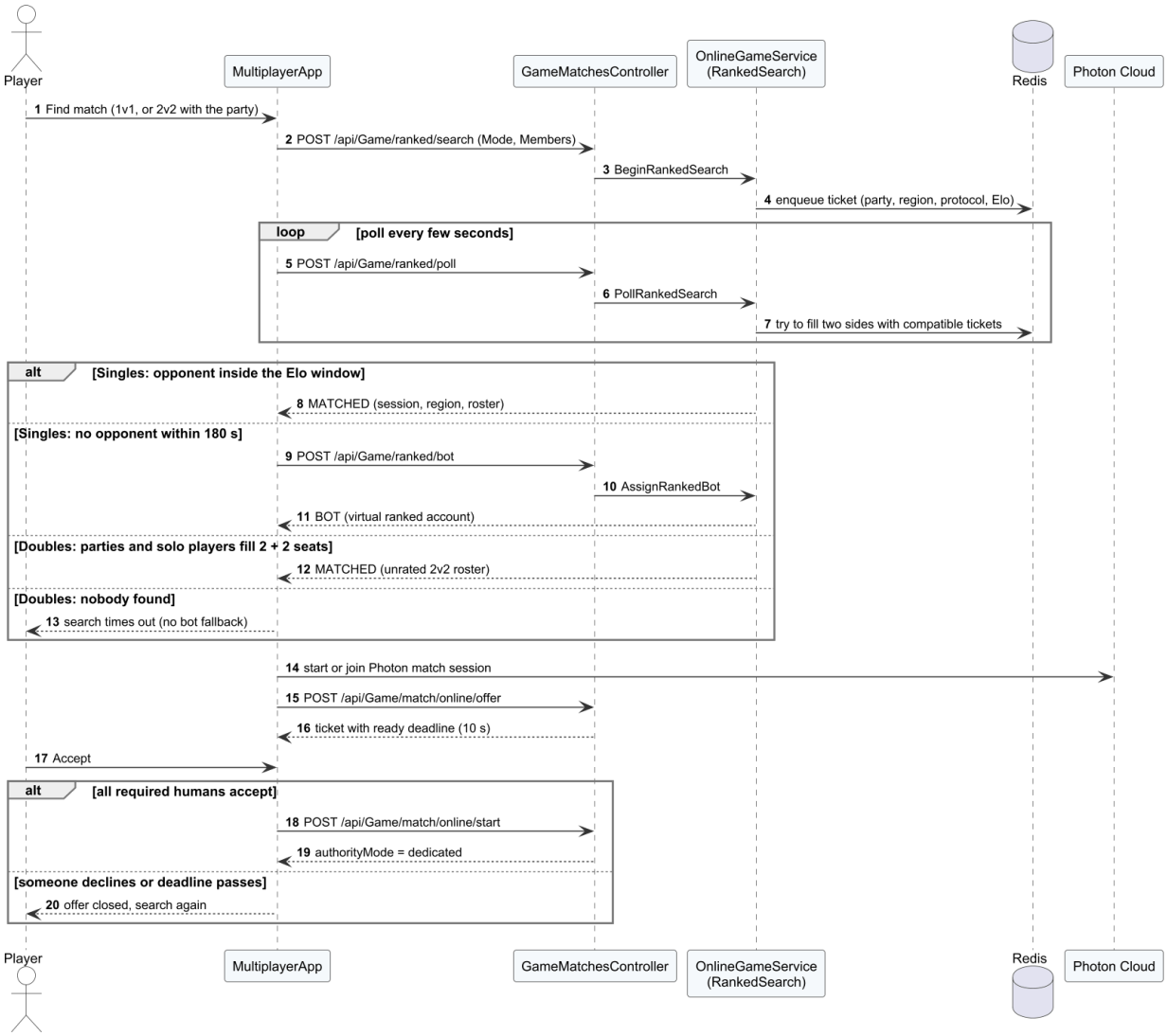


Figure 25. Sequence diagram – Find a ranked or quick doubles match

3.5.5 Invite a friend to a room

The invitation is rejected when the friend is not in the lobby, is blocked or the room is full; otherwise the friend polls and accepts it.

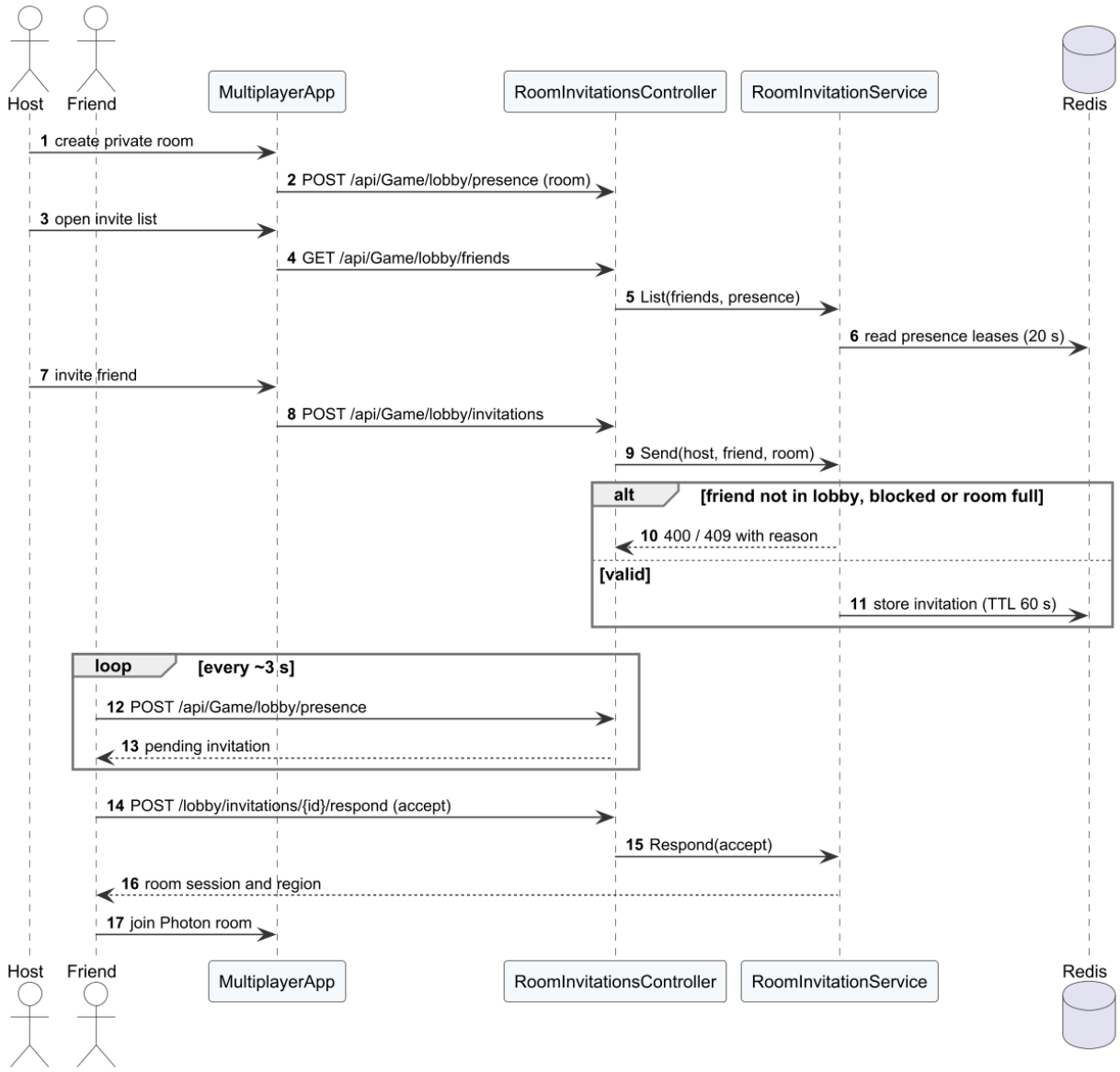


Figure 26. Sequence diagram – Invite a friend to a room

3.5.6 Allocate the dedicated server and play

The server waits for work on /api/Server/poll. When the slot is busy the client waits. When the slot is reserved the API signs tickets; the client polls every 500 ms until the server is ready; during play the server sends heartbeats and at the end telemetry, replay and the result.

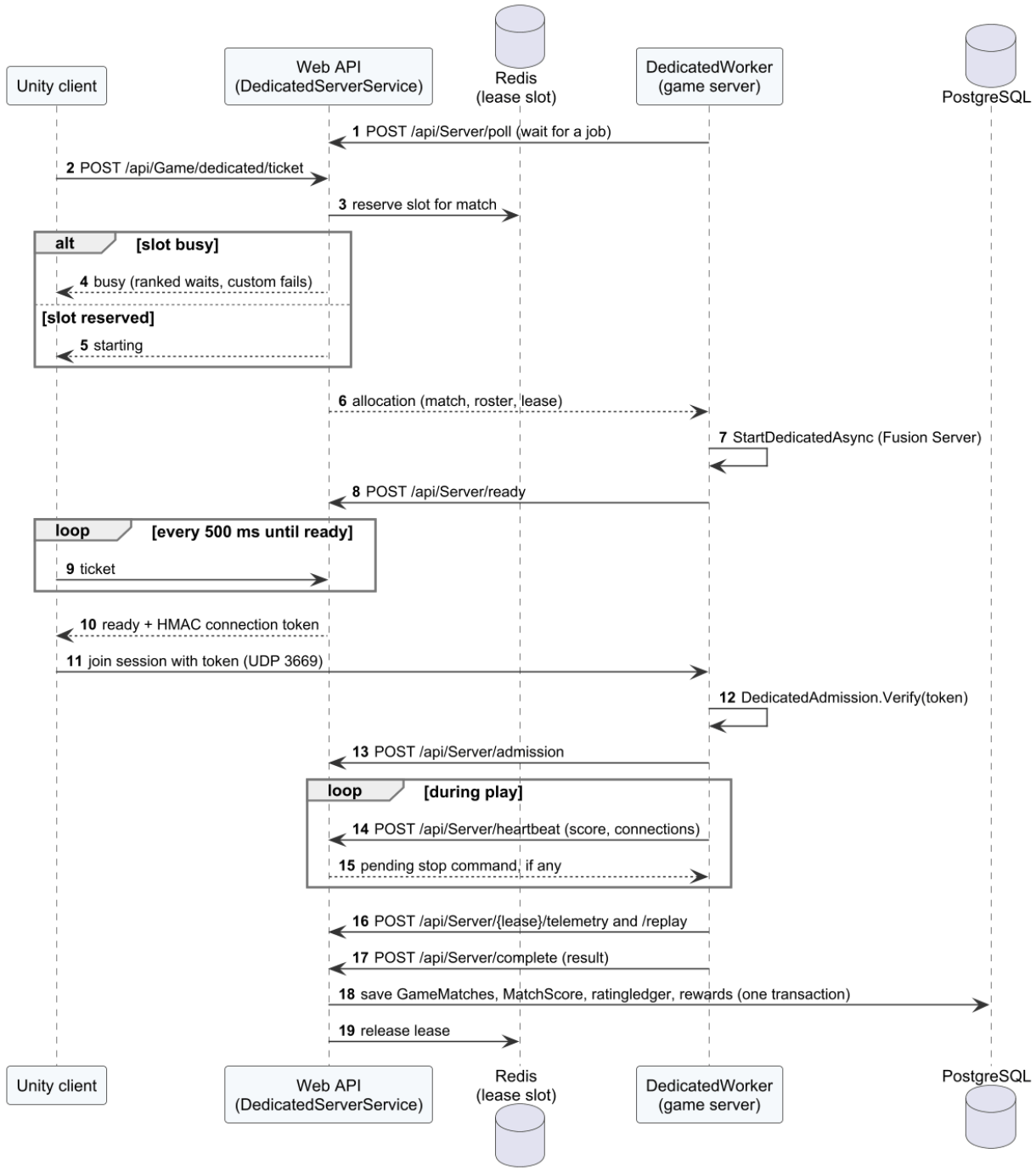


Figure 27. Sequence diagram – Allocate the dedicated server and play

3.6 AI Coaching

AI coaching implements FR-19 and NFR-05/NFR-06. Telemetry is uploaded by CoachingRuntime (drills and bot matches) or DedicatedWorker (online matches). HybridCoachingService stores it and creates an analysis job; HybridAiWorker calls the FastAPI service, which runs the rule engine and, when approved, the tier model.

Table 28. AI coaching parameters

Parameter	Value	Source / meaning
Model features	8	mean_abs_timing, late_rate, early_rate, kitchen_rate, momentum_rate, target_rate, recovery_rate, input_error_rate

Parameter	Value	Source / meaning
Tiers	beginner, intermediate, advanced	tier.py
Minimum confidence / opportunities	0.65 / 20	Below these values no tier is returned
Finding acceptance	precision ≥ 0.80, recall ≥ 0.70	calibration.py, NFR-06
Model approval	real data, ≥ 10 holdout players per tier, macro-F1 ≥ 0.70 and ≥ baseline + 0.05	train.py
Recommendations	at most 3	engine.py

3.6.1 Class Diagram

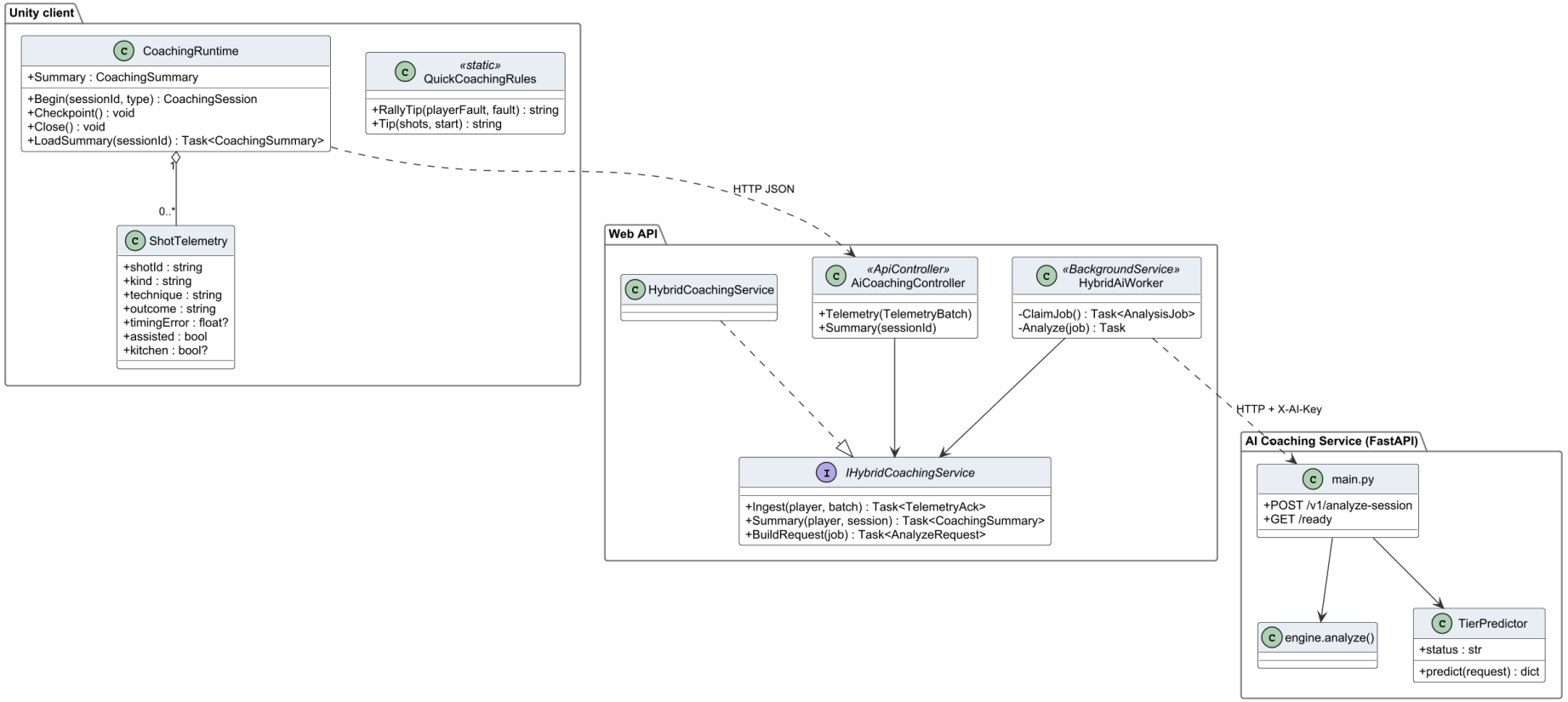


Figure 28. Class diagram – AI coaching

3.6.2 Class Specifications

Table 29. Class specifications – AI coaching

Class	Responsibility	Main attributes	Main operations
QuickCoachingRules «static»	Client-side quick tip shown when offline or before the AI result arrives	–	+ RallyTip(playerFault, fault) : string + Tip(shots, start) : string
CoachingRuntime	Uploads telemetry, polls the analysis job and shows the coaching popup	+ Summary : CoachingSummary	+ Begin(sessionId, type) : CoachingSession + Checkpoint() : void + Close() : void + LoadSummary(sessionId) : Task<CoachingSummary>
ShotTelemetry	Per-shot record: timing, contact, landing, position and errors	+ shotId : string + kind : string + technique : string + outcome : string + timingError : float? + assisted : bool + kitchen : bool?	–
AiCoachingController «ApiController»	REST endpoints for telemetry upload and coaching results	–	+ Telemetry(TelemetryBatch) + Summary(sessionId)
IHybridCoachingService «interface»	Contract of telemetry ingestion and coaching retrieval	–	+ Ingest(player, batch) : Task<TelemetryAck> + Summary(player, session) : Task<CoachingSummary> + BuildRequest(job) : Task<AnalyzeRequest>
HybridCoachingService	Stores telemetry, creates analysis jobs and returns results	–	–
HybridAiWorker «BackgroundService»	Sends pending jobs to the AI service with retries and stores the result	–	- ClaimJob() : Task<AnalysisJob> - Analyze(job) : Task
main.py	FastAPI entry point: /v1/analyze-session and health	+ POST /v1/analyze-session + GET /ready	–
engine.analyze()	Rule-based analysis producing findings, summary and recommendations	–	–
TierPredictor	Loads the approved model and predicts the tier with a confidence threshold	+ status : str	+ predict(request) : dict

3.6.3 Analyse a session

The worker marks the job processing and calls the service; on success the result is stored and the job completed, on failure the job is retried and finally marked failed.

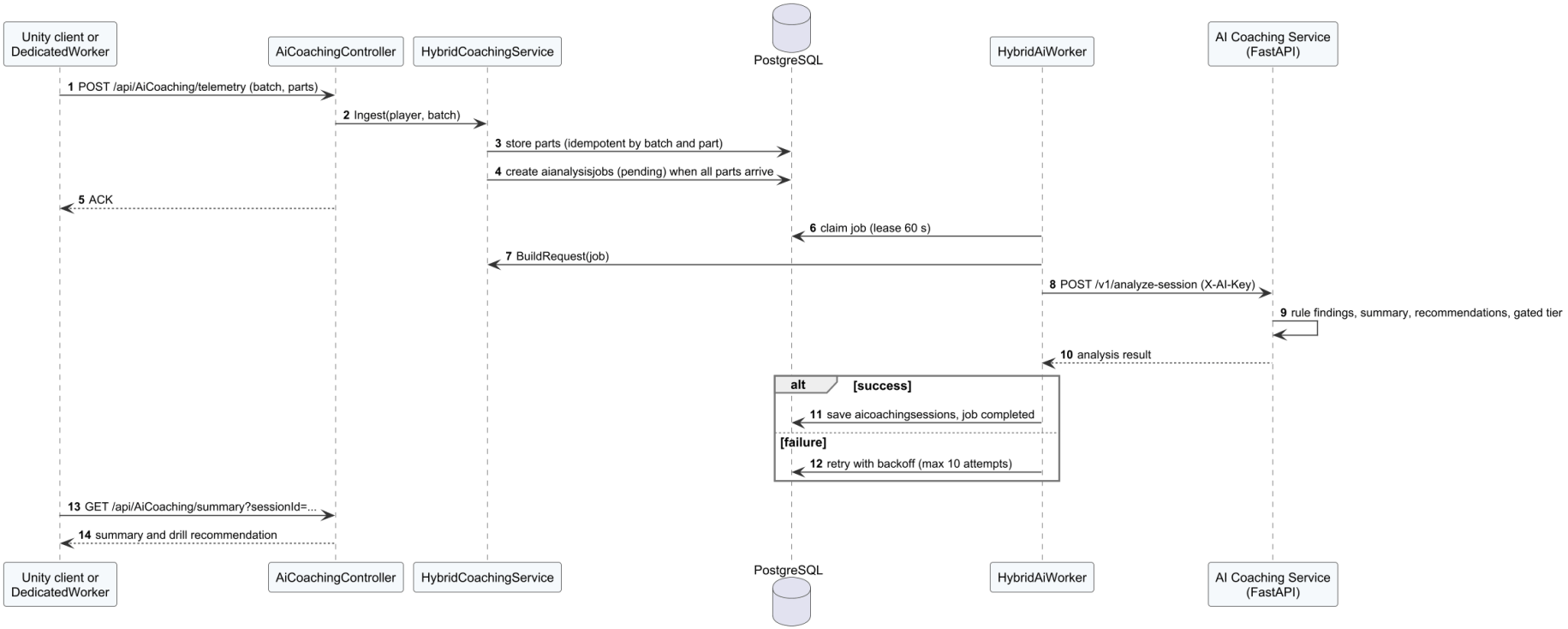


Figure 29. Sequence diagram – Analyse a session

3.7 Progress, Economy, Social and Replays

This group implements FR-20 to FR-26. The client screens share ProgressionRuntime. On the server, OnlineGameService settles rewards and purchases, OnlineSocialService manages friends and the leaderboard, and IMatchReplayService stores replays; GameRepository performs the EF Core queries and row locks inside the unit of work.

3.7.1 Class Diagram

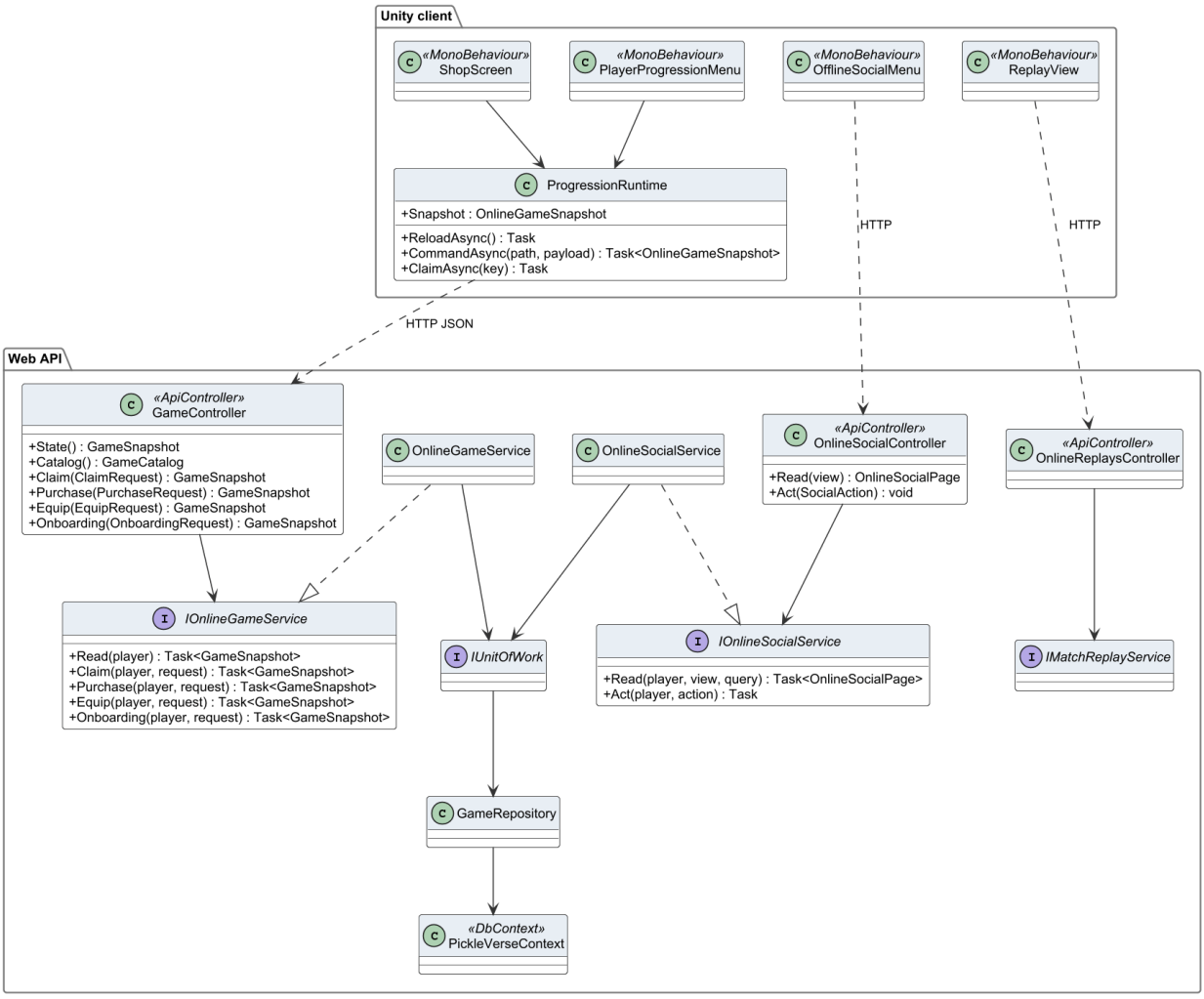


Figure 30. Class diagram – progress, economy, social and replays

3.7.2 Class Specifications

Table 30. Class specifications – progress, economy and social

Class	Responsibility	Main attributes	Main operations
ProgressionRuntime	Client-side cache of profile, wallet, catalogue and the synchronisation queue	+ Snapshot : OnlineGameSnapshot	+ ReloadAsync() : Task + CommandAsync(path, payload) : Task<OnlineGameSnapshot> + ClaimAsync(key) : Task
PlayerProgressionMenu «MonoBehaviour»	Progress, missions and history popups	–	–
OfflineSocialMenu «MonoBehaviour»	Friends and leaderboard drawer	–	–
ShopScreen «MonoBehaviour»	Shop scene: list, preview and purchase	–	–
ReplayView «MonoBehaviour»	Replay viewer: download and playback controls	–	–

Class	Responsibility	Main attributes	Main operations
GameController «ApiController»	REST endpoints for profile, wallet, catalogue, purchase and claims	–	+ State() : GameSnapshot + Catalog() : GameCatalog + Claim(ClaimRequest) : GameSnapshot + Purchase(PurchaseRequest) : GameSnapshot + Equip(EquipRequest) : GameSnapshot + Onboarding(OnboardingRequest) : GameSnapshot
OnlineSocialController «ApiController»	REST endpoints for friends, blocking and leaderboard	–	+ Read(view) : OnlineSocialPage + Act(SocialAction) : void
OnlineReplaysController «ApiController»	REST endpoints for replay upload and download	–	–
IOnlineGameService «interface»	Contract of progression, economy and match use cases	–	+ Read(player) : Task<GameSnapshot> + Claim(player, request) : Task<GameSnapshot> + Purchase(player, request) : Task<GameSnapshot> + Equip(player, request) : Task<GameSnapshot> + Onboarding(player, request) : Task<GameSnapshot>
IOnlineSocialService «interface»	Contract of social use cases	–	+ Read(player, view, query) : Task<OnlineSocialPage> + Act(player, action) : Task
IMatchReplayService «interface»	Contract of replay storage	–	–
OnlineGameService	Server implementation of training, match settlement, matchmaking and economy use cases	–	–
OnlineSocialService	Friend requests, friendships, blocks and leaderboard cache	–	–
IUnitOfWork «interface»	Gives services access to repositories and one shared transaction	–	–
GameRepository	EF Core queries and locks for matches, progression and economy	–	–
PickleVerseContext «DbContext»	EF Core DbContext mapping the 51 tables	–	–

3.7.3 Purchase a cosmetic

The command ID makes the purchase idempotent: an already processed request returns the stored receipt; otherwise ownership and balance are checked and the debit, ledger entry and ownership are written in one transaction.

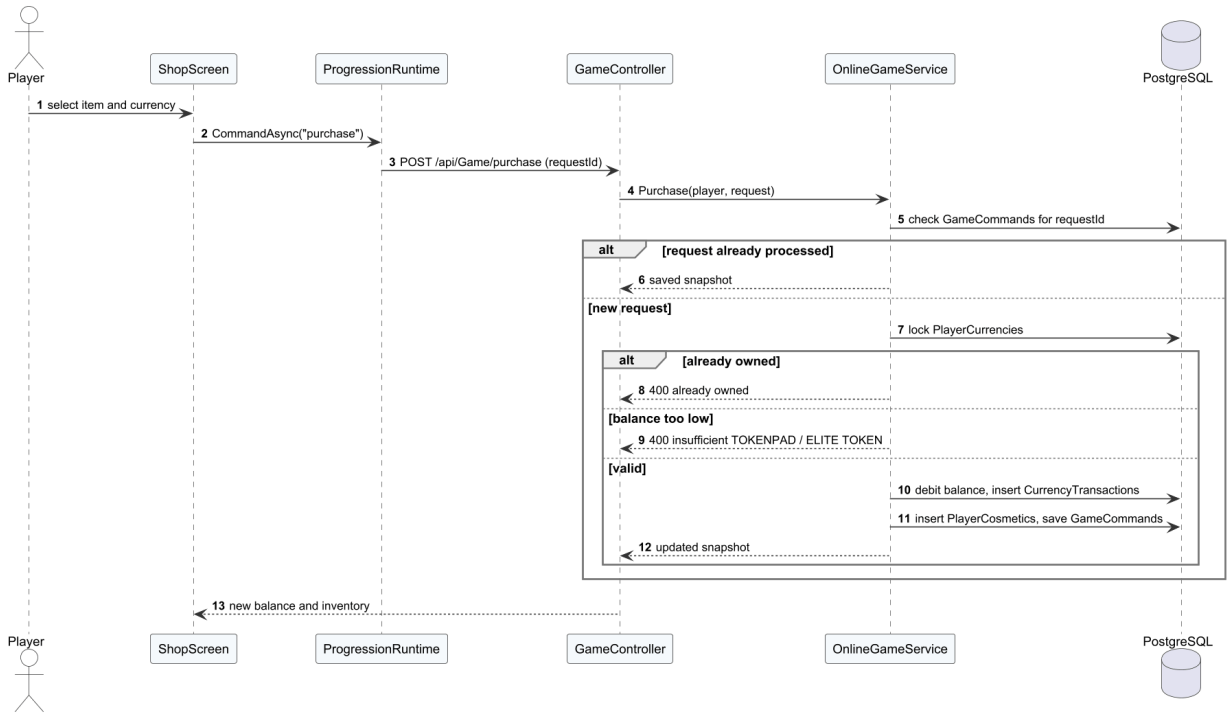


Figure 31. Sequence diagram – Purchase a cosmetic

3.8 Administration and Moderation

Administration implements FR-27 to FR-33. Every admin controller requires the Admin role; services write an AdminAudits record in the same transaction as the change.

3.8.1 Class Diagram

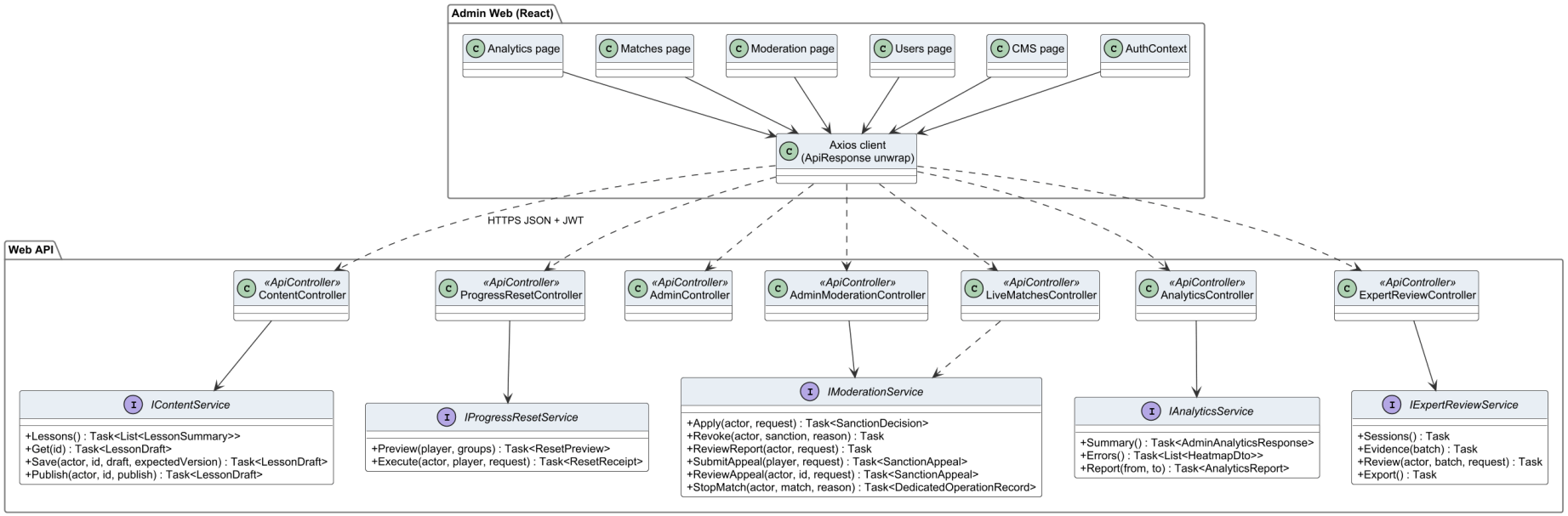


Figure 32. Class diagram – administration web and admin API

3.8.2 Class Specifications

Table 31. Class specifications – administration

Class	Responsibility	Main attributes	Main operations
AuthContext	Admin web session state and role check	–	–
CMS page	Lesson, drill, item and mission forms	–	–
Users page	Account search and actions	–	–
Moderation page	Reports, sanctions and appeals	–	–
Matches page	Live matches and stop dialog	–	–
Analytics page	Charts for the selected date range	–	–
Axios client (ApiResponse unwrap)	HTTP client that adds the JWT and unwraps the ApiResponse envelope	–	–
ContentController «ApiController»	Admin endpoints for lessons and catalogue	–	–
ProgressResetController «ApiController»	Admin endpoints for reset preview and execution	–	–
AdminController «ApiController»	Admin endpoints for users, roles, sanctions and e-mail activation	–	–
AdminModerationController «ApiController»	Admin endpoints for reports, appeals and match stop	–	–
LiveMatchesController «ApiController»	Admin endpoints for live and finished matches	–	–
AnalyticsController «ApiController»	Admin endpoints for analytics summary and error heat map	–	–
ExpertReviewController «ApiController»	Admin endpoints for expert labelling and dataset export	–	–
IContentService «interface»	Contract of content management	–	+ Lessons() : Task<List<LessonSummary>> + Get(id) : Task<LessonDraft> + Save(actor, id, draft, expectedVersion) : Task<LessonDraft> + Publish(actor, id, publish) : Task<LessonDraft>
IProgressResetService «interface»	Contract of progress reset	–	+ Preview(player, groups) : Task<ResetPreview> + Execute(actor, player, request) : Task<ResetReceipt>
IModerationService «interface»	Contract of moderation use cases	–	+ Apply(actor, request) : Task<SanctionDecision> + Revoke(actor, sanction, reason) : Task + ReviewReport(actor, request) : Task + SubmitAppeal(player, request) : Task<SanctionAppeal> + ReviewAppeal(actor, id, request) : Task<SanctionAppeal> + StopMatch(actor, match, reason) : Task<DedicatedOperationRecord>
IAnalyticsService «interface»	Contract of analytics queries	–	+ Summary() : Task<AdminAnalyticsResponse> + Errors() : Task<List<HeatmapDto>> + Report(from, to) : Task<AnalyticsReport>
IExpertReviewService «interface»	Contract of expert review and export	–	+ Sessions() : Task + Evidence(batch) : Task + Review(actor, batch, request) : Task + Export() : Task

3.8.3 Report, sanction and appeal

A player reports; the administrator accepts the report and applies a sanction; the restricted player appeals; the administrator accepts or declines the appeal.

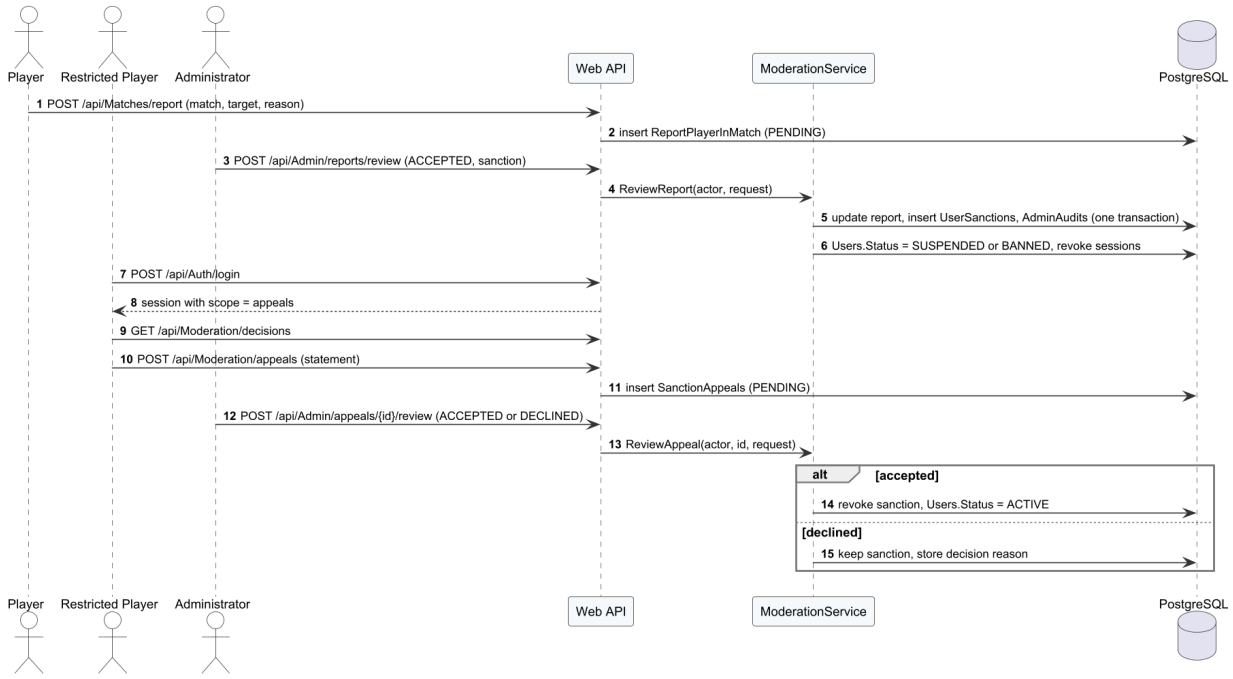


Figure 33. Sequence diagram – Report, sanction and appeal

3.8.4 Reset player progress

The dialog previews the impact; execution is refused when the revision changed after the preview.

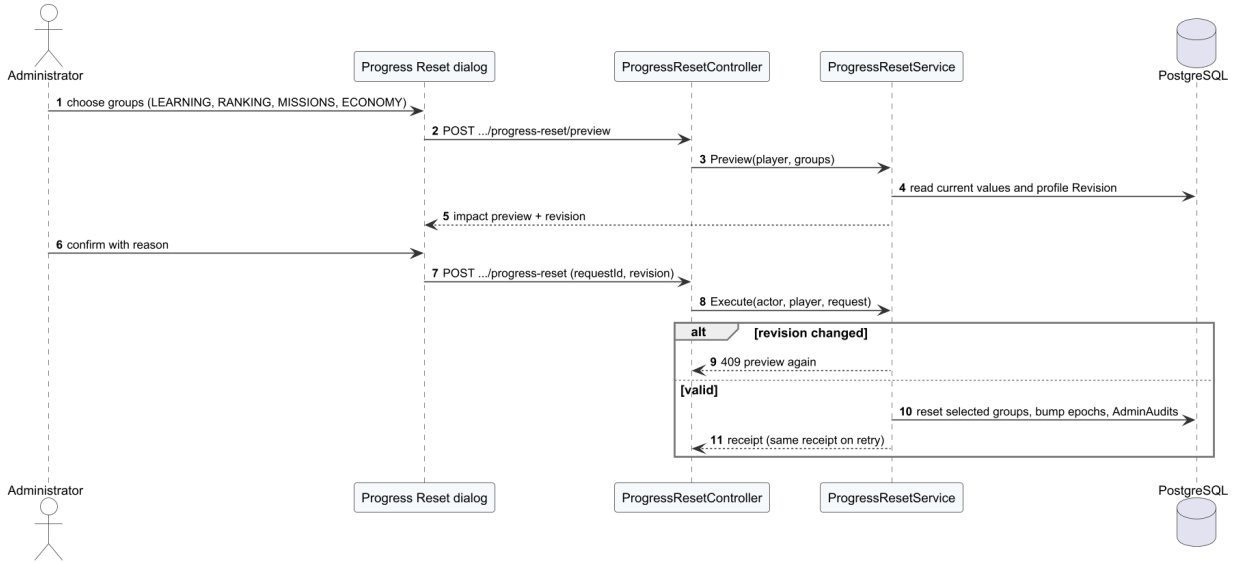


Figure 34. Sequence diagram – Reset player progress

3.8.5 Publish a lesson

Publishing is refused when the expected version changed; otherwise the lesson becomes PUBLISHED and a new catalogue version is produced.

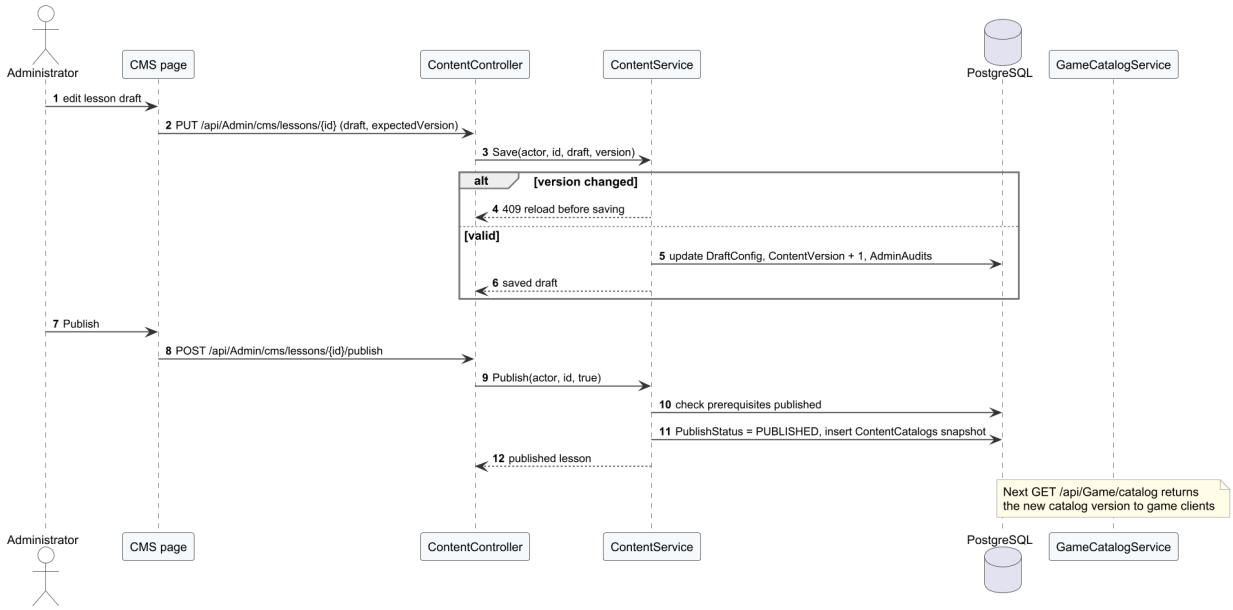


Figure 35. Sequence diagram – Publish a lesson

3.8.6 Stop a live match

The stop request is stored as a dedicated operation; the server reads it with the next heartbeat, ends the match as interrupted and acknowledges.

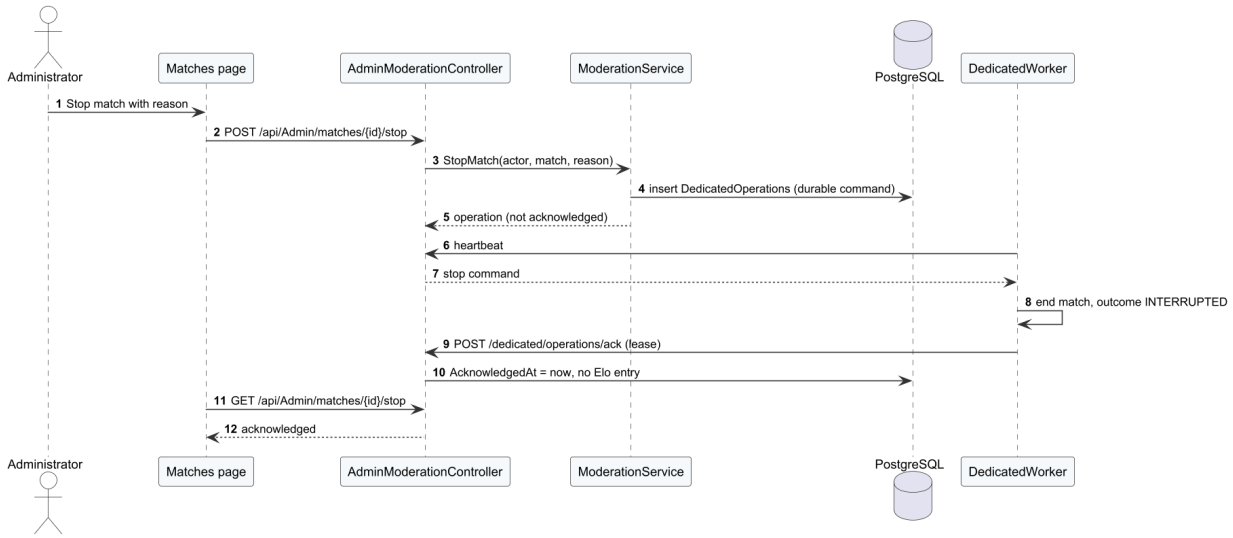


Figure 36. Sequence diagram – Stop a live match

3.9 API Design

The Web API exposes 129 REST endpoints in 28 controller classes (24 source files). The endpoint list below is extracted from the controller attributes by build/api_extract.py, so routes and access levels are those of the code on 12/11/2026. The column Used by shows which program calls each route (game client, dedicated game server or admin web), found by searching the client sources; code compiled only for the old Netcode prototype is ignored. Access: Anonymous (no token), Player (any signed-in account with game scope), Player, Restricted (also allowed for a restricted account with appeals scope), Admin (role Admin), Service key (dedicated server only).

Table 32. API conventions

Item	Design
Base path and format	/api/<Controller>/...; JSON (UTF-8); enum values serialised as strings
Envelope	Success: ApiResponse<T> {success: true, message, data}. Error: ApiResponse {success: false, message, code, errors}
Status codes	400 validation, 401 missing or expired token, 403 role or scope not allowed, 404 not found, 409 conflict or stale version, 429 rate limit, 503 dependency unavailable, 500 internal (no details)
Authentication	Bearer JWT (HS256) with claims sub/NameIdentifier, email, role (Player, Admin or Restricted), access_scope (game or appeals) and auth_version; refresh tokens rotate
Rate limiting	Policy auth: 20 requests per minute per IP on authentication endpoints; rejection returns 429 with Retry-After: 60
Idempotency	Commands carry a request/command ID stored in GameCommands or receipts
Service calls	Dedicated server: service key header and HMAC-signed tickets; AI service: X-AI-Key header
Documentation	OpenAPI document generated at runtime with the ApiResponse schema

Table 33. Web API endpoints (129)

Controller	Method	Route	Action	Access	Used by
Achievements	GET	/api/Achievements	GetMyAchievements	Player	Not called
Achievements	POST	/api/Achievements/{achievementId}/claim	ClaimReward	Player	Not called
Admin	POST	/api/Admin/cms/cosmetics	CreateCosmetic	Admin	Admin web
Admin	POST	/api/Admin/cms/drills	CreateDrill	Admin	Admin web
Admin	POST	/api/Admin/cms/lessons	CreateLesson	Admin	Admin web
Admin	POST	/api/Admin/cms/missions	CreateMission	Admin	Admin web
Admin	GET	/api/Admin/matches/recent	GetRecentMatches	Admin	Admin web
Admin	GET	/api/Admin/reports	GetReports	Admin	Admin web
Admin	POST	/api/Admin/reports/review	ReviewReport	Admin	Admin web
Admin	GET	/api/Admin/users	GetUsers	Admin	Admin web
Admin	POST	/api/Admin/users/assign-role	AssignRole	Admin	Admin web
Admin	POST	/api/Admin/users/ban	BanUser	Admin	Admin web
Admin	POST	/api/Admin/users/{playerId}/reset-elo	ResetPlayerElo	Admin	Admin web
Admin	POST	/api/Admin/users/{userId}/unban	UnbanUser	Admin	Admin web
Admin	POST	/api/Admin/users/{userId}/verify-email	ManualVerifyUserEmai l	Admin	Admin web
AdminModeration	GET	/api/Admin/appeals	Appeals	Admin	Admin web
AdminModeration	POST	/api/Admin/appeals/{id:guid}/review	Review	Admin	Admin web
AdminModeration	GET	/api/Admin/matches/{match:guid}/stop	StopOperation	Admin	Admin web
AdminModeration	POST	/api/Admin/matches/{match:guid}/stop	Stop	Admin	Admin web
AdminModeration	GET	/api/Admin/sanctions	Sanctions	Admin	Admin web
AiCoaching	POST	/api/AiCoaching/log-error	LogTechniqueError	Player	Not called
AiCoaching	GET	/api/AiCoaching/mastery	GetMastery	Player	Not called
AiCoaching	GET	/api/AiCoaching/recommendation	GetRecommendation	Player	Not called
AiCoaching	GET	/api/AiCoaching/summary	Summary	Player	Game client
AiCoaching	POST	/api/AiCoaching/telemetry	Telemetry	Player	Game client
Analytics	GET	/api/Admin/analytics/error-heatmap	Errors	Admin	Admin web
Analytics	GET	/api/Admin/analytics/report	Report	Admin	Admin web
Analytics	GET	/api/Admin/analytics/summary	Summary	Admin	Admin web
AppActivity	POST	/api/Game/activity/checkpoint	Activity	Player	Game client
AppActivity	POST	/api/Game/activity/open	OpenActivity	Anonymous	Game client
Auth	POST	/api/Auth/forgot-password	ForgotPassword	Anonymous	Admin web, Game client
Auth	POST	/api/Auth/login	Login	Anonymous	Admin web, Game client
Auth	POST	/api/Auth/logout	Logout	Player	Game client
Auth	POST	/api/Auth/refresh-token	RefreshToken	Anonymous	Game client
Auth	POST	/api/Auth/register	Register	Anonymous	Game client
Auth	POST	/api/Auth/resend-otp	ResendOtp	Anonymous	Game client
Auth	POST	/api/Auth/reset-password	ResetPassword	Anonymous	Admin web, Game client
Auth	POST	/api/Auth/verify-email	VerifyEmail	Anonymous	Game client
Content	GET	/api/Admin/cms/lessons	List	Admin	Admin web
Content	GET	/api/Admin/cms/lessons/{id:int}	Get	Admin	Admin web
Content	PUT	/api/Admin/cms/lessons/{id:int}	Save	Admin	Admin web
Content	POST	/api/Admin/cms/lessons/{id:int}/publish	Publish	Admin	Admin web
Content	POST	/api/Admin/cms/lessons/{id:int}/unpublish	Unpublish	Admin	Admin web
Dedicated	POST	/api/Game/dedicated/ticket	Ticket	Player	Game client
DedicatedWorker	POST	/api/Server/admission	AuthorizeAdmission	Service key	Game server
DedicatedWorker	POST	/api/Server/complete	Complete	Service key	Game server
DedicatedWorker	POST	/api/Server/heartbeat	Heartbeat	Service key	Game server
DedicatedWorker	POST	/api/Server/operations/ack	Acknowledge	Service key	Game server
DedicatedWorker	POST	/api/Server/poll	Poll	Service key	Game server
DedicatedWorker	POST	/api/Server/ready	Ready	Service key	Game server
DedicatedWorker	POST	/api/Server/release	Release	Service key	Game server

Controller	Method	Route	Action	Access	Used by
DedicatedWorker	POST	/api/Server/{lease}/replay	Replay	Service key	Game server
DedicatedWorker	POST	/api/Server/{lease}/replay/chunks/{index:int}	Chunk	Service key	Game server
DedicatedWorker	POST	/api/Server/{lease}/telemetry	Telemetry	Service key	Game server
Economy	POST	/api/Economy/buy	BuyCosmetic	Player	Not called
Economy	GET	/api/Economy/currencies	GetCurrencies	Player	Not called
Economy	POST	/api/Economy/equip	EquipCosmetic	Player	Not called
Economy	GET	/api/Economy/my-inventory	GetMyCosmetics	Player	Not called
Economy	GET	/api/Economy/shop	GetShopItems	Player	Not called
ExpertReview	GET	/api/Admin/coaching/dataset/coaching	CoachingDataset	Admin	Admin web
ExpertReview	GET	/api/Admin/coaching/dataset/tier	TierDataset	Admin	Admin web
ExpertReview	GET	/api/Admin/coaching/demo-sessions	DemoSessions	Admin	Admin web
ExpertReview	GET	/api/Admin/coaching/demo-sessions/{sessionId}	DemoEvidence	Admin	Admin web
ExpertReview	GET	/api/Admin/coaching/sessions	Sessions	Admin	Admin web
ExpertReview	GET	/api/Admin/coaching/sessions/{batch:guid}	Evidence	Admin	Admin web
ExpertReview	POST	/api/Admin/coaching/sessions/{batch:guid}/review	Review	Admin	Admin web
Friends	GET	/api/Friends	GetFriends	Player	Not called
Friends	POST	/api/Friends/block/{targetUserId}	BlockUser	Player	Not called
Friends	POST	/api/Friends/request	SendFriendRequest	Player	Not called
Friends	POST	/api/Friends/respond	RespondFriendRequest	Player	Not called
Friends	DELETE	/api/Friends/{friendId}	RemoveFriend	Player	Not called
Game	GET	/api/Game/catalog	Catalog	Player	Game client
Game	POST	/api/Game/claim	Claim	Player	Game client
Game	POST	/api/Game/equip	Equip	Player	Game client
Game	POST	/api/Game/onboarding	Onboarding	Player	Game client
Game	POST	/api/Game/purchase	Purchase	Player	Game client
Game	GET	/api/Game/state	State	Player	Game client
Game	POST	/api/Game/tutorial	Tutorial	Player	Game client
GameMatches	POST	/api/Game/match/bot/complete	FinishBot	Player	Game client
GameMatches	POST	/api/Game/match/bot/start	StartBot	Player	Game client
GameMatches	POST	/api/Game/match/online/heartbeat	HeartbeatOnline	Player	Game client
GameMatches	POST	/api/Game/match/online/leave	LeaveOnline	Player	Game client
GameMatches	POST	/api/Game/match/online/offer	OfferOnline	Player	Game client
GameMatches	POST	/api/Game/match/online/prepare	PrepareOnline	Player	Game client
GameMatches	POST	/api/Game/match/online/report	ReportOnline	Player	Game client
GameMatches	POST	/api/Game/match/online/start	StartOnline	Player	Game client
GameMatches	POST	/api/Game/ranked/bot	Bot	Player	Game client
GameMatches	POST	/api/Game/ranked/cancel	CancelSearch	Player	Game client
GameMatches	POST	/api/Game/ranked/poll	Poll	Player	Game client
GameMatches	POST	/api/Game/ranked/room-ready	PublishRoom	Player	Game client
GameMatches	POST	/api/Game/ranked/search	Search	Player	Game client
GameTraining	POST	/api/Game/training/checkpoint	Checkpoint	Player	Game client
GameTraining	POST	/api/Game/training/complete	Complete	Player	Game client
GameTraining	POST	/api/Game/training/start	Start	Player	Game client
GameTraining	POST	/api/Game/training/sync	SyncTraining	Player	Game client
LiveMatches	GET	/api/Admin/matches/live	Live	Admin	Admin web
Matches	POST	/api/Matches/finish	FinishMatch	Player	Not called
Matches	GET	/api/Matches/my-history	GetMyMatchHistory	Player	Not called
Matches	POST	/api/Matches/report	ReportPlayer	Player	Game client
Matches	GET	/api/Matches/{id:guid}/report-targets	ReportTargets	Player	Game client
Missions	GET	/api/Missions	GetMyMissions	Player	Not called
Missions	POST	/api/Missions/{missionId}/claim	ClaimReward	Player	Not called
Moderation	GET	/api/Moderation/appeals	Appeals	Player, Restricted	Game client
Moderation	POST	/api/Moderation/appeals	Appeal	Player, Restricted	Game client
Moderation	GET	/api/Moderation/decisions	Decisions	Player, Restricted	Game client
OnlineReplays	GET	/api/Game/replays	List	Player	Game client
OnlineReplays	GET	/api/Game/replays/{match:guid}	Read	Player	Game client
OnlineReplays	PUT	/api/Game/replays/{match:guid}	Save	Player	Game client
OnlineReplays	GET	/api/Game/replays/{match:guid}/chunks/{index:int}	Chunk	Player	Game client
OnlineReplays	PUT	/api/Game/replays/{match:guid}/chunks/{index:int}	Upload	Player	Game client
OnlineSocial	GET	/api/Game/social	Read	Player	Game client
OnlineSocial	POST	/api/Game/social	Act	Player	Game client
Profiles	GET	/api/Profiles/leaderboard	GetLeaderboard	Anonymous	Not called
Profiles	GET	/api/Profiles/me	GetMyProfile	Player	Game client
Profiles	PUT	/api/Profiles/me	UpdateProfile	Player	Game client
Profiles	POST	/api/Profiles/sync-time	SyncTime	Player	Not called
Profiles	GET	/api/Profiles/{playerId}	GetProfileById	Anonymous	Game client
ProgressReset	POST	/api/Admin/users/{player:guid}/progress-reset	Reset	Admin	Admin web
ProgressReset	POST	/api/Admin/users/{player:guid}/progress-reset/preview	Preview	Admin	Admin web
Replays	POST	/api/Replays/save	SaveReplay	Player	Not called
Replays	GET	/api/Replays/{matchId}	GetReplay	Player	Admin web
RoomInvitations	GET	/api/Game/lobby/friends	Friends	Player	Game client

Controller	Method	Route	Action	Access	Used by
RoomInvitations	POST	/api/Game/lobby/invitations	Send	Player	Game client
RoomInvitations	POST	/api/Game/lobby/invitations/{id}/respond	Respond	Player	Game client
RoomInvitations	POST	/api/Game/lobby/presence	Presence	Player	Game client
TechniqueReferences	GET	/api/Admin/cms/techniques/references	References	Admin	Admin web
TechniqueReferences	POST	/api/Admin/cms/techniques/{id:int}/references	Create	Admin	Admin web
Training	GET	/api/Training/lessons	GetLessons	Player	Not called
Training	POST	/api/Training/submit-drill	SubmitDrill	Player	Not called

24 endpoints are marked Not called. They belong to the first resource-style controllers (Achievements, Economy, Friends, Matches, Missions, Profiles, Replays, Training) and to three AiCoaching routes; the game now uses the command-style /api/Game routes for the same data. They still require authentication and pass through the same services, but they are not tested by the clients and are recorded as issue I-12 of the project tracking workbook: they will be removed or documented as public API before release.

3.10 Security Design

Table 34. Security design

Concern	Design
Passwords	BCrypt hash with salt; 6–128 characters; never logged
Tokens	JWT HS256, 2-hour access token; refresh token 14 days stored as SHA-256 hash, rotated on use; auth_version in the token is incremented to revoke all sessions (password reset, ban)
Codes	OTP and reset codes stored as HMAC, 10-minute expiry, 5 attempts, 60-second resend cooldown
Authorisation	Role-based ([Authorize(Roles = "Admin")]) plus access scope: restricted accounts receive scope appeals and can call only moderation endpoints; resource ownership is checked in services (own matches, replays, telemetry)
Transport	HTTPS with TLS 1.2+ behind the reverse proxy. Since 13/11/2026 the staging API and admin web are served only over HTTPS on port 443 (Let's Encrypt certificate, TLS 1.3; port 80 redirects and port 8080 is closed) and the game client uses https://tad-min.io.vn (commit be556a1; Report 5, D-18 fixed)
Secrets	Read from environment variables and the server .env file. Commit be556a1 (13/11/2026) emptied the development credentials in appsettings.json and removed the fallback JWT key from Program.cs: the API refuses to start without a key of at least 32 bytes. The published images are also built from sanitised settings (deploy/api/sanitize_settings.py). The old values remain in the Git history and must be rotated (Report 5, D-17 partly fixed)
Abuse protection	Rate limiting on authentication; server-authoritative online results; idempotent commands; validation of telemetry ownership; HMAC tickets bind a player to a match and slot
CORS	AdminWebPolicy allows only the admin web origins; production serves the admin web and API under the same origin through the reverse proxy
Data protection	Other players never receive e-mail addresses; audit log for administrative actions; since be556a1 the deployment takes a PostgreSQL dump before every release and verifies it with pg_restore (kept under .deploy/backups on the host); there is no scheduled daily backup or off-host copy yet (Report 2, risk 11)
Data at rest	Passwords are BCrypt hashes, refresh tokens SHA-256 hashes and codes HMACs. Database files, backups and replays are not encrypted with AES-256 as the proposal stated; disk encryption of the production host is planned and the deviation is recorded in Report 1 (LI-10)
Error handling	GlobalExceptionHandler maps AppException types to status codes and hides internal details of unexpected exceptions

4. Workflows and State Changes

4.1 Activity Diagrams

Each review workflow is owned by one team member, as committed in the Review 2 form BM04 (W01 Learn and sync a drill – Trần Nguyễn Hiếu Thuận; W02 Play a bot and get a tip – Nguyễn Gia Huy; W03 Play online and save the result – Nguyễn Văn Thành Đạt; W04 Review a report and publish a lesson – Huỳnh Trung Kiên), and is drawn as a UML activity diagram with one swimlane per participant, decisions and exception branches. W04 has two independent parts and therefore two diagrams.

W01 – Learn and sync a drill (owner: Trần Nguyễn Hiếu Thuận)

The player signs in or opens offline training; a beginner is sent to the tutorial; the player practises a drill; the attempt is applied online or queued offline and synchronised when the network returns.

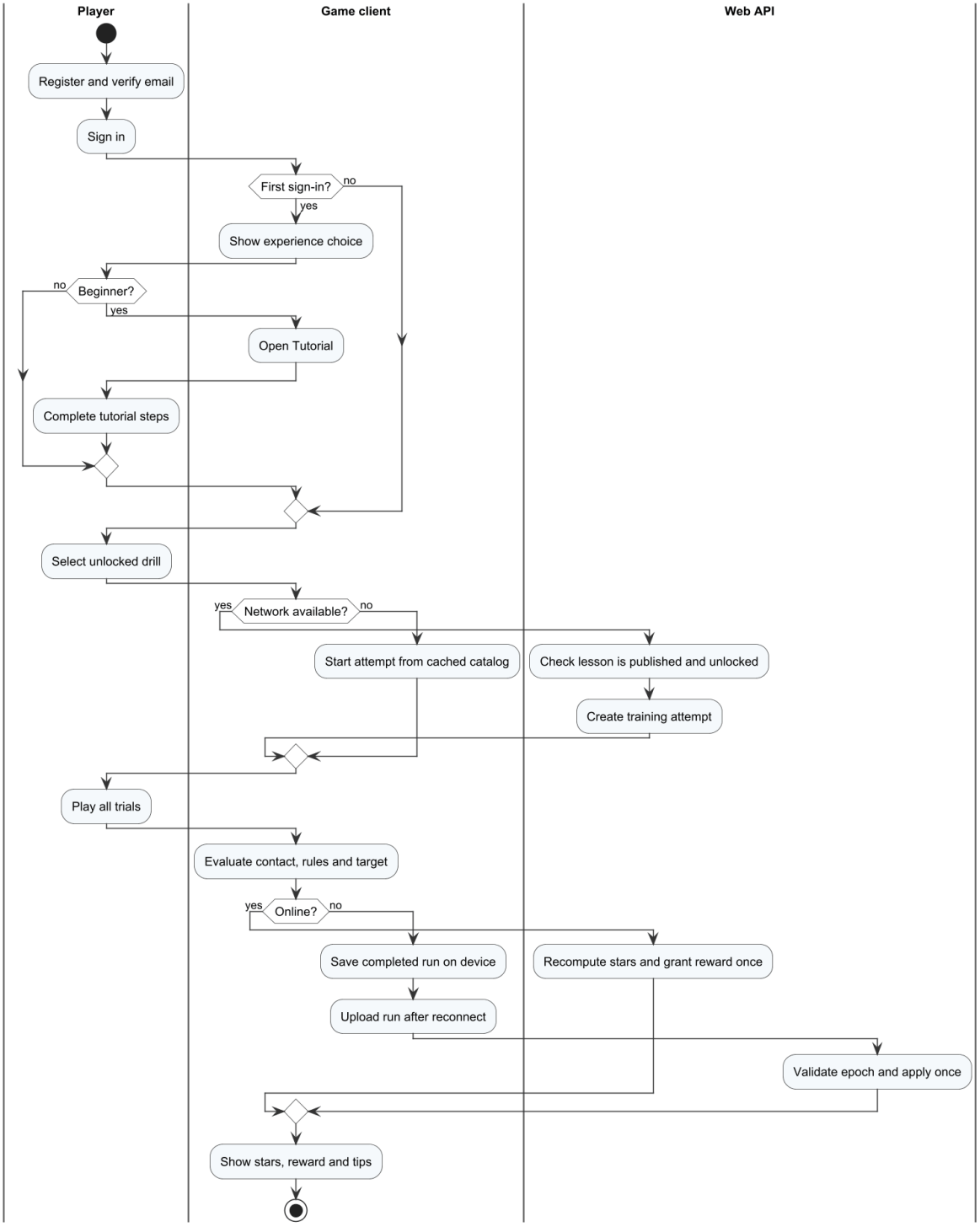


Figure 37. W01 – Learn and sync a drill (owner: Trần Nguyễn Hiếu Thuận)

W02 – Play a bot and get a tip (owner: Nguyễn Gia Huy)

The player sets up a bot match and plays rallies until a side wins; the API settles the result, the AI service analyses the telemetry and the player reads the tip and opens the recommended drill.

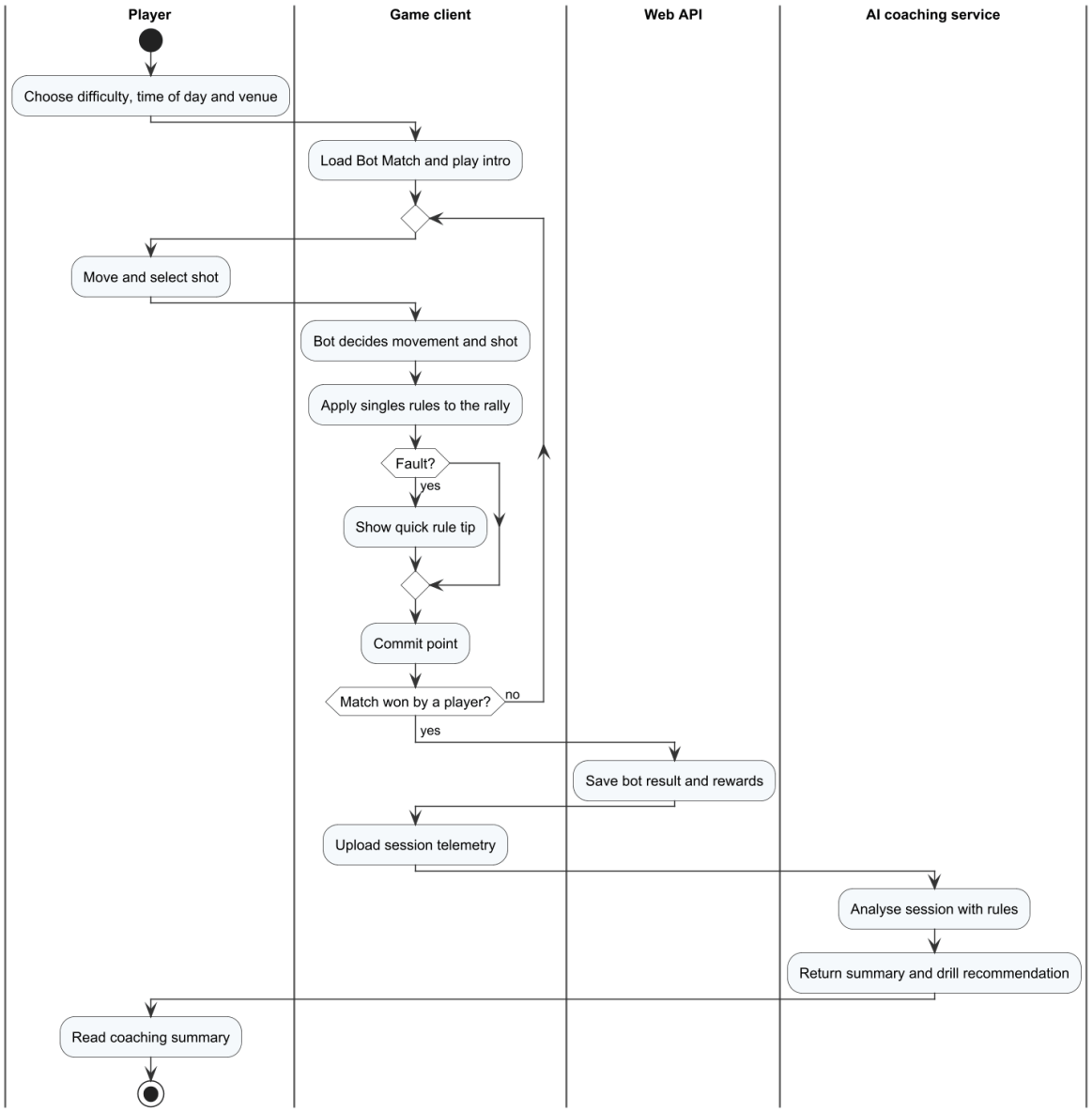


Figure 38. W02 – Play a bot and get a tip (owner: Nguyễn Gia Huy)

W03 – Play online and save the result (owner: Nguyễn Văn Thành Đạt)

The player creates a private room, starts quick doubles (alone or with a party of two) or searches a ranked singles match; only ranked singles has the bot fallback after 180 seconds; all players must accept; the dedicated server runs the match unless it is stopped or the lease is lost, and the API saves the result once.

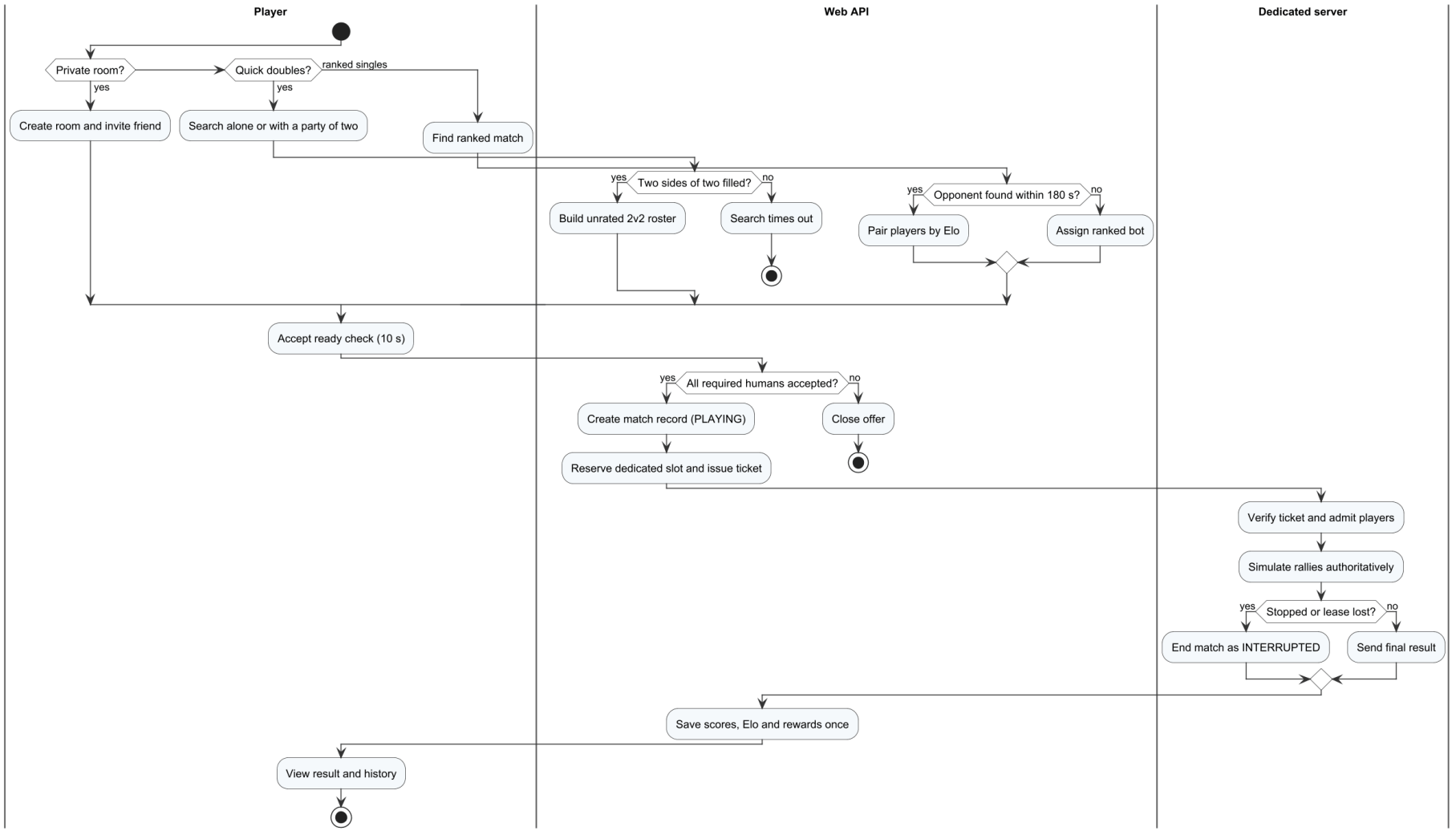


Figure 39. W03 – Play online and save the result (owner: Nguyễn Văn Thành Đạt)

W04 – Review a report and publish a lesson, part 1: report, sanction and appeal (owner: Huỳnh Trung Kiên)

A player reports an opponent; the administrator reviews and may sanction; the restricted player may appeal; the administrator decides the appeal.

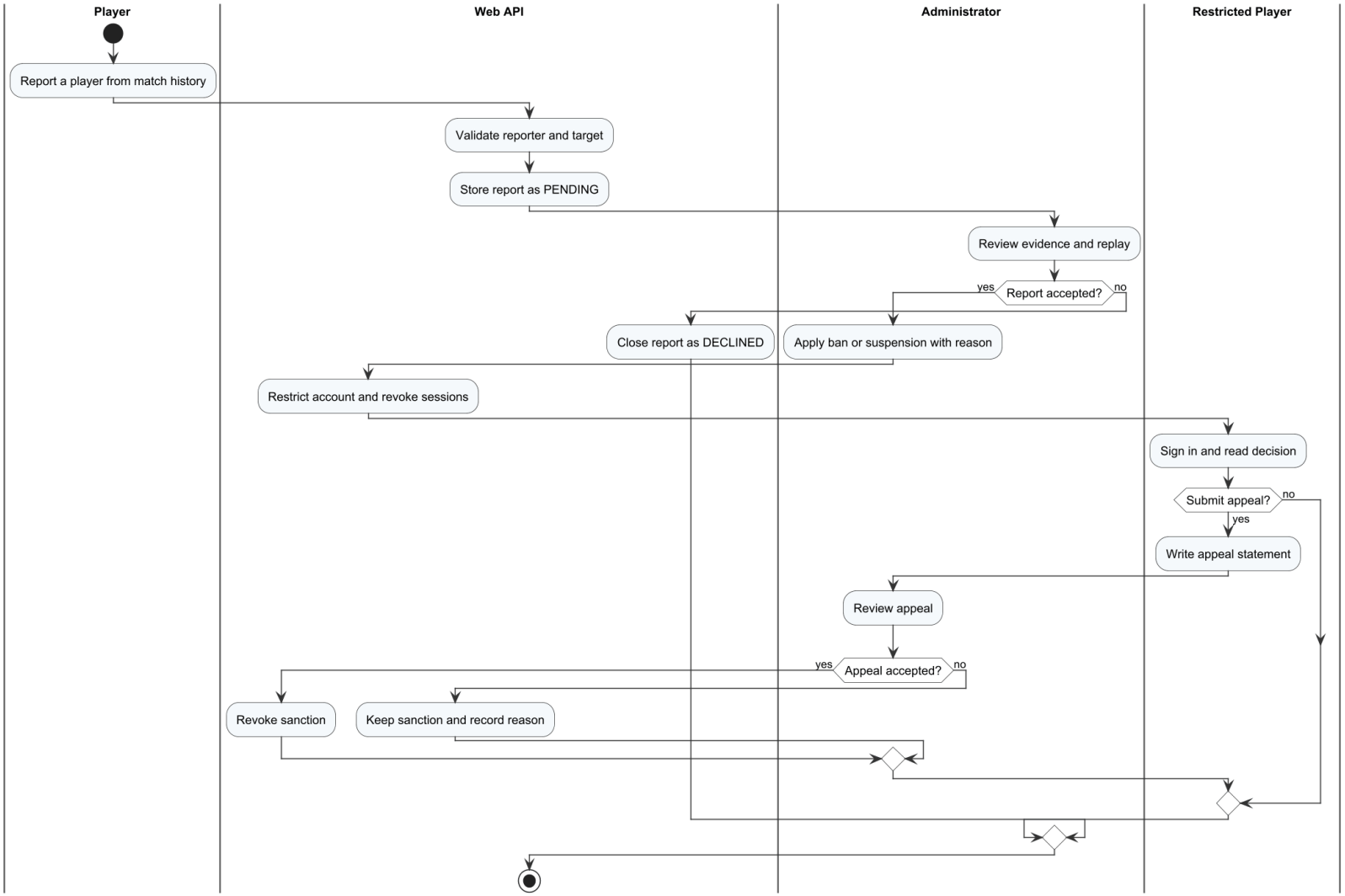


Figure 40. W04 – Review a report and publish a lesson, part 1: report, sanction and appeal (owner: Huỳnh Trung Kiên)

W04 – Review a report and publish a lesson, part 2: publish a lesson (owner: Huỳnh Trung Kiên)

The administrator edits and publishes a lesson; the API checks the expected version and that prerequisites are published; clients receive the new catalogue version.

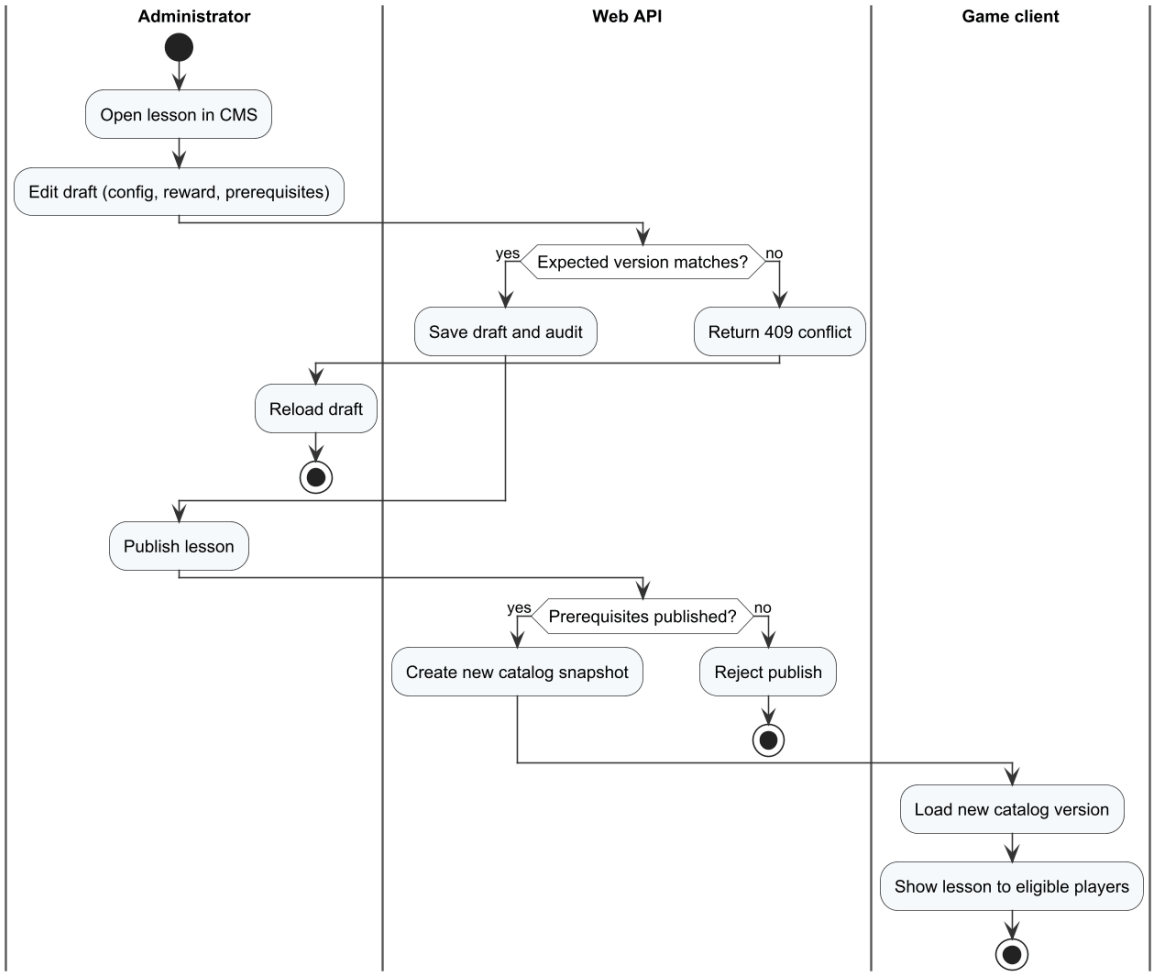


Figure 41. W04 – Review a report and publish a lesson, part 2: publish a lesson (owner: Huỳnh Trung Kiên)

4.2 State Machine Diagrams

The following state machines specify every entity or runtime object with a life cycle. Each transition is labelled event [guard] / action.

SM-01 Account status

ACTIVE, SUSPENDED and BANNED; suspensions end by expiry or revocation, and an accepted appeal revokes the sanction.

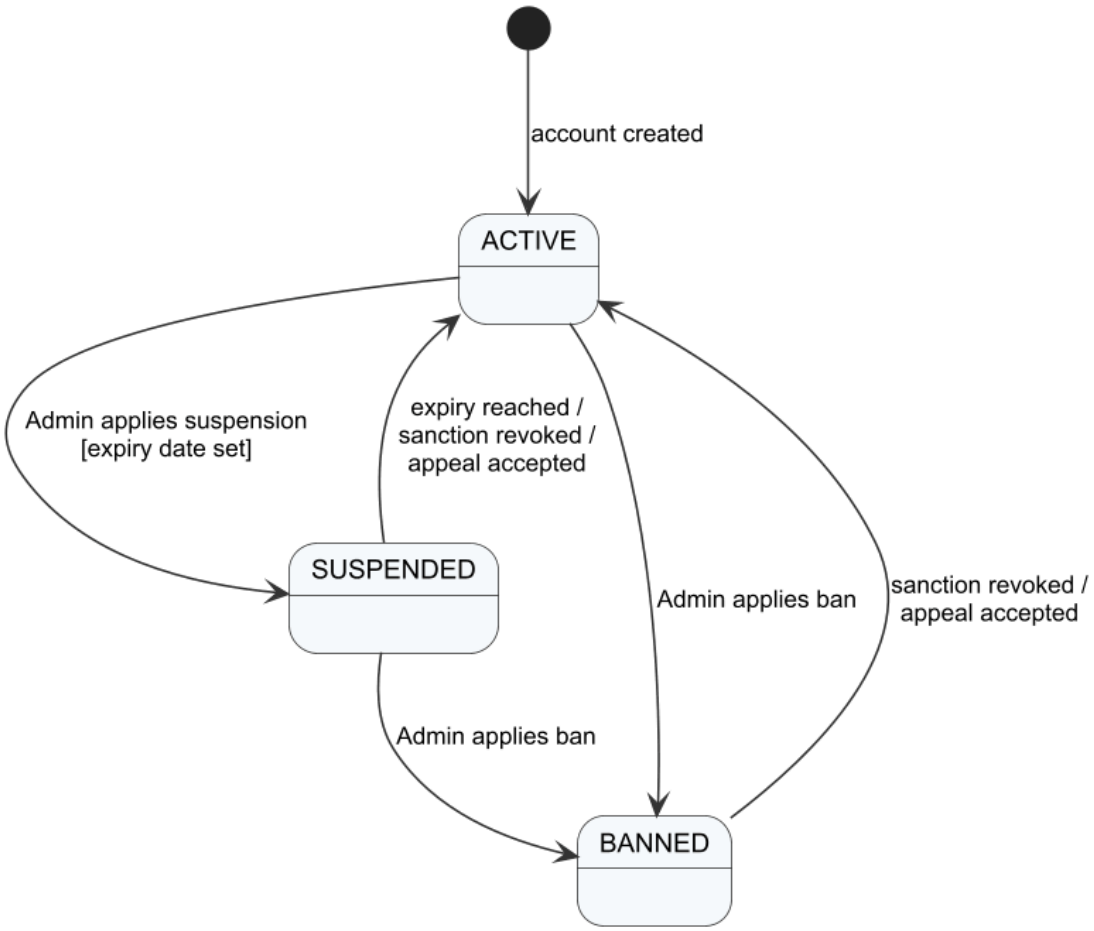


Figure 42. SM-01 Account status

SM-02 E-mail verification code

Pending, verified, expired and locked after 5 wrong attempts; a resend creates a new pending code.

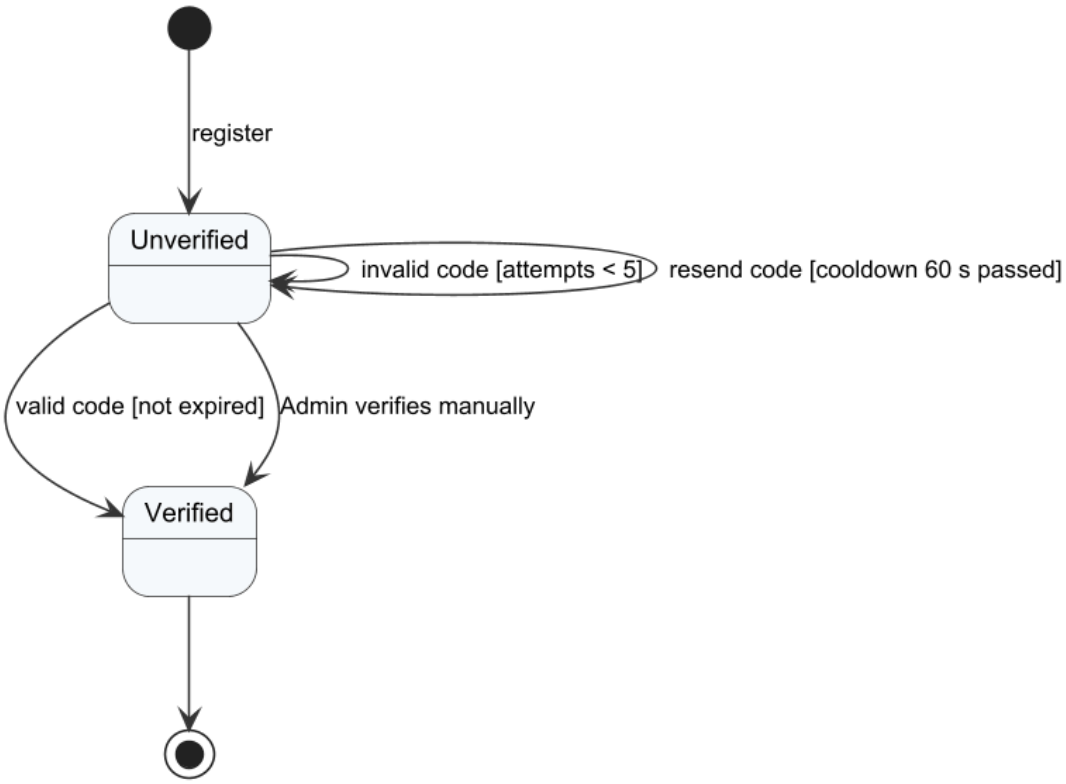


Figure 43. SM-02 E-mail verification code

SM-03 Training attempt and receipt

Briefing, practice, result; the receipt is APPLIED, DUPLICATE or REJECTED.

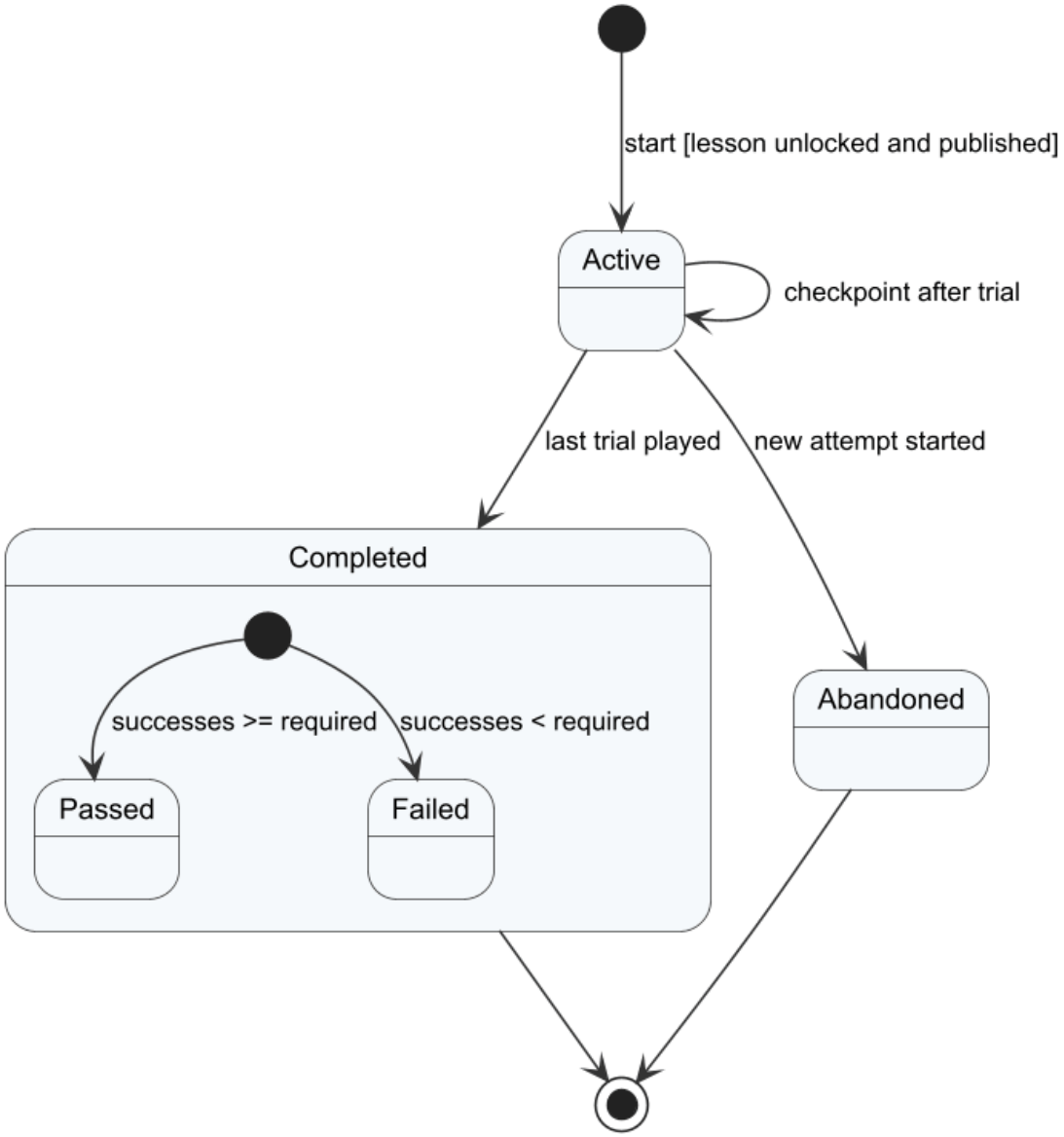


Figure 44. SM-03 Training attempt and receipt

SM-04 Offline queue item

Queued, sending, applied, duplicate or rejected; network failure returns the item to the queue.

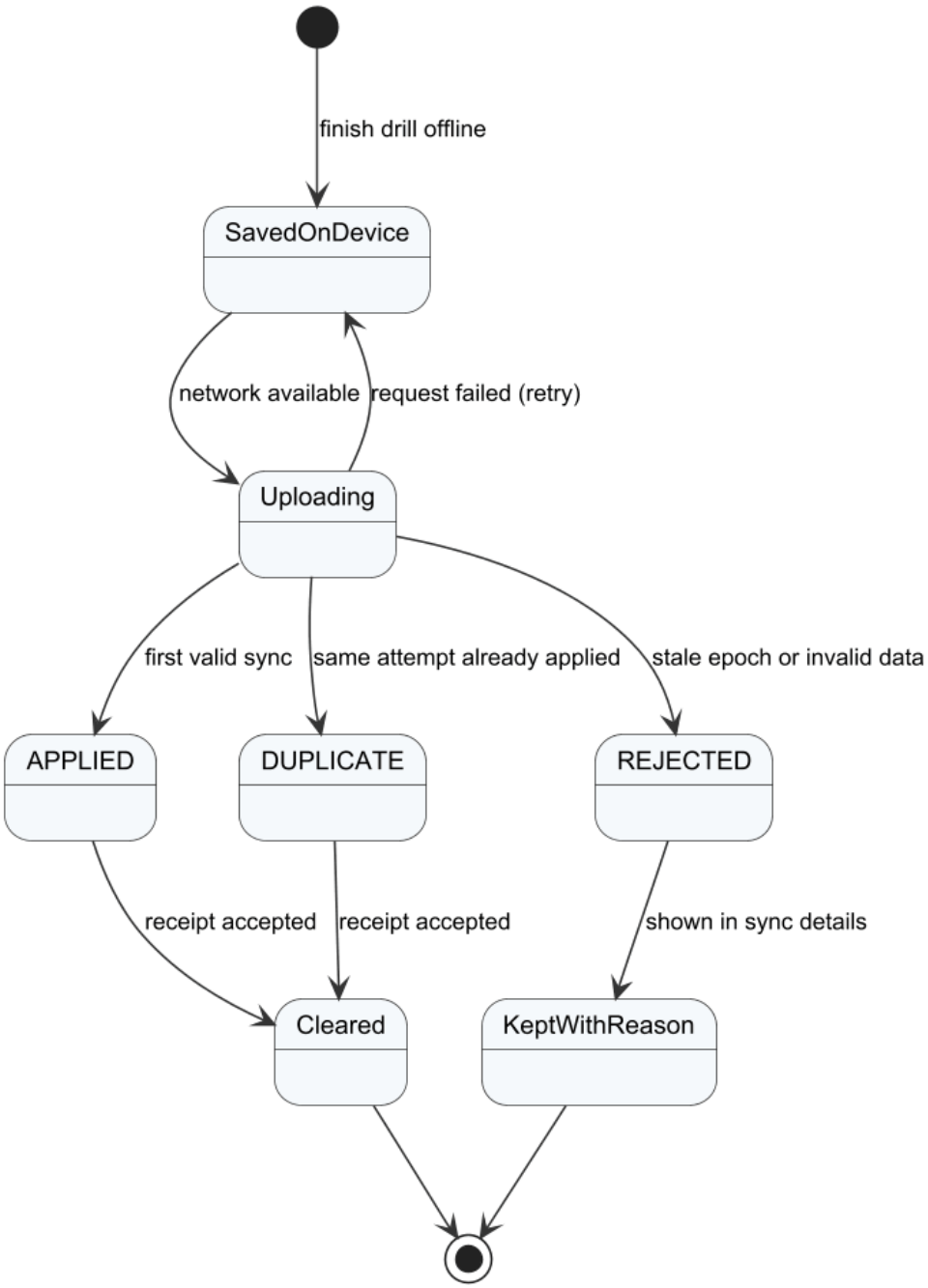


Figure 45. SM-04 Offline queue item

SM-05 Match phase (local match)

Intro, Cinematic, PreServe, Rally, Settling, BetweenPoints, BetweenGames and MatchOver. Time-outs were removed from play, so the Timeout value of the enum is never entered.

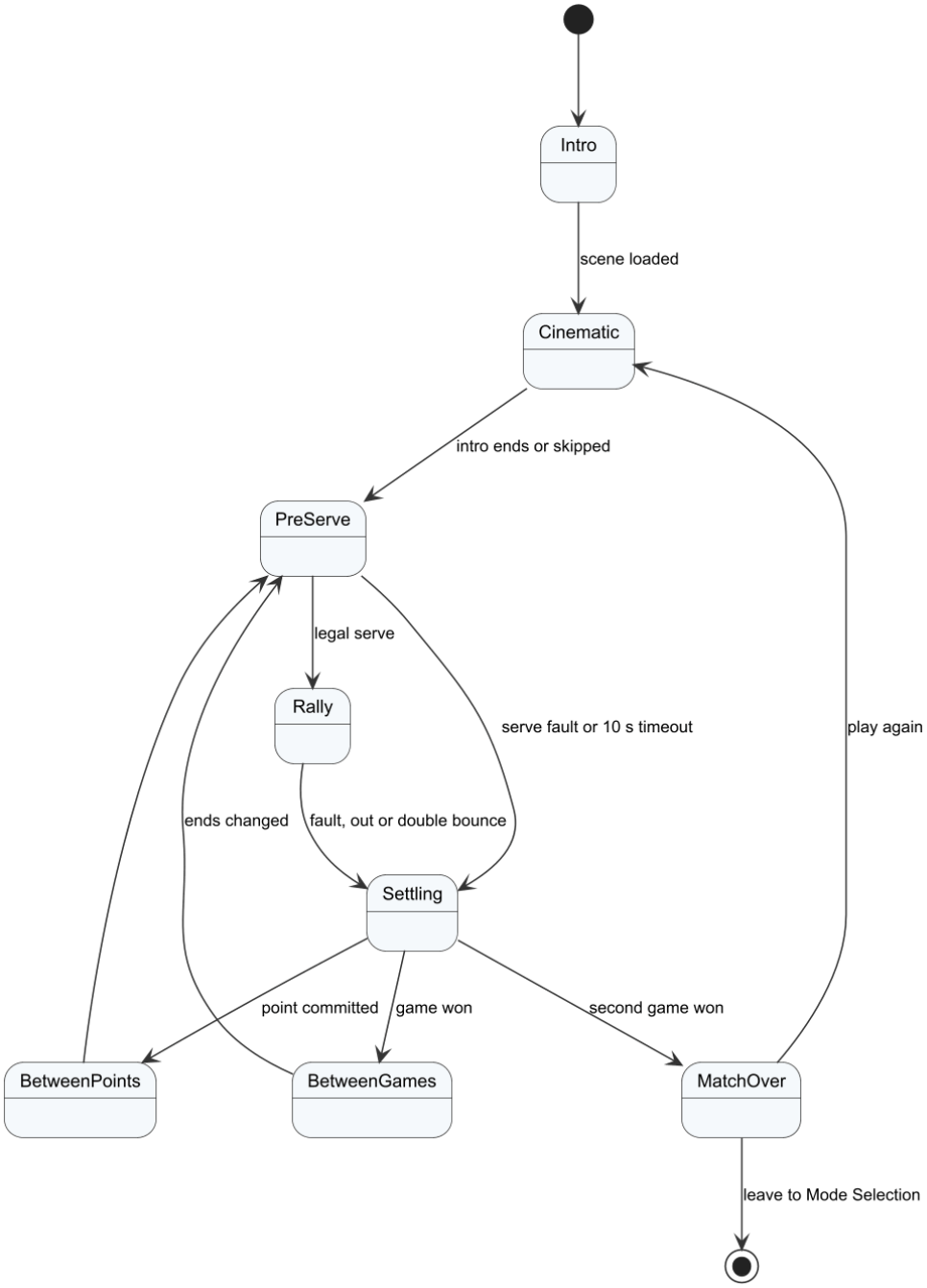


Figure 46. SM-05 Match phase (local match)

SM-06 Matchmaking search

SEARCHING ends in MATCHED, BOT (ranked singles fallback only), CANCELLED or TIMEOUT; a declined offer returns the other player to SEARCHING.

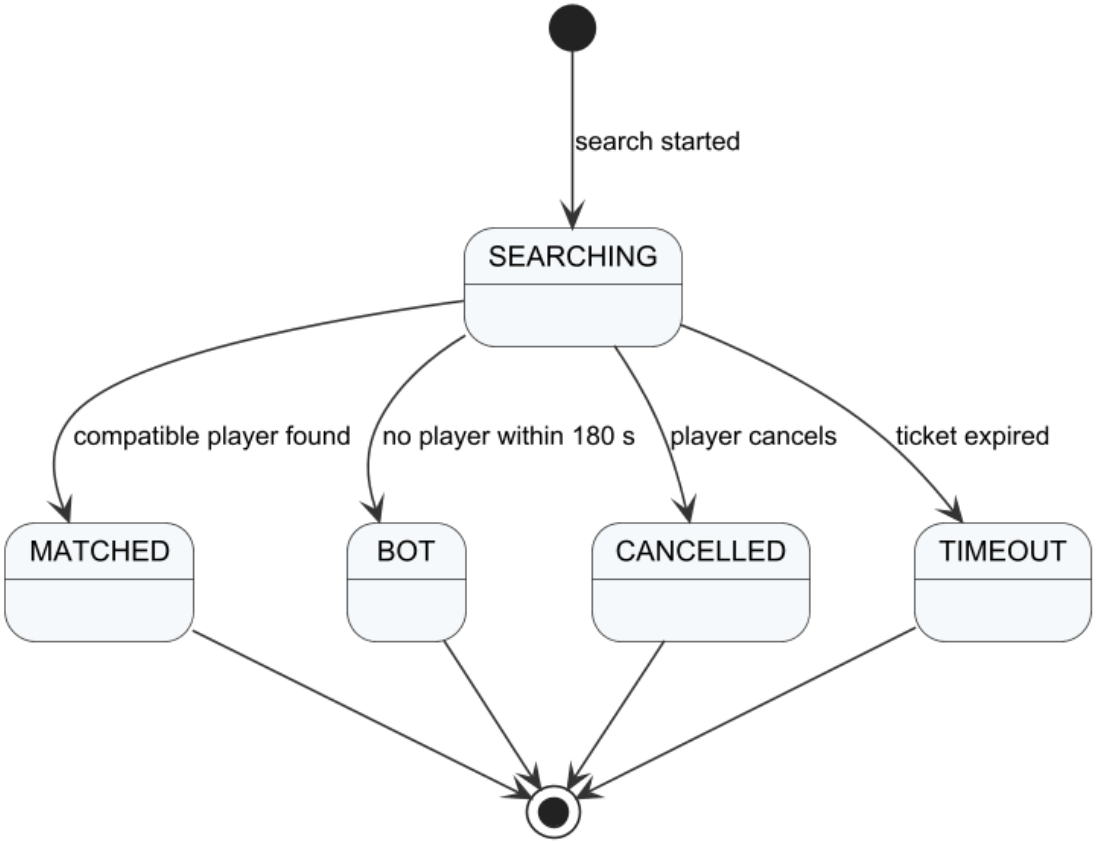


Figure 47. SM-06 Matchmaking search

SM-07 Online room phase

idle, searching, assigning-bot, ready, loading, playing, interrupted, closed and rejected.

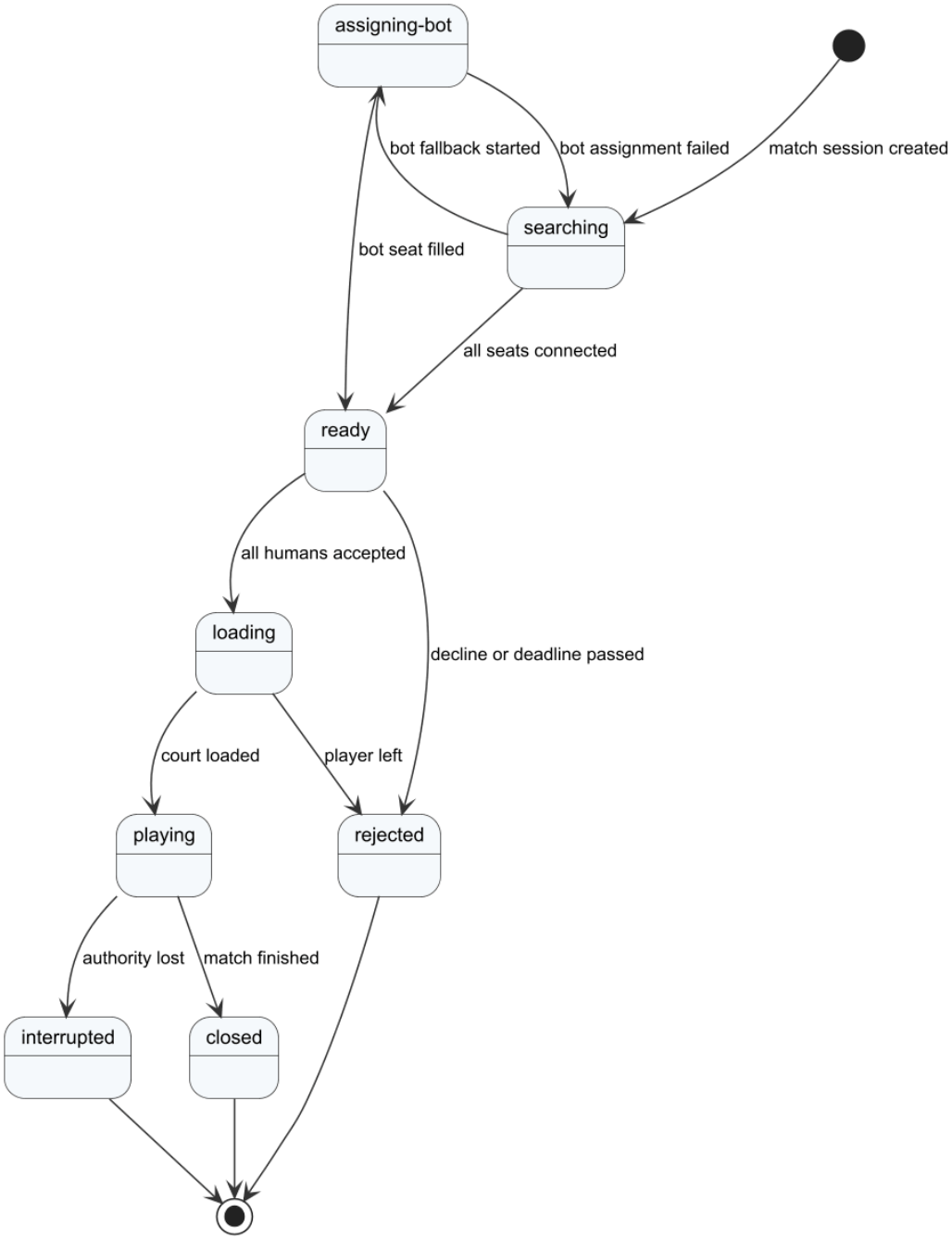


Figure 48. SM-07 Online room phase

SM-08 Game match

PREPARING, PLAYING, FINISHED and CANCELLED; interruption and administrator stop end the match without Elo.

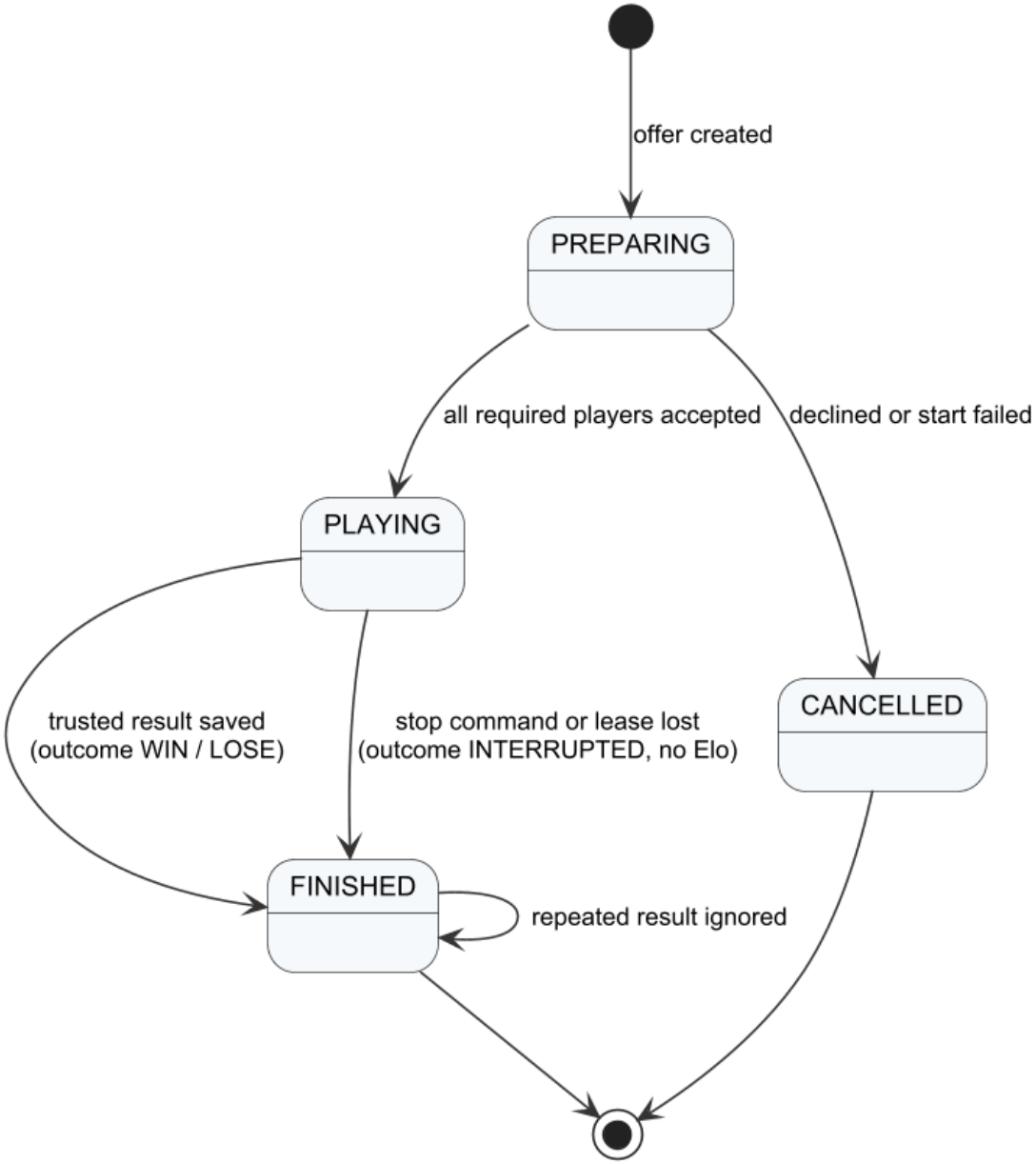


Figure 49. SM-08 Game match

SM-09 Dedicated server lease

queued, starting, ready, finishing, pending and busy; an expired lease releases the slot.

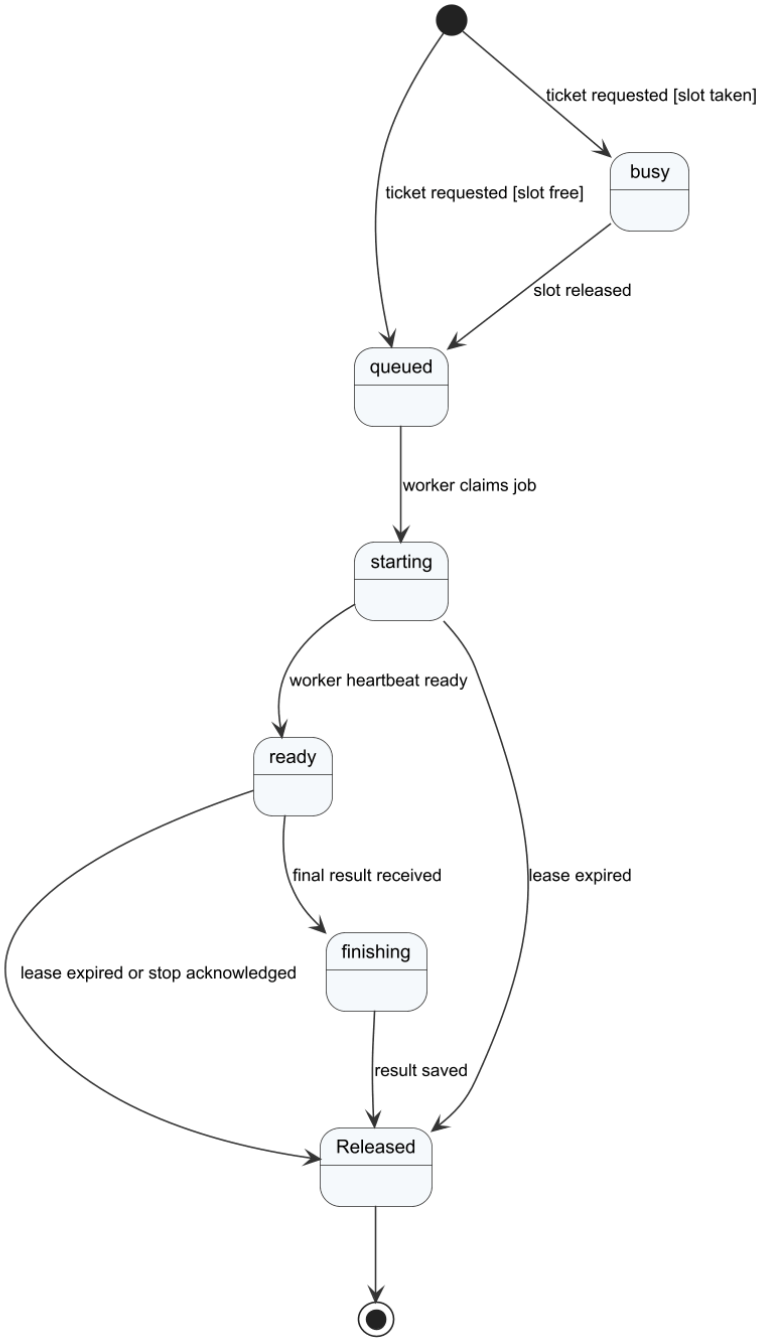


Figure 50. SM-09 Dedicated server lease

SM-10 Lesson publication

DRAFT, PUBLISHED and UNPUBLISHED.

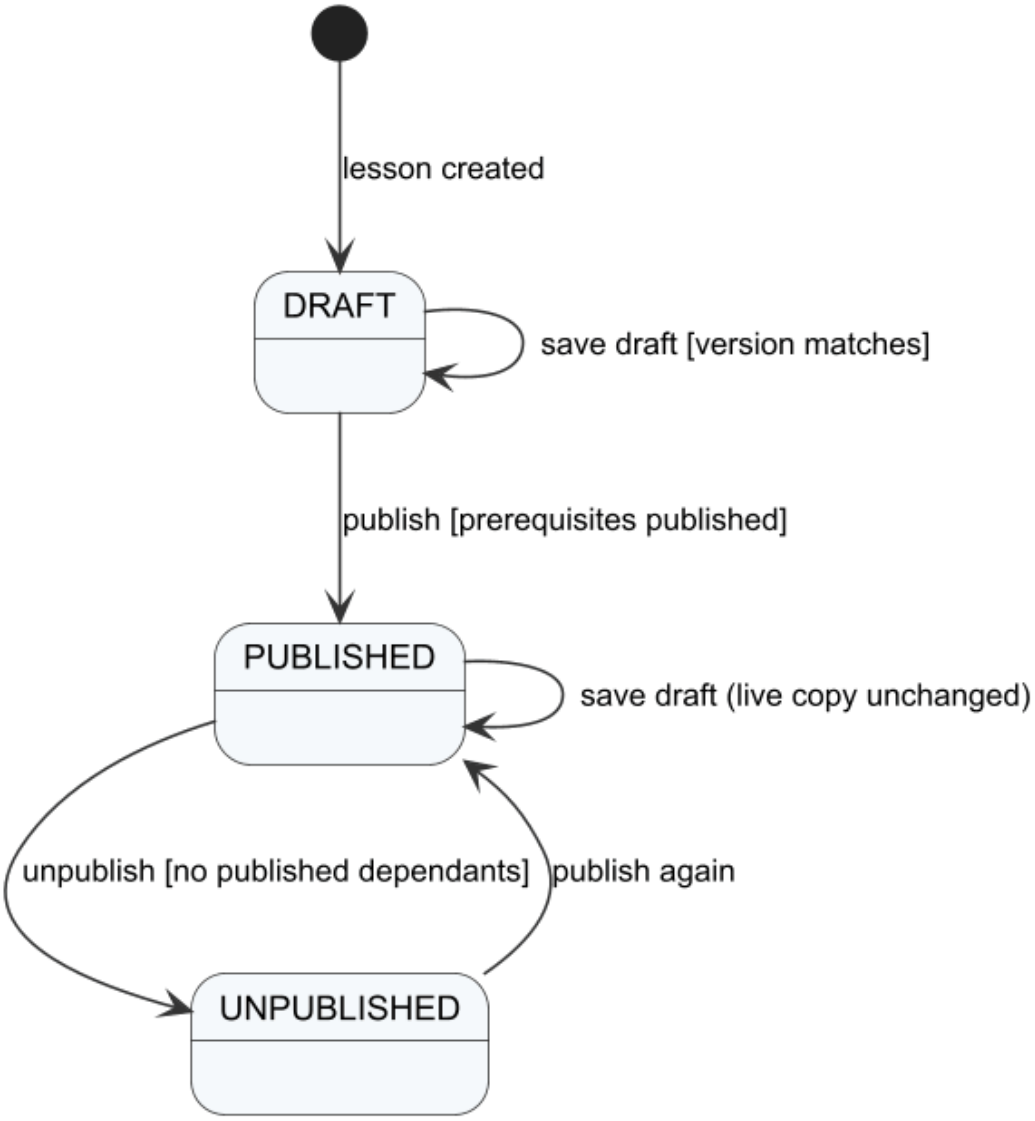


Figure 51. SM-10 Lesson publication

SM-11 Player report

PENDING, CHECKING, ACCEPTED and DECLINED.

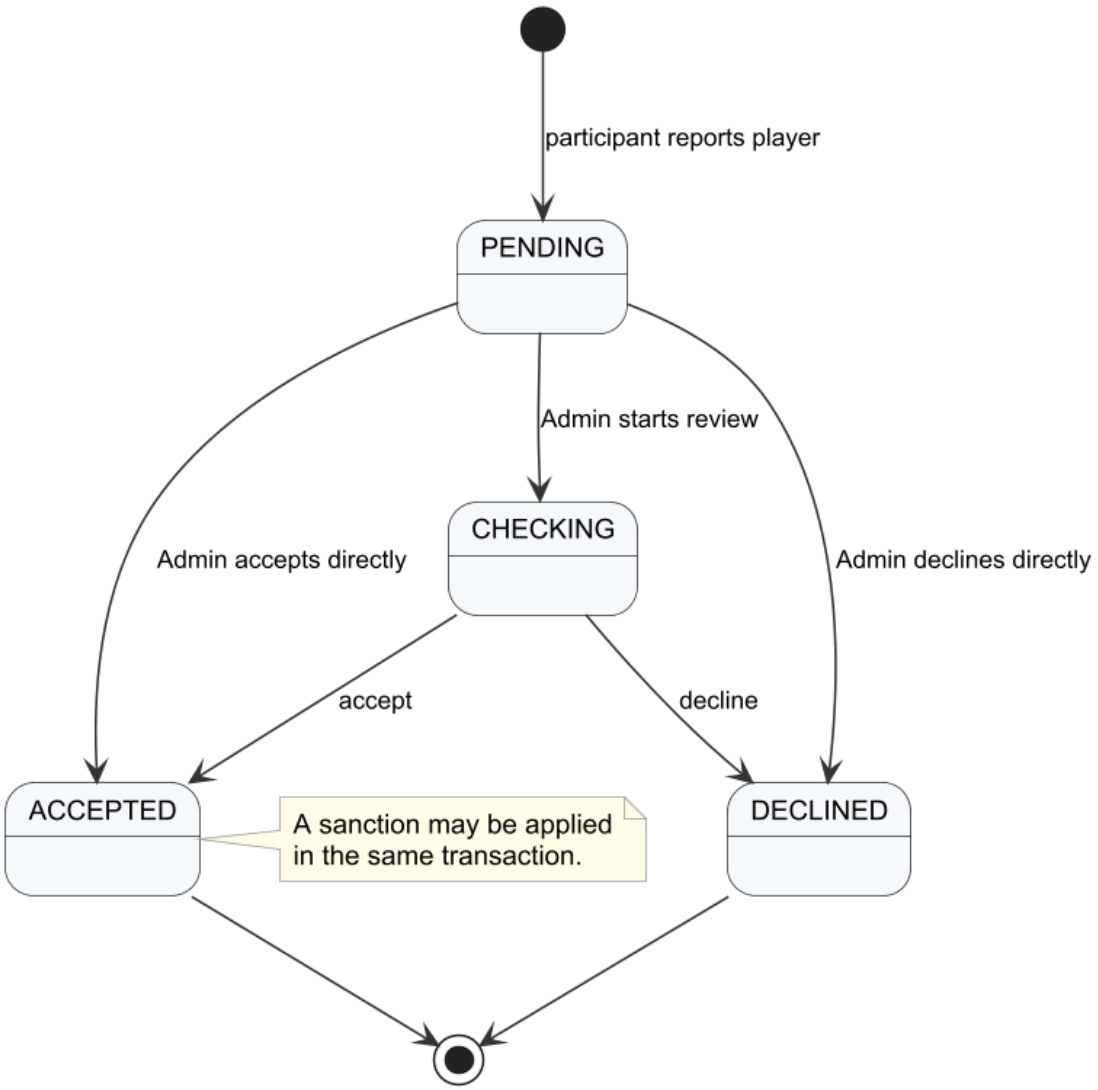


Figure 52. SM-11 Player report

SM-12 Sanction appeal

PENDING, CHECKING, ACCEPTED (sanction revoked) and DECLINED.

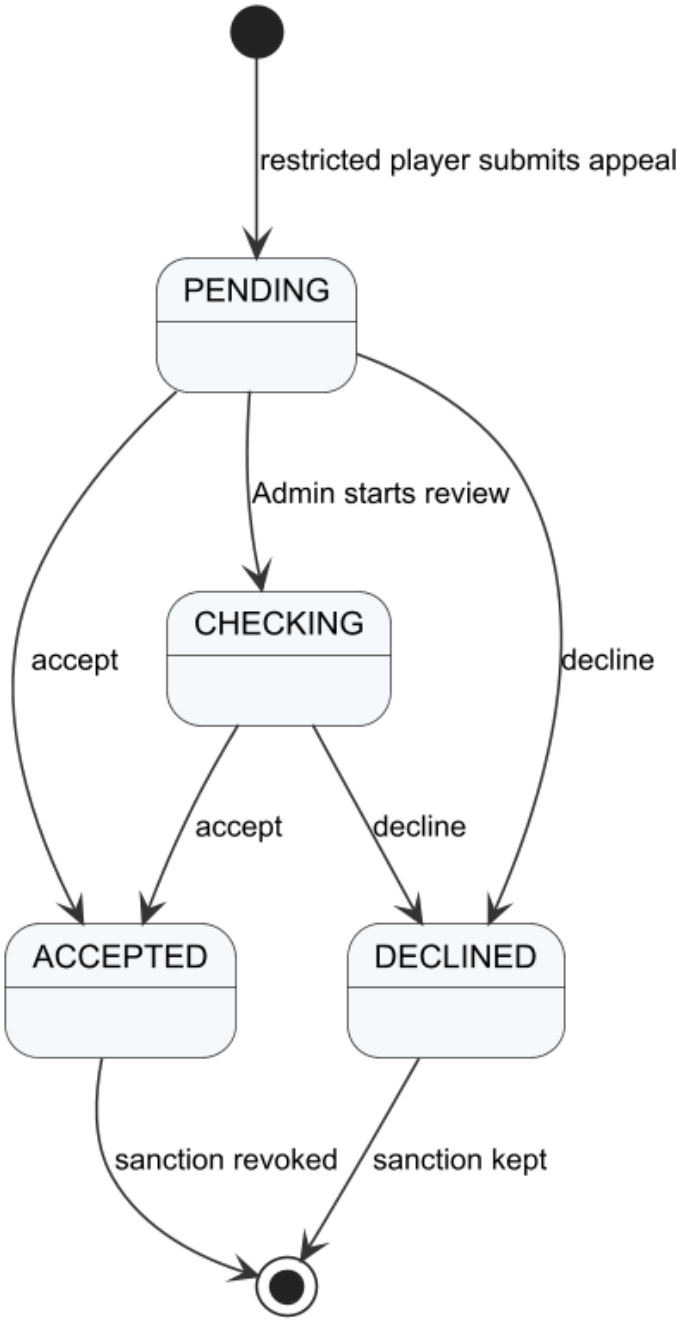


Figure 53. SM-12 Sanction appeal

SM-13 AI analysis job
pending, processing, completed and failed with retries.

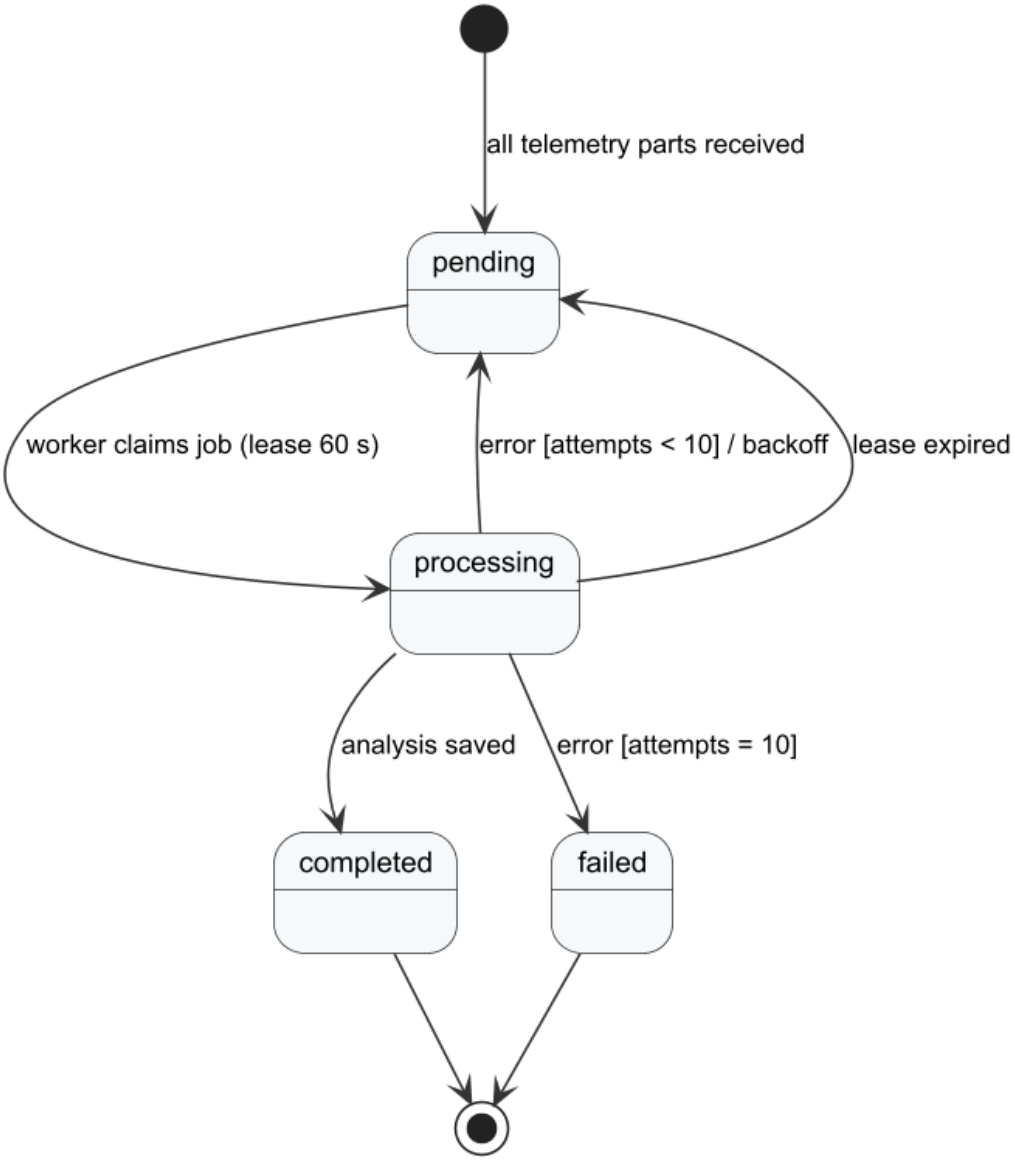


Figure 54. SM-13 AI analysis job

SM-14 Friend request

PENDING, ACCEPTED and DECLINED; blocking removes the friendship.

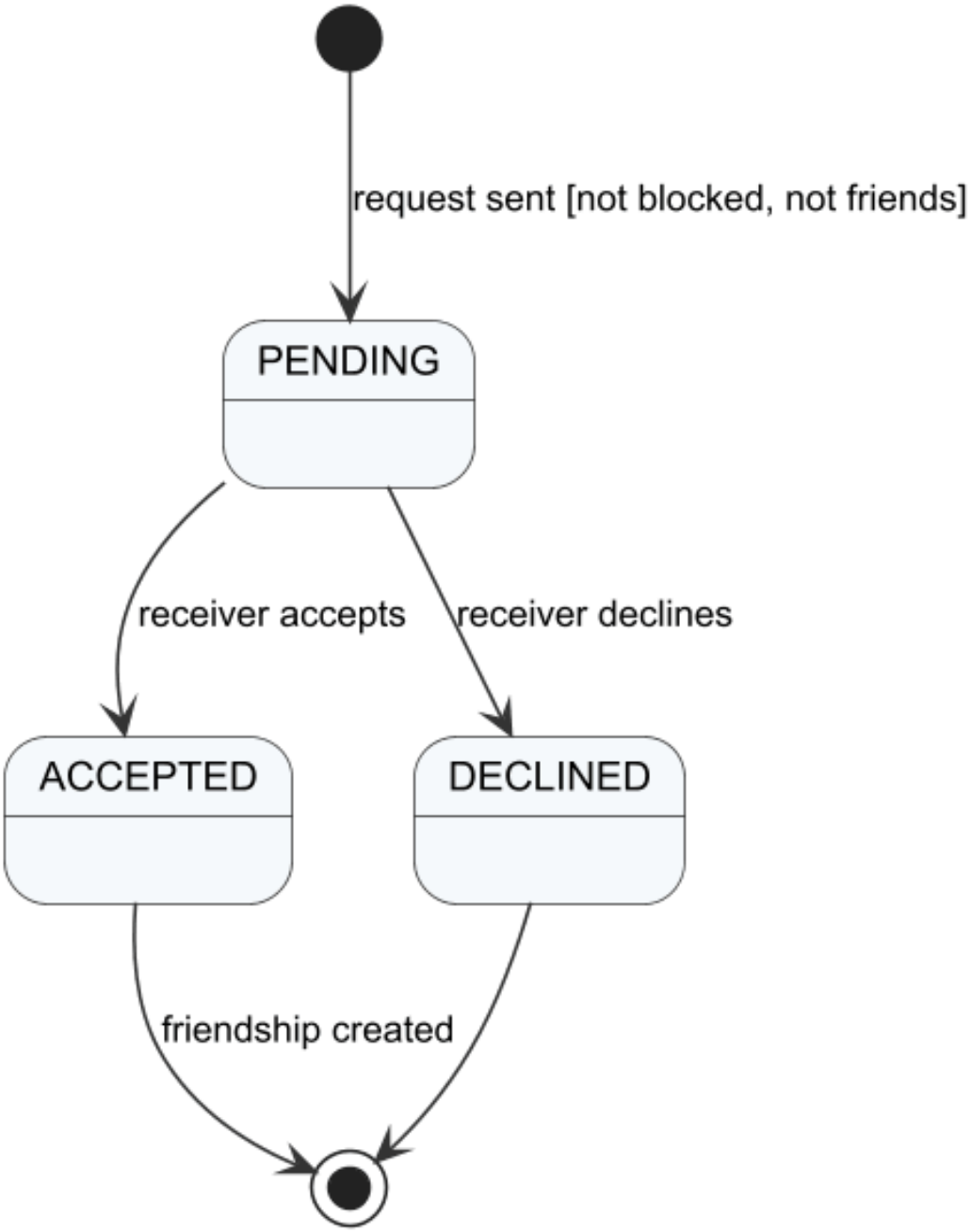


Figure 55. SM-14 Friend request

V. Software Testing Documentation

1. Scope of Testing

The target of test is PickleVerse release 1.0 as specified in Report 3: the ten features FE-01 to FE-10 with their functional requirements FR-01 to FR-33, the business rules BR-01 to BR-51, and the non-functional requirements NFR-01 to NFR-12. Testing covers the Unity game client (Windows build and Unity Editor), the dedicated game server, the Web API with its workers, the AI coaching service and the administration web application deployed on the staging server.

In scope. All screens and non-screen functions of Report 3, the gameplay rules and physics parameters of Report 4 §3.2, online matchmaking and dedicated play, data integrity of rewards and results, role-based access, and the measurable quality attributes NFR-01 to NFR-12.

Out of scope. iOS builds and real-money payment (Report 1, LI-02 and LI-06); load beyond the API capacity target, because the staging deployment has one dedicated-server slot (LI-07); third-party services themselves (Photon Cloud, Gmail SMTP), which are only tested through their integration points; and the tier model's accuracy until enough expert-labelled data exists (NFR-06).

Table 1. Test levels: responsibility, inputs, focus and exit criteria

Level	In charge	Inputs and time	Focus	Acceptance (exit) criteria
Unit	Each module owner; review by Trần Nguyễn Hiếu Thuận	Source code, BR-xx; before every merge	Rules, scoring, Elo, input decoding, economy, offline store, matchmaking policies (Unity EditMode, pytest)	All unit tests pass; every public rule method has at least one normal, abnormal and boundary case
Integration	Nguyễn Văn Thành Đạt	API build, Database.sql seed, PostgreSQL and Redis; each package milestone	API contracts, database writes, dedicated tickets and leases, training sync receipts, role and scope checks	All IntegrationChecks pass on a fresh database; no contract difference between Unity, API and AI service
System	Trần Nguyễn Hiếu Thuận with the feature owners	Staging deployment, test accounts and seed data; Rounds 1–3 (§3.3)	Every FR through the user interface, state transitions, NFR measurements	≥ 90% of system cases passed, all FRs covered by a passed case, no open Critical/High defect
Acceptance	Huỳnh Trung Kiên with the supervisor	Release candidate, demo scenarios W01–W04	End-to-end workflows of the four members	All four workflows pass in front of the supervisor

Constraints and assumptions.

- The staging server runs one dedicated-server container, so online cases that need two simultaneous matches cannot be executed.
- The GitHub Actions pipeline passes since 13/11/2026 (EV-08) and runs the IntegrationChecks, the FastAPI tests and the admin web type check for the parts that a push changes, but it has no Unity test step. The Unity EditMode suite is therefore still run with the Unity CLI on a team PC and the result file is stored with the commit hash; its last run is on main @ 5963501, before the client address changed in be556a1.
- An Android APK was built on 12/11/2026 (EV-10), but no Android phone has been used for testing yet. That APK predates the HTTPS switch, so the Android cases (TC-N03, TC-N10) need a new APK from main @ d9200cc and a reference device in Round 2.
- Tests that need several players (rooms, friends, reports) require two or more accounts and devices at the same time.

- The admin web moved to https://tad-min.io.vn on 13/11/2026 and was unreachable between the deployments of 01:18 and 14:34 that day; the admin web cases (TC-F15 to TC-F18, TC-W04) are run against the new address in Round 2.
- A dedicated test mailbox and seed accounts (for example a player at level 24 or with 9 passed lessons) must be added to Database.sql before Round 2.
- The Photon free plan limits concurrent users; group playtests are scheduled so that the limit is not reached.

2. Test Strategy

2.1 Testing Types

Table 2. Testing types

Type	Objective	Technique	Completion criteria
Functional	Verify every FR and BR	Black-box: equivalence partitions and boundary values from BR-xx; decision tables for scoring and Elo	Each FR has ≥ 1 passed case; all Main cases passed
Gameplay rules and physics	Verify serve, faults, scoring, contact timing and bounce	Automated EditMode tests on the engine-independent rule classes; scripted rallies	All rule tests pass; every fault type has a test
User interface and usability	Verify layout, messages and learnability	Checklist per screen against Report 3 §3.1 and the message list; moderated usability test (NFR-01)	No overlapping or truncated elements; NFR-01 met
Integration / API contract	Verify API, database and service contracts	IntegrationChecks console harness on a fresh database; shared JSON fixtures in contracts/	All checks pass on the release commit
Security	Verify authentication, authorisation and secret handling	Negative API tests with wrong roles and scopes, token expiry, rate limit; secret scan	NFR-09 checklist satisfied; no secret in the repository
Performance and load	Verify NFR-02 to NFR-05	Unity Profiler on reference devices; Fusion RTT statistics; k6 load test on the API	All measured values within the NFR limits
Network resilience	Verify reconnection and interruption	Disconnect a client for 20 s and 60 s; drop the dedicated server heartbeat	Behaviour matches BR-28 and BR-29
Compatibility	Verify target platforms	Install and smoke test on Windows 10/11 and Android 8+ (ARM64); admin web on Chrome and Edge	NFR-10 met
AI quality	Verify coaching findings and the gated model	Holdout evaluation against expert labels	NFR-06 thresholds met, otherwise the model stays disabled
Exploratory playtest and balance	Find defects that scripted cases miss; tune bot difficulty	Time-boxed sessions with a charter; win rate of players of known level against each bot level	Each session logged; balance targets agreed with the supervisor
Regression	Keep fixed defects fixed	Re-run the automated suites and the cases linked to closed defects in every round	No reopened defect

2.2 Test Levels

Table 3. Test types per level

Type of Tests	Unit	Integration	System	Acceptance
Functional	X	X	X	X
Gameplay rules and physics	X		X	X
User interface and usability			X	X
Integration / API contract		X		
Security	X	X	X	
Performance and load		X	X	
Network resilience	X		X	
Compatibility			X	X
AI quality	X		X	
Exploratory playtest and balance			X	X

Type of Tests	Unit	Integration	System	Acceptance
Regression	X	X	X	

2.3 Supporting Tools

Table 4. Supporting tools

Purpose	Tool	Vendor/In-house	Version
Unity unit tests (EditMode, NUnit)	Unity Test Framework	Unity Technologies	1.8.0
Run tests and capture screens from the command line	Unity CLI with the Pipeline package	Unity Technologies	CLI 1.0.0-beta.17, Pipeline 0.9.0-exp.1
Python unit tests	pytest (with httpx)	Open source	8.x
API and database integration checks	IntegrationChecks console harness	In-house	main @ d9200cc (run by the CI pipeline)
Manual API calls	Swagger UI	Swashbuckle (open source)	10.3.1
API load test (NFR-04)	k6	Grafana Labs (open source)	latest stable, planned
Frame-rate measurement	Unity Profiler	Unity Technologies	6000.6.2f1
Network statistics	Photon Fusion statistics and dashboard	Exit Games	Fusion 2.1.3
Test case and result management	Microsoft Excel (Report5 workbooks)	Microsoft	Microsoft 365
Continuous integration	GitHub Actions (plan, checks, build, publish and deploy jobs)	GitHub	Hosted runner ubuntu-24.04; runs 27, 28 and 30 of 13/11/2026 passed every job

3. Test Plan

3.1 Human Resources

Table 5. Human resources

Worker/Doer	Role	Specific Responsibilities/Comments
Trần Nguyễn Hiếu Thuận	Test lead	Owns the test plan and workbooks; runs system rounds; tracks defects; tests the account, onboarding, offline and social screens (F01, F02, F12, F14); owner of workflow W01
Nguyễn Gia Huy	Tester, gameplay owner	Unit tests of rules, physics, training and bot; playtests and balance sessions; runs the Unity EditMode suite; F03, F05, F11, F13; owner of workflow W02, support for W01 and W03
Nguyễn Văn Thành Đạt	Tester, backend and network owner	IntegrationChecks, API security and load tests, online and dedicated cases (F04, F06–F10); owner of workflow W03, support for W02
Huỳnh Trung Kiên	Tester, leader	Admin web cases (F15–F18), acceptance session with the supervisor, release decision; owner of workflow W04
Mr. Lại Đức Hùng	Supervisor	Witnesses the acceptance scenarios and approves the release criteria

3.2 Test Environment

Table 6. Test environment

Purpose	Tool	Provider	Version
Staging server	Ubuntu with Docker Engine and Docker Compose (API, AI service, dedicated server, PostgreSQL, Redis, S3 storage, web) behind a Caddy reverse proxy; API and admin web on https://tad-min.io.vn since 13/11/2026	Team VPS tad-min.io.vn	Ubuntu 24.04; images published by the CI run of main
Game client under test	Unity Editor and Windows build	Unity Technologies	6000.6.2f1

Purpose	Tool	Provider	Version
Test PC	Windows PC with keyboard and mouse	Team member	Windows 11 Home
Android device	Android phone (ARM64) for NFR-03 and NFR-10; APK of EV-10	Team member	Android 8.0 or later (to be recorded in Round 2)
Real-time service	Photon Cloud	Exit Games	Fusion 2.1.3
Database and cache	PostgreSQL, Redis	Docker images postgres:16-alpine and redis:7.4-alpine	16 / 7.4
Admin web browser	Google Chrome, Microsoft Edge	Google, Microsoft	Latest stable

3.3 Test Milestones

Table 7. Test milestones

Milestone Task	Start Date	End Date
Round 1 – automated suites and first system pass on staging	09/11/2026	13/11/2026
Prepare seed accounts, test mailbox and analytics seed	16/11/2026	18/11/2026
Round 2 – all system cases; retest of the fixed defects and fix verification of D-01 to D-21	16/11/2026	27/11/2026
Android install, smoke and frame-rate test on a reference phone (NFR-03, NFR-10)	16/11/2026	27/11/2026
Usability test with 10 students (NFR-01)	23/11/2026	27/11/2026
Load, latency and frame-rate measurements (NFR-02 to NFR-05)	23/11/2026	29/11/2026
Uptime probe for NFR-07 (14 days)	30/11/2026	13/12/2026
Round 3 – regression on the release candidate	30/11/2026	06/12/2026
Acceptance test W01–W04 with the supervisor	07/12/2026	11/12/2026

4. Test Cases

Unit test cases are documented in **Report5_Unit Test.xlsx** as decision tables (conditions, confirmations, type N/A/B and result for each UTCID) for ten core functions, each mapped to the NUnit test that implements it. Integration, system and acceptance test cases are documented in **Report5_Test Report.xlsx**, one sheet per feature group, with ID, description, procedure with concrete data, expected result, pre-conditions, three test rounds and a note that names the evidence or the reason why a case is pending.

The number of cases follows the Review 2 formula: 18 committed features × 3 cases (54) + 4 workflows + 42 state transitions (three per state machine SM-01 to SM-14) + 12 NFRs = 112 cases. Six cases are added: two for the expert review tool (FE-10), which is not in the Review 2 feature list, three for FR-03 to FR-05, which the Review 2 feature F01 did not detail, and one for quick doubles (TC-F07-04), which F07 describes only in general, giving **118 cases**.

Table 8. Test cases by sheet and level

Sheet	Scope	Cases	Unit	Integration	System	Acceptance	Automated
FE01 Account	FE-01 Account and onboarding	6	1	0	5	0	1
FE02 Training	FE-02 Tutorial, training and offline	9	4	0	5	0	4
FE03 Bot	FE-03 Bot practice	3	2	0	1	0	2
FE04 Online	FE-04 Online matches	13	8	0	5	0	8
FE05 Coaching	FE-05 AI coaching	3	1	0	2	0	1
FE06 Progress	FE-06 Progress, rewards and shop	6	4	1	1	0	5
FE07 Social	FE-07 History, social and ranking	6	2	1	3	0	2
FE08-09 Admin	FE-08/FE-09 Moderation and administration	12	1	2	9	0	2
FE10 Expert	FE-10 Expert review	2	0	1	1	0	1
Workflows	End-to-end workflows W01–W04	4	0	0	0	4	0
States	State transitions SM-01 to SM-14	42	11	27	4	0	22
NFR	Non-functional requirements	12	0	0	12	0	0

Sheet	Scope	Cases	Unit	Integration	System	Acceptance	Automated
Total		**118**	**34**	**32**	**48**	**4**	**48**

4.1 Requirement Traceability

Table 9. Traceability from functional requirements to test cases

Requirement	Test cases	Cases	Passed	Coverage status
FR-01	TC-F01-01, TC-F01-03, TC-W01	3	1	Covered
FR-02	TC-F01-01, TC-F01-02	2	1	Covered
FR-03	TC-F01-04	1	0	Not yet covered
FR-04	TC-F01-05	1	0	Not yet covered
FR-05	TC-F01-06	1	1	Covered
FR-06	TC-F02-01, TC-F02-02, TC-W01	3	1	Covered
FR-07	TC-F03-01, TC-F03-03	2	2	Covered
FR-08	TC-F03-01, TC-F03-02, TC-W01	3	2	Covered
FR-09	TC-F14-01, TC-F14-02, TC-F14-03, TC-F06-03, TC-W01	5	2	Covered
FR-10	TC-F11-01	1	1	Covered
FR-11	TC-F02-03	1	1	Covered
FR-12	TC-F05-01, TC-F05-03, TC-F09-02	3	3	Covered
FR-13	TC-F05-01, TC-F05-02, TC-F05-03, TC-W02	4	4	Covered
FR-14	TC-F07-01, TC-F07-02, TC-F07-03, TC-F07-04, TC-F10-02, TC-W03	6	4	Covered
FR-15	TC-F08-01, TC-F08-02, TC-F08-03	3	2	Covered
FR-16	TC-F09-01, TC-F09-02, TC-W03	3	2	Covered
FR-17	TC-F09-03	1	0	Not yet covered
FR-18	TC-F09-01, TC-F10-01, TC-F10-02, TC-F10-03, TC-W02, TC-W03	6	5	Covered
FR-19	TC-F06-01, TC-F06-02, TC-F06-03, TC-W02	4	3	Covered
FR-20	TC-F04-03	1	1	Covered
FR-21	TC-F13-01, TC-F13-02, TC-F13-03	3	3	Covered
FR-22	TC-F04-01	1	1	Covered
FR-23	TC-F04-02	1	1	Covered
FR-24	TC-F11-02, TC-F11-03	2	1	Covered
FR-25	TC-F12-01, TC-F12-02	2	1	Covered
FR-26	TC-F12-03	1	1	Covered
FR-27	TC-F17-01, TC-W04	2	0	Not yet covered
FR-28	TC-F17-03, TC-W04	2	0	Not yet covered
FR-29	TC-F15-01, TC-F15-02, TC-F15-03	3	0	Not yet covered
FR-30	TC-F17-01, TC-W04	2	0	Not yet covered
FR-31	TC-F16-01, TC-F16-02, TC-F16-03, TC-W04	4	1	Covered
FR-32	TC-F17-02, TC-F18-01, TC-F18-02, TC-F18-03	4	1	Covered
FR-33	TC-X01-01, TC-X01-02	2	1	Covered

26 of the 33 functional requirements are covered by at least one passed case after Round 1 and the CI run of 13/11/2026 (EV-08); FR-03, FR-04, FR-17, FR-27, FR-28, FR-29, FR-30 have no passed case yet. NFR-01 to NFR-12 are traced one-to-one to TC-N01 to TC-N12: 1 of the 12 have been executed (TC-N09 failed), none has passed.

5. Test Reports

5.1 Evidence

Table 10. Test evidence

ID	Run	Date	Build / commit	Result	Location
EV-01	Unity EditMode suite through the Unity CLI (Pipeline test run)	11/11/2026	main @ 7f8ee32	515 tests, 515 passed, 0 failed, 6.19 s	Doc/Project_report/Report/evidence/editmode_results_20261111.json
EV-02	Playtest on the staging server (offline training, signed-in menu, bot match, AI coaching, ranked 1v1 with bot fallback)	10/11/2026	main @ 82f6a7a client; staging images of 06/11/2026	See Report5_Test Report.xlsx	screenshots/01–28, evidence/playtest_log_20261110.md
EV-03	UI check of the register, verification, password-reset screens and the tutorial	11/11/2026	main @ 7f8ee32	See TC-F01-03, TC-F02-01	screenshots/29–34
EV-04	Match history of the team test accounts (two ranked 1v1 attempts)	09/11/2026	staging	Both interrupted	screenshots/12_history.png
EV-05	Team-recorded IntegrationChecks on disposable PostgreSQL databases (12 areas)	06/11/2026	df5f671	Reported as passed; logs not in the repository, so not counted	Review3_Evidence.md (file names and SHA-256)
EV-06	Unity EditMode suite through the Unity CLI after the UI and camera fixes (adds the camera framing test)	11/11/2026	main @ 5963501	516 tests, 516 passed, 0 failed, 5.96 s	Doc/Project_report/Report/evidence/editmode_results_20261111_5963501.json
EV-07	Unity Play Mode check of the fixed defects D-03, D-04, D-05, D-07 and D-19 (menu, mode selection, bot setup, bot match, training HUD)	11/11/2026	main @ 5963501	All five behave as expected; to be retested on staging in Round 2	screenshots/05, 11, 16, 17, 19
EV-08	GitHub Actions run 30 of the workflow "Build, publish and deploy PickleVerse": Backend integration checks (12 IntegrationChecks groups on PostgreSQL 16 and Redis 7.4), FastAPI tests, admin web type check, Unity Linux dedicated-server build, publish with smoke test, deployment over SSH. Runs 27 and 28 passed on main @ 1dfefe9	13/11/2026 14:20–14:34	main @ d9200cc	All nine jobs and all their steps succeeded (14 min)	evidence/ci_runs_20261113.json; github.com/Tad-Min/PickleVerse/actions/runs/43385071385
EV-09	Transport and secret check: HTTPS, redirects and open ports of the staging host; credential values in the tracked settings files. Run at 11:30 and again at 16:21 after commit be556a1 was deployed	13/11/2026	staging; main @ 1dfefe9 and d9200cc	16:21: API and admin web only over HTTPS, port 8080 closed, settings files clean; old values remain in the Git history	evidence/staging_check_20261113.md
EV-10	Android build with Unity batch mode and inspection of the APK manifest (aapt2, apksigner)	12/11/2026	main @ 5963501	APK built (190.5 MB, IL2CPP, ARM64, API 26–36); not installed on a device; cleartext HTTP forbidden (D-21, fixed in be556a1)	evidence/android_build_20261112.md

5.2 Unit Test Results

Table 11. Unity EditMode results by assembly (EV-06)

Test assembly	Tests	Passed	Failed
Audio.Tests	23	23	0
FrontEnd.Tests	4	4	0
HybridBot.Tests	15	15	0
LocalMatch.Tests	157	157	0
Multiplayer.Tests	101	101	0
Progression.Tests	125	125	0

Test assembly	Tests	Passed	Failed
Replays.Tests	16	16	0
Services.Tests	31	31	0
Shop.Tests	33	33	0
Social.Tests	11	11	0
Total	**516**	**516**	**0**

The table above shows the latest run (EV-06, 516 tests); the Round 1 run EV-01 on main @ 7f8ee32 had 515 tests, all passed, and EV-06 adds the camera framing regression test of D-03. The ten decision tables of Report5_Unit Test.xlsx contain 66 unit test cases (normal, abnormal and boundary), all passed in EV-01 and EV-06. The other automated suites run in the GitHub Actions pipeline. In run 30 on main @ d9200cc (EV-08) the jobs "Backend integration checks" (twelve IntegrationChecks groups: client-security, api-contract, schema, status-enums, content, training-sync, progress-reset, moderation, analytics, expert-review, dedicated and hybrid, each on a disposable PostgreSQL 16 database with Redis 7.4), "FastAPI tests" (pytest of the AI service) and "Admin web type check" all succeeded. A check group stops with an error at its first failed assertion, so a successful step means that every assertion passed. The backend integration checks have passed in every run since run 20 of 11/11/2026 (twelve runs); the FastAPI tests failed once, in run 29, and passed again in run 30. The earlier team runs of 06/11/2026 (EV-05) stay uncounted because their logs are not in the repository. The Unity EditMode suite is not part of the pipeline yet, and the job logs are kept by GitHub, not in the repository; the step results are stored in evidence/ci_runs_20261113.json. Code coverage has not been measured yet; the Unity Code Coverage package will be enabled in Round 2.

5.3 System and Integration Test Results

Table 12. Results by module after Round 1 and the CI run of 13/11/2026

Module	Cases	Passed	Failed	Pending
FE-01 Account and onboarding	6	3	0	3
FE-02 Tutorial, training and offline	9	7	0	2
FE-03 Bot practice	3	3	0	0
FE-04 Online matches	13	8	2	3
FE-05 AI coaching	3	2	1	0
FE-06 Progress, rewards and shop	6	6	0	0
FE-07 History, social and ranking	6	4	0	2
FE-08/FE-09 Moderation and administration	12	2	0	10
FE-10 Expert review	2	1	0	1
End-to-end workflows W01–W04	4	2	0	2
State transitions SM-01 to SM-14	42	26	0	16
Non-functional requirements	12	0	1	11
Total	**118**	**64**	**4**	**50**

Table 13. Release criteria on 13/11/2026

Metric	Value	Target	Status
Test coverage (executed / total)	57.6% (68 / 118)	100% before release	Not met
Pass rate of executed cases	94.1% (64 / 68)	≥ 90%	Met for executed cases
Test successful coverage (passed / total)	54.2%	≥ 90%	Not met
Automated cases	40.7% (48 / 118)	≥ 50%, run in CI	Not met (14 of them run in CI)
Functional requirements covered by a passed case	26 / 33	33 / 33	Not met
Open Critical / High defects	0 / 2 (D-09, D-17)	0 / 0	Not met
Non-functional requirements measured and met	1 measured, 0 met	12 / 12	Not met

5.4 Defects

Table 14. Defects found in Round 1 and in the checks of 11–13/11/2026

ID	Summary	Type	Severity	Found by	Owner	Status
D-01	Offline banner overlaps the titles of Progress, the training HUD and the lesson list	UI	Low	TC-F14-01	Thuận	Open
D-02	Offline: the analysis area stays at "Đang phân tích…" forever	Functional	Medium	TC-F06-03	Đạt	Open
D-03	Key-hint box (shot cue) covers the player character in training, bot and online matches	UI	Low	TC-F03-01	Huy	Fixed (5963501)
D-04	"Quyết định & Khiếu nại" button overlaps the PLAY button and the profile panel	UI	Medium	TC-F01-02	Thuận	Fixed (31a7c2e)
D-05	Fourth bot difficulty shows the technical name "BhTree"	UI text	Low	TC-F05-01	Huy	Fixed (d2fc5be)
D-06	Bot card says "Không cần kết nối mạng" but bot matches are disabled offline	UI text	Low	TC-F14-01	Thuận	Open
D-07	Male/female choice is not applied in bot match, training and tutorial (always the female character)	Functional	Medium	TC-F05-01	Huy	Fixed (6d1ed37)
D-08	New players start at Elo 100 while the admin reset-Elo default is 2000	Business rule	Medium	TC-F10-01	Đạt	Open
D-09	Ranked 1v1 between two humans ended "Gián đoạn" twice, without replay	Functional	High	TC-F09-01	Đạt	Open
D-10	"Xem phân tích" button on the match result is oversized and covers the centre of the screen	UI	Low	TC-F05-01	Thuận	Open
D-11	Coaching popup mixes session-wide (2/86) and per-technique (1/16) figures	Usability	Low	TC-F06-01	Đạt	Open
D-12	Elite Token has no earning source, so GEMSTONE can only be granted at level 25	Requirement	Medium	TC-F04-01	Kiên	Open
D-13	Hard bot is beaten 2–0 by the auto-player with short rallies; balance not validated	Balance	Medium	TC-F05-01	Huy	Open
D-14	After a normal ranked result the lobby shows "Mất kết nối Photon."	UI message	Low	TC-W03	Đạt	Open
D-15	Ranked bot offered about 160 s after the search started instead of 180 s	Functional	Low	TC-F07-01	Đạt	Open
D-16	Sync-UnityTests.ps1 fails to parse in Windows PowerShell 5.1 (UTF-8 file without BOM)	Tooling	Medium	Unit test run	Thuận	Open
D-17	Development credentials committed in appsettings.json and a fallback JWT key in Program.cs	Security	High	TC-N09	Đạt	Partly fixed
D-18	Staging API and admin web are served over HTTP instead of HTTPS	Security	High	TC-N09	Đạt	Fixed (be556a1)
D-19	Settings tile is pushed below the screen after the Decisions and Appeals tile was added to the menu	UI	Low	TC-F01-02	Huy	Fixed (9dd994c)
D-20	Level 25 needs 60,000 XP in total (n × 200 per level) while a match gives 5–10 XP, so the GEMSTONE paddle (BR-37) is effectively unreachable	Requirement	Medium	TC-F04-03	Kiên	Accepted
D-21	The Android APK declares usesCleartextTraffic=false (manifest written by the App UI package) while BackendApiSettings used http://tad-min.io.vn:8080, so API calls from an Android phone are expected to be refused	Configuration	High	TC-N10	Huy	Fixed (be556a1)

Round 1 (09–11/11/2026) and the follow-up checks of 11–13/11/2026 found 21 defects: 4 High, 8 Medium and 9 Low. 7 of them (D-03, D-04, D-05, D-07, D-18, D-19, D-21) are fixed in code; the fix commit is shown in the Status column. The five user-interface fixes were checked in Unity Play Mode (EV-07), the HTTPS switch of D-18 was verified on staging (EV-09), and D-21 is fixed by the same commit but not yet verified with a new Android build. Fixed defects stay in the Fixed state until the Round 2 retest. 13 defects still need work, including 2 High defects (D-09, D-17).

- **D-17 – Partly fixed (13/11/2026).** Commit be556a1 emptied the credential values in appsettings.json and removed the fallback JWT key from Program.cs (the API now refuses to start without a key of at least 32 bytes). The old values are still readable in the Git history, so they must be rotated on staging and the rotation confirmed.

- **D-20 – Accepted (13/11/2026).** Team decision of 13/11/2026: accepted as designed, the XP curve is not changed; closed without a fix.

Severity follows Report 3 NFR-08: Critical means loss of player data or inability to sign in or play; High means a main workflow or a security requirement fails; Medium means a requirement is partly wrong or a rule is inconsistent; Low means a cosmetic or wording problem. The defect life cycle (Open → Fixed → Verified → Closed) is tracked in PickleVerse_Project_Tracking.xlsx.

5.5 Analysis and Next Actions

- **Engine-level quality is good.** All 516 EditMode tests pass (EV-06), and every system case that could be executed for the single-player features (training, bot match, coaching, progress screens) passed, apart from the offline analysis status (D-02).
- **Online play between humans is the main risk.** The only human-versus-human ranked attempts (EV-04) were interrupted (D-09), which also blocks the reconnection, report and appeal cases. Reproducing D-09 with server logs is the first task of Round 2 (owner: Đạt, by 18/11/2026).
- **Security improved on 13/11/2026 but is not closed.** Commit be556a1 moved the game client, the API and the admin web to https://tad-min.io.vn, closed port 8080 and removed the credentials from the settings file and the fallback JWT key from the code; EV-09 confirms the transport part (D-18 fixed). The removed values are still readable in the Git history, so D-17 stays partly fixed and TC-N09 failed until the team rotates them on staging and records the rotation; the rest of the NFR-09 checklist and a secret scan in the pipeline are planned for Round 2 (owner: Đạt).
- **Coverage grew through the pipeline, and is now limited by test data.** The CI run EV-08 moved 14 cases to Passed and three more requirements (FR-20, FR-32, FR-33) to Covered. 50 cases are still pending, mainly because seed accounts, a test mailbox and a second player were not available; 16 of them are partly covered by a CI check and still need the missing step named in their note. The seed data will be added to Database.sql before Round 2 (owner: Thuận, by 18/11/2026).
- **Automation is below the 50% target.** 48 of 118 cases are automated: 34 by the Unity EditMode suite, which is still run by hand, and 14 by the IntegrationChecks of the CI validate job. Next: add the EditMode suite as a GameCI unity-test-runner step that reuses the licence secrets of the dedicated-server build, and add the check groups that the job does not run yet (admin authorisation, shared replays, social, password recovery) so that the remaining state-transition cases can be linked to them.
- **7 defects are fixed but not yet closed.** D-03, D-04, D-05, D-07 and D-19 were fixed on 11/11/2026 and checked in Play Mode; D-18 and D-21 were fixed by be556a1 on 13/11/2026. All are closed only after the Round 2 retest on the staging build. D-20 (60,000 XP for the GEMSTONE level) was accepted as designed on 13/11/2026 and is closed without a fix; D-12 (no Elite Token source) is still open.
- **Android is built but untested.** An APK was built from main @ 5963501 on 12/11/2026 (EV-10) and its inspection found D-21; no Android device has run the game yet. A new APK from main @ d9200cc, which uses the HTTPS address, is needed for TC-N03 and TC-N10 in Round 2.
- **The delivery pipeline works.** Of the 32 runs, 29 did not pass: two configuration failures on 05/11, 18 runs that did not start while the GitHub account was locked for a billing issue from 06/11 to 11/11, six dedicated-server build failures, one failed FastAPI test and two failed deployment steps (one of them a manual re-run on 13/11 at 16:07). Runs 27, 28 and 30 of 13/11/2026 completed checks, build, publication and deployment in about 15 minutes each, and run 30 deployed the release that is on staging now.

VI. Release Package & User Guides

1. Deliverable Package

Table 1 lists the items of the release package of PickleVerse 1.0. Source code and documents are versioned in the Git repository https://github.com/Tad-Min/PickleVerse; this package corresponds to the commit main @ d9200cc (13/11/2026) for the software and to the documents of 13/11/2026 (Reports 1–5 v1.3, Report 6 v1.2 and Report 7 assembled from them). Since main @ 5963501 (11/11/2026) the commits repaired the delivery pipeline, moved the client and the server to HTTPS and removed the credentials from the settings file; the gameplay code is unchanged. The final release will be tagged v1.0 at milestone M08.

Table 1. Deliverable package of PickleVerse 1.0

No.	Deliverable Item	Description
1	Project Schedule / Tracking	Doc/Project_report/Report/PickleVerse_Project_Schedule.xml (MS Project XML: 82 tasks, 4 resources, 61 assignments, % complete as of 13/11/2026) and PickleVerse_Project_Tracking.xlsx (sheets WBS, Backlog, Issues, Defects, Q&A)
2	Project Backlog	Sheet Backlog of the tracking workbook: 21 product and iteration backlog items for Iterations 4 to 6 with priority, owner, estimate and status
3	Source Codes	Unity client and dedicated server: My project/ (Unity 6000.6.2f1, 13 runtime assemblies); Web API: src/Backend (ASP.NET Core .NET 10, 4 projects); AI coaching service: src/AiService (Python 3.12, FastAPI); admin web: src/Frontend (React 18, TypeScript, Vite); tests: src/Testing; deployment: deploy/, docker-compose*.yml, .github/workflows/dockerhub.yml
4	Database Script(s)	src/Database.sql – single script with the schema (51 tables, 71 foreign keys) and seed data (roles, currencies, 15 lessons, 12 techniques, 3 paddles, 17 missions, 10 achievements, 10 ranked bot accounts) executed in one transaction
5	Final Report Document	Report7_Final Project Report.docx, which consolidates Reports 1–6; the individual Reports 1–6 (.docx) and their PDF exports in Doc/Project_report/Report/pdf
6	Test Cases Document	Report5_Test Documentation.docx, Report5_Test Report.xlsx (118 cases with the results of Round 1 and of the CI run of 13/11/2026) and Report5_Unit Test.xlsx (10 functions, 66 decision-table cases); evidence files in Doc/Project_report/Report/evidence
7	Defects List	Sheet Defects of the tracking workbook (21 defects with severity, owner and status: 7 fixed and awaiting retest, 1 accepted as designed and 13 open, one of them partly fixed); also listed in the Test Report workbook
8	Issues List	Sheet Issues of the tracking workbook (15 issues with priority, owner, open and close dates; 4 closed)
9	Slide	Defense presentation – prepared for milestone M10 (Week 15, 14–20/12/2026)
10	Server images	Docker Hub repository tadminm/pickelverse with four images per release: api-, ai-, server- (Unity Linux dedicated server) and web-<release>. Since 13/11/2026 the pipeline builds the images that a push changes, publishes them with the tag <commit>-<run>-<attempt> and deploys all four by digest; the four earlier staging releases (08–10/11/2026) were built by hand and tagged <component>-manual-<yyyymmdd-hhmmss>
11	Game client builds	Windows 64-bit build (PickleVerse.exe, Mono) and Android build (PickleVerse.apk, IL2CPP, ARM64) produced from the Unity project for the release; the scene list and settings are defined in My project/ProjectSettings. An Android APK was built on 12/11/2026 from main @ 5963501 (190.5 MB, signed with the debug key); it predates the HTTPS switch and has not been tested on a physical device, so the release APK has to be built again from main @ d9200cc (Report 1, LI-11; Report 5, D-21)
12	Design sources	PlantUML sources and rendered diagrams (Doc/Project_report/Report/diagrams), the editable draw.io file PickleVerse_UseCase_ConceptualERD.drawio and the report build scripts (Doc/Project_report/Report/build)

2. Installation Guides

2.1 System Requirements

Table 2. System requirements

Component	Requirement
Player PC	Windows 10 or 11, 64-bit; DirectX 11 graphics card; recommended Intel Core i5 (10th generation) or equivalent, GTX 1650 or equivalent, 16 GB RAM (reference device of NFR-03); keyboard and mouse; broadband Internet for online features; outbound UDP allowed for Photon and port 3669

Component	Requirement
Player phone	Android 8.0 (API level 26) or later, ARM64 processor; touch screen; Wi-Fi or 4G connection for online features
Administrator	Latest Google Chrome or Microsoft Edge on a desktop screen of at least 1366 × 768; an account with the Admin role
Server host	Linux amd64 (Ubuntu 24.04 used for staging) with Docker Engine and Docker Compose v2 that supports up `--wait-timeout`; recommended 4 vCPU and 8 GB RAM; the game-server container is limited to 2 CPUs and 2 GB RAM by default; open ports 80/tcp (admin web), 8080/tcp (Web API) and 3669/udp (dedicated game server)
External services	Photon Cloud application for Fusion 2.1.3 (App ID in the Fusion settings of the Unity project); a Gmail account with an app password for SMTP (STARTTLS, port 587)
Developer PC	Unity Hub with Unity 6000.6.2f1 (Windows, Linux Dedicated Server and Android build support); .NET 10 SDK; Python 3.12; Node.js 22 with npm; Docker Desktop; Git; Java 17 only for rendering the report diagrams

2.2 Installation Instruction

2.2.1 Install the game on Windows

1. Copy the Windows release folder (PickleVerse_Windows) to the PC, for example to C:\Games\PickleVerse. The folder contains PickleVerse.exe, PickleVerse_Data and the Unity runtime files; keep them together.
2. Run PickleVerse.exe. When Windows Defender Firewall asks, allow the game on private networks so that UDP traffic to Photon and to the game server is possible.
3. The Login screen opens (Figure 1). The first sign-in needs an Internet connection; afterwards training can also be started offline from the same PC.

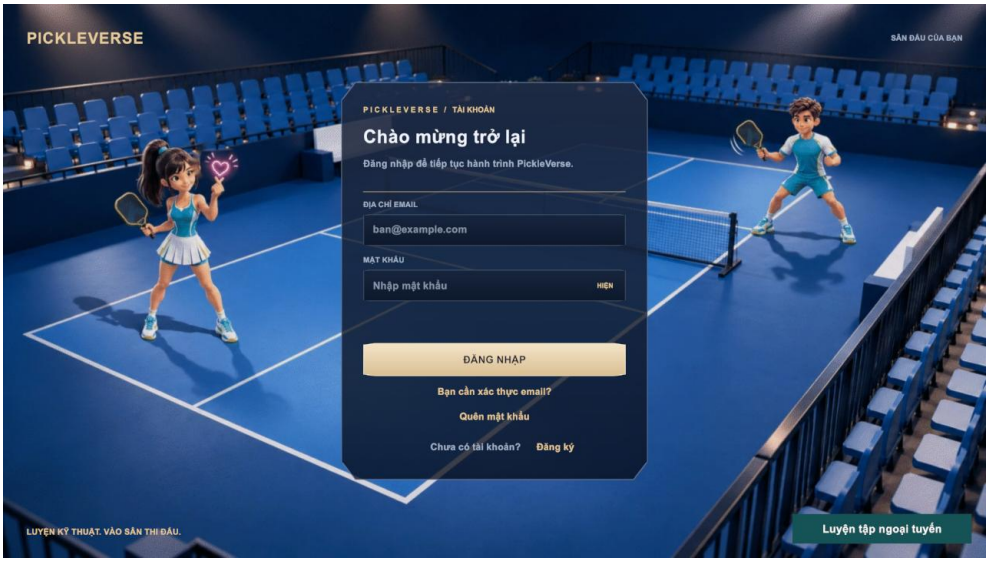


Figure 1. First screen after installation – Login

2.2.2 Install the game on Android

1. Copy PickleVerse.apk to the phone or download it from the release location shared by the team.
2. In Settings → Security (or Apps → Special access), allow installing apps from the file manager or browser used to open the APK.
3. Open the APK and select Install. Start PickleVerse from the app drawer; the game runs in landscape mode with touch controls (joystick and shot buttons).

Note: on 13/11/2026 the Android build has not yet been installed and tested on a physical phone; these steps are the standard Android procedure and will be confirmed in test Round 2 (TC-N10). The APK built on 12/11/2026 still contains the former http:// address and forbids cleartext HTTP, so it

cannot sign in; use an APK built from main @ d9200cc or later, which calls https://tad-min.io.vn (Report 5, D-21). The offline training mode does not need the server.

2.2.3 Deploy the server (CI/CD pipeline and manual release)

The deployment is automatic: every push to the main branch runs the GitHub Actions workflow "Build, publish and deploy PickleVerse" (.github/workflows/dockerhub.yml) with the stages below. **Current status:** the pipeline passes since 13/11/2026 (runs 27, 28 and 30), and run 30 deployed the staging release that is running now. The 29 other runs did not pass: configuration failures on 05/11/2026, the billing lock of the GitHub account from 06/11 to 11/11, failures of the dedicated-server build on 11–12/11, one failed FastAPI test and two failed deployment steps. A push that changes only documents builds and deploys nothing, and a commit whose message contains [skip ci] does not start the workflow at all. If the pipeline is unavailable, the four images can still be built on a developer PC, pushed to Docker Hub with the tag <component>-manual-<yyyymmdd-hhmmss> and started on the host by hand with the same .env file, as was done on 08–10/11/2026.

1. **Plan** – reads the release that is deployed on the host and compares it with the pushed commit to select the checks and the images that the changed files need (deploy/ci/plan-release.py).
2. **Checks** – as selected by the plan: the twelve IntegrationChecks groups against PostgreSQL 16 and Redis 7.4 service containers, the FastAPI tests and the type check of the admin web. The job "Validate" passes only when every selected check passed.
3. **Build Unity Linux Dedicated Server** – builds the dedicated server with GameCI (unity-builder) using the Unity licence stored in the repository secrets.
4. **Publish** – removes the credentials from the API settings files, builds and pushes the changed images among api, ai, server and web to Docker Hub tadminm/pickelverse with the tag <component>-<commit>-<run>-<attempt>, then starts the four image digests with Docker Compose on the runner and checks the health endpoints (smoke test).
5. **Deploy** – over SSH (deploy/ci/ssh-deploy.sh): locks the release on the host, takes a PostgreSQL dump into .deploy/backups and checks it with pg_restore, copies docker-compose.linux.yml and src/Database.sql to the deployment root, writes the four image digests into the host .env file, runs `docker compose up -d --wait` and records the deployed release.

Before the first deployment the administrator of the host prepares the deployment root and creates the host `.env` file from deploy/linux/runtime.env.example and installs Caddy with deploy/linux/Caddyfile, which serves the public HTTPS address, forwards /api and /health/ready to the API and every other path to the admin web. The variables are listed in Table 3; secret values are never committed to the repository. The workflow itself needs the repository secrets SSH_HOST, SSH_PORT, SSH_PASSWORD, SSH_KNOWN_HOSTS, DOCKERHUB_TOKEN, UNITY_EMAIL, UNITY_PASSWORD and UNITY_LICENSE (or UNITY_SERIAL), and the variables PKV_DEPLOY_ROOT (or PKV_DATA_ROOT), SSH_USER and DOCKERHUB_USERNAME of the GitHub environment "development".

Table 3. Server environment variables (host .env)

Variable	Purpose
PKV_API_IMAGE, PKV_AI_IMAGE, PKV_WEB_IMAGE, PKV_GAME_SERVER_IMAGE	Image references by digest; the deploy job writes them
PKV_DATA_ROOT	Absolute path of the host data directories (PostgresData, RedisData, S3Data, AiService/models, GameServerData)
POSTGRES_USER, POSTGRES_PASSWORD	User name and password of the PostgreSQL database
JWT_KEY	Signing key of the access tokens (at least 32 random characters)
SMTP_USERNAME, SMTP_APP_PASSWORD, SMTP_FROM_EMAIL	Gmail account, app password and sender address for OTP e-mails
S3_ACCESS_KEY, S3_SECRET_KEY	Credentials of the S3-compatible replay storage
AI_SERVICE_KEY	Shared key between the Web API worker and the AI coaching service (X-AI-Key)

Variable	Purpose
PKV_GAME_SERVER_KEY	Service key used by the dedicated server to call the Web API
API_PORT, WEB_PORT	Local ports of the API (8080) and the admin web (3000); both are bound to 127.0.0.1 and published by Caddy
PGADMIN_EMAIL, PGADMIN_PASSWORD, PGADMIN_BIND, PGADMIN_PORT	Account and local address of the pgAdmin container (127.0.0.1:15050)
POSTGRES_PORT, POSTGRES_MAX_POOL_SIZE, REDIS_PORT, S3_PORT	Internal ports and the database connection pool size
PKV_GAME_CPUS, PKV_GAME_MEMORY	Resource limits of the dedicated game server (default 2 CPUs, 2 GB)
AI_COACHING_CALIBRATION_SHA256	Hash of the approved coaching calibration file; the service refuses a different file

2.2.4 Initialise the database and create the first administrator

1. On first start, the PostgreSQL container runs src/Database.sql on the empty data volume. It creates all 51 tables and the seed data in one transaction. An existing volume is never re-initialised; use a new volume name to start from an empty database in development.
2. To use an existing PostgreSQL server instead, create an empty database and run `psql -v ON_ERROR_STOP=1 -d <database> -f src/Database.sql` once.
3. The script creates no administrator account. Register a normal account in the game, verify its e-mail, then grant it the Admin role once with SQL: `UPDATE Users SET RoleId = (SELECT RoleId FROM Roles WHERE RoleName = 'Admin') WHERE Email = '<e-mail>';`. Further administrators are assigned from the Users page of the admin web (FR-29).
4. Check the deployment: `GET https://<host>/health/ready` must return HTTP 200 with the status ready, and https://<host>/ must open the admin web and accept the administrator account. Ports 8080 and 3000 are bound to 127.0.0.1 and must not answer from another computer.

2.2.5 Set up a local development environment

1. Clone the repository and copy `.env.example` to `.env`; fill JWT_KEY, the SMTP values and AI_SERVICE_KEY.
2. Start the back-end services with `docker compose up -d --build`. The development stack exposes PostgreSQL on 127.0.0.1:15432, pgAdmin on 15050, S3 storage on 19000, Redis on 16379 and the Web API on 8080; the AI service is reachable only inside the Docker network.
3. Admin web: in src/Frontend run `npm install` and `npm run dev`, then open http://localhost:5173.
4. Unity: open the folder My project with Unity 6000.6.2f1 from Unity Hub. Select Assets/Resources/BackendApiSettings.asset and set **Base Url** to the API address (http://localhost:8080 for local development; the committed setting is https://tad-min.io.vn since 13/11/2026). Open the scene Assets/Scenes/UI/Login.unity and press Play.
5. Tests: run the Unity EditMode tests in Window → General → Test Runner (or `unity command test run --mode editor` with the Unity CLI), `python -m pytest -q` in src/AiService, and `dotnet run --project src/Testing/Backend` for the integration checks.

3. User Manual

3.1 Overview

PickleVerse has two applications. Players use the game client on Windows or Android to learn the rules, train techniques, play bot and online matches, receive AI coaching and follow their progress. Administrators use the admin web application to manage accounts, training content, matches, moderation, analytics and expert labelling. Figure 2 shows how a player moves between the screens of the game; the four sections that follow describe the four end-to-end workflows committed at Review 2, each owned by one team member.

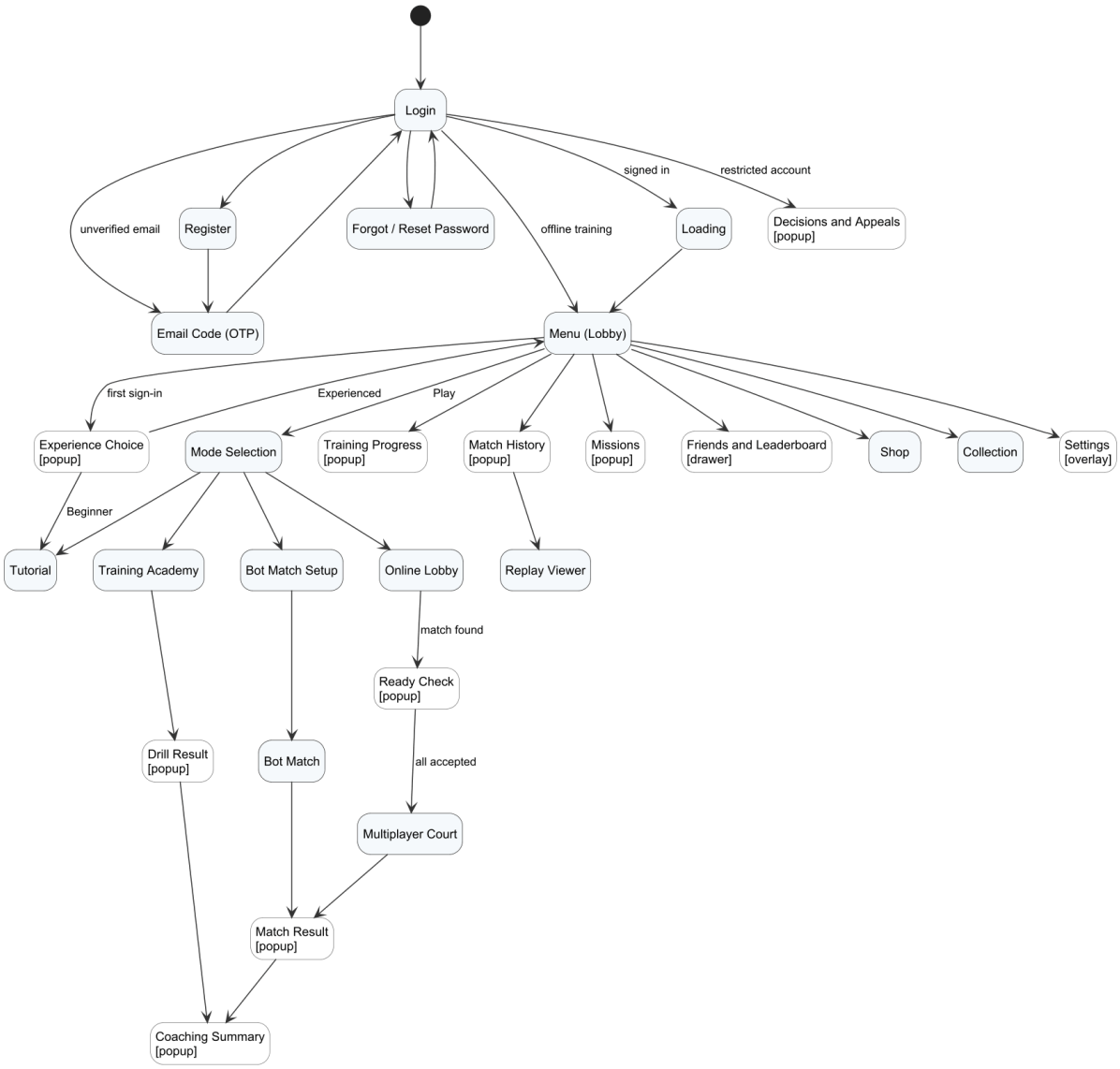


Figure 2. Screens of the game client and the navigation between them

Table 4. Workflows of the user manual

Workflow	Purpose	Main screens	Owner (BM04)
W01 – Learn and sync a drill	Create an account, learn the rules and practise a drill online or offline	Login, Register, Tutorial, Training Academy, Drill Result	Trần Nguyễn Hiếu Thuận
W02 – Play a bot and get a tip	Play a singles match against a bot and read the AI coaching	Mode Selection, Bot Match Setup, Bot Match, Match Result, Coaching	Nguyễn Gia Huy
W03 – Play online and save the result	Play ranked singles, quick doubles or a private room with friends on the dedicated server; the result is saved once	Online Lobby, Ready Check, Multiplayer Court, History	Nguyễn Văn Thành Đạt
W04 – Review a report and publish a lesson (administrator)	Handle reports and appeals and publish lessons; also manage users, matches and analytics	Admin web: Dashboard, Users, Moderation, CMS, Matches, Analytics	Huỳnh Trung Kiên

Controls. Table 5 summarises the default controls. Keys can be changed in Settings → Controls; the size, opacity, left-handed mode and position of the touch controls are set in the same tab. During a match the shot cue under the player shows the keys of the shot being prepared, and the camera keeps the player above the cue (PC) or the touch controls (phone).

Table 5. Default controls

Action	PC (keyboard)	Android (touch)
Move / run	W, A, S, D / hold Left Shift	Drag the joystick on the left / hold the CHẠY button on the right
Drive (and serve)	J – hold and release	Hold and release DRIVE
Dink	K – hold and release	Hold and release DINK
Volley	L – hold and release	Hold and release VOLLEY
Drop / Smash / Reset	J+K / J+L / K+L	While holding, drag up: DINK → DROP, DRIVE → SMASH, VOLLEY → RESET
Add power / aim cross-court	Hold U / hold I while releasing the shot	While holding, drag right (LỰC) / drag left (CHÉO); for the serve drag DRIVE left or right
Cancel the prepared shot	Backspace	Drag down (HỦY)
Settings / help / confirm	Esc / F1 / Enter	CÀI ĐẶT button / help button / tap

The shot is decided when the first shot key is released, so press the key early and release it just before the ball reaches the paddle. A release that is too early or too late produces the feedback "Bạn thả nút hơi sớm/muộn".

3.2 Workflow W01 – Learn and Sync a Drill

Purpose. A new player creates an account, learns the court, rules and controls with the tutorial, then practises the techniques with graded drills. Training also works offline after the first sign-in, and offline results are synchronised once when the game is online again. Figure 3 shows the activity flow.

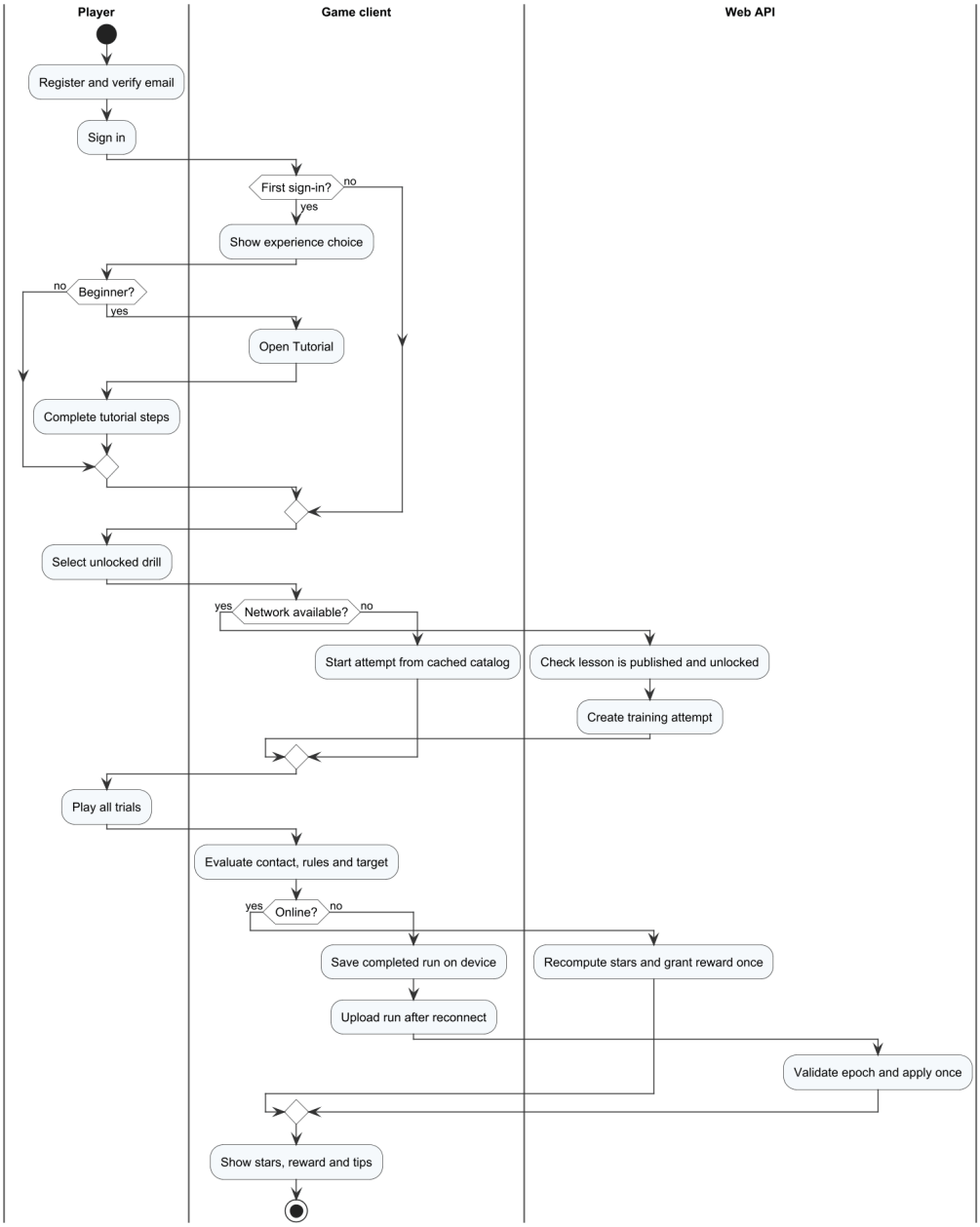


Figure 3. Workflow W01 – Learn and sync a drill

Step 1 – Register and verify the e-mail

1. On the Login screen select **Đăng ký** (Register).
2. Enter the e-mail address, a password of at least 6 characters and the same password again, then select **TẠO TÀI KHOẢN**. Invalid input is shown in red under the form (Figure 5).
3. Open the mailbox (also check the spam folder) and copy the six-digit code. Enter it on the verification page and select **XÁC THỰC**. If the code does not arrive, select **Gửi lại mã xác thực** after 60 seconds.

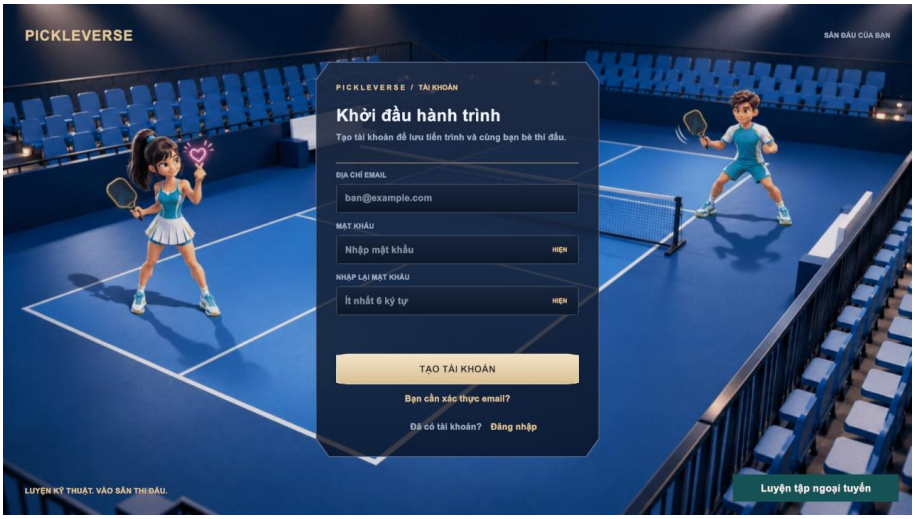


Figure 4. Register screen

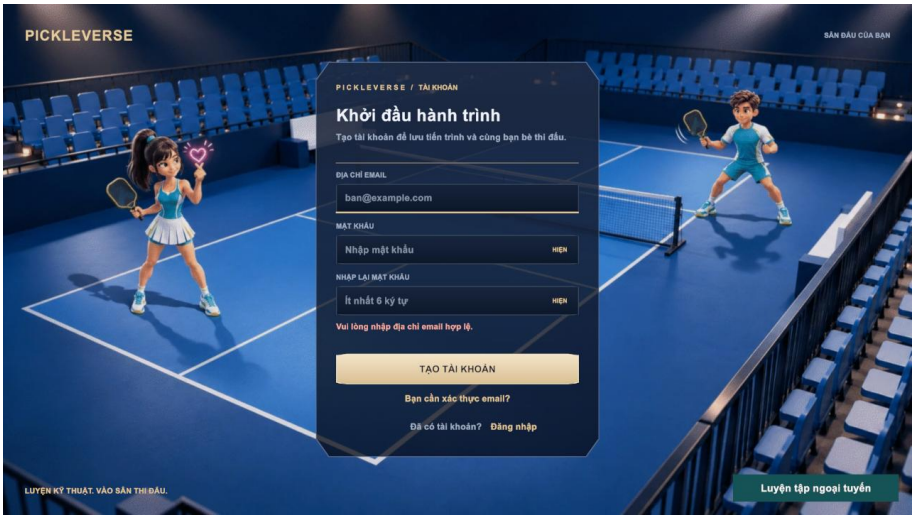


Figure 5. Validation message on the Register screen

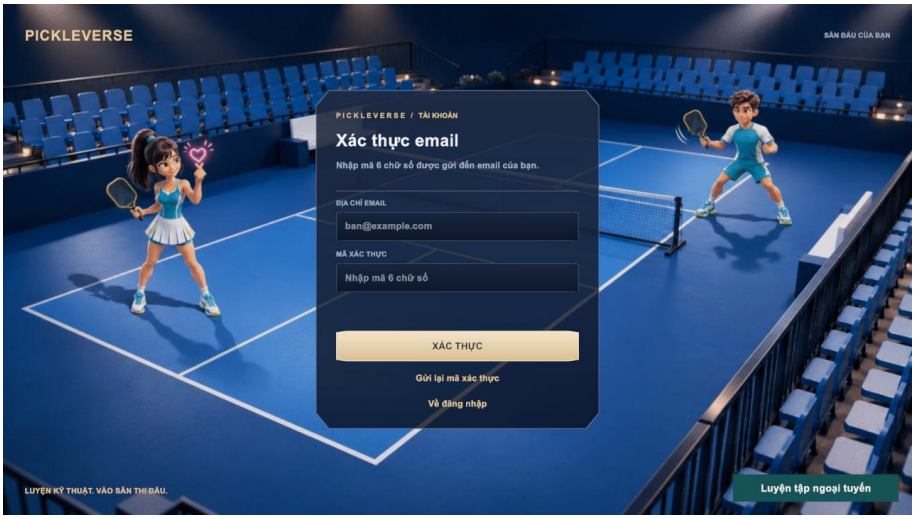


Figure 6. E-mail verification with the six-digit code

Step 2 – Sign in and choose the experience level

1. Enter the e-mail and password and select **ĐĂNG NHẬP**. Forgotten passwords are reset with **Quên mật khẩu** (Figure 7): enter the e-mail, then the code received and a new password.
2. At the first sign-in the game asks "Bạn đã quen với pickleball chưa?". Choose **Người mới** to start the tutorial or **Đã thành thạo** to go to the Menu. The choice cannot be changed later.

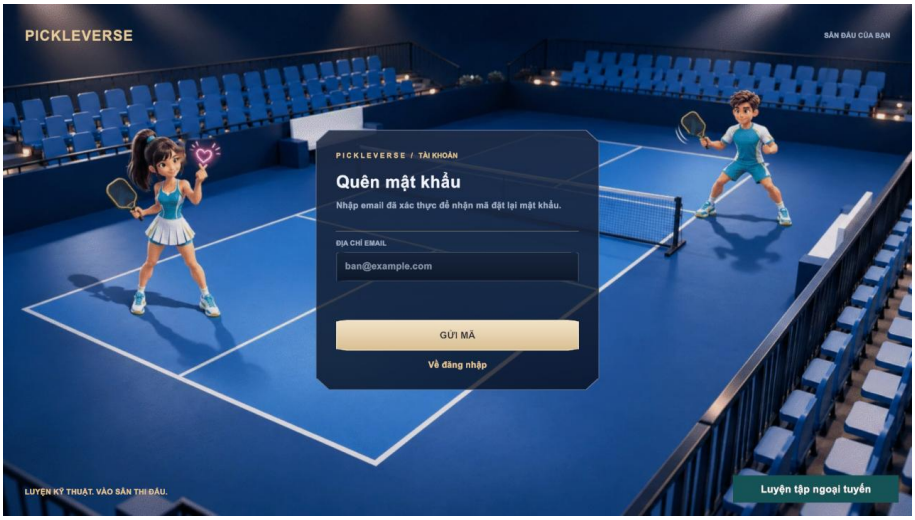


Figure 7. Forgot password screen

Step 3 – Complete the tutorial

The tutorial has 17 short steps about the court, faults, the two-bounce rule, the non-volley zone and the shot keys. Read each step, select **TIẾP TỤC** (Enter) to continue, **XEM LẠI** to replay the demonstration or **ĐỂ SAU** to leave and finish later from Mode Selection.

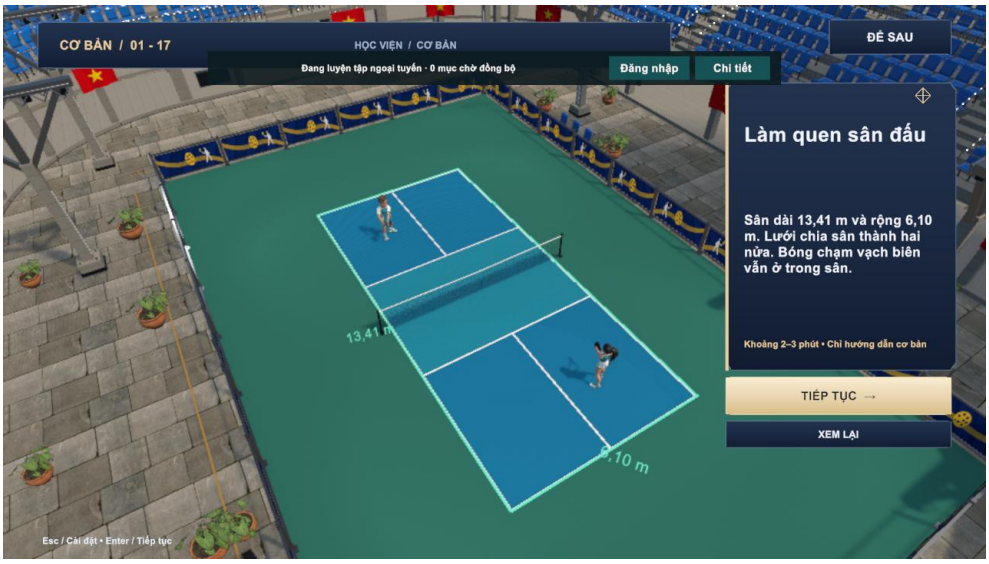


Figure 8. Tutorial – step 1 of 17

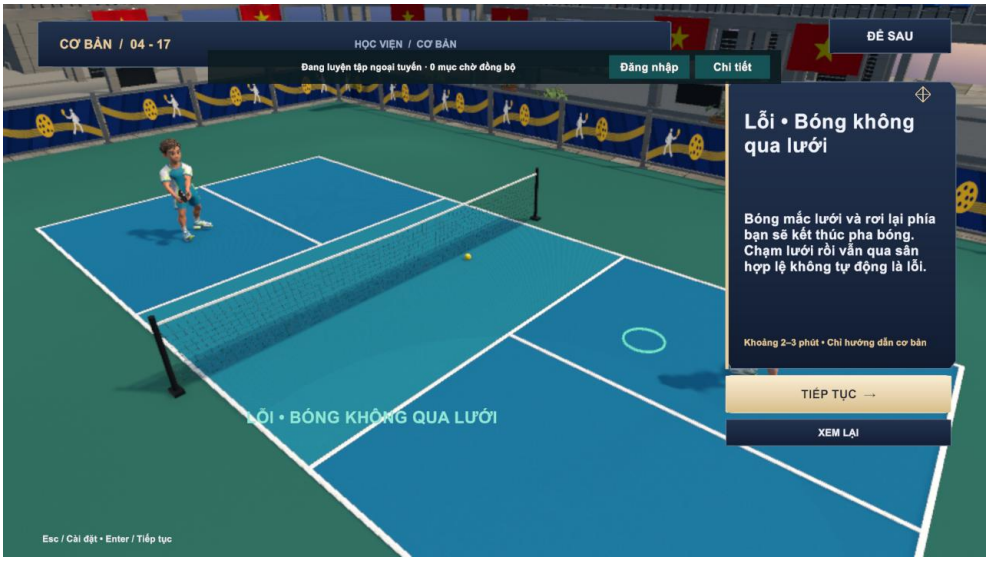


Figure 9. Tutorial – fault when the ball does not cross the net

Step 4 – Practise a drill

1. From the Menu select **CHƠI NGAY → Luyện tập**. The academy lists 15 lessons in the groups Basic, Control, Advanced and Doubles; locked lessons open after their prerequisite is passed.
2. Select **LUYỆN TẬP** on a lesson to read the briefing: target (for example 6 of 8 serves), first-pass reward and the star rules. Select **BẮT ĐẦU**.
3. Hit the shots shown by the coach panel. After the last shot the result shows the number of successful shots, the accuracy, the mean timing error, the practice time and the most frequent error with a tip. Stars: 1 star when the target is reached, 2 stars at 80%, 3 stars at 90% with timing ≤ 0.16 s.
4. Select **LUYỆN LẠI** to retry, **BÀI TIẾP THEO** for the next lesson, or **DANH SÁCH BÀI** for the catalogue. A failed attempt costs nothing.

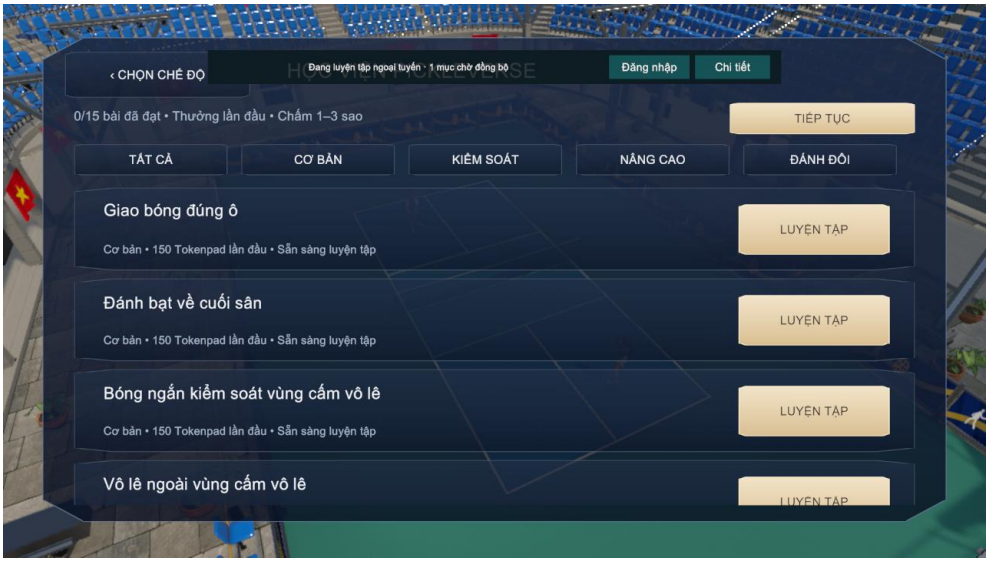


Figure 10. Lesson catalogue

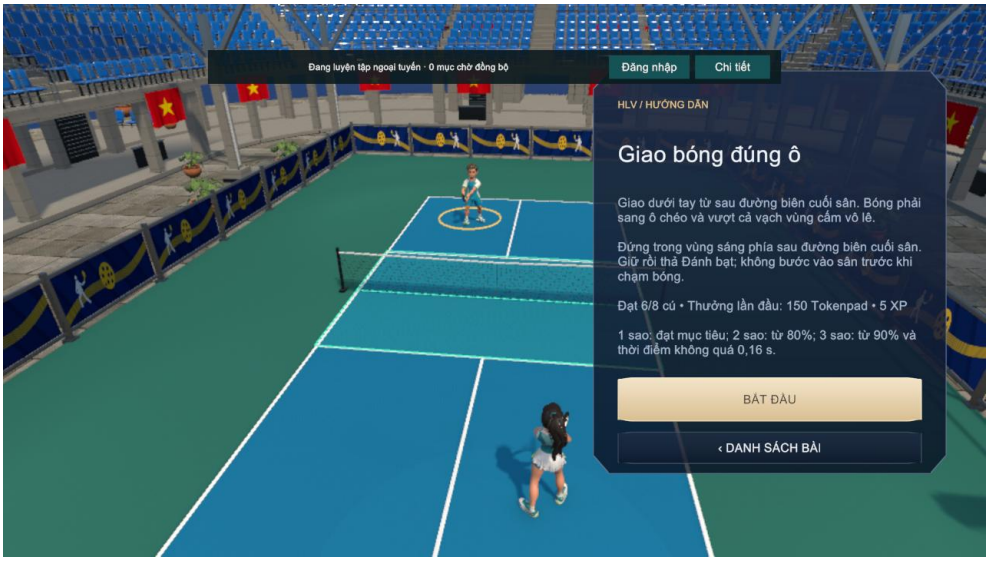


Figure 11. Lesson briefing with target and star rules

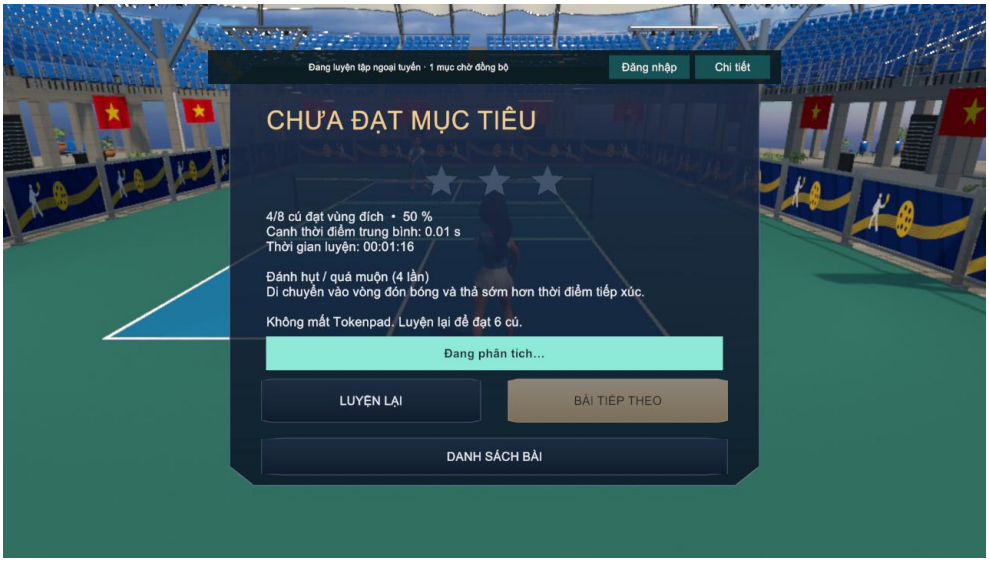


Figure 12. Drill result

Step 5 – Train offline and synchronise

1. Without a connection, open the game and select **Luyện tập ngoại tuyến** on the Login screen. This button appears only on a device where the account has signed in before.
2. Practise as in Step 4. The banner "Đang luyện tập ngoại tuyến · n mục chờ đồng bộ" shows how many results wait for synchronisation; bot and online matches and the shop are disabled.
3. When the Internet is available again, select **Đăng nhập** on the banner. Pending results are sent and applied once; **Chi tiết** shows rejected items, if any.

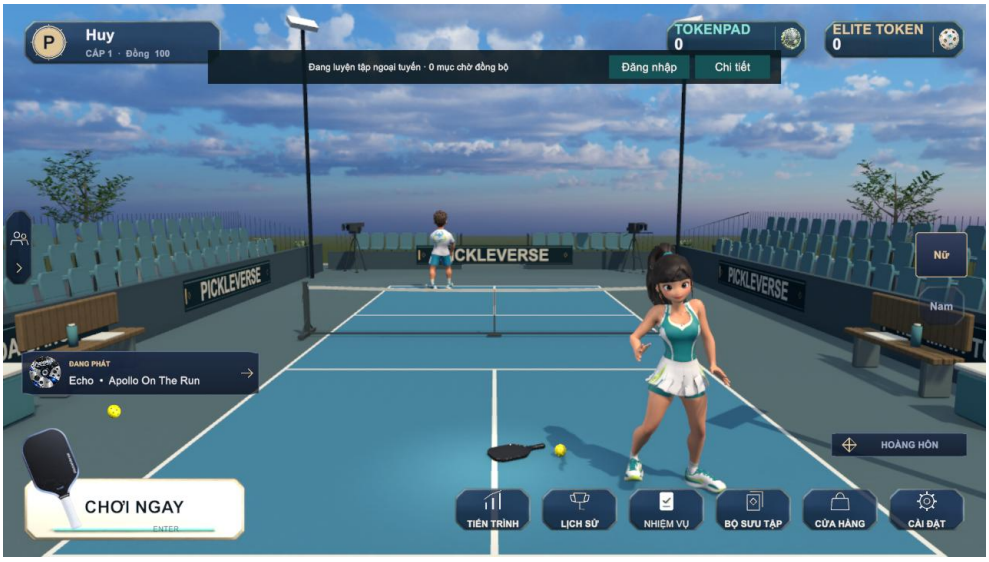


Figure 13. Menu in offline training mode

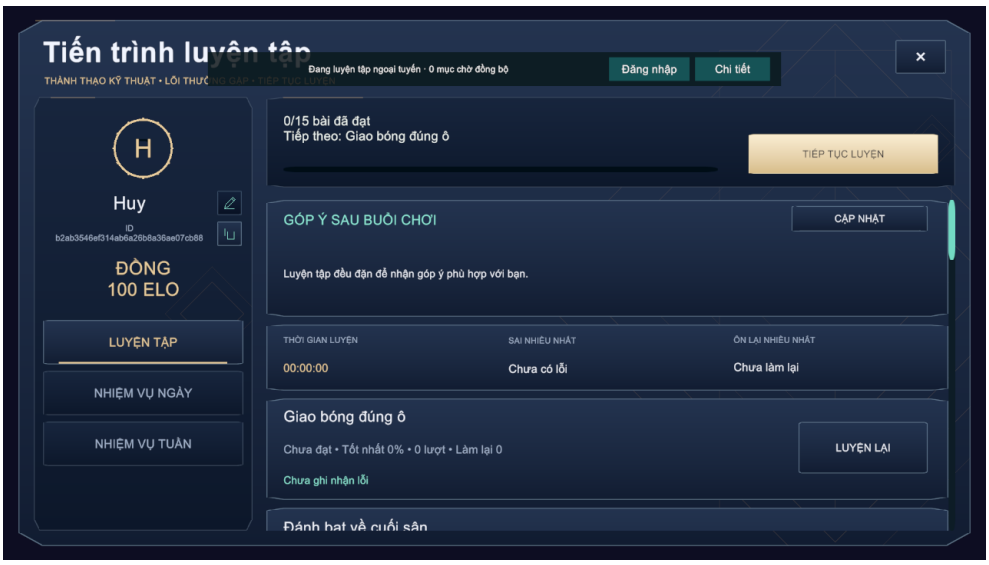


Figure 14. Training progress after practising

3.3 Workflow W02 – Play a Bot and Get a Tip

Purpose. The player applies the techniques in a singles match against a computer opponent and receives AI coaching based on the shots of the match. Figure 15 shows the flow.

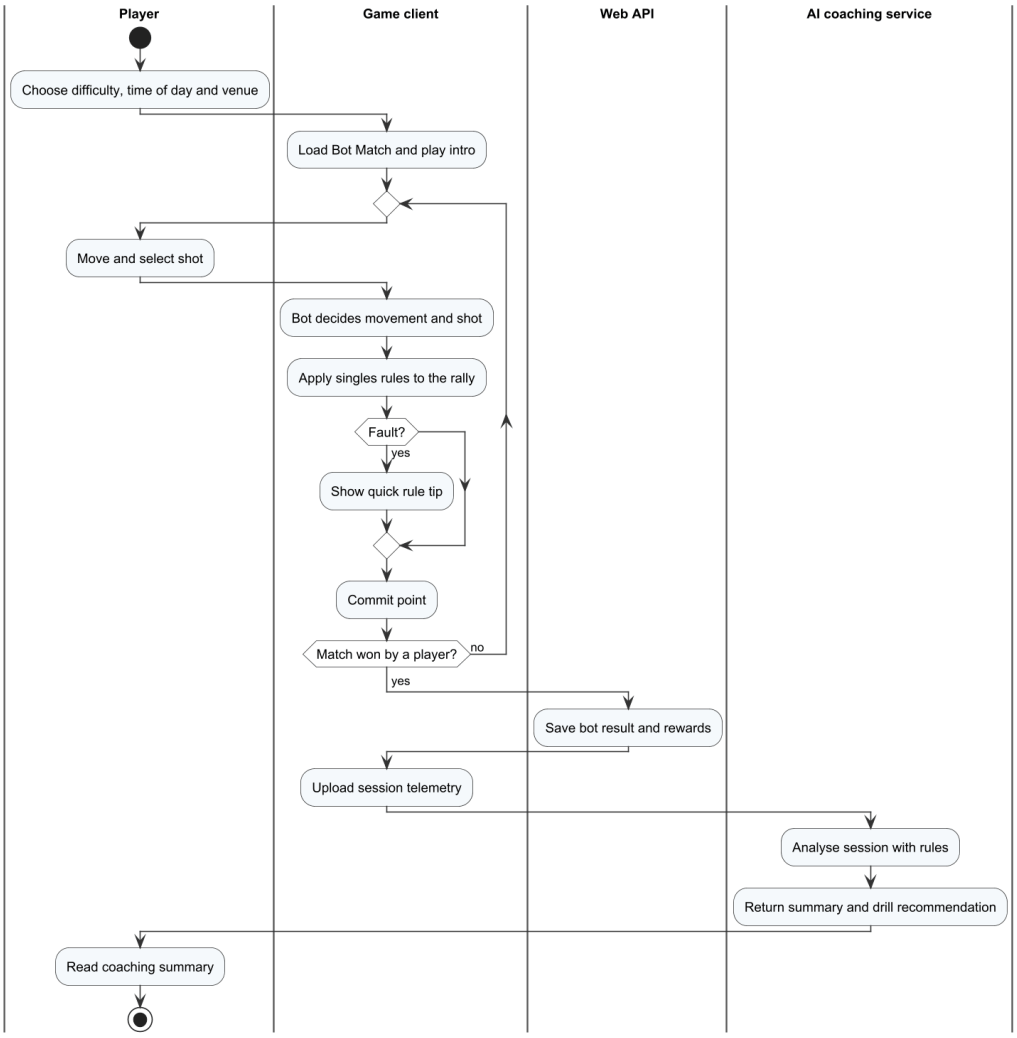


Figure 15. Workflow W02 – Play a bot and get a tip

Step 1 – Set up the match

1. In the Menu select **CHƠI NGAY → Đấu với bot**.
2. Choose the difficulty (Dễ, Vừa, Khó, or **Tự động**, the adaptive option that follows your Elo), the time of day (Ban ngày / Ban đêm) and the court (Chuyên nghiệp / Ngoài trời), then select **VÀO SÂN**.

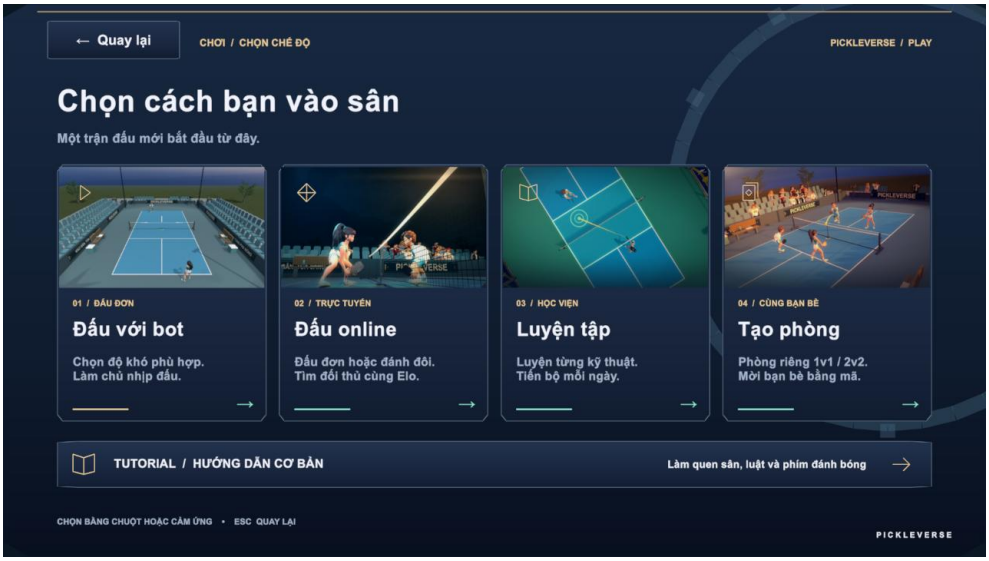


Figure 16. Mode selection

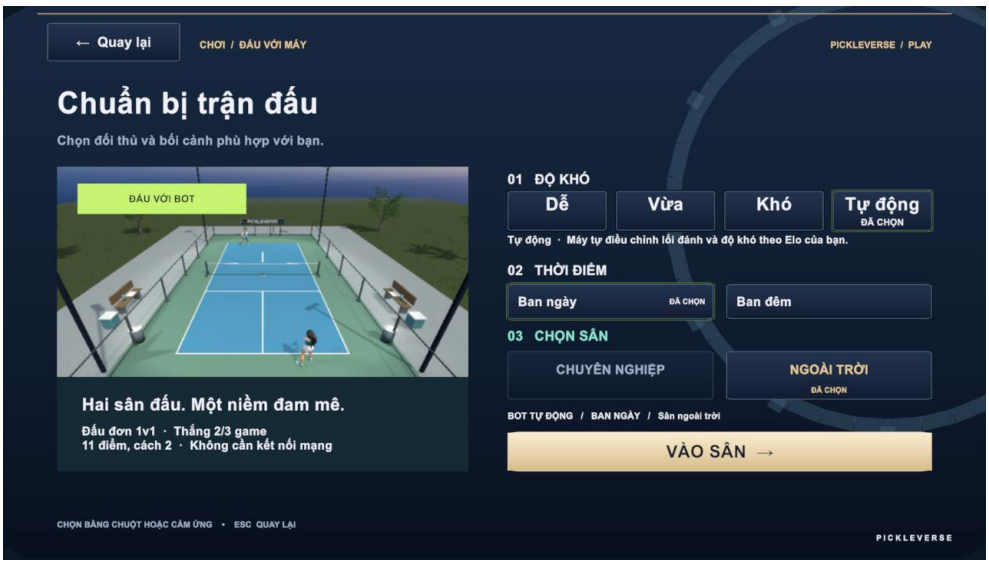


Figure 17. Bot match setup

Step 2 – Play the match

1. After the introduction, serve within 10 seconds: hold J (DRIVE on a phone) behind the baseline and release to serve diagonally over the non-volley zone. The hint on the left and the shot cue under the player show what to press.
2. Let the return and the third shot bounce (two-bounce rule); afterwards volley with L only outside the non-volley zone. Faults appear as a banner, for example "Lỗi • Bóng ra ngoài".
3. A game is won at 11 points with a 2-point lead and the match at two games; in the third game the players change ends at 6 points. There are no time-outs; Esc opens the settings and F1 the help.

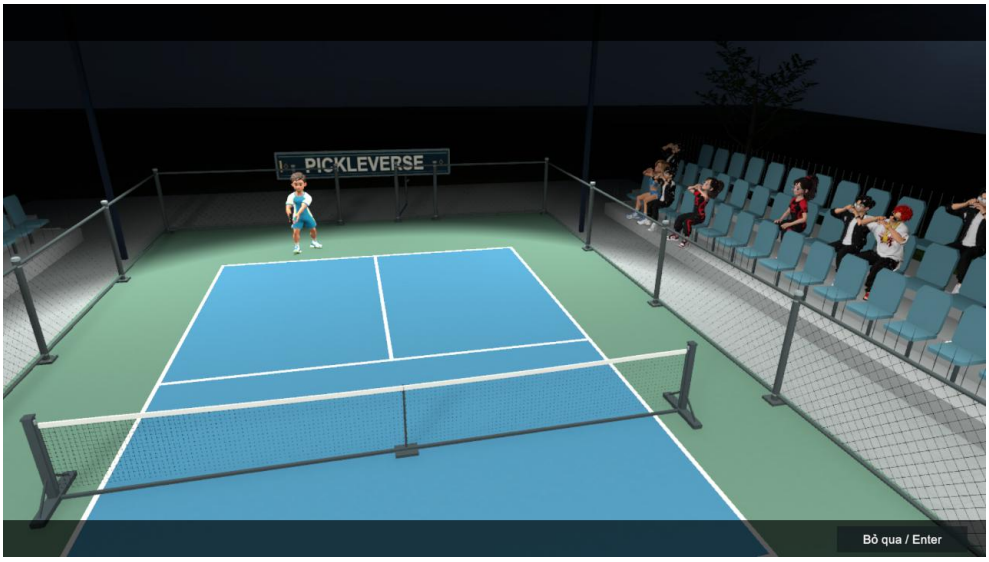


Figure 18. Bot match – introduction



Figure 19. Bot match – serving; the shot cue under the player shows J + I (cross-court serve)

Step 3 – Read the result and the coaching

1. The result shows the winner and the game score. A win gives 100 Tokenpad and 10 XP, a loss 50 Tokenpad and 5 XP; practice bot matches do not change Elo.
2. Select **Xem phân tích** to open the coaching popup: number of shots, contacts and serves, the main errors and up to three recommended drills. Select **Luyện ngay** to open the recommended lesson.
3. Open **LỊCH SỬ** in the Menu to see the match in the history list.

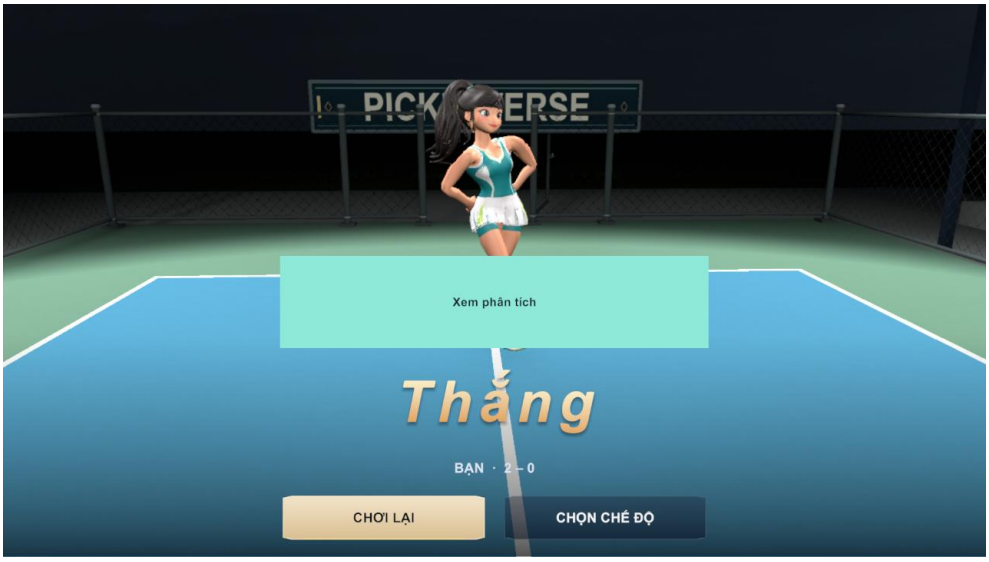


Figure 20. Match result

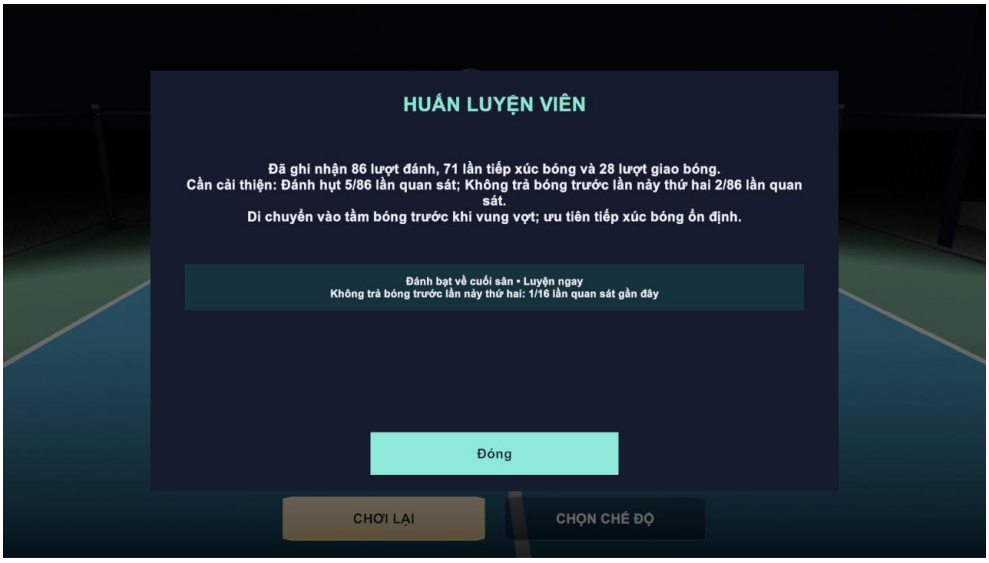


Figure 21. AI coaching summary with a recommended drill



Figure 22. Match history

3.4 Workflow W03 – Play Online and Save the Result

Purpose. The player plays against other people: a ranked singles match that changes Elo, an unrated quick doubles match, or a private singles or doubles room with friends. Online matches run on the dedicated server, which decides the score, and the result is saved once. Figure 23 shows the flow.

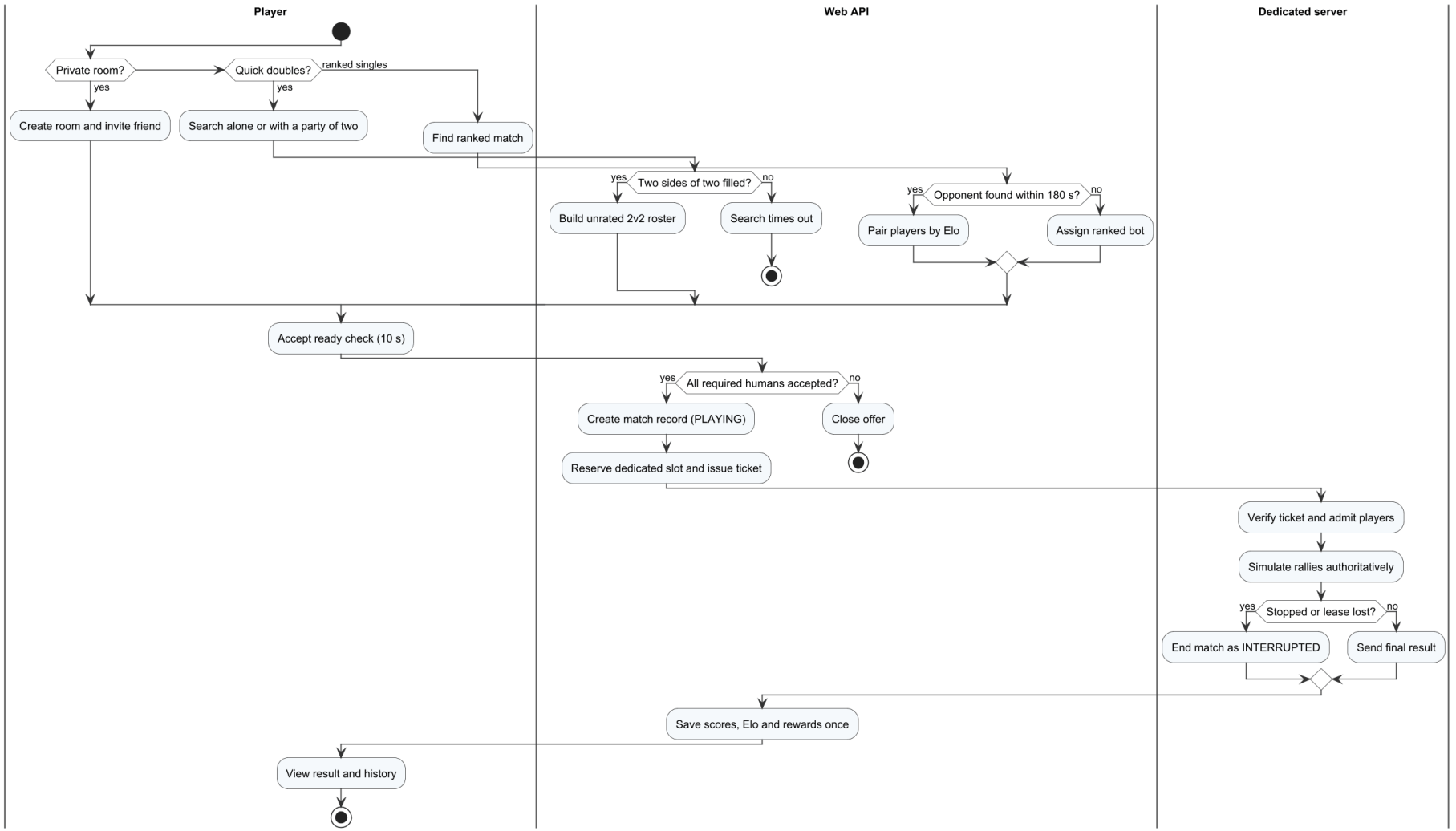


Figure 23. Workflow W03 – Play online and save the result

Step 1 – Find a ranked match

1. Select **CHƠI NGAY** → **Đấu online** → **ĐẤU ĐƠN 1v1**, then **Tìm trận**. The lobby shows "Đang tìm đối thủ trực tuyến…" and the waiting time; **Hủy** stops the search.
2. When an opponent with a similar Elo is found, accept within 10 seconds. If nobody is found within 180 seconds, a ranked bot is offered instead.
3. If the dedicated server is busy, the lobby shows "Máy chủ đang bận • Đang chờ lượt thi đấu…" and continues automatically when the server is free.

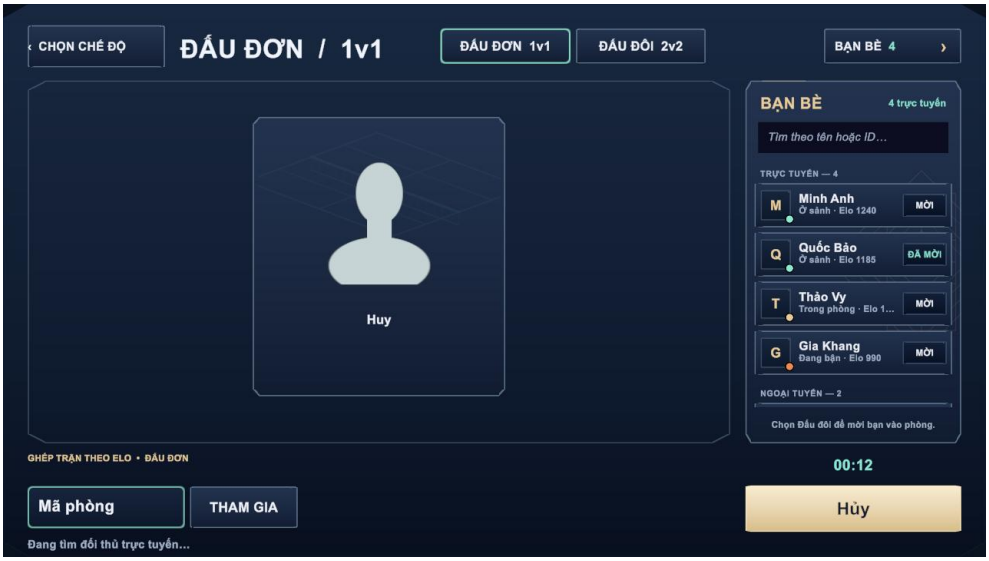


Figure 24. Searching for a singles opponent (layout preview rendered in the Unity Editor)

Step 2 – Play quick doubles or with friends in a private room

1. Quick doubles: select **ĐẤU ĐÔI 2v2** in the online lobby and **Tìm trận**, alone or after a friend has joined your party. The game fills two teams of two with other searching players; there is no bot, and the match does not change Elo. If no teams are found, the search ends with "Hết thời gian tìm trận. Hãy thử lại."
2. Private room: select **Tạo phòng** in Mode Selection and choose 1v1 or 2v2. Share the room code with **SAO CHÉP**, or open the friends column on the right of the lobby and invite a friend who is in the lobby.
3. Friends type the code in **Mã phòng** and select **THAM GIA BẰNG MÃ**, or accept the invitation. The host starts the match. Private matches do not change Elo.

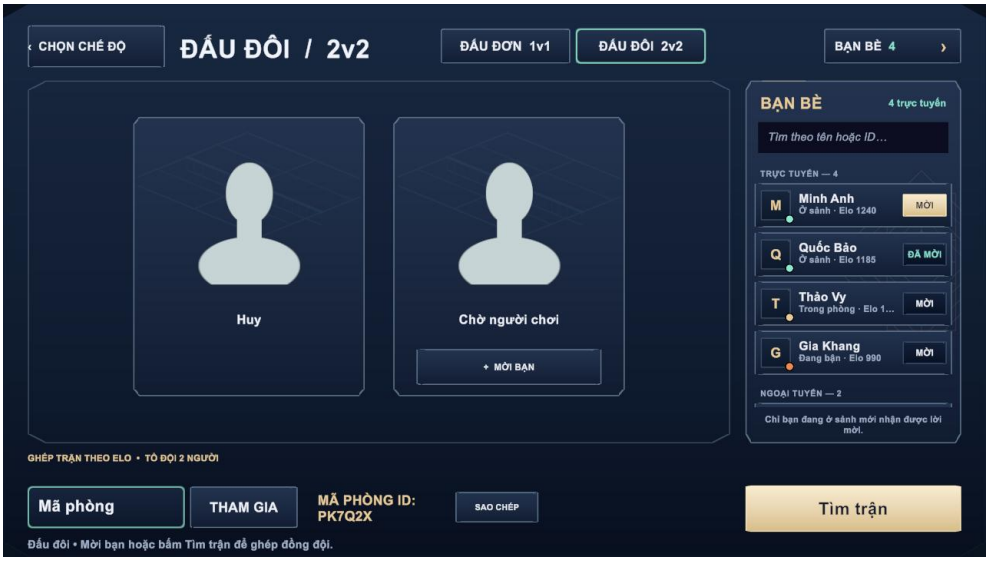


Figure 25. Online lobby – doubles room with the room code and the friends column (layout preview rendered in the Unity Editor with sample friend data)



Figure 26. Friends drawer used to find and invite friends

Step 3 – Play, reconnect and see the result

1. Play as in a bot match. If the connection drops, the game reconnects automatically for up to 30 seconds and you continue with the server score.
2. If you leave or cannot reconnect in time, the match ends as interrupted and gives no Elo or reward.
3. At the end the result shows the Elo change for ranked matches. The match appears in **LỊCH SỬ**; from there you can watch the replay or report a player (**Tố cáo**).



Figure 27. Multiplayer court

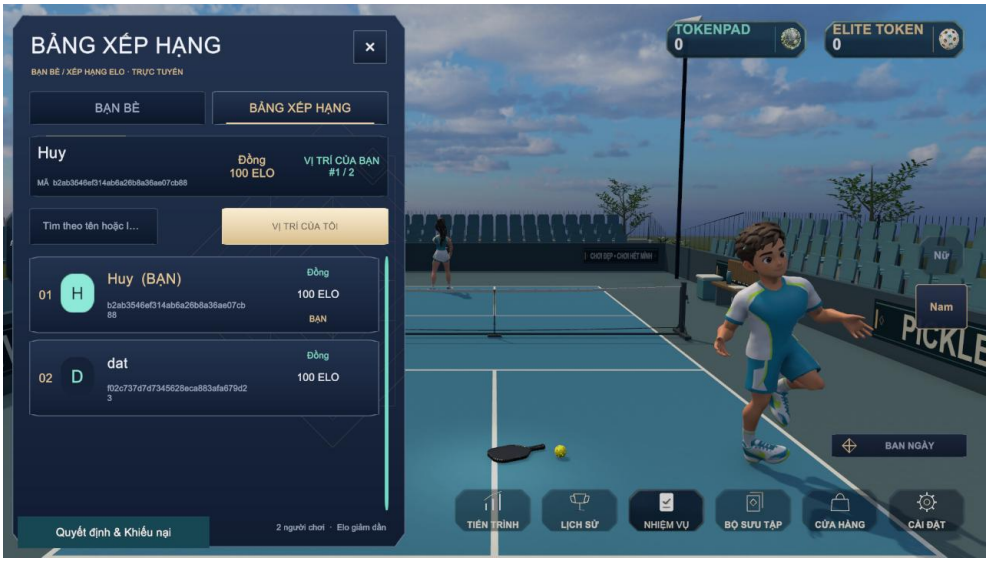


Figure 28. Elo leaderboard

3.5 Workflow W04 – Review a Report and Publish a Lesson (Administrator)

Purpose. An administrator keeps the community and the training content healthy: handles player reports and appeals, manages accounts and roles, publishes lessons, monitors matches and reads the analytics. Figures 29 and 30 show the moderation and content flows. Every administrative change is recorded in the audit log.

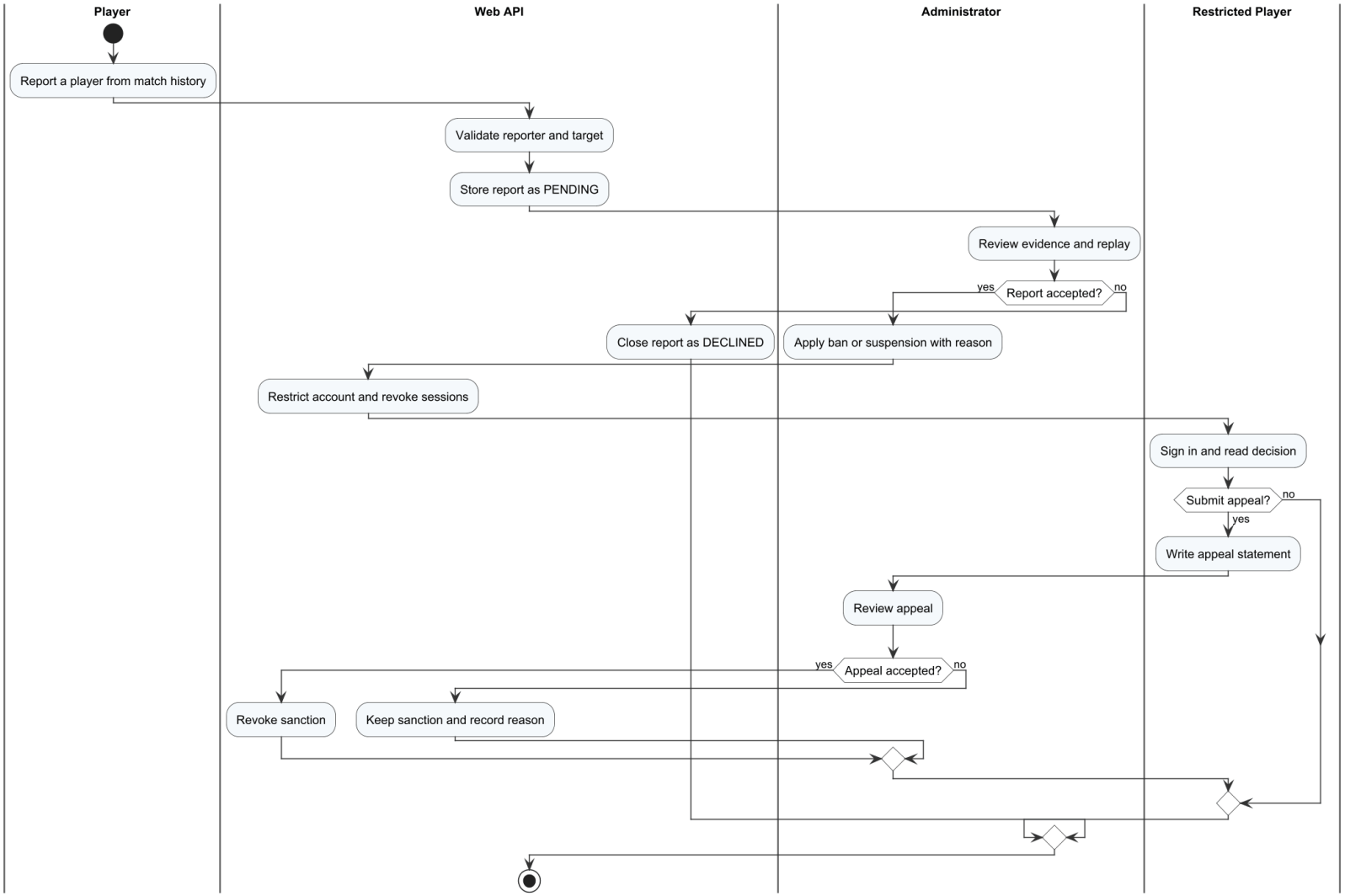


Figure 29. Workflow W04, part 1 – Report, sanction and appeal

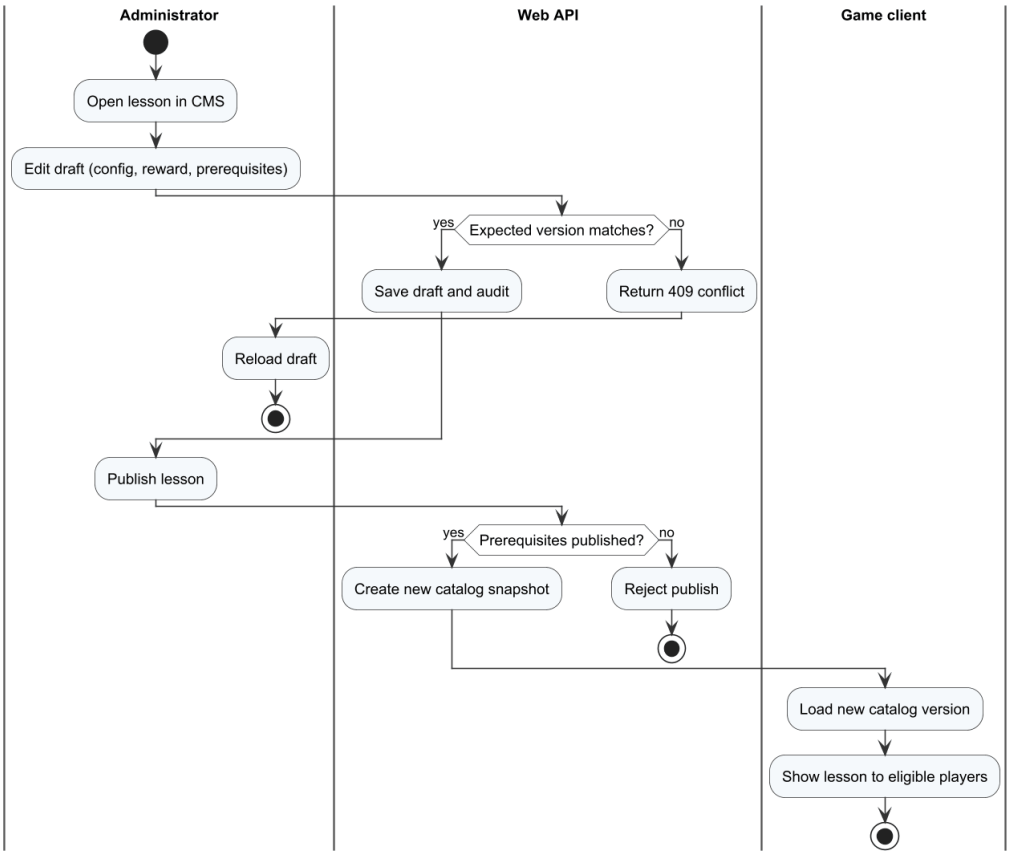


Figure 30. Workflow W04, part 2 – Publish a lesson

Step 1 – Sign in to the admin web

Open the admin web address (https://tad-min.io.vn on staging since 13/11/2026), enter the e-mail and password of an Admin account and select **Xác thực & Truy cập**. Accounts without the Admin role are rejected. The dashboard shows the number of users, today's active players, matches, online time and pending reports.

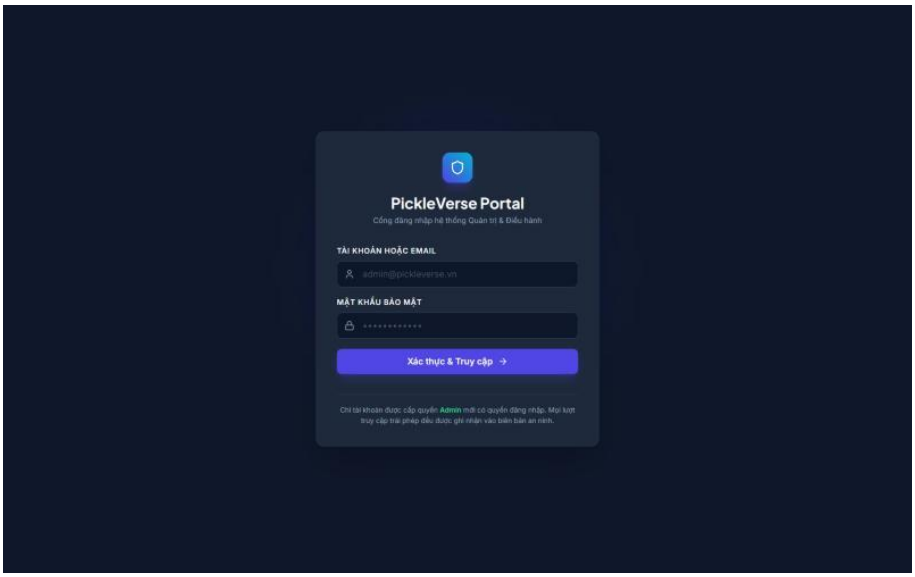


Figure 31. Admin web login

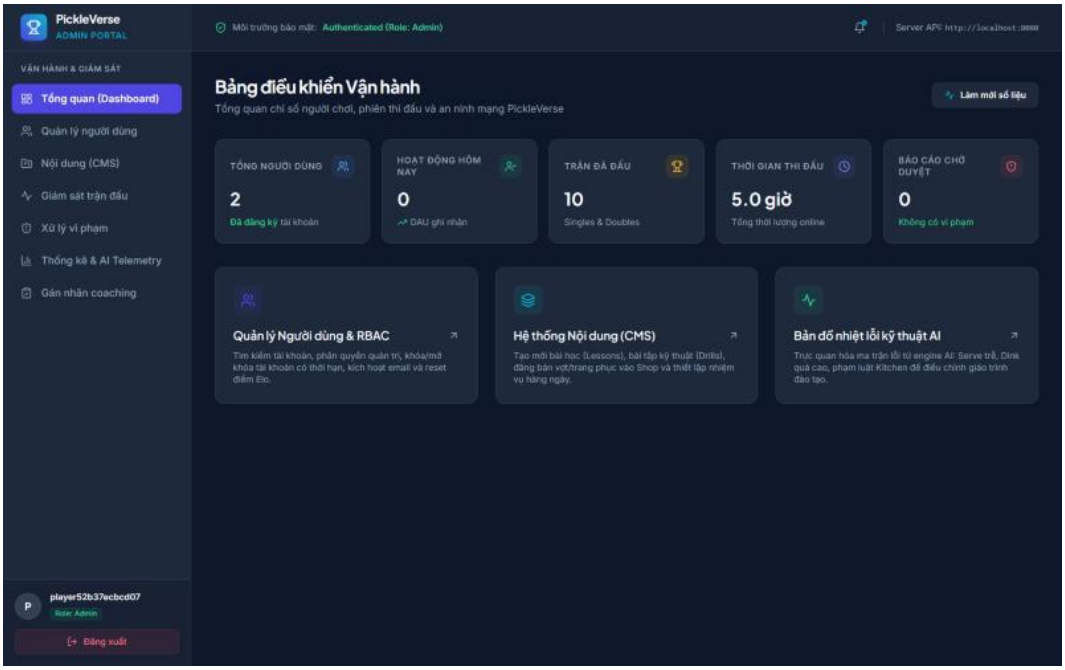


Figure 32. Admin dashboard

Step 2 – Manage user accounts

1. Open **Quản lý người dùng**. Search by name, e-mail or ID and filter by status (ACTIVE, SUSPENDED, BANNED).
2. Use the action icons of a row to assign a role, activate an e-mail manually, apply or revoke a sanction (a suspension needs a future expiry date and a reason of 3–2,000 characters) or reset player progress.
3. Progress reset: select the groups LEARNING, RANKING, MISSIONS or ECONOMY, check the preview of the impact and confirm. Reset is refused while the player is searching or playing.

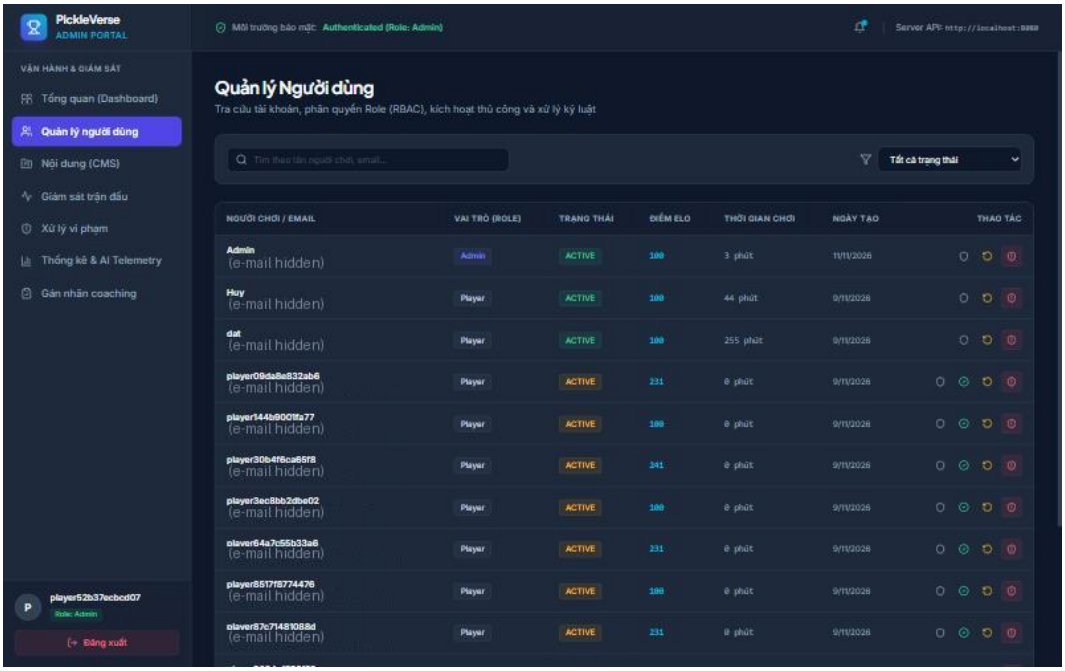


Figure 33. Users page (e-mail addresses hidden in this figure)

Step 3 – Review reports and appeals

1. Open **Xử lý vi phạm**. Pending reports are listed with the reporter, the reported player, the match and the reason.
2. Open a report, download the match evidence or replay, choose **Chấp nhận tố cáo** (optionally with a suspension or ban) or **Bác tố cáo**, write the reason and select **Lưu quyết định**.
3. In the appeals section, review the statement of a restricted player. Accepting an appeal revokes only the sanction it concerns.

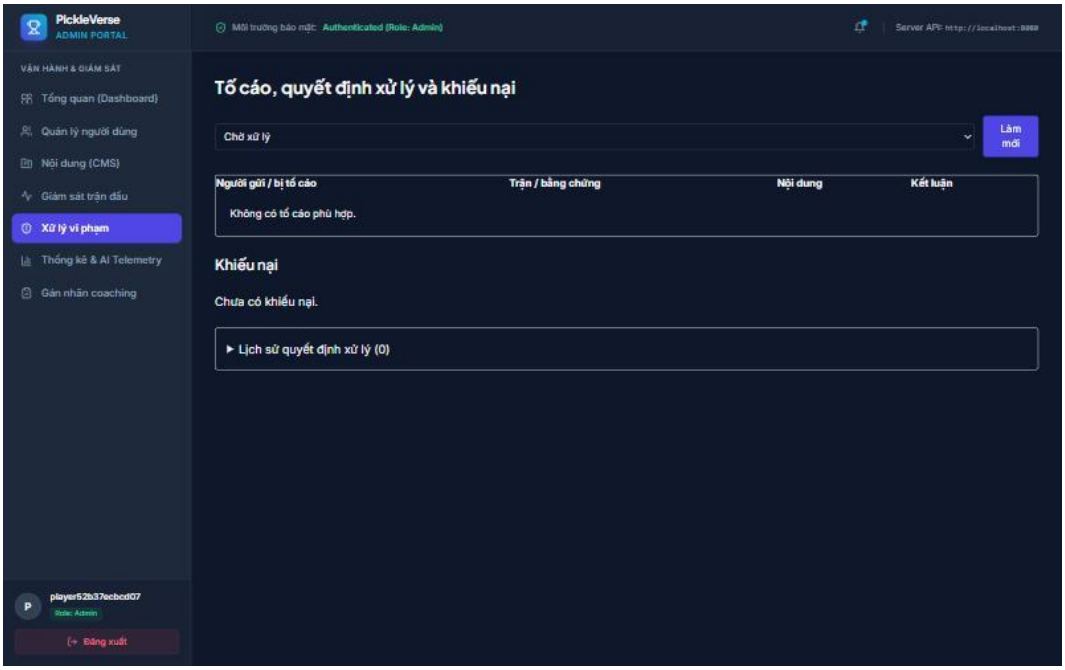


Figure 34. Reports, decisions and appeals

Step 4 – Manage lessons and content

1. Open **Nội dung (CMS)**. The left list shows the 15 lessons with their status and version.
2. Select a lesson to edit the code, name, technique template, group, number of shots, target, rewards, minimum level and target radius, then select **Lưu bản nháp**.
3. Select **Xuất bản bản đã lưu** to publish the draft for new attempts, or **Gỡ xuất bản** to hide the lesson. Attempts already in progress keep their version.
4. Use the other tabs to add a drill, publish a cosmetic item in the shop, or create a daily or weekly mission.

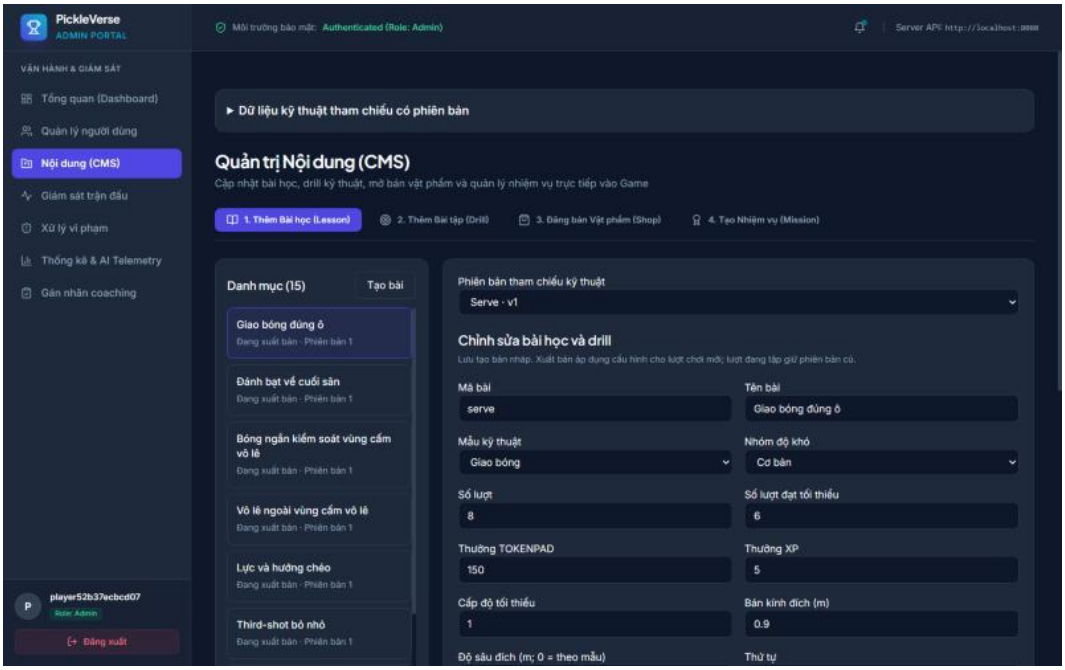


Figure 35. Content management – lesson editor

Step 5 – Monitor matches

Open **Giám sát trận đấu**. **Trận đang hoạt động** lists live matches refreshed every 5 seconds with the score and the connection reported by the dedicated server. To stop a live match select **Dừng trận có lý do**, enter the reason and confirm; the match ends as interrupted without Elo or reward. **Lịch sử** shows finished matches with kind, games won and end time.

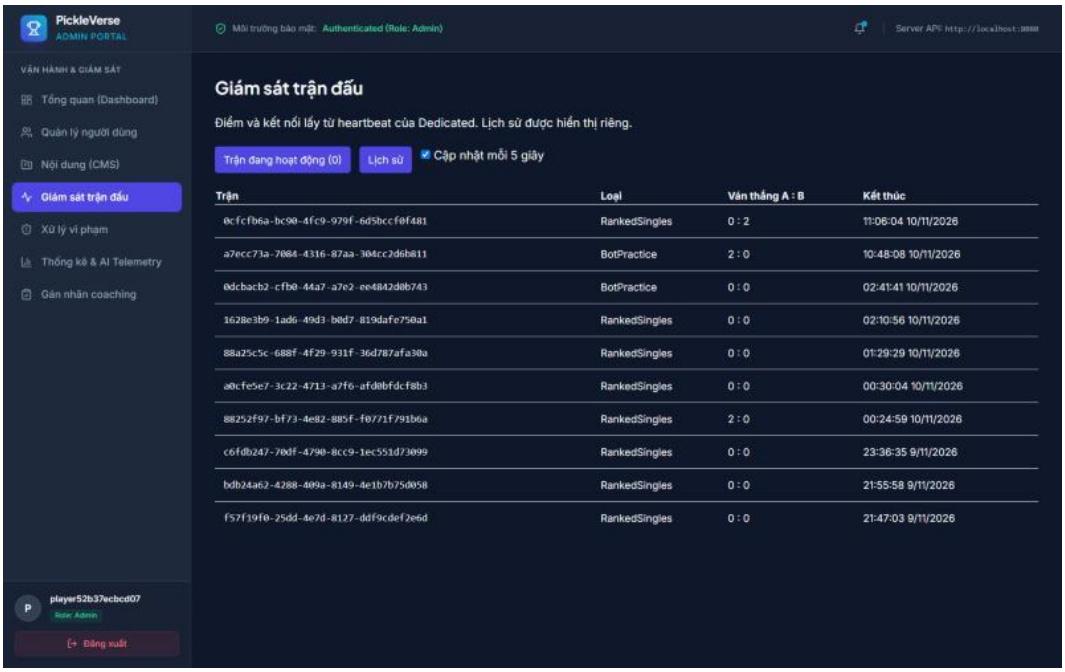


Figure 36. Match monitoring – history

Step 6 – Read analytics and label coaching sessions

1. Open **Thống kê & AI Telemetry**, choose the date range (Vietnam time, at most 366 days) and select **Làm mới**. The page shows registrations, daily players, completed matches, active hours,

retention, popular lessons with pass rate, the technique-error heat map and matchmaking health.

2. Open **Gán nhãn coaching** to act as an expert reviewer. Select a synchronised session, inspect the shot-by-shot evidence, set the observed level, the error labels and the data group, then save. **Xuất dataset** exports the labelled data used to evaluate the AI coaching.

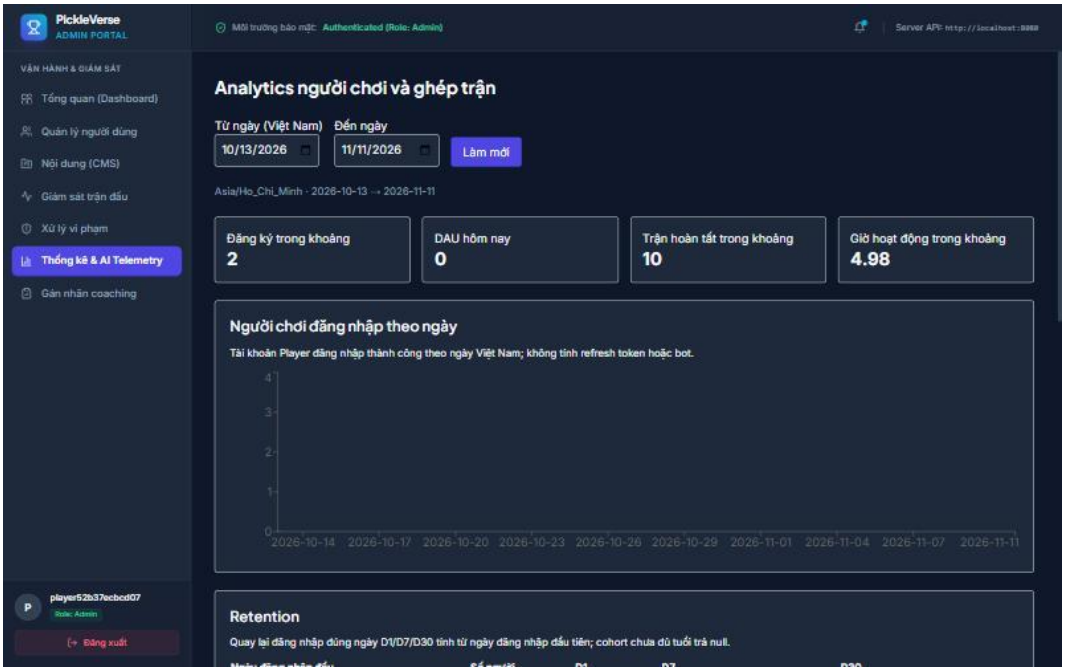


Figure 37. Player and matchmaking analytics

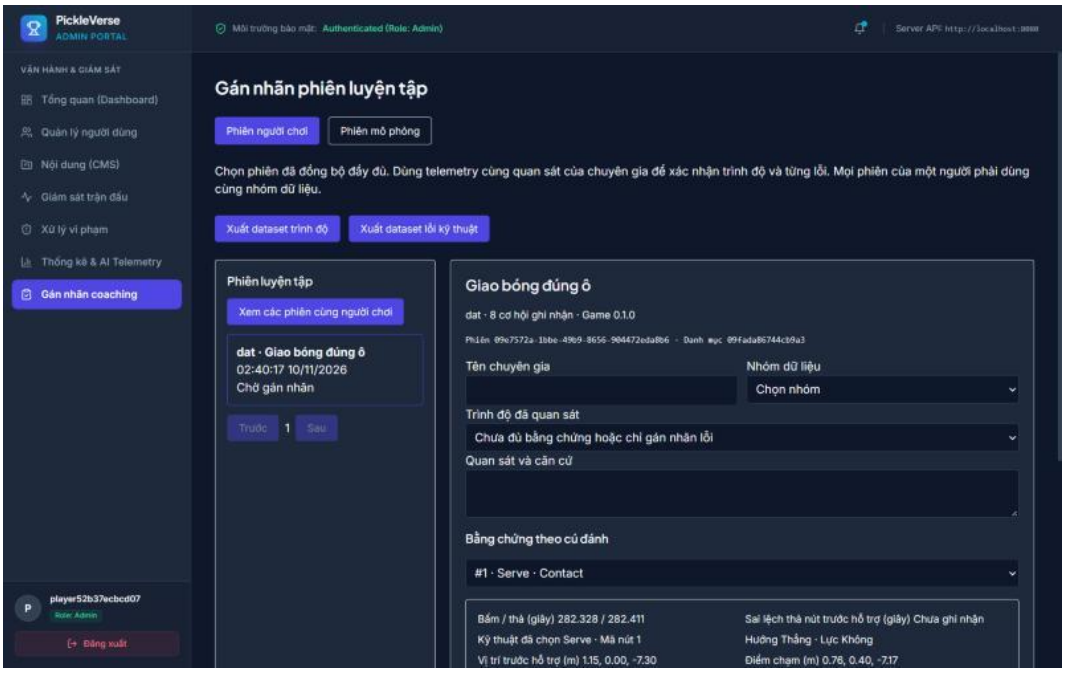


Figure 38. Expert labelling of a training session

3.6 Other Player Functions

Table 6. Other player functions

Function	How to use
Missions and streak	Menu → NHIỆM VỤ: 7 daily and 10 weekly missions; select NHẬN THƯỞNG when a mission is complete and NHẬN THƯỞNG TỔNG for the set bonus. Missions renew at 07:00 Vietnam time; play at least one match per day to keep the streak. A 3-day streak completes a weekly mission and a 5-day streak unlocks an achievement.
Progress	Menu → TIẾN TRÌNH: passed lessons, practice time, the most frequent error and, for each lesson, the stars, the best accuracy and the four most recent results (Gần đây).
Shop	Menu → CỬA HÀNG: preview a paddle and buy it with Tokenpad or Elite Token (if a price exists in that currency).
Collection	Menu → BỘ SƯU TẬP: owned paddles; select TRANG BỊ to use a paddle in matches.
Friends and leaderboard	Friend drawer on the left of the Menu: search by name or ID, send or answer requests, remove or block players; the Bảng xếp hạng tab ranks players by Elo.
Replay	LỊCH SỬ → replay button of a match: pause, seek and change speed from 0.5× to 2×. Only players of the match can watch it.
Report a player	LỊCH SỬ → Tố cáo on a finished match: choose the player and enter the reason (up to 254 characters).
Decisions and appeals	Menu → "!" KHIẾU NẠI tile next to TIẾN TRÌNH: shows sanctions on the account; a restricted player can send one appeal (at least 10 characters) per sanction.
Settings	Menu → CÀI ĐẶT: Tổng quan (sign out), Âm thanh, Đồ họa, Điều khiển (key binding and touch layout). Select ÁP DỤNG VÀ LƯU to keep the changes.



Figure 39. Missions and daily streak

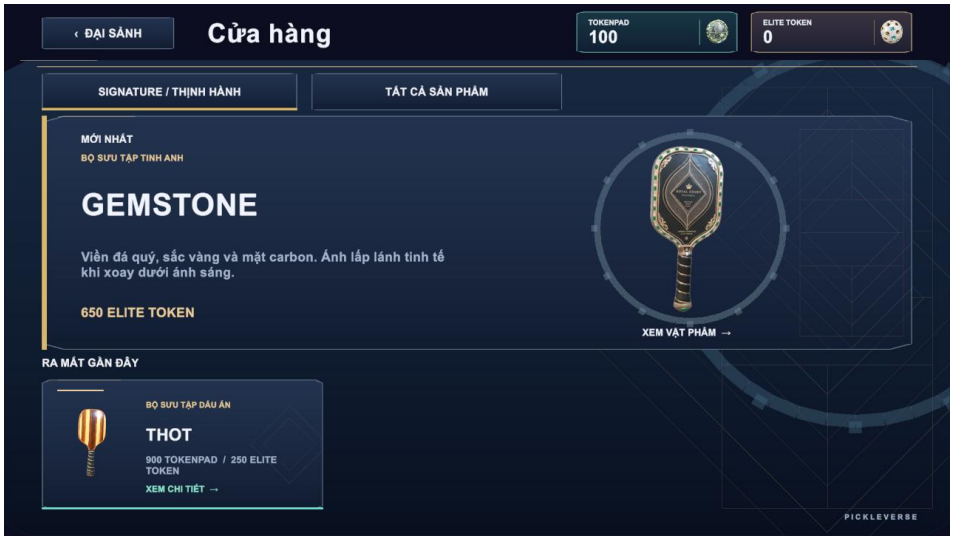


Figure 40. Shop

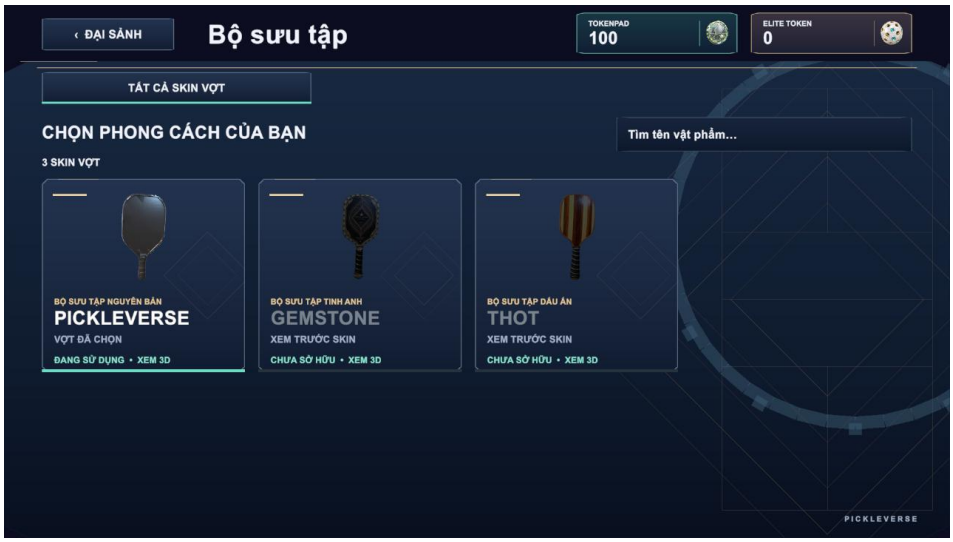


Figure 41. Collection



Figure 42. Settings

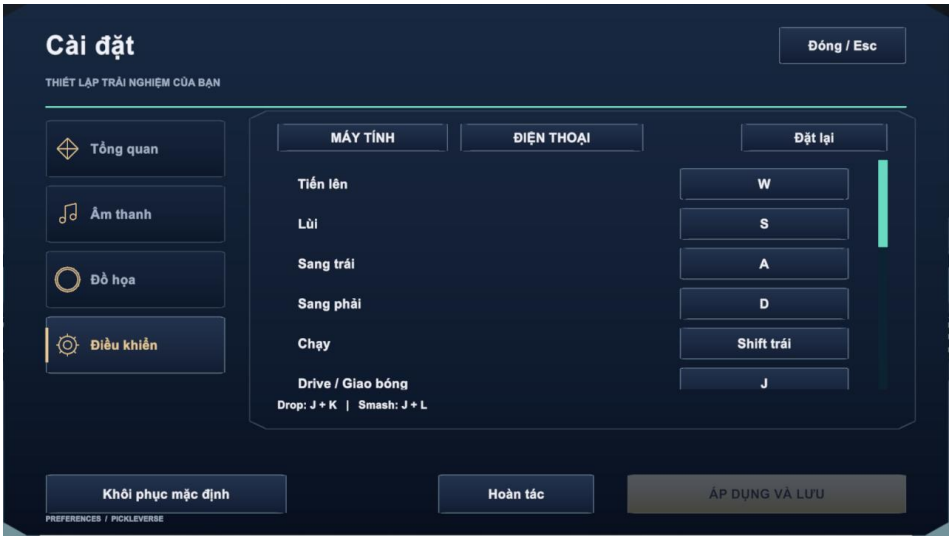


Figure 43. Settings – key binding

3.7 Troubleshooting

Table 7. Troubleshooting

Message or symptom	Cause	What to do
No verification e-mail	The e-mail is delayed or in the spam folder; SMTP is unavailable	Check the spam folder; select Gửi lại mã xác thực after 60 seconds; if it still fails, ask an administrator to activate the account manually
"Email chưa được xác thực…"	The account was registered but not verified	Open Bạn cần xác thực email? and enter the code
"Email hoặc mật khẩu không chính xác."	Wrong e-mail or password	Check the input or use Quên mật khẩu
"Tài khoản đang bị hạn chế…"	The account is suspended or banned	Read the decision and send an appeal from Quyết định & Khiếu nại
"Không kết nối được Photon…"	No Internet or UDP blocked by a firewall	Check the connection and allow the game in the firewall
"Máy chủ đang bận • Đang chờ lượt thi đấu…"	The dedicated server is running another match	Wait in the lobby; the match starts when the server is free
Offline results not synchronised	The device has not signed in since training offline	Select Đăng nhập on the offline banner; check Chi tiết for rejected items
"Số dư tiền tệ không đủ…"	Not enough Tokenpad or Elite Token	Earn Tokenpad by playing matches, lessons and missions